पाइपिंग स्ट्रेस एनालिसिस के लिए मानक विनिर्देश

STANDARD SPECIFICATION FOR PIPING STRESS ANALYSIS

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Abbreviations:

ANSI	:	American National Standards Institute
API	:	American Petroleum Institute
ASME	:	American Society of Mechanical Engineers
ASTM	:	American Society for Testing and Materials
DLF	:	Dynamic Load Factor
EJMA	:	Expansion Joint Manufacturers Association
MR	:	Material Requisition
NEMA	:	National Electrical Manufacturers Association
MSS	:	Manufacturers Standardization Society
SIF	:	Stress Intensification Factor
WNC	:	Weight No Content
WRC	:	Welding Research Council

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CONTENTS

1.0 SCOPE ……..……....4

2.0 CODES AND STANDARDS……..……....…..……4

3.0 REFERENCE DOCUMENTS ……..……5

4.0 STRESS ANALYSIS CRITERIA……..……..……....5

5.0 STRESS ANALYSIS PARAMETERS……..…..……6

6.0 MAXIMUM PIPE SPAN AND DEFLECTION.....7

7.0 PIPE STRESS ANALYSIS AND SUPPORTING……..……....7

8.0 WIND ANALYSIS……..……..……..……....8

9.0 SEISMIC ANALYSIS……..……..……..……..……..9

10.0 COMPOSITE ANALYSIS……..……....……9

11.0 SPECIFIC REQUIREMENTS……..…..……9

12.0 SOFTWARE USED……..……...……11

13.0 CAESAR II MODELING ……..……...……..11

14.0 SPRING.....……..……..……..12

15.0 UNIT FILE.....……..……..……..13

16.0 LOAD CASES.....……..……..……..……13

17.0 DOCUMENTATION (for Stress Critical Lines only)……..……..……..……..14

18.0 ATTACHMENTS……..……..……..……..……..15

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1.0 SCOPE

This specification covers the minimum design requirements for carrying out complete piping flexibility/stress analysis for above ground piping systems in accordance with the code requirements of ASME B31.3 Process Piping.

2.0 CODES AND STANDARDS

The following latest editions of codes and standards, but not limited to shall apply in the design and analysis of the piping systems covered under this specification:

S.No.	Description	Standards / Codes	Edition
1.0	International Codes & Standards		
1.1	Process Piping	ASME B31.3	Latest
1.2	Pipe Flanges and Flanged Fittings	ASME B16.5	Latest
1.3	Large Diameter Steel Flanges	ASME B16.47	Latest
1.4	Stress Intensification Factors (i-factors), Flexibility Factors (k-Factors), and their determination for metallic piping components	ASME B31J	Latest
1.5	Expansion Joints Manufacturers' Association	EJMA Standards	Latest
1.6	Sizing, Selection & Installation of Pressure Relieving Devices in Refineries	API RP 520	Latest
1.7	Fired Heater	API Std. 560	Latest
1.8	Centrifugal Pumps for Petroleum, Heavy Duty Chemical and Gas Industry Service	API Std. 610	Latest
1.9	Positive Displacement Pumps - Reciprocating	APT Std. 674	Latest
1.10	Steam Turbines for Mechanical Drive Service	ANSI/NEMA SM 23	Latest
1.11	Centrifugal Compressors for Petroleum, Chemical and Gas Industry Service	API Std. 617	Latest
1.12	Reciprocating compressors for Petroleum, chemical and gas industry Services	API Std. 618	Latest
1.13	Recommended Practices for Machinery Installation and Installation Design	API RP 686	Latest
1.14	Design and Construction of Large, Welded, Low-pressure Storage Tanks	API Std. 620	Latest
1.15	Welded Tanks for Oil Storage	API Std. 650	Latest
1.16	Air Cooled Heat Exchanger	API Std. 661	Latest
1.17	Pipe Hangers and Supports - Materials, Design, Manufacture, Selection, Application, and Installation	MSS SP 58	Latest
2.0	EIL Standards		
2.1	Table of Basic Span	7-44-0506	Latest
2.2	Maximum Spacing of Guides for Vertical and Horizontal Pipes	7-44-0507	Latest

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3.0 REFERENCE DOCUMENTS

The following documents shall be used for stress analysis

- Piping GAD / Isometrics
- Piping clip drawings, if required
- Piping Engineering Design basis
- EIL Pipe Support Standards
- Procedure for Stress Analysis, 5-1643-4001 and documents referred therein
- Check list for Flexibility Analysis, 9-1643-9903
- P&ID, Process Line list, Process inputs like Slug data
- Mechanical Equipment Data Sheet
- Piping Material Specification
- Allowable Nozzle loads for Vessels, Columns, and Exchangers etc.
- Piping 3D model
- Scope of Pulsation and Vibration study in the form of Vendor marked-up P&ID indicating lines & equipment

4.0 STRESS ANALYSIS CRITERIA

4.1 Piping flexibility and Stress

The flexibility and stress analysis shall be performed to ensure that the piping arrangement (with its equipment nozzles, supports & restraints) is safe at various stages of piping life-period.

The various stages of piping life-period are : as-installed (WNC), hydro-test, dry-out (steam out), start-up, operation, upset operation, shutdown etc.

The minimum criteria for defining the minimum acceptable flexibility are:

- Stress in piping system,
- Deflection/ movement of piping,
- Induced forces & moments on the connected equipment and special items,
- Natural frequency of piping system
- Induced loading on supports/ structures
- Unintentional disengagement of piping from its supports
- Interference between adjacent lines due to thermal expansion or contraction of the piping system
- Excessive sag in piping systems, particularly those requiring slopes for drainage.

The minimum criteria for pipe support design shall be:

- Pipe supports shall be allocated such as not to block access or headroom clearance.
- Anchor loads, thermal loads, and friction loads for piping laid on pipe racks should be provided for in structural design.
- Pipe supports shall be selected considering the line design temperatures. It shall be ensured that extreme temperatures are not transmitted to supporting structural members.

4.2 Stress Range Calculations

For stress range calculation, whichever of the below listed equations produces the maximum temperature differential shall be used:
Maximum design temp – Minimum installation temp
Maximum installation temp – Minimum design temp

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Maximum design temp – Minimum design temp

4.3 Stress Critical Lines

A line is selected as a Stress Critical Line if it falls under the category of "Lines Requiring Mandatory Formal Analysis" as defined in Engineering Design Basis for Piping. Formal stress analysis shall be done using Caesar II software.

All non-critical lines (having size more than half of connecting critical line size) that tie-in into the critical lines should be considered as critical up to the first cross-guide/anchor (on that line), if they have significant effect on the main line flexibility.

Over and above, the stress critical line list shall also comprise the lines which are deemed critical by licensor.

Non critical lines shall be reviewed for proper supporting and flexibility. Reference shall be made to Stress nomographs, table of basic span, maximum spacing for guide etc. where deemed necessary for these lines.

4.4 Stress Analysis Index Format

Stress critical lines, selected according to criteria described under Clause No. 4.3 shall be indexed as per format no. 3-1643-0701 (Stress Analysis Index).

All lines in this list shall be formally computer analysed and documented.

5.0 STRESS ANALYSIS PARAMETERS

5.1 Stress Analysis Temperature

All lines shall be analysed at analysis temperature. In the absence of analysis temperature, lines shall be analysed at operating temperature if difference between operating & design temperatures is less than or equal to 30 deg. C, otherwise it shall be analysed at design temperature. However, analysis of flare line & calculation of thermal growth of tall equipments shall be done as per Process input.

However, the other alternate temperatures (as to take care of the occasional conditions, such as start-up, shutdown, regeneration, depressurising and other special conditions) should also be considered as per Process information.

All hydrocarbon lines shall be analysed at steam out temperature where steam-out temperature is more than the design temperature. Steam out temperature shall be taken as per Line list. In case of non-availability of Line list, it shall be taken as 30°C less than the saturation temperature of type of steam.

5.2 Site Data

Following temperatures shall be used unless otherwise specified in project specific site data.
Installation temperature = 21°C for restraint loads and stress analysis of hot lines.

5.3 Friction Coefficients

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The effects of friction on supports and strain sensitive equipment shall be considered based on the following coefficient

Steel to Steel (flat contact)	0.3
Steel to PTFE	0.1
Steel to Graphite	0.15
Steel to Rod (line contact)	0.25
SS304 Roller	0.05
SS304 Spherical Ball	0.05

6.0 MAXIMUM PIPE SPAN AND DEFLECTION

The Table of Basic Span and Maximum spacing of Guides for Vertical and Horizontal Pipes shall be as per 7-44-0506 & 7-44-0507 respectively.

Longitudinal pipe movements not to exceed 150MM. For axial movement exceeding 100MM, shoe length shall be increased correspondingly.

Vertical deflections in piping system in horizontal run between two adjacent supports due to sustained loading = 15MM (max) for all lines except steam lines. For steam lines the above deflection limit shall be 3MM (max).

7.0 PIPE STRESS ANALYSIS AND SUPPORTING

All piping systems shall be properly supported taking into account of the following points:

7.1 Piping stress analysis shall be as per ASME B 31.3

7.2 Analysis shall include, but not be limited to the following; self-weight, expansion, contraction, differential settlement, relief valve reactions, frequency check, pressure, wind, seismic effect, hydraulic surge, slug flow and other mechanical loading, if required, and a combination of these based on ASME B31.3.

Analysis of lines connected with the existing lines in the revamp units shall include supporting's of existing line up to the nearest possible two supports after cross guide/ anchor location given in existing lines.

7.3 Nozzle Loadings shall fall within the allowable loads as per codes and /or as specified by equipment designer/ vendor. Thermal movement at the equipment nozzle shall be obtained from the equipment designer/vendor or equipment shall be modelled based on vendors drawing. Calculated nozzle loads wherever exceeds, shall be sent to equipment designer/vendor for approval, if piping design changes are not possible.

7.4 Pipe supports shall be able to withstand the loads determined by the piping analysis.
Stress Engineer shall evaluate all pipe support loads including friction forces due to thermal expansion using good engineering judgement. Adjustments shall be made to the piping system and model such that the pipe supports loads are within a reasonable uniformity throughout the piping system.

7.5 Piping Stress engineer will co-ordinate between the Piping Layout Design/Structural and Static & Mechanical Equipment Division to ensure that all piping loads associated with these areas are fully accounted for.

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7.6 Standard supports shall be used for the supporting. If standard supports are not meeting requirements then job specific support shall be used. Low friction pads, branch reinforcement pads for stress, special support gaps (other than 3mm) shall be called out in the isometrics.

7.7 Flare lines: Proper thermal analysis temperature shall be established including the possibility of temperature gradient along the line before providing expansion loops. Efforts shall be made to minimize the number of loops.

Flare line between knock out drum and water seal drum shall be designed for pressure fluctuations and adequately supported to avoid vibrations.

7.8 Flange Leakage:

Flange leakage check shall be performed for lines required as per Engineering Design Basis for Piping.

The combined effect of pressure and piping loads (resultant bending) shall be checked against the standard pressure rating of flange. Following CAESAR II in built flange check application may be used for such checking,

- Pressure Equivalent method based on ASME B 16.5 pressure temperature table
- ASME BPVC Sec VIII Div 1 Appendix 2 method
- Flange Leakage Evaluation based on NC 3658.3 Method

7.9 Use expansion joints when space or pressure drop limitation does not permit pipe bends. Bellows/ expansion joints are the least preferred means for providing flexibility. The use of expansion joints shall be minimised. Expansion joints and all other nonstandard items, as snubbers etc., are subjected to Sr. Group Leader's approval.

7.10 Directional stops and guides shall be used to protect terminal equipment or to direct the expansion to the bends, expansion loops, and expansion joints.

Pipe guides should be used to prevent buckling of long-run piping. They are also to be used on both sides of the loops to avoid shifting of line sideways. Due care to be taken to divert the thermal/ other deflection away so as to avoid interference with the other piping or structures.

7.11 Support Lift-Up Stress Check Philosophy:

In systems where supports may be active in some conditions and inactive in others, the maximum value of sustained stress, considering all support conditions, shall be checked.

For the lift up case, the stress analysis shall also be carried out without considering the lifted supports and the sustained stresses in this case shall be checked with allowable stresses.

7.12 STRESS INTENSIFICATION FACTOR:

Stress Intensification factor shall be used as per ASME B31J.

8.0 WIND ANALYSIS

The effect of wind on a piping system is to produce a set of vibratory/dynamic loads acting along different directions depending on the configuration of the piping. However, the design wind load for a piping system is often calculated on the basis of a statically applied design pressure(s) that can be expected over the design life of the plant/unit.

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Wind analysis shall be done as per Design Basis for Wind Loads on Piping issued by Engineering & Technology Development Department (ETDD).
All lines should have a sufficient number of horizontal restraints/guides to resist the wind load. Additional horizontal supports may be required to reduce excessive horizontal deflection of pipe due to wind load.

9.0 SEISMIC ANALYSIS

Seismic analysis shall be done as per Design Basis for Seismic Design Requirements for Piping issued by Engineering & Technology Development Department (ETDD).

10.0 COMPOSITE ANALYSIS

Composite Analysis is generally carried out for systems where the Structural/Vessel stiffness and response may affect the Piping Stress analysis results significantly.

The requirements of Composite Stress Analysis shall be decided in consultation with Engineering & Technologyl Development Department (ETDD) on case to case bases, for critical and important system only, like FCC R&R section, CCR Cyclemax Reactor System etc. where the relative stiffness of structure and /or equipment may affect the response.

11.0 SPECIFIC REQUIREMENTS

11.1 Fired Heaters

The allowable forces, moments and movements at the nozzles should be finalised with Heater Vendor during engineering as per API 560. Slug forces shall be considered in the analysis if applicable.

A clear battery limit anchor (designed by heater vendor) shall be specified. Portion of piping between battery limit (B/L) and heater shall be under the scope of the heater vendor. Loads coming from piping outside heater B/L shall be informed to heater vendor for designing the anchor.

Pipe support cleats should be provided on the heater structure for applied loading. Platforms should be designed to accommodate loads coming from the pipe supports. The loads need to be informed to heater vendor during engineering.

11.2 Air Cooler

The allowable nozzle loads given by the Static & Mechanical Equipment Division (SMED) shall serve as the maximum limit. Calculated loads wherever exceeds the allowable load shall be sent to Vendor for approval, if piping design changes are not possible.

11.3 Pumps

The allowable loads at Pump nozzles shall be as follow.

Pump Type	Allowable Nozzle Loads
API 610	2 times of API 610 as per Appendix F or 2 times of API 610 subject to confirmed by Vendor
ANSI pumps	As per Vendor Allowable

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API Reciprocating Pumps	As per Vendor Allowable
Others	As per Vendor Allowable

For one pump working and another at stand-by. The following three different cases shall be considered:

- Both pumps working at operating temperature and pressure.
- Pump A working at operating temperature and pressure, Pump B stand-by
- Pump B working at operating temperature and pressure, Pump A stand-by

For two pumps operating and another at stand-by. The following four different cases shall be considered:

- Pumps A & B working at operating temperature and pressure, Pump C stand-by
- Pumps A & C working at operating temperature and pressure, Pump B stand-by
- Pumps B & C working at operating temperature and pressure, Pump A stand-by
- All three pumps working at operating temperature and pressure.

The pipe to/from the stand-by pump to tee can be taken as the average of the temperature at which the line is being analysed and the ambient temperature if the check valve is fitted with a by-pass to facilitate a 'warm up'.

If a by-pass is not provided, the temperature of the dead leg for the stand-by pump from the tee to the block valve, for both suction and discharge lines, shall be considered to be as the average of the temperature at which the line is being analysed and the ambient temperature. The temperature of the piping from the block valve to the pump shall also be taken as ambient.

Cold alignment check shall be carried out as per Vendor recommended values or as per API RP 686 for top suction & top discharge pumps as part of flexibility analysis and needful supports shall be provided.

To ensure the proper alignment with nozzle as per vendor recommendation/ API RP 686, the first support shall be provided as near as possible to pump suction/ discharge nozzle.

Where springs are located near to pump nozzles a separate nozzle loading check (WNC+H) is required to cover the short term condition where the springs are active, but the line is empty.

11.4 Pressure Vessel and Tower

The allowable nozzle loads provided by Static and Machinery Equipment Division (SMED) shall serve as the maximum limit. In case the given allowable load cannot be met through eql file checking, nozzle loading shall be checked by WRC module in caesar and the actual loads shall be given to SMED for their review and acceptance.

11.5 Shell and Tube Heat Exchangers

The allowable nozzle loads given by the Static and Machinery Equipment Division (SMED) shall serve as the maximum limit. In case the given allowable load cannot be met, the same shall be conveyed to SMED for approval.

11.6 Reactor

The allowable nozzle loads shall be as specified in MR for reactors.

11.7 Reciprocating Machinery

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Calculated nozzle loads shall be maintained within vendor allowable at vendor interface. Vendor has to define the scope for Pulsation and Vibration study and perform Pulsation and Vibration study as per the scope and based on static analysis of the lines done by EIL. Recommendations of the pulsation study shall be incorporated in Isometric drawings.

11.8 Package Units

The tie-in point displacements and maximum allowable loads on the terminal point at battery limit shall be furnished by package vendor. The calculated loads shall be within the allowable limit. Wherever loads are exceeding the allowable limit and further reduction is not possible, these loads shall be sent to vendor for approval.

11.9 Turbine and Centrifugal Compressor

The Tie-in point displacements shall be furnished by the Turbine/ Centrifugal Compressor vendor. Allowable loads shall be as per the standards given below if the Manufacturer's allowable loads are not available.

For Turbines: allowable loads shall be as per NEMA SM 23.

For Centrifugal Compressor: Allowable loads shall be 1.85 times of allowable loads given in NEMA SM 23 as specified in API 617.

Calculated loads wherever exceeding shall be sent to Vendor for approval, if piping design changes are not possible.

Cold alignment check shall be carried out as per Vendor recommended values or as per API RP 686 for Compressor Piping as part of flexibility analysis and needful supports shall be provided.

11.10 Safety Relief valve blow off (Open to Atmosphere)

Reaction forces from Relief valve shall be obtained from vendor. However if during preliminary stage of analysis data is not available then Relief valve discharge forces shall be calculated in accordance with API RP 520 Part II. But later on it must be corrected for forces received from vendor. Consider 'DLF' equal to 2.

11.11 Jacketed Piping

Jacketed piping to be analysed along with Core piping.

Calculated stresses at the junction of core and jacket pipe shall be corrected by adding axial stresses due to differential expansion of core and jacket pipe. Corrected stress shall be within the allowable stress.

12.0 SOFTWARE USED

CAESAR II software shall be used for the formal piping stress analysis. Refer Annexure-A for materials for which allowable stresses need to be checked and aligned with ASME B31.3.

13.0 CAESAR II MODELING

Generally, starting node shall be 10 and in steps of 10, preferably in the direction of flow. Trunnion shall be in 20000 series and Equipment in 10000 series

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- Model the equipment as per MDS (e.g. Diameter, wall thickness, material as on Equipment drawing, with temperatures as per the load case).

- The nozzles to be modelled as a rigid from the centre line of the Equipment to the outside of the shell & the nozzle neck as per MDS (diameter and wall thickness). At the shell-nozzle junction, a connecting node to be modelled. The result is that movements of the equipment and its nozzles are calculated via the equipment model and its rigid elements, the nozzle loads are known at the connecting nodes.

- Modeling of equipment connected to multiple system should be modeled in one system and the same to be used for all connected systems.

- All items influencing stiffness of the system significantly have to be modelled

- The use of Trunnion shall be minimized.The trunnion would be modelled with SIF (i & o) calculated and pad would be provided if stress failure is observed. Option of Increasing trunnion size /reduction of length can also be considered.
  Strength check of trunnion can also be done by utilizing other methods of design of pipe attachment.

- Normally Liberal stress shall not be used.

- Support modelling

  Resting supports: +Y with friction: 0.3 (default)
  Guide/ Cross Guide Support: 3 mm gap (default) with friction: 0.3 (default)

14.0 SPRING

Wherever possible the use of spring supports shall be minimized by careful consideration of support location and alternative pipe routing.

- The spring should be selected for the operating condition.

- The spring load variation shall not be more than 15% for the operating condition.

- Maximum travel out of all conditions like design, operating, upset, startup, WNC etc. to be checked.

- For all the spring hangers on the liquid lines, the WNC (Weight No Content) case must be checked. In WNC case the nozzle loads must be within the allowable limits.

- Highly loaded guides and stops should be avoided in combination with springs, as spring will not operate properly.

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15.0 UNIT FILE

Review Current Units

ITEM	Internal Units		Constant		User Units	ITEM	Internal Units		Constant		User Units
Length	inches	*	25.4	=	mm.	Fluid Den.	lbs./cu.in.	*	27680	=	kg/cu.m.
Force	pounds	*	0.453592	=	Kg.	Transl.	lbs./in.	*	0.017858	=	Kg./mm.
Mass-dynamics	pounds	*	0.4536	=	kg.	Rotl. Stiff.	in-lb/deg	*	0.011521	=	Kg.m./deg.
Moment-input	in.-lb.	*	0.011521	=	Kg.m.	Unif. Load	lb./in.	*	0.017858	=	Kg./mm.
Moment-output	in.-lb.	*	0.011521	=	Kg.m.	G Load	g's	*	1	=	g's
Stress	lbs./sq.in.	*	1	=	lb./sq.in.	Wind Load	lbs./sq.in.	*	6894.76	=	N./sq.m.
Temp. Scale	degrees F	*	0.5556	=	C	Elevation	inches	*	0.0254	=	m.
Pressure	psig	*	0.070307	=	Kg./sq.cm.	Cmpd Lng	inches	*	25.4	=	mm.
Elastic Modulus	lbs./sq.in.	*	1	=	lb./sq.in.	Diameter	inches	*	25.4	=	mm.
Pipe Density	lbs./cu.in.	*	27680	=	kg/cu.m.	Thickness	inches	*	25.4	=	mm.
Insulation Den.	lbs./cu.in.	*	27680	=	kg/cu.m.	Nominals				=	ON

Units File Label: EILR1

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16.0 LOAD CASES

The various Load cases, but not limited to the same, considered in Caesar II to check stresses, loads, displacements etc. in piping system shall be the following:

Coordinate system is as follows: North = (-X)
West = (+Z)
Vertical up = (+Y)

	Load Case	Type	Remarks
1)	WW+HP+H	HYD	hydro test Load case
2)	W+T1+P1+H	OPE	operating load case at Operating Temp
3)	W+P1+H	ALT-SUS	alternate sustained stress based on operating Temp
4)	W+T2+P2+H	OPE	operating load case at Design Temp
5)	W+P2+H	ALT-SUS	alternate sustained stress based on Design Temp
6)	W+T1+P1+F1+H	OPE	operating case at Operating Temp with Force
7)	W+T1+P1+U1+H	OPE	U1 +X direction seismic load
8)	W+T1+P1+U2+H	OPE	U2 +Y direction seismic load
9)	W+T1+P1+U3+H	OPE	U3 +Z direction seismic load
10)	W+T1+P1-U1+H	OPE	U1 -X direction seismic load
11)	W+T1+P1-U2+H	OPE	U2 -Y direction seismic load
12)	W+T1+P1-U3+H	OPE	U3 -Z direction seismic load
13)	W+T1+P1+WIN1+H	OPE	WIN1 +X direction wind load
14)	W+T1+P1+WIN2+H	OPE	WIN2 -X direction wind load
15)	W+T1+P1+WIN3+H	OPE	WIN3 +Z direction wind load
16)	W+T1+P1+WIN4+H	OPE	WIN4 -Z direct ion wind load
17)	W+P1+H	SUS	sustained case operating Temp

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18)	W+P2+H	SUS	sustained case Design Temp
19)	WNC+H	SUS	sustained case
20)	L2-L17	EXP	expansion case for operating temperature
21)	L4-L18	EXP	expansion case for Design temperature
22)	L7-L2	OCC	seismic case net deflection (+X dir.)
23)	L8-L2	OCC	seismic case net deflection (+y dir.)
24)	L9-L2	OCC	seismic case net deflection (+z dir.)
25)	L10-L2	OCC	seismic case net deflection (-x dir.)
26)	L11-L2	OCC	seismic case net deflection (-y dir.)
27)	L12-L2	OCC	seismic case net deflection (-z dir.)
28)	L13-L2	OCC	wind case net deflection (+X dir.)
29)	L14-L2	OCC	wind case net deflection (-X dir.)
30)	L15-L2	OCC	wind case net deflection (+Z dir.)
31)	L16-L2	OCC	wind case net deflection (-Z dir.)
32)	L17+L22	OCC	scalar
33)	L17+L23	OCC	scalar
34)	L17+L24	OCC	scalar
35)	L17+L25	OCC	scalar
36)	L17+L26	OCC	scalar
37)	L17+L27	OCC	scalar
38)	L17+L28	OCC	scalar
39)	L17+L29	OCC	scalar
40)	L17+L30	OCC	scalar
41)	L17+L31	OCC	scalar

P1-Operating Pressure
T1-Operating Temperature
P2-Design Pressure
T2-Design Temperature
HP- Hydro test Pressure

W – Dead Weight
WW – Water Weight
WIN – Wind Load
WNC – Weight No Content
U - Uniform Load (Seismic)

F1- PSV force, Slug Force, Water Hammer
H - Hanger load
L – Load Case
SUS, EXP, OCC, HYD, OPE – Various load types, viz., sustained, expansion, occasional, hydrotest, operating etc.

Notes:

- The above load cases are indicative and shall be modified based on the applicability.
- Code Stresses to be checked in Sustain, Sustain+Occasional, Hydro test and Expansion cases
- Loadings and Displacements to be checked in Operating(all ope cases), Sustain, Sustain+Occasional, Hydro test cases
- The load cases 22 to 31 represent the case of only an occasional load and do not have a code stresses comparison.
- The load cases 32 to 41 represent the occasional case combined with load case due to pressure and dead weight and have a code stress comparison.
- In the case of cold lines, minimum design condition shall also be considered.
- Sustained Stresses for Lift-up Supports (hot sustain) during operating condition, cases 3 and 5 shall also be checked. The primary stress in each position of the piping system must be evaluated. An alternate SUS load case uses the restraint status from the previous OPE load case to evaluate the stresses induced by primary loads.

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17.0 DOCUMENTATION (for Stress Critical Lines only)

The final report / stress package folder shall include all pertinent information that shall include but not be limited to the following.

1 Top sheet, containing Project name, project title, Stress System number etc.
2 Check list for Flexibility Analysis, 9-1643-9903
3 Basis of analysis
4 Line list
5 Piping Material Specification
6 P&ID marked up
7 Equipment's Drawing with allowable loads
8 Node marked AFC Isometrics with supports type and locations specified
9 Specialty items' Material Requisition (springs, expansion joints, etc.) if applicable
10 Caesar II model plot
11 Trunnion temperature Calculation sheet
12 Trunnion strength Check Report (if any)
13 Soft copies of CAESAR II analysis files
14 Flange leakage calculations, PSV calculation sheet if any
15 Nozzle Evaluation Report if checked in WRC 107/WRC 297/ Nozzle pro/API610/NEMA SM23/or any other communication related to nozzle load acceptance from other departments / vendor
16 Conclusion

18.0 ATTACHMENTS

6-44-0091-A1 List of piping materials requiring checking of CAESAR Allowable stresses with ASME B31.3

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List of Pipe/Tube materials having thickness or diameter dependent allowable stresses as per ASME B31.3 is given below, However refer B31.3 for the exhaustive list.

Sr no	Material	ASTM Material Specification	Thickness/dia	Product form
1	Carbon Steel	A671 CD70 K12437	≤ 64 mm Thk.	Pipe & tube
2		A671 CK75 K02803	≤ 25 mm Thk.	
3			> 25 mm Thk.	
4		A672 D70 K12437	≤ 64 mm Thk.	
5		A672 N75 K02803	≤ 25 mm Thk.	
6			> 25 mm Thk.	
7	Alloy Steel	A335 P91	≤ 75 mm Thk.	Pipe
8		A691 Gr.91	≤ 75 mm Thk.	
9	Stainless Steel	A249 S31254	≤ 5 mm Thk.	Tube
10			> 5 mm Thk.	
11		A312 S31254	≤ 5 mm Thk.	Pipe
12			> 5 mm Thk.	
13		A358 N08367	≤ 5 mm Thk.	Wld. pipe
14			>5 mm Thk.	
15		A358 S31254	≤ 5 mm Thk.	Wld. pipe
16			>5 mm Thk.	
17		A813 N08367	≤ 5 mm Thk.	Wld. pipe
18			>5 mm Thk.	
19		A814 N08367	≤ 5 mm Thk.	Wld. pipe
20			>5 mm Thk.	
21		A312 TP321 S32100	≤ 10 mm Thk.	Smls. pipe
22			>10 mm Thk.	
23		A312 TP321H S32109	≤ 10 mm Thk.	
24			>10 mm Thk.	
25		A376 TP321 S32100	≤ 10 mm Thk.	
26			>10 mm Thk.	
27		A376 TP321H S32109	≤ 10 mm Thk.	
28			>10 mm Thk.	
29		A312 N08367	≤ 5 mm Thk.	Smls. & wld. Pipe
30			>5 mm Thk.	

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Sr no	Material	ASTM Material Specification	Thickness/dia	Product form
31	Duplex Stainless steel	A928 S32003	≤ 5 mm Thk.	Pipe & tube
32			>5 mm Thk.	
33		A790 S32906	<10 mm Thk.	Smls. & wld. pipe
34			≥ 10 mm Thk.	
35		A790 S82441	<10 mm Thk.	
36			≥ 10 mm Thk.	
37		A789 S32003	≤ 5 mm Thk.	Smls. & wld. tube
38			>5 mm Thk.	
39		A789 S32101	≤ 5 mm Thk.	Smls. & wld. tube
40			>5 mm Thk.	
41		A789 S82441	<10 mm Thk.	Smls. & wld. tube
42			≥ 10 mm Thk.	
43		A790 S32003	≤ 5 mm Thk.	Smls. & wld. pipe
44			>5 mm Thk.	
45		A790 S32101	≤ 5 mm Thk.	Smls. & wld. pipe
46			>5 mm Thk.	
47	Ni & Ni Alloys	B161 N02200	≤ 125 mm O.D.	Pipe & tube
48			>125 mm O.D.	
49		B161 N02201	≤ 125 mm O.D.	
50			>125 mm O.D.	
51		B165 N04400	≤ 125 mm O.D.	
52			>125 mm O.D.	
53		B167 N06600	≤ 125 mm O.D.	
54			>125 mm O.D.	
55		B167 N06690	≤ 125 mm O.D.	
56			>125 mm O.D.	
57	Ni Alloys	B675 N08367	≤ 5 mm Thk.	Pipe & tube
58			>5 mm Thk.	
59		B690 N08367	≤ 5 mm Thk.	
60			>5 mm Thk.	
61		B725 N02200	≤ 125 mm O.D.	
62			>125 mm O.D.	
63		B725 N02201	≤ 125 mm O.D.	
64			>125 mm O.D.	
65		B725 N04400	≤ 125 mm O.D.	
66			>125 mm O.D.	
67		B804 N08367	≤ 5 mm Thk.	
68			>5 mm Thk.	

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Sr no	Material	ASTM Material Specification	Thickness/dia	Product form
69	Copper Alloys	B466 C71000	≤ 114 O.D.	Pipe & tube
70		B467 C70600 W050	≤ 114 O.D.	
71			>114 O.D.	
72		B467 C70600 W061	≤ 114 O.D.	
73			>114 O.D.	
74		B467 C71500 W061	≤ 114 O.D.	
75			>114 O.D.	
76		B467 C71500 WO50	≤ 114 O.D.	
77			>114 O.D.	

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तेल एवं गैस उत्पादन के लिए सॉर सर्विस में इस्तेमाल होने वाले कार्बन स्टील अवयवों हेतु सामग्री अपेक्षा

MATERIAL REQUIREMENTS FOR CARBON STEEL COMPONENTS USED IN SOUR SERVICE FOR OIL AND GAS PRODUCTION

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
3	30.03.26	REVISED & ISSUED AS STANDARD SPECIFICATION	NS	PVC	PS	NK
2	05.07.19	REAFFIRMED AND RE-ISSUED	SM	DD	SG	RKT
1	07.10.11	REVISED & ISSUED AS STANDARD SPECIFICATION	VC	SG	PPL	DM
0	07.10.04	REVISED & ISSUED AS STANDARD SPECIFICATION	GM	VRK	KKM	SKG

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Abbreviations:

ANSI	:	American National Standards Institute
API	:	American Petroleum Institute
ASME	:	American Society of Mechanical Engineers
ASTM	:	American Society for Testing and Materials
ISO	:	International Organization for Standardization
NACE	:	National Association of Corrosion Engineers

SMMS Standards Committee:

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CONTENTS

1.0 SCOPE 4
2.0 REFERENCED CODES / STANDARDS 4
3.0 GENERAL REQUIREMENTS 6
4.0 MATERIAL SPECIFICATION AND PRODUCTION TESTS 7
5.0 HARDNESS TEST 8
6.0 CORROSION TESTS 8
7.0 QUALITY SURVEILLANCE AND INSPECTION DURING PRODUCTION 9
8.0 QUALIFICATION OF MANUFACTURERS/ SUPPLIERS OF SOUR SERVICE MATERIALS 10
9.0 SPECIAL REQUIREMENTS FOR WELDING AND FABRICATION FOR SOUR SERVICE 10
TABLE-I 12
TABLE – II 12
TABLE – III 13
TABLE - IV 13
TABLE – V 14

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1.0 SCOPE

1.1 This specification lays down the requirements related to the chemical composition, manufacture, fabrication and testing requirements for carbon steel components intended to be used in sour service environment. These requirements are specified in order to make the carbon steel component resistant to the various forms of material damage in a sour environment such as Sulfide Stress Cracking (SSC), Hydrogen Induced Cracking (HIC), Stress Oriented Hydrogen Induced Cracking (SOHIC), Blistering etc. These requirements can be used for resisting Alkaline Stress Corrosion Cracking (ASCC) also. This specification is applicable for offshore and onshore oil and gas production facilities. Pipeline field welding is not included in this specification.

1.2 The service medium is defined as "sour" when the partial pressure of H_2S in a wet (water as a liquid) gas phase of a gas, gas condensate or crude oil system is equal to or exceeds 0.0003 MPa (0.05 Psi).

2.0 REFERENCED CODES / STANDARDS

Standard Number	Title
NACE MR 0175 / ISO 15156	Petroleum and natural gas industries — Materials for use in H2S-containing environments in oil and gas production
NACE RP 0472	Methods and Controls to Prevent In-Service Environmental Cracking of Carbon Steel Weldments in Corrosive Petroleum Refining Environments
NACE TM 0177	Laboratory Testing of Metals for Resistance to Sulfide Stress Cracking and Stress Corrosion Cracking in H2S Environments
NACE TM 0284	Evaluation of Pipeline and Pressure Vessel Steels for Resistance to Hydrogen-Induced Cracking
ASTM A 20	Standard Specification for General Requirements for Steel Plates for Pressure Vessels
ASTM A 105	Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A 106	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

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Standard Number	Title
ASTM A 216	Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
ASTM A 234	Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High-Temperature Service
ASTM A 370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A 516	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service
ASTM A 578	Standard Specification for Straight-Beam Ultrasonic Examination of Rolled Steel Plates for Special Applications
ASTM A 694	Standard Specification for Carbon and Alloy Steel Forgings for Pipe Flanges, Fittings, Valves, and Parts for High-Pressure Transmission Service
ASTM A 770	Standard Specification for Through-Thickness Tension Testing of Steel Plates for Special Applications
ASTM E 18	Standard Test Methods for Rockwell Hardness of Metallic Materials
ASTM E 45	Standard Test Methods for Determining the Inclusion Content of Steel
ASTM E 92	Standard Test Methods for Vickers Hardness and Knoop Hardness of Metallic Materials
API 5L	Specification for Line Pipe
API 6A	Specification for Wellhead and Tree Equipment

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Standard Number	Title
ASME SEC II PART C	Specifications for Welding Rods, Electrodes, and Filler Metals
ASME SEC V	Nondestructive Examination
ASME SEC VIII-DIV.1	Rules for Construction of Pressure Vessels
ASME SEC IX	Welding, Brazing, and Fusing Qualifications
ANSI B 16.34	Valves Flanged, Threaded, and Welding End
ANSI B 31.3	Process Piping

All ASTM Standard designations shall be applicable for corresponding ASME designations, which would be read as ASME SA XXX instead of ASTM A XXX.

3.0 GENERAL REQUIREMENTS

3.1 The steel for sour service shall be manufactured by either basic oxygen or electric arc furnace route and shall be fully killed and fine grained.

3.2 All wrought products shall be produced and certified by the manufacturer as HIC resistant steel and shall be free of elongated inclusions.

3.3 All rolled products such as plates and hot rolled coils used for components and equipment shall be of low sulfur and low phosphorus content, shall be vacuum degassed and calcium treated for sulfide shape control.

3.4 Plates shall be supplied in normalized, normalized and tempered or thermo-mechanically treated condition, unless otherwise agreed in writing.

3.5 Manufacturer shall take care to ensure that all products are free of low temperature transformation microstructures such as bainite bands or islets of martensite.

3.6 HIC test shall be conducted only for steel plates and products made out of plates.

3.7 All items are required to conform to applicable material specification and chemical composition as detailed in Tables I to V.

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4.0 MATERIAL SPECIFICATION AND PRODUCTION TESTS

4.1 Special Requirements

4.1.1 The various types of carbon steel sour service material component shall conform to the recommended product specifications and shall meet the additional requirements and modifications mentioned herein against each product specification.

4.1.2 The chemical composition analysis and mechanical tests shall be conducted for each heat and for each heat treatment batch of finished product in accordance with the respective product specifications and shall also meet the requirements mentioned in the various Tables referred to against each type of product.

4.1.3 The hardness testing shall be conducted on all products as mentioned in clause No.5.0 of this specification and shall meet the requirements laid down.

4.1.4 Corrosion tests, wherever specified against each type of product shall be conducted to meet the acceptance limits mentioned, unless specifically permitted otherwise in writing by the owner.

4.1.5 Non-destructive testing of the finished products shall be in accordance with the governing product specification and as modified in this specification in the corresponding clauses or as mentioned in any of the documents attached with the material requisition.

4.2 Product specific requirements

4.2.1 **Seamless Pipes**

ASTM A106 Gr. B and API 5L Gr. B seamless pipes shall be used. The chemical composition shall be as per Table I.

4.2.2 **Welded Pipes**

API 5L Gr. B and API 5L Gr.X-52/60 longitudinal submerged arc welded pipes shall be used. The chemical composition shall be as per Table II.

4.2.3 **Fittings (elbow, tee, reducer, cap, weldolet, socket etc.)**

Seamless fittings shall be ASTM A234 Gr. WPB. The chemical composition requirements for seamless fittings shall be as per Table I.

4.2.4 **Flanges**

4.2.4.1 The following product specifications shall be applicable for forged flanges:

ASTM A105
ASTM 694 Gr. F52
API 6A Type 2
API 6A Type 4

The chemical composition shall be as per Table III.

4.2.4.2 All forging shall be normalized in the finished condition.

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4.2.5 **Valves (ball, gate, globe, check, needle etc.)**

4.2.5.1 Valve body shall be of forged ASTM A105 or cast ASTM A216 Gr. WCB. Forgings shall be acceptable in place of castings but not vice versa. All other components of the valve shall be as per the applicable valve data sheet.

Chemical composition of forged valves body and of cast valve bodies shall be as per Table III & Table IV respectively.

4.2.5.2 Non-Destructive Testing

All valve castings shall be subjected to radiography. The radiography procedure, the area to be subjected to radiography, and the acceptance criteria shall be as per ANSI B16.34. In addition, dye penetrant testing (DPT) or magnetic particle testing (MPT) shall also be conducted.

4.2.6 Plates for pressure vessels, heat exchangers, piping and other process equipment.

4.2.6.1 Plates shall conform to ASTM A516 Gr.60. Maximum ultimate tensile strength & yield strength shall not exceed 15.0 ksi more than the minimum specified values.

4.2.6.2 Chemical analysis of plates used in sour service shall be carried out both on heat and on product as per ASTM A20 and meet the requirements laid own in Table II.

4.2.6.3 For plates including and above 25 mm, through-thickness tensile testing shall be carried out in accordance with ASTM A770, with the minimum reduction in area being 35% as determined in accordance with ASTM A370, on one plate per heat per heat-treatment batch.

4.2.6.4 No repair welding shall be permitted on plates.

4.2.6.5 HIC test shall be conducted on one sample per heat per heat treatment batch of plates and other rolled products.

5.0 HARDNESS TEST

5.1 Hardness measurement shall be conducted on products/items/weldments of all materials.

5.2 Hardness test shall be conducted on a sample of each finished product. The products that do not meet the hardness limits shall be rejected. For small products that cannot be hardness tested individually, the manufacturer shall conduct tests on a random basis on samples selected from production run or stored batches to check the compliance.

5.3 The maximum acceptable limit of hardness shall be RC 22 or equivalent.

6.0 CORROSION TESTS

6.1 Selection of Corrosion Test Samples

6.1.1 Test specimens shall be taken from the finished product. Where the test specimen cannot be extracted from the finished product, a representative test specimen shall be taken from the same heat.

6.1.2 In the case of welded products, the test piece shall include the parent metal, weld metal and the heat affected zones (HAZ).

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6.2 HIC test (test duration: 96 hours)

6.2.1 This test shall be performed on a set of three specimens representing each production batch/heat of rolled products in accordance with NACE TM0284 with the following acceptance criteria:

(i) crack length ratio (CLR) ≤ 10.0%
(ii) crack sensitivity ratio (CSR) ≤ 1%
(iii) crack thickness ratio ≤ 1.5 %.

6.2.2 In case of failure of any one of the specimens, three more specimens from the same product shall be retested and all the specimens shall meet the acceptance criteria.

6.2.3 HIC testing shall be required only for plates or rolled products. This shall be conducted for pre-qualification of the mill and as well as on the batch certification of each heat.

6.2.4 When the fabrication of the components from the previously HIC tested plates involves no heavy wall thickness reduction or material flow, such as bending, spinning or welding, no fresh HIC test would be required on the final product. However, for fabrication of products where heavy reduction of thickness and heavy flow of material is encountered while forming such as like forging, extrusion or drawing, metallographic examination of the final product shall be required to be carried out to ensure absence of elongated inclusions.

6.3 SSC test

6.3.1 SSC test shall be conducted only for the purpose of pre-qualification of the mills and the welding consumables and not to be used for batch testing. Mill qualification by any reputed third party inspection agency like Lloyds, BV, CEIL, DNV or TUV etc., would be acceptable.

6.3.2 SSC test shall be conducted as per NACE TM0177, with solution A,and shall meet the requirements of NACE MR0175/ISO 15156. The test duration shall be 720 hours. The applied stress values as per table B.1 and acceptance criteria as per TM0177.

6.3.3 Mills, which have been previously qualified for the supply of materials and welding consumables shall be permitted to issue the test certificates which will be considered acceptable. However, in case of doubt, the owners/purchasers can get these tests done and shall bear cost of testing, provided such tests meet the requirements. In case, the re-tested material fails to conform to the requirement, the cost of testing will be borne by the manufacturers/suppliers.

7.0 QUALITY SURVEILLANCE AND INSPECTION DURING PRODUCTION

The manufacturers/suppliers shall perform all the tests and inspection procedures mentioned in this specification, under a third party inspection agency witness to ensure conformance. Company's duly authorized representative shall have unhindered access to the material being subjected to inspection, for witnessing of the sample selection, test specimen preparation and to the performance of tests.

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8.0 QUALIFICATION OF MANUFACTURERS/SUPPLIERS OF SOUR SERVICE MATERIALS

8.1 General

All materials for sour service application to this specification can be supplied only by pre-qualified manufacturers. Whenever the term "pre-qualified manufacturer" appears, it refers only to the manufacturer who is pre-qualified by the company based upon the qualification criteria specified in NACE MR0175/ISO15156.

8.2 Pre-qualification criteria for manufacturers/suppliers with prior experience track record

The manufacturers/suppliers who have proven track record of supplying materials for offshore sour service applications to any of the process licensors such as UOP/IFP/EXXON/BP/CHEVRON or any of the reputed operating companies/consultants like ONGC/IOCL/HPCL/BPCL/EIL will be considered as 'pre-qualified manufacturers", on production of the relevant purchase orders and test certificates certified by a witnessing third party inspection agency such as DNV/LLOYDS/BV/ABSTECH/TUV/SGS/CEIL for the conformance of the materials to the respective materials specification of the mentioned operators/engineering consultants.

8.3 Pre-qualification criteria for new manufacturers/suppliers without track record

The manufacturers/suppliers who have no track record of supplying sour service materials to any of the process licensors or the reputed operating companies/consultants as mentioned in Clause 8.2 above, shall be considered as new manufacturers and shall be subjected to the elaborate pre-qualification by any one of the reputed third party inspection agencies like DNV/LLOYDS/BV/ABSTECH/TUV/SGS/CEIL as per the requirements of NACE MR0175/ISO 15156 (latest edition).

9.0 SPECIAL REQUIREMENTS FOR WELDING AND FABRICATION FOR SOUR SERVICE

9.1 Welding Procedure Qualification

9.1.1 Each particular welding procedure shall be separately qualified for the job. Previously qualified procedures, if they were qualified under EIL or a reputed engineering consultant/operator, may be submitted for review and approval.

9.1.2 Welding consumable used shall be of a grade that would produce a weld metal equivalent to the base metal used for the project and shall meet the chemical composition requirements of Table V.

9.2 Welding Consumable Qualification

Welding consumable shall conform to a standard classification and of a type and make approved by the company or by any of the reputed third party inspection agencies like Lloyds, BV, CEIL, DNV, TUV, etc. New welding consumables shall be subject to qualification testing to check their conformance to the classifications to which they belong.

9.3 The consumable qualification testing shall include all the tests specified by ASME SEC.II Part C as well as the SSC tests, chemical composition analysis and hardness testing as per the requirements of this specification.

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9.4 Acceptance criteria for mechanical and corrosion tests shall be as those applicable for the base metal. These tests shall be carried out on the undiluted all weld metal.

9.5 The test coupons for the welding procedure qualification shall be extracted from the plates/ pipes having the maximum values of carbon equivalent, maximum inclusion content and maximum thickness, which are intended to be used for the fabrication on the job.

9.6 Tests to be Conducted on Procedure Qualification Testing (PQT)

9.6.1 Chemical composition of the weldment shall be carried out to check its conformance to Table- V. Microstructural examination at a magnification of 200x shall be conducted to assure complete absence of un-tempered martensite.

9.6.2 Mechanical tests shall include tensile testing, impact testing and bend testing as per ASME Sec. IX with acceptance criteria as per the governing code of fabrication. Each procedure qualification test sample shall be subjected to detailed hardness survey, with Vickers hardness tester, with 5 kg load, of the weld heat affected zone and base metal in accordance with ASTM E-92. The maximum permissible hardness shall be equivalent to RC 22.

9.6.3 **Corrosion Tests**

Only SSC test shall be conducted on the procedure qualification test samples as mentioned in clause 6.3 on a sample that contains the weld, parent material and the heat affected zone. Alternatively, a four point bend specimen SSC test for 500 hour can also be accepted for procedure qualification.

9.6.4 100% radiographic examination of the test welds and production welds shall be carried out as per ASME Sec. V, with acceptance criteria as per ASME Sec. VIII Div I, UW-51. Ultrasonic testing can be accepted in lieu of radiographic examination if approved by company.

9.6.5 Dye penetrant testing (DPT) or magnetic particle testing (MPT) shall be carried out for the fillet and groove welds, wherever radiography is not feasible.

9.6.6 Welder qualification tests

Welders shall be qualified at the fabrication yard/barge, employing the qualified welding procedures. The welder test specimens shall be subjected to radiographic examination, DPT/MPT and cross-sectional hardness survey, with the same procedures adopted for the PQT and with acceptance criteria mentioned in ASME Sec. IX.

9.6.7 **Post Weld Heat Treatment (PWHT)**

All the weld joints irrespective of thickness shall be given a post weld heat treatment. The recommended temperature range for PWHT is 595-650°C with a close control. Other aspects such as rate of heating, holding time, etc. for the PWHT shall be as per ASME Sec.VIII, Div. I /ANSI B31.3 requirements.

PWHT may be waived off for socket welds and butt joints of valve size 1.5" and below as high temperature may damage the soft seat. Welding procedure for such joints shall demonstrate the cross-section hardness of the weld do not exceed RC22.

PWHT of components made out of Thermo Mechanically controlled plates (TMCP) is not required. However, hardness of weldment shall not exceed RC22.

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TABLE-I

Chemical Composition Requirements for Seamless Pipes & Fittings (ASTM A106 Gr. B /API 5L Gr. B)

Element	Limits
C	0.14% max.
Si	0.10%-0.35%
S	0.01% max
P	0.02% max.

Note: The unspecified elements Mn, Cu, Ni, Cr, Mo, V etc. shall be as per the limits of ASTM A106 Gr. B.

TABLE – II

Chemical Composition Requirements for Plates and Rolled Products of Carbon Steel (line pipes/pressure vessels/heat exchangers/storage vessels) [ASTM A516 Gr. 60/A234 Gr.WPB (Welded)/API 5L Gr B (SAW)/API 5L X52/60 (SAW)]

Element	Limits
C	0.20% max.
Si	0.15%-0.35%
S	0.003% max.
P	0.020% max.
N	0.01% max.
Ca/S	2-3%
Al/N	2% min
B	0.002% max.

Note :

1. All other unspecified elements like Mn, Cu, Ni, Cr, Mo, V, Nb, Ti, etc. shall be as per ASTM A516 and A20.
2. Steel shall be produced as HIC resistant steel with inclusion content and morphology control and shall be free of elongated manganese sulfide inclusion.
3. The ratio Ca/S is not applicable when S ≤ 0.0015%. In such cases, Ca upto 50 ppm shall be permitted.

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TABLE – III

Chemical Composition Requirements for Forged Flanges (ASTM A105, ASTM A694 Gr.F52, API 6A (Type 2 & 4)

Element	Limits
C	0.25% max.
S	0.01% max.
P	0.02% max.
Note: All other elements shall be as specified in the basic specification. The forgings shall be produced with inclusion count and morphology control.	

TABLE - IV

Chemical Composition Requirements for Castings (ASTM A216 Gr. WCB)

ELEMENT	LIMITS
C	0.20 % max..
Si	0.10%-0.35%
S	0.02% max
P	0.02% max.
Note: All other unspecified elements shall be as per the basic specification ASTM A216.	

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TABLE – V

Chemical Composition Requirements for Welding Consumables

Element	Limits
C	0.15 % max.
Mn	1.40% max.
P	0.020% max.
S	0.012% max.
Cr+Mo	0.50 % max.
Ni+Cu	0.20 % max
Ni	0.20% max.
Si	As per AWS specification.
CE	0.40 max.

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विक्रेता कार्यशाला में सकारात्मक सामग्री पहचान के लिए मानक विनिर्देश

STANDARD SPECIFICATION FOR POSITIVE MATERIAL IDENTIFICATION (PMI) AT SUPPLIER'S WORKS

Rev. No	Date	Purpose	Prepared by	Checked by	Standards Committee Convenor	Standards Bureau Chairman
					Approved by	
4	23.12.21	REVISED AND RE-ISSUED	HKM	CS	RK	S Majumdar
3	19.09.16.	REVISED AND RE-ISSUED	TKK	HP	RKS	RN
2	20.10.11	REVISED AND RE-ISSUED	RKS	SCG	AKC	DM
1	15.07.08	REVISED AND RE-ISSUED	NKR	SSL	SKP	VC
0	07.12.00	ISSUED AS STANDARD SPECIFICATION	AKC	AKB	AKB	MI

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Abbreviations:

API	:	American Petroleum Institute
AS	:	Alloy Steel
HIC	:	Hydrogen Induced Cracking
LSTK	:	Lump Sum Turn Key
NACE	:	National Association of Corrosion Engineers
PMI	:	Positive Material Identification
RTJ	:	Ring Type Joint
Sch	:	Schedule
SS	:	Stainless Steel
TPI or TPIA	:	Third Party Inspection Agency

Inspection Standards Committee

Convenor: Mr. Rajeev Kumar

Members: Mr. Rajesh Sinha
Mr. Himangshu Pal
Mr. R. Muthuramalingam (RPO Representative)
Mr. Chandrashekhar
Mr. Avdhesh Agrawal
Mr. P V Satyanarayana (Engg. Representative)
Mr. Mahendra Mittal

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CONTENTS

1.0 SCOPE 4
2.0 REFERENCE DOCUMENTS 4
3.0 DEFINITIONS 4
4.0 PMI EXAMINATION 5
5.0 ACCEPTABLE METHODS FOR PMI 5
6.0 EXTENT OF PMI EXAMINATION 6
7.0 ACCEPTANCE CRITERIA 6
8.0 REJECTION CRITERIA 7
9.0 RECORDING AND DOCUMENTATION 7
10.0 MARKING 7

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1.0 SCOPE

1.1 This specification applies to the requirements for Positive Material Identification (PMI) to be performed at the Supplier's works on Metallic Materials procured either directly by the Owner/EIL/ LSTK contractor or indirectly through the sub-Suppliers.

1.2 This specification covers the procedures and methodology to be adopted to assure that the chemical composition of the material is consistent with the material specifications as specified in purchase documents using 'Alloy Analyzer' at the time of final inspection before dispatch.

1.3 The scope of this specification shall include but shall not be limited to Positive Material Identification (PMI) to be performed on Materials listed below:

1.3.1 For alloy Steel materials as below:

- Alloy Steel Pipes including Clad Pipes
- Alloy Steel Flanges & Forgings
- Alloy Steel Fittings including Clad Fittings
- Alloy Steel Fasteners
- Alloy Cast & Forged steel valves
- Alloy Steel Instrumentation Items (Control Valves, Safety Valves etc.)
- Longitudinal Pipe & Fittings Welds.
- Gaskets (for Ring Type Joints)

1.3.2 For Carbon Steel materials as below:

- All Carbon Steel Piping items under NACE or HIC or H2 or Wet Hydrogen Sulfide (H_2S), Hydrofluoric acid (HF), Sulfuric acid (H_2SO_4) services etc.
- Carbon Steel flanges and valves (Rating 900# and above)
- LTCS items

Following items shall be excluded from scope of PMI examination.

- Gaskets other than for Ring Type Joints
- Internal Components of Valves

1.4 All grades of material supplies including Stainless Steels shall be liable for PMI test at site. In case of any defective materials being found at site, the Supplier shall be responsible to effect replacement of such defective materials at project site without any delays to the satisfaction of EIL site RCM (Resident Construction Manager).

2.0 REFERENCE DOCUMENTS

2.1 API Recommended Practice 578 (Third Edition, Feb 2018) - Material Verification Program for New and Existing Alloy Piping Systems.

3.0 DEFINITIONS

3.1 **Supplier:** Any Supplier or Manufacturer on whom an order is placed for the supply of referred items. This definition shall also include any sub-Supplier or manufacturer on whom a sub-order is placed by the Supplier.

3.2 **Inspection Lot:** A group of items offered for inspection covered under same size, Heat and Heat treatment lot.

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3.3 **Alloy Material:** Any metallic material (including welding filler materials) that contains alloying elements such as Chromium, Nickel, Molybdenum or Vanadium, which are intentionally added to enhance mechanical or physical properties and/or corrosion resistance.

4.0 PMI EXAMINATION

4.1 The Supplier shall submit a procedure of PMI to comply with the requirements of this Specification. Approval of PMI Procedure shall be obtained from Owner / EIL / TPIA prior to commencing manufacture / inspection of product.

4.2 PMI examination of materials is independent of any certification, markings or colour coding that may exist and is aimed at verifying that the alloy used are as per specified grades.

4.3 The Supplier shall identify all incoming alloy materials and maintain full traceability of all alloy
materials, including all off-cuts. Transfer of identification marks shall be undertaken prior to cutting to ensure maintenance of identification on off-cuts.

4.4 The Supplier shall ensure that all alloy materials are segregated and stored in separately identified locations to prevent the mix up of materials of different alloy specifications or alloy material with carbon steel. Non ferro-magnetic materials shall be segregated at all times from ferro-magnetic materials.

4.5 PMI examination is subject to surveillance inspection by Owner / EIL / TPIA.

5.0 ACCEPTABLE METHODS FOR PMI

5.1 The method used for PMI examination shall provide a quantitative determination of the alloying elements like Chromium, Nickel, Molybdenum or Vanadium in Alloy Steel items.

5.2 Instruments or methods used for PMI examination shall be able to provide quantitative, recordable, elemental composition results for positive identification of alloying elements present.

5.3 The acceptable instruments for alloy analyzer shall be either "Portable X-Ray fluorescence" or "Optical Emission" type each capable of verifying the percentage of alloy elements within specified range.

5.4 Chemical spot testing, magnets, alloy sorters and other methods using eddy current or triboelectric testing methods are not acceptable for PMI examination.

5.5 The PMI instrument used shall have the sensitivity to detect the alloying elements in the specified range.

5.6 All PMI instruments shall have been serviced within a 6 month period of the time of use to verify the suitability of batteries, sources etc., and the data of the last service shall be stated on the PMI Report Form (Sample enclosed).

5.7 Each analyzer must be calibrated according to the manufacturer's specification at the beginning and end of each shift. Instrument must be checked against known standard for each alloy type to be inspected during the shift.

5.8 Certified samples, with full traceability, of a known alloy materials shall be available for use as a random spot check on the instrument calibration.

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5.9 The surfaces to be examined shall be prepared by light grinding or abrasive paper and solvent cleaner. Evidence of Arc burn resulting from examination shall be removed by light grinding or abrasive paper. No permanent marks, which are injurious to the usage of product in service, are acceptable.

5.10 Alloy Steel ring type joint Gaskets shall be inspected by using portable X-Ray fluorescence instrument.

5.11 Testing shall be done as per the procedures outlined by the manufactures of alloy analyzer being used. Modification of these procedures if any must be approved by Owner/EIL.

5.12 The persons performing PMI shall demonstrate their capabilities to the satisfaction of Owner/EIL/TPIA visiting engineer. If the Supplier has qualified operator on their rolls, he may perform the examination. Otherwise PMI examination shall be sub-contracted to an independent testing agency approved by EIL.

5.13 Whenever material is identified as not meeting requirements by the visiting engineer a rejection note shall be issued.

6.0 EXTENT OF PMI EXAMINATION

Following sampling plans shall be applicable for PMI examination of various items.

A. Flanges, Fittings Valves, RTJ Gaskets - 100% by Supplier and 10% @ random witness by EIL/TPIA

B. Pipes - 100% (for pipes procured from traders)
2 random samples drawn from each Size/Heat/Lot (for pipes procured directly from mills)

C. Fasteners -

Lot Size	**Sample Size for** EIL/TPIA	Sample size for Supplier
Upto 100	2% (Min 2)	Min 10%
101 to 500	1% (Min 3)	Min 10%
501 and above	0.5% (Min 5)	Min 10%

Note:

a. For Welded Pipes and Fittings and flanges like Fig 8, Spacer and blinds etc. PMI shall be performed on Base Metal as well as weldments.

7.0 ACCEPTANCE CRITERIA

7.1 Base Metal

PMI test results showing presence of characteristic elements upto 10% less than the minimum specified value in the material specification and upto 10% more than the maximum specified value in the material specification shall be acceptable.

7.2 Deposited Weld Metal

For deposited weld metal using welding consumables matching with base metals, the recorded presence of characteristic elements upto 12.5% less than the minimum specified value in the

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welding consumable specification and upto 12.5% more than the maximum specified value in the welding consumable specification shall be acceptable.

8.0 REJECTION CRITERIA

8.1 If PMI test results fall outside the acceptable range as specified in 7.0 above, the supplier shall obtain a quantitative check analysis performed by a laboratory acceptable to Owner / EIL / TPIA for a complete chemical analysis. Alternatively, the item can be tested with a spark analyser for verification. Results of this analysis shall be submitted to Owner / EIL / TPIA for final decision.

Decision of Owner / EIL / TPIA shall be final in this regard.

8.2 If any sample drawn to PMI test on the basis of percentage selection as per 6.0 above, fails to meet specification requirements, 100% of items of lot shall be tested for PMI by supplier. Any failure thereafter during sample check shall mean rejection of whole lot.

9.0 RECORDING AND DOCUMENTATION

The results of PMI examination shall be recorded in a Report Format as enclosed with this specification.

10.0 MARKING

10.1 All alloy materials tested by PMI shall be identified using either of the following methods by indicating "PMI OK"

a) Bar Code/Hologram Sticker
b) A low stress stamp marking

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	POSITIVE MATERIAL IDENTIFICATION REPORT BULK MATERIALS								Page of
Project:			Client						Job No.
PMI Report No.			Supplier/Sub-Supplier						
Purchase Order No.			Testing Agency						
Purchase Requisition No:			PMI Location						
Bulk Item Type (as per Requisition)									
Material Specification/Grade Number of items in Lot									
Requisition Item No./ Description			Major content, Weight Percent						Remarks Accept/Reject
Element	Cr	Ni	Mo	V	Ti *	Cb/ Nb **	Mn ***		
Specified Range									
Actual observations 1.									
2.									
3.									
4.									
5.									
6.									
7.									
8.									
Instrument Type / ID Last Service Date		Inspection Agency							Witnessed By

* To be reported in case of SS321 Material
** To be reported in case of SS347 Material
*** To be reported in case of LTCS Material

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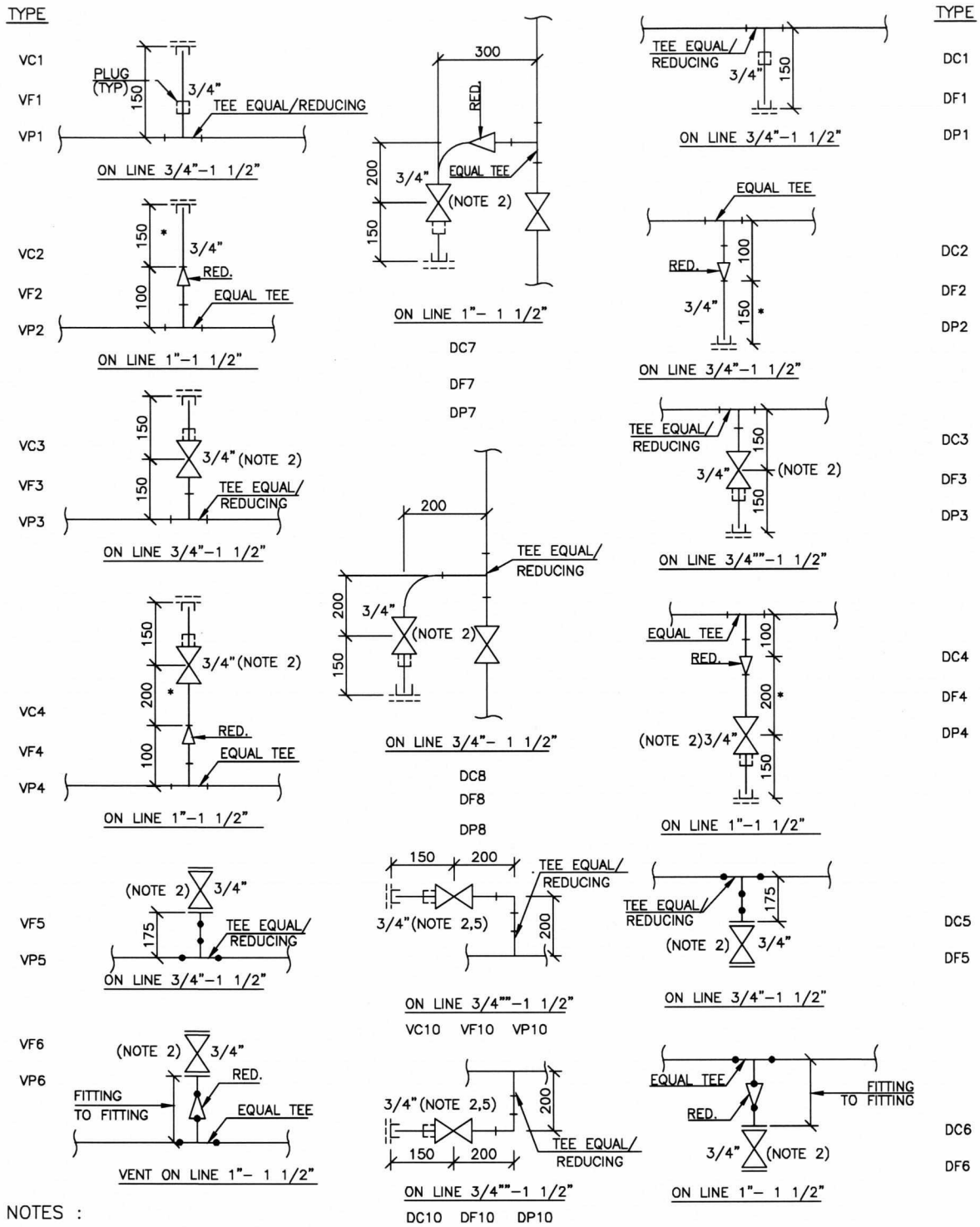

NOTES :

1. DIMENSIONS INDICATED ARE VALID FOR 50mm(MAX) INSULATION. FOR HIGHER INSULATION THICKNESS, INCREASE DIMENSIONS AS REQUIRED. DIMENSIONS MARKED '*' ARE MAXIMUM AND MAY BE REDUCED TO SUIT.
2. VALVES TO BE PROVIDED SHALL BE SIMILAR TO LINE ISOLATION VALVE (GATE, BALL OR PLUG VALVE ETC., WITH FLGD, S.W. OR SCR'D ENDS) UNLESS OTHERWISE SPECIFIED IN PMS, BRANCH–OFFS LIKE TEES(EQUAL OR REDUCING)/HALF COUPLINGS S.W. OR SCR'D)/SOCKOLETS/WELDOLETS AND END CONNECTIONS LIKE CAP/PLUG/FLANGE & BLIND FLANGE SHALL BE AS PER PIPING MATERIAL SPECIFICATION.
3. LEGEND : V = VENT; D = DRAIN; C = CAP; F = FLANGE; RED. = REDUCER, COUPLING OR SWAGE; P = PLUG.
4. PLUGGED END OF VALVE OR FITTING SHALL BE THREADED.
5. VALVE STEM ORIENTATION TO BE AT OR ABOVE 45° ABOVE THE HORIZONTAL POSITION IN LIQUID CRYOGENIC SERVICE.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
6	20.06.22	REVISED & ISSUED AS STANDARD	PK	SH	GB	SM
5	06.12.19	REAFFIRMED & ISSUED AS STANDARD	SG	SH	MI	RKT

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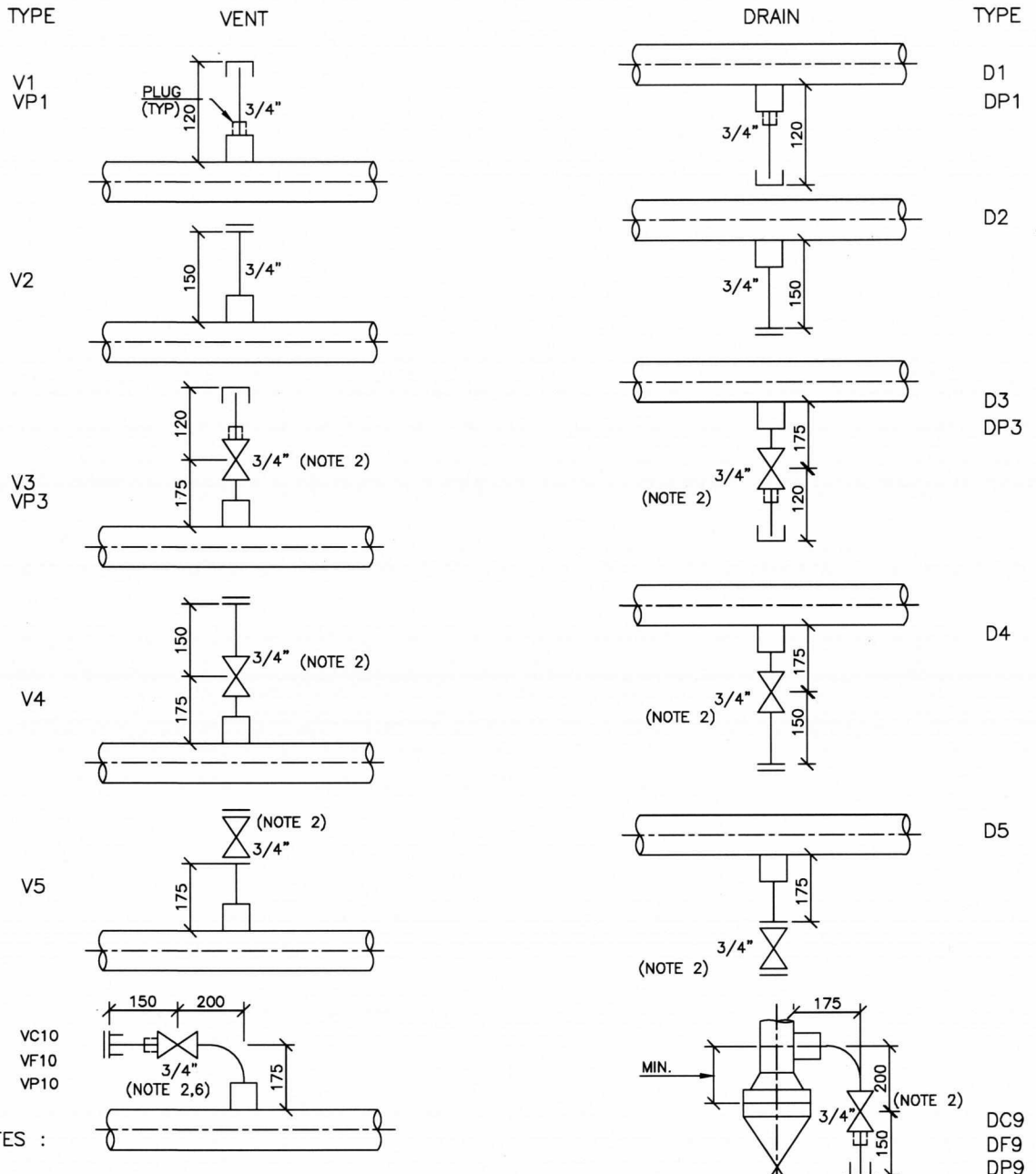

NOTES :

1. DIMENSIONS INDICATED ARE VALID FOR 75mm (MAX) INSULN. THICKNESS. FOR HIGHER INSULATION THICKNESS INCREASE DIMENSION AS REQUIRED
2. VALVES TO BE PROVIDED SHALL BE SIMILAR TO LINE ISOLATION VALVE (GATE, BALL OR PLUG VALVE ETC., WITH FLGD, SW OR SCR'D ENDS) UNLESS OTHERWISE SPECIFIED IN PMS.
   BRANCH-OFFS LIKE HALF CPLG(SW OR SCR'D)/SOCKOLET/WELDOLET & END CONNECTIONS LIKE CAP/PLUG/FLANGE & BLIND FLANGE SHALL BE AS PER PIPING MATERIAL SPECIFICATIONS.
3. VENTS/DRAINS CAN BE PROVIDED ON FLAT SIDE OF ECCENTRIC REDUCER ON SIZES 4" & ABOVE
4. LEGEND V=VENT, D=DRAIN, C=CAP, F=FLANGE, P=PLUG
5. PLUGGED END OF VALVE OR FITTING SHALL BE THREADED
6. FOR LIQUID CRYOGENIC SERVICE VALVE STEM ORIENTATION TO BE AT OR ABOVE 45° ABOVE HORIZONTAL POSITION.

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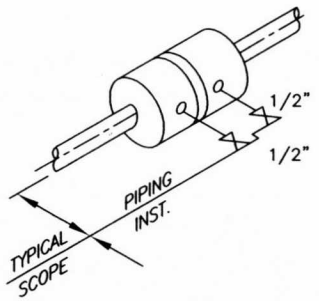
ORIENTATION OF ORIFICE TAPS (CLASS 600 & BELOW)

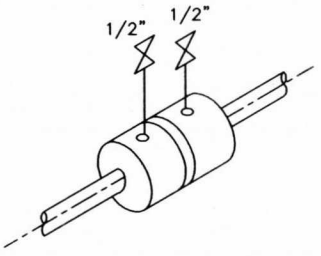
FOR LIQUID

DETAIL– ①



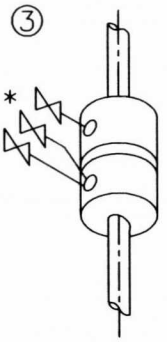
FOR GAS LINES & STEAM LINES (NOTE–10)

DETAIL– ②



FOR VERTICAL LINES

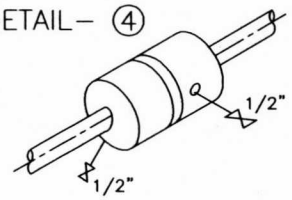
DETAIL– ③



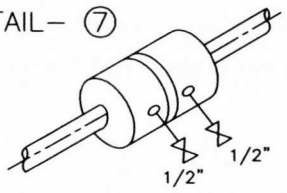
* FOR STEAM & CONDENSIBLES, LOWER BLOCK VALVE SHALL BE RAISED TO THE LEVEL OF THE UPPER BLOCK VALVE

FOR LIQUID LINES ON PIPE RACK WITH OPERATING TEMP > = 0°C

DETAIL– ④ NOTE–5

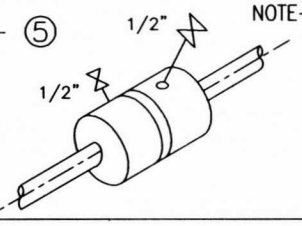


DETAIL– ⑦

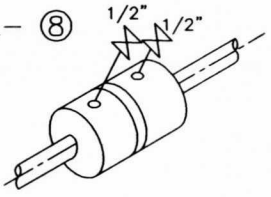


FOR STEAM & GAS LINES ON PIPE RACK & LIQUID LINES WITH TEMP BELOW 0°C

DETAIL– ⑤ NOTE–5

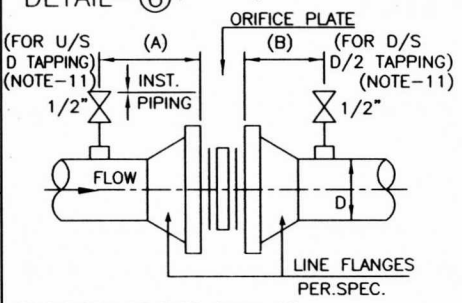


DETAIL– ⑧



FOR TAPS ON PIPE RACK (150# & 300#)

DETAIL– ⑥ (NOTE–3)



ORIENTATON OF TAPS SHALL BE
LIQUID LINES WITH OPERATING TEMP. >= 0°C : 45° FROM VERTICAL DOWNWARDS
STEAM, GAS & LIQUID LINES WITH OP.TEMP < 0°C : 45° FROM VERTICAL UPWARDS

INSTRUMENT SCOPE

A. ORIFICE FLANGES ASSEMBLY WITH BOLTS/NUTS AND GASKETS
B. IMPULSE LINE STARTING AFTER THE TAPPING VALVE

PIPING SCOPE

A. TAPPING NIPPLE, COUPLINGS AND VALVES

NOTES :

1. ORIFICE FLANGES IN HORIZONTAL LINES SHALL BE PREFERED.
2. TAPS SHALL BE ORIENTED TO SUIT LOCATION OF INSTRUMENT.
3. TAPPING ON PIPE AS PER DETAIL–6 SHALL BE CONSIDERED IN ACCORDANCE WITH THE COUPLING POSITIONS FROM UPSTREAM AND DOWNSTREAM SIDE FLANGE FACES FOR LINE OF :–
   A. RATING 150# AND SIZE >=16"
   B. RATING 300# AND SIZE >=20"
   FOR ALL OTHER LINE SIZES AND RATINGS. TAP SHALL BE CONSIDERED ON ORIFICE FLANGE FOR ORIFICE ASSEMBLIES BY INSTRUMENTATION.
4. IN CASE OF VERTICAL LINES FLOW SHALL BE UPWARDS FOR LIQUIDS CONTAINING VAPOUR & DOWNWARDS FOR WET GASES & STEAM.
5. THE ORIENTATION OF TAPPING IN CASE OF DETAILS 4 & 5 SHALL BE AS FOLLOWS:
   FOR LIQUID LINES WITH OPERATING TEMPERATURE >= 0°C :
   4,8 &16 BOLT FLANGE – AT 45° TO VERTICAL POINTING DOWNWARDS
   12 BOLT FLANGE – AT 60° TO VERTICAL POINTING DOWNWARDS
   20 BOLT FLANGE – AT 54° TO VERTICAL POINTING DOWNWARDS.
   FOR LIQUID LINES WITH OPER. TEMP. < 0°C AND STEAM AND GAS LINES :
   4,8 & 16 BOLT FLANGE – AT 45° TO VERTICAL POINTING UPWARDS,
   12 BOLT FLANGE – AT 60° TO VERTICAL POINTING UPWARDS
   20 BOLT FLANGE – AT 54° TO VERTICAL POINTING UPWARDS.
6. DETAILS 4 & 5 SHALL NOT BE USED FOR PRE–FABRICATED HOOK–UPS.
7. FOR CLEARANCE REQUIREMENT AROUND ORIFICE FLANGE REFER EIL STD. 7–44–0504
8. FOR DETAILS OF ORIFICE TAPPINGS AND ORIENTATION, REFER EIL STD. 7–52–0005.
9. IN CASE OF PRE–FABRICATED HOOKUP, NIPPLE & ISOLATION VALVE SHALL BE SUPPLIED BY PREFABRICATION VENDOR.
10. FOR STEAM LINES, DETAIL–1 CAN BE CONSIDERED IN CASE OF SPACE CONSTRAINTS OR AS PER LICENSOR REQUIREMENT.
11. FOR DIMENSIONS A & B REFER STD. 7–52–0022.
12. FOR LIQUID CRYOGENIC SERVICE VALVE STEM ORIENTATION TO BE AT OR ABOVE 45° ABOVE HORIZONTAL POSITION.

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6	20.06.22	REVISED & ISSUED AS STANDARD	PK	SH	GB	SM
5	24.01.20	REVISED & ISSUED AS STANDARD	SG	SH	MI	RKT

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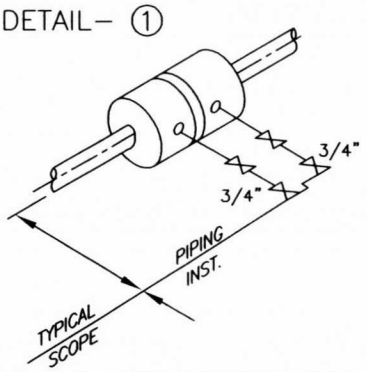
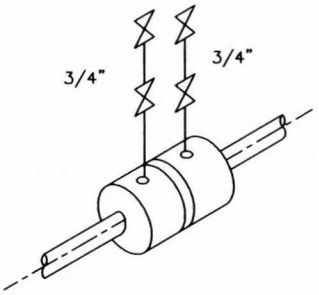
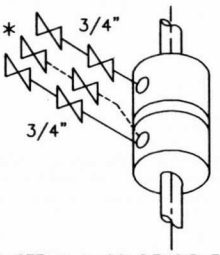
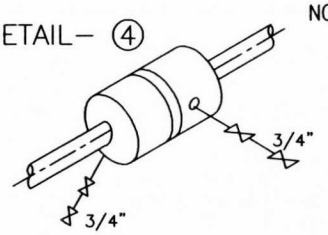
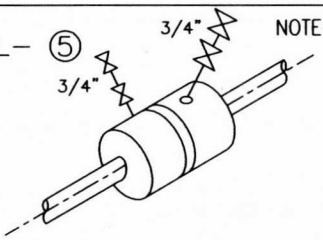
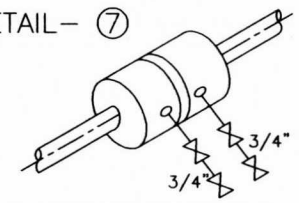
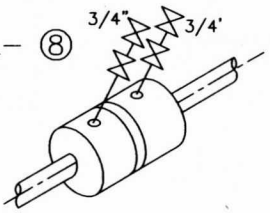
FOR LIQUID	FOR GAS LINES & STEAM LINES (NOTE-10)	FOR VERTICAL LINES
DETAIL– ① 	DETAIL– ② 	DETAIL– ③  * FOR STEAM & CONDENSIBLES, LOWER BLOCK VALVE SHALL BE RAISED TO THE LEVEL OF THE UPPER BLOCK VALVE
FOR LIQUID LINES ON PIPE RACK WITH OPERATING TEMP > = 0°C	FOR STEAM & GAS LINES ON PIPE RACK & LIQUID LINES WITH TEMP BELOW 0°C	
DETAIL– ④ NOTE-5 	DETAIL– ⑤ NOTE-5 	
DETAIL– ⑦ 	DETAIL– ⑧ 	

INSTRUMENT SCOPE

A. ORIFICE FLANGES ASSEMBLY WITH BOLTS/NUTS AND GASKETS
B. IMPULSE LINE STARTING AFTER THE TAPPING VALVE

PIPING SCOPE

A TAPPING NIPPLE, COUPLINGS AND VALVES

NOTES :

1. ORIFICE FLANGES IN HORIZONTAL LINES SHALL BE PREFERED.
2. TAPS SHALL BE ORIENTED TO SUIT LOCATION OF INSTRUMENT.
3. DELETED.
4. IN CASE OF VERTICAL LINES FLOW SHALL BE UPWARDS FOR LIQUIDS CONTAINING VAPOUR & DOWNWARDS FOR WET GASES & STEAM.
5. THE ORIENTATION OF TAPPING IN CASE OF DETAILS 4 & 5 SHALL BE AS FOLLOWS:
   FOR LIQUID LINES WITH OPERATING TEMPERATURE >= 0°C :
   4,8 &16 BOLT FLANGE – AT 45° TO VERTICAL POINTING DOWNWARDS
   12 BOLT FLANGE – AT 60° TO VERTICAL POINTING DOWNWARDS
   20 BOLT FLANGE – AT 54° TO VERTICAL POINTING DOWNWARDS.
   FOR LIQUID LINES WITH OPER. TEMP. < 0°C AND STEAM AND GAS LINES :
   4,8 & 16 BOLT FLANGE – AT 45° TO VERTICAL POINTING UPWARDS,
   12 BOLT FLANGE – AT 60° TO VERTICAL POINTING UPWARDS
   20 BOLT FLANGE – AT 54° TO VERTICAL POINTING UPWARDS.
6. DETAILS 4 & 5 SHALL NOT BE USED FOR PRE–FABRICATED HOOK–UPS.
7. FOR CLEARANCE REQUIREMENT AROUND ORIFICE FLANGE REFER EIL STD. 7–44–0504
8. FOR DETAILS OF ORIFICE TAPPINGS AND ORIENTATION. REFER EIL STD. 7–52–0005.
9. IN CASE OF PRE–FABRICATED HOOKUP, NIPPLE & ISOLATION VALVE SHALL BE SUPPLIED BY PREFABRICATION VENDOR.
10. FOR STEAM LINES, DETAIL–1 CAN BE CONSIDERED IN CASE OF SPACE CONSTRAINTS OR AS PER LICENSOR REQUIREMENT.
11. FOR LIQUID CRYOGENIC SERVICE VALVE STEM ORIENTATION TO BE AT OR ABOVE 45° ABOVE HORIZONTAL POSITION.

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5	24.01.20	REVISED & ISSUED AS STANDARD	SG	SH	MI	RKT

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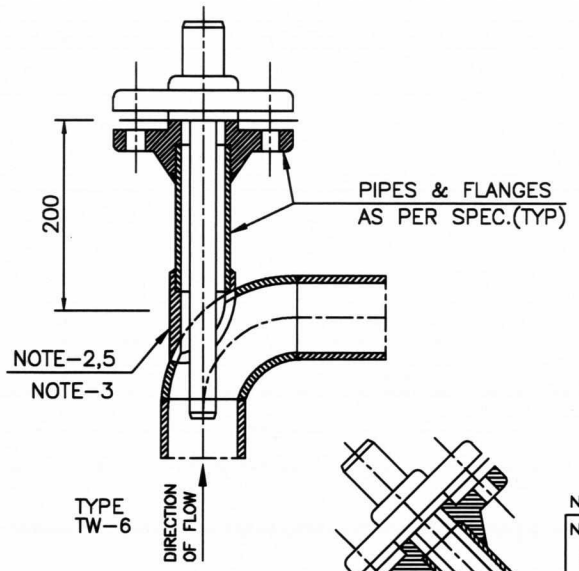

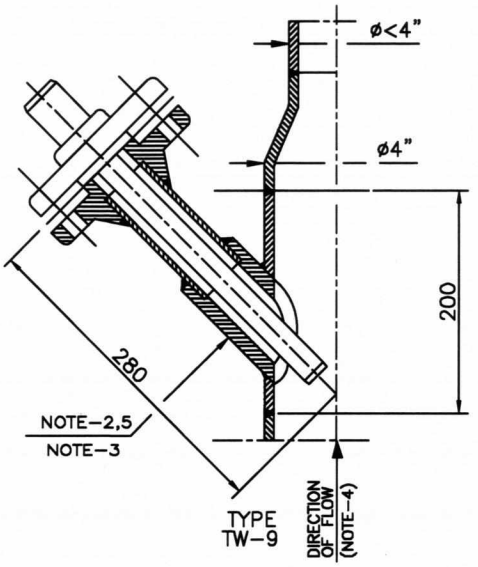

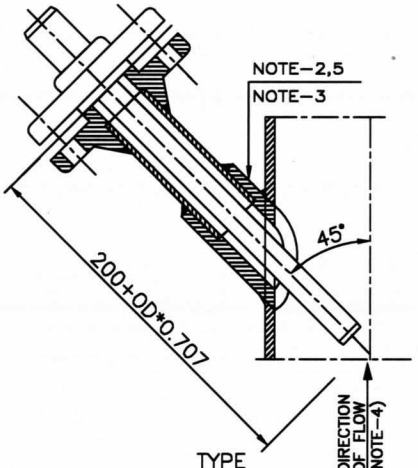

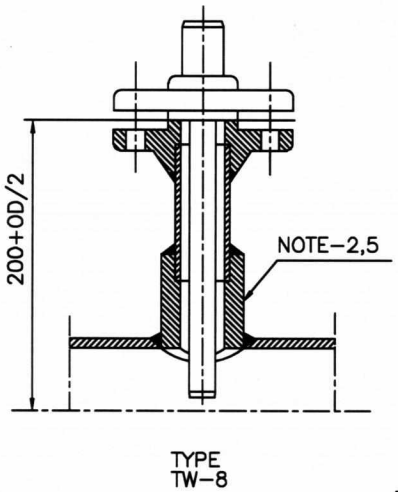

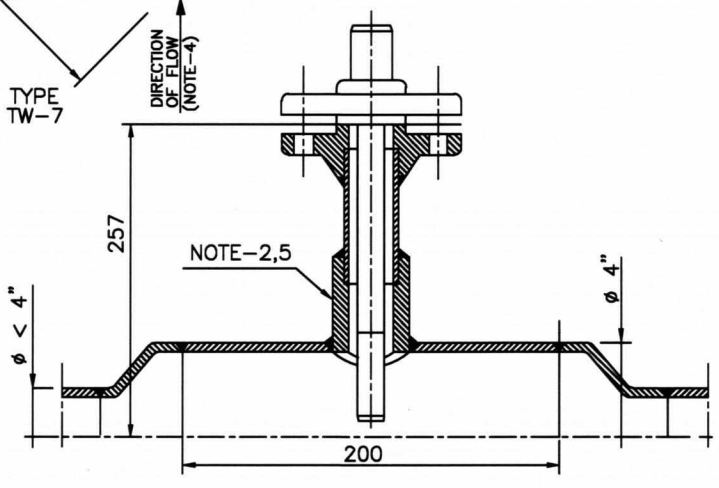

LINE DIA.	MINIMUM CLEARANCE FOR REMOVAL
4"	630
6"	630
8"	670
10"	670
12"	670
14"	670
16"	670
18"	670
20" AND LARGER	750
VESSELS	750

NOTES:–

1. BOLTS, NUTS AND GASKETS BY PIPING.
2. BRANCH FITTING SHALL BE AS PER PMS
3. IF BRANCH FITTING IS A COUPLING, IT SHALL BE OF SPECIAL LENGTH.
4. THE ARRANGEMENT CAN BE USED FOR DOWNWARD FLOW ALSO, IF CONFIRMED BY INSTRUMENTATION.
5. SIZE AND DETAIL OF TAPPING SHALL BE AS PER APPLICABLE SPECIFICATION FOR CLADDED/LINED/JACKETED PIPING.

OD : OUTSIDE DIA IN MM
TYPE TW–6 : ELBOW MIN. 4" DIA. OR LARGER
TYPE TW–7 : VERTICAL LINE 4" DIA. OR LARGER
TYPE TW–8 : HORIZONTAL LINE 4" DIA. OR LARGER
TYPE TW–9 : VERTICAL LINE DIA. LESS THAN 4"
TYPE TW–10: HORIZONTAL LINE DIA. LESS THAN 4"

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Approved by
6	20.06.22	REVISED & ISSUED AS STANDARD	PK	SH	GB	SM
5	24.01.20	REVISED & ISSUED AS STANDARD	SG	SH	MI	RKT

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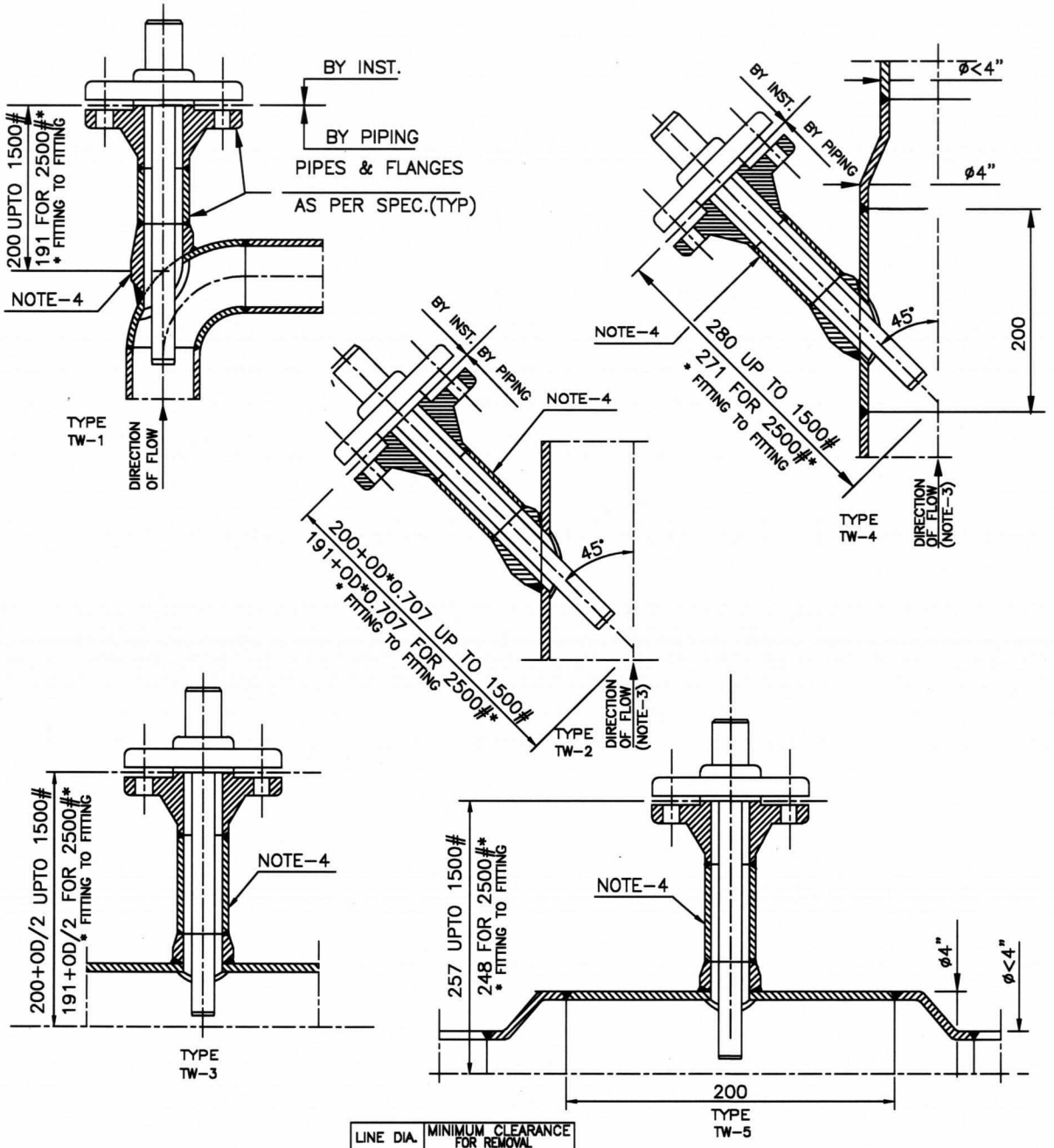

LINE DIA.	MINIMUM CLEARANCE FOR REMOVAL
4"	630
6"	630
8"	670
10"	670
12"	670
14"	670
16"	670
18"	670
20" & ABV	750
VESSELS	750

NOTES:-

1. BOLTS, NUTS AND GASKETS BY PIPING.
2. BRANCH FITTING SHALL BE AS PER PMS.
3. THE ARRANGEMENT CAN BE USED FOR DOWNWARD FLOW ALSO, IF CONFIRMED BY INSTRUMENTATION.
4. SIZE AND DETAIL OF TAPPING SHALL BE AS PER APPLICABLE SPECIFICATION FOR CLADDED/LINED/JACKETED PIPING.

OD : OUTSIDE DIA IN MM

TYPE TW-1 : ELBOW MIN. 4" ø OR LARGER
TYPE TW-2 : VERTICAL LINE 4" ø OR LARGER
TYPE TW-3 : HORIZONTAL LINE 4" ø OR LARGER
TYPE TW-4 : VERTICAL LINE DIA. LESS THAN 4"
TYPE TW-5 : HORIZONTAL LINE DIA. LESS THAN 4"

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
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5	24.01.20	REVISED & ISSUED AS STANDARD	SG	SH	MI	RKT

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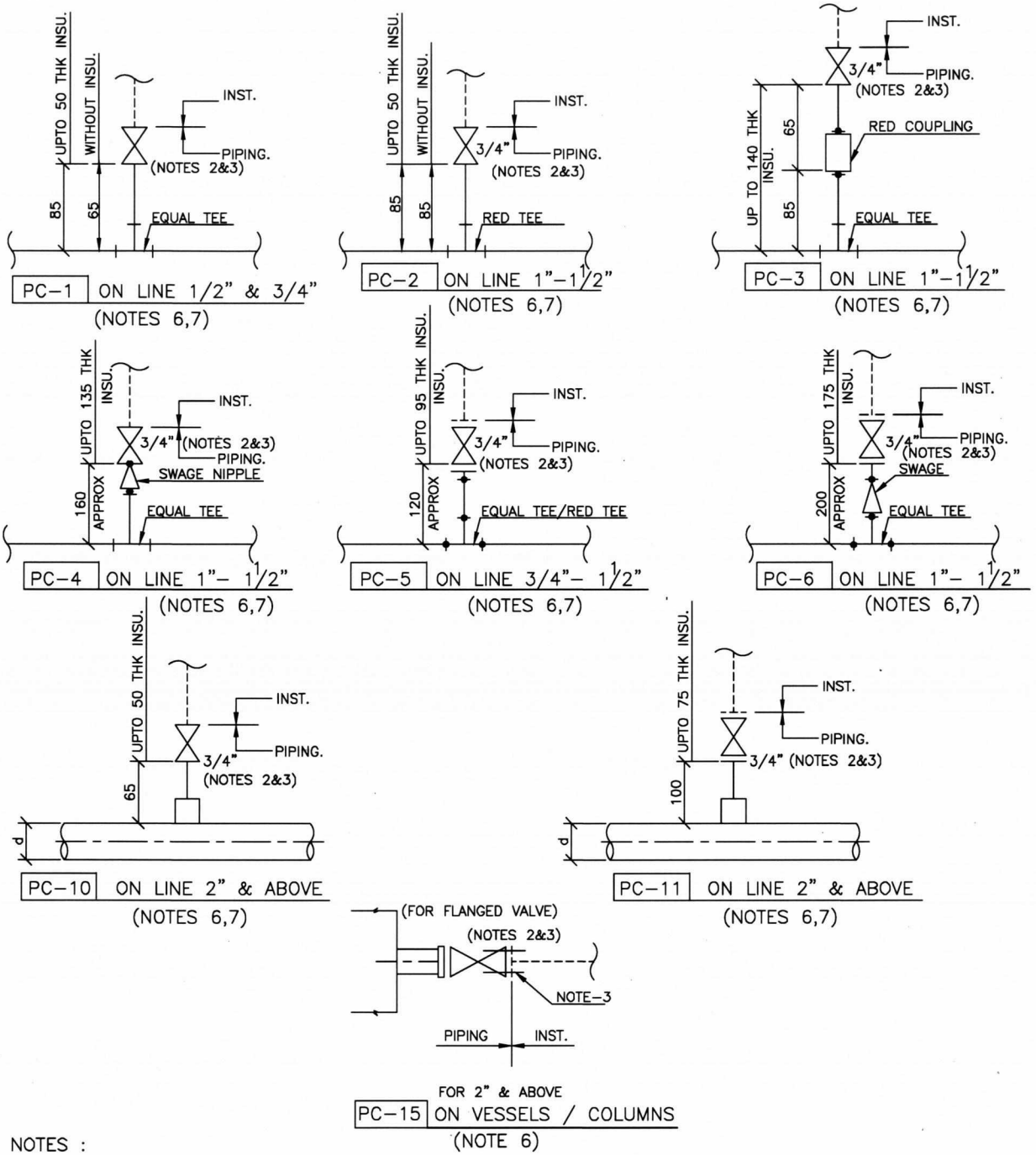

NOTES :

1. THE INDICATED DIMENSIONS ARE MINIMUM WHICH ALSO COVER INSULATION TO THE EXTENT SHOWN ABOVE. IN CASE OF HIGHER THICKNESS OF INSULATION THAN INDICATED THE DIFFERENCE SHALL BE ADDED IN THE DIMENSIONS SHOWN ABOVE.
2. PRESSURE TAPPING SHALL BE PROVIDED WITH VALVE SIMILAR TO LINE ISOLATION VALVE (GATE, BALL OR PLUG VALVE ETC., WITH FLGD, S.W. OR SCR'D ENDS) UNLESS OTHERWISE SPECIFIED IN PMS, TEE(EQUAL OR REDUCING)/HALF COUPLING(S.W. OR SCR'D)/STUB IN/SOCKOLET/WELDOLET SHALL BE AS PER PIPING MATERIAL SPECS. DOUBLE ISOLATION VALVE SHALL BE USED FOR CLASS 900 & ABOVE.
3. IN CASE OF FLGD VALVES BOLTING & GASKET ON BOTH SIDES OF VALVE SHALL BE IN PIPING SCOPE
4. IN CASE OF TAPPING PROVIDED OTHER THAN INDICATED IN THIS STD FOR LAYOUT REASONS DETAILED DIMENSIONS WILL BE CALLED OUT.
5. IN CASE OF PRE-FABRICATED HOOKUP, ISOLATION VALVE WITH NIPPLE SHALL BE SUPPLIED BY PREFABRICATION VENDOR HOWEVER, 3/4" WELDOLET/SOCKOLET/HALF COUPLING/EQUAL TEE/RED. TEE ETC. AS PER PMS SHALL BE IN PIPING SCOPE.
6. VALVE STEM ORIENTATION TO BE AT OR ABOVE 45° ABOVE THE HORIZONTAL POSITION IN LIQUID CRYOGENIC SERVICE.
7. FOR LIQUID CRYOGENIC SERVICE ADDITIONAL 45°/90° ELBOW TO BE INTRODUCED TO ENSURE REQUIREMENT GIVEN IN NOTE-6 IS MET.

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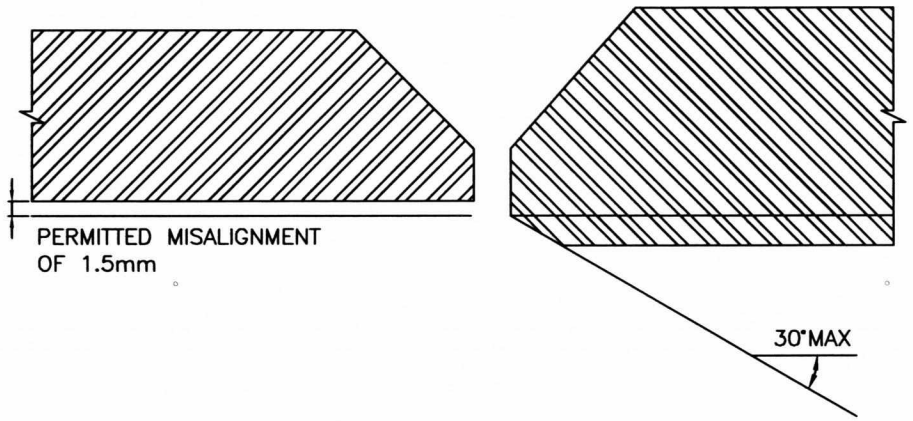

THICKER PIPE TAPER BORED TO ALIGN

(AS PER ASME B31.3)

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
6	28.09.23	REAFFIRMED AND ISSUED AS STANDARD	ABA	SH	GB	SM
5	10.09.18	REAFFIRMED AND ISSUED AS STANDARD	PK	SH	MI	RKT

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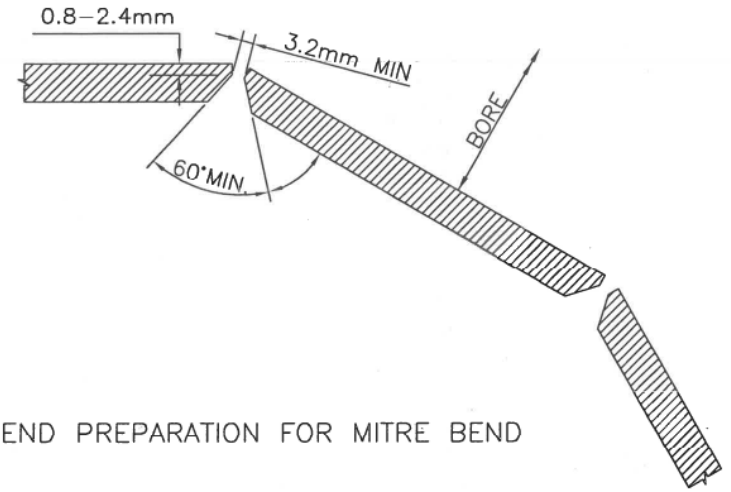
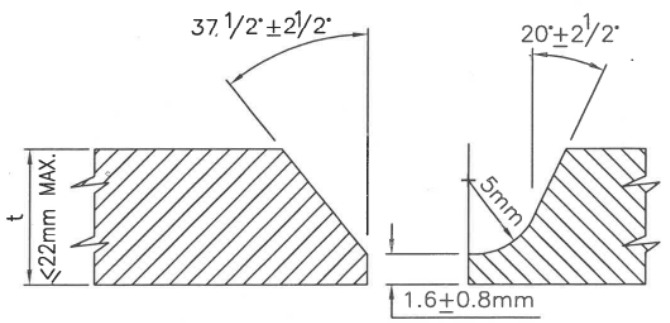
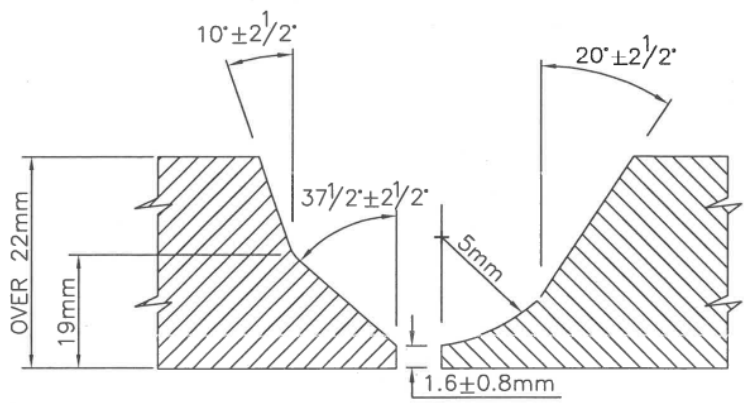


END PREPARATION FOR MITRE BEND



(A) WALL THICKNESS 5–22mm INCLUSIVE
(AS PER ASME B31.3)



(B) WALL THICKNESS OVER 22mm
(AS PER ASME B31.3)

NOTES :

1. SQUARE ENDS MAY BE USED FOR WALL THICKNESS LESS THAN 5mm
2. PIPE ENDS SUPPLIED WITH BEVEL ANGLE OF $30^{\circ}{}^{+5^{\circ}}_{0}$ NEED NOT BE BEVELED FURTHER.

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6	04.03.25	REAFFIRMED & ISSUED AS STANDARD	PK	SH	GB	MN
5	06.12.19	REAFFIRMED & ISSUED AS STANDARD	SG	SH	MI	RKT

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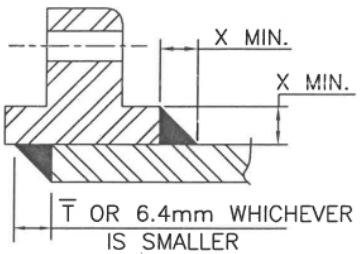
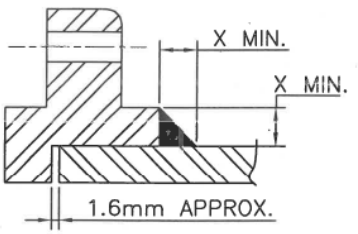


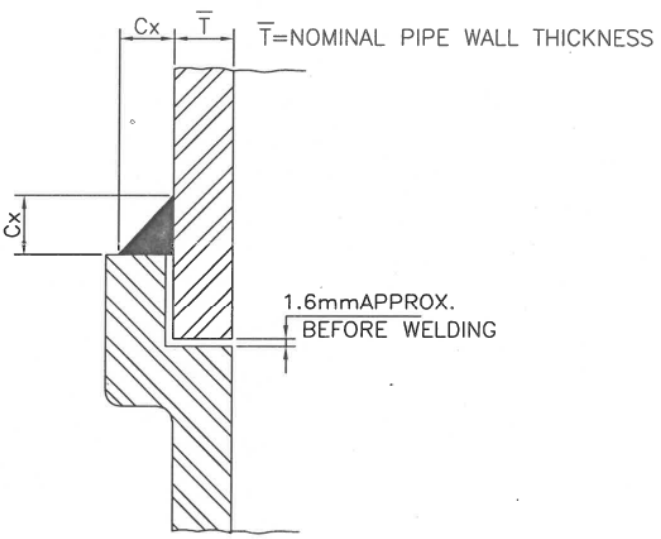
SLIP ON FLANGE



SOCKET WELDED FLANGE

X MIN. = $1.4\overline{T}$ OR THE THICKNESS OF HUB, WHICHEVER IS SMALLER.

$\overline{T}$ = NOMINAL PIPE WALL THICKNESS



SOCKET WELDING FITTINGS

Cx (MIN)=$1.09\overline{T}$ BUT NOT LESS THAN 3.2mm

FILLET WELD DETAILS
(AS PER ASME B31.3)

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
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5	06.12.19	REAFFIRMED & ISSUED AS STANDARD	SG	SH	MI	RKT

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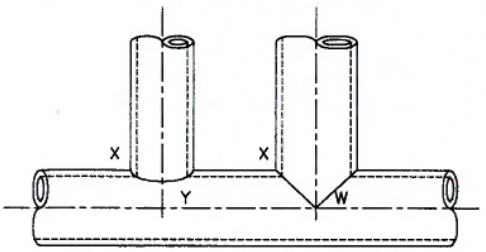
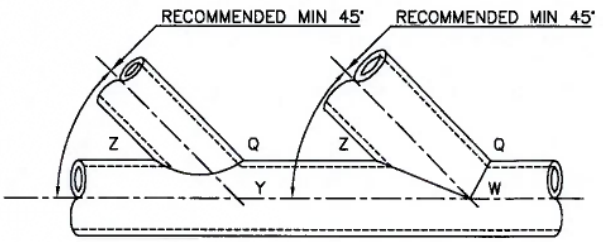
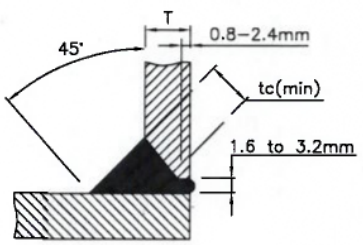
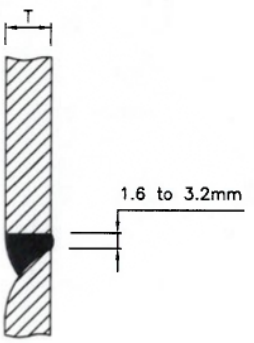




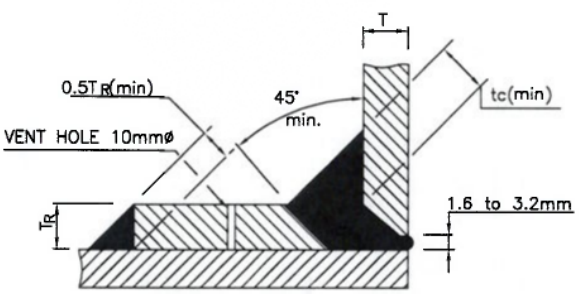


tc: MINIMUM OF 0.7T & 6.4mm

SECTION X AT CROTCH OF RIGHT ANGLE BRANCH

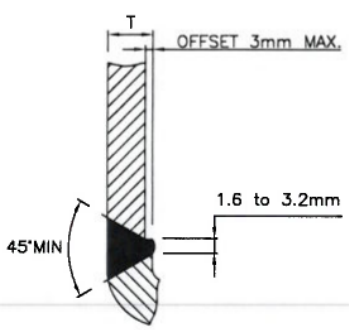


SECTION Y AT FLANK

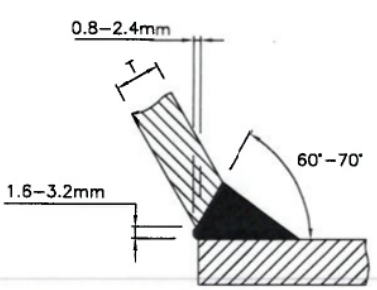


tc: MINIMUM OF 0.7T & 6.4mm

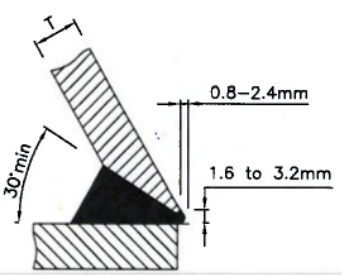
DETAIL FOR REINFORCED BRANCH



SECTION W AT FLANK OF EQUAL BRANCH



SECTION Q AT BACK OF SLOPING BRANCH



SECTION Z AT CROTCH OF SLOPING BRANCH

NOTES :–

1. THIS TYPE IS TO BE USED WHEN BRANCH IS EQUAL TO OR ONE SIZE SMALLER THAN MAIN PIPE.
2. REFER STANDARD 7–44–0484 FOR DIMENSIONAL DETAILS.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
6	10.10.25	REAFFIRMED & ISSUED AS STANDARD	SRG	PK	SH	MN
5	23.09.20	REAFFIRMED & ISSUED AS STANDARD	SG	SH	GB	SM

Approved by

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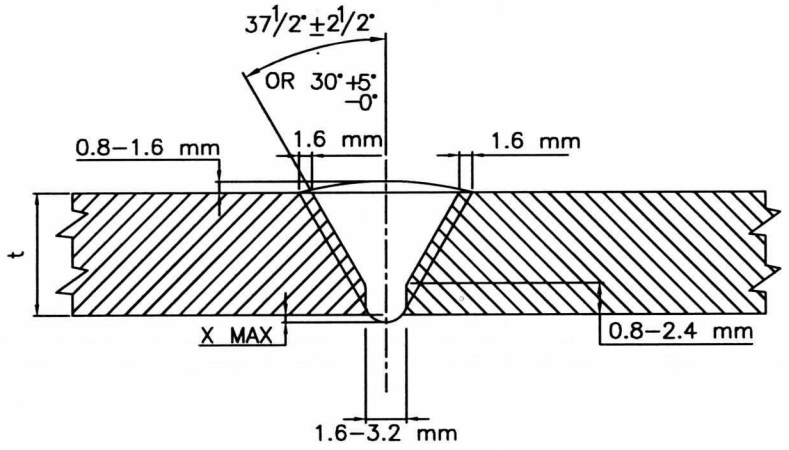

4.8 mm ≤ t ≤ 19.1 mm

X MAX = 1.5 mm FOR PIPES 2" AND BELOW
3.0 mm FOR PIPES ABOVE 2"

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
6	28.09.23	REAFFIRMED AND ISSUED AS STANDARD	ABA	SH	GB	SM
5	10.09.18	REAFFIRMED AND ISSUED AS STANDARD	PK	SH	MI	RKT

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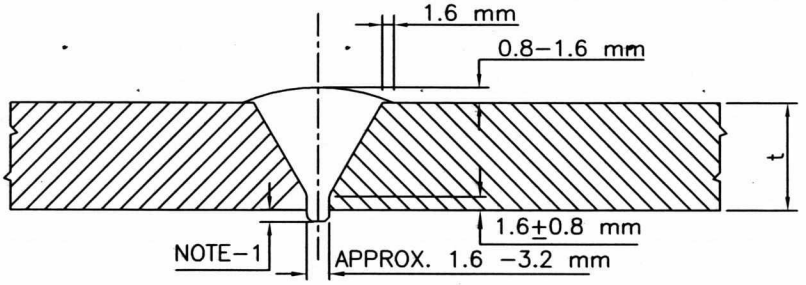
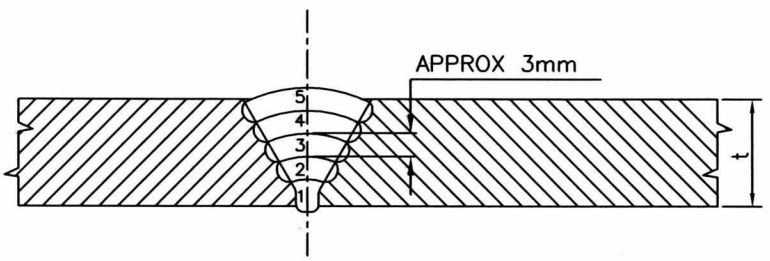


STANDARD 'V' BEVEL
BUTT JOINT



SEQUENCE OF WELDS

$4.8 \text{ mm} \leq t \leq 19.1 \text{ mm}$

NOTES :

1. MAXIMUM INTERNAL WELD PROTRUSION SHALL BE AS PER ASME B31.3

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
6	22.01.24	REAFFIRMED & ISSUED AS STANDARD	PK	SH	~~GB~~	MN
5	31.12.18	REVISED & ISSUED AS STANDARD	PK	SH	MI	RKT

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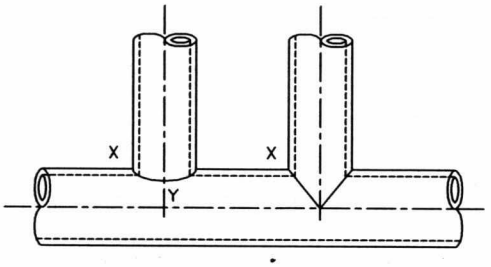
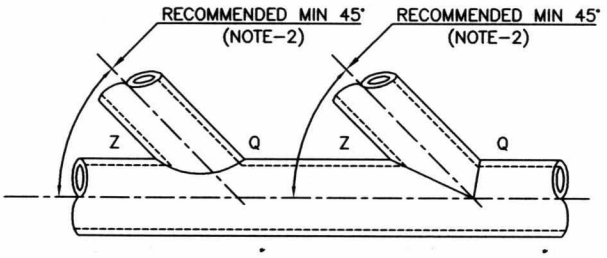
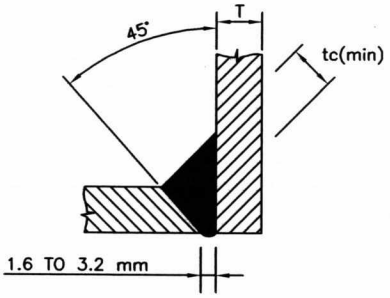
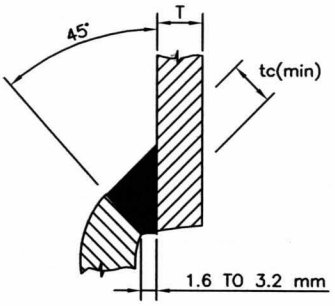






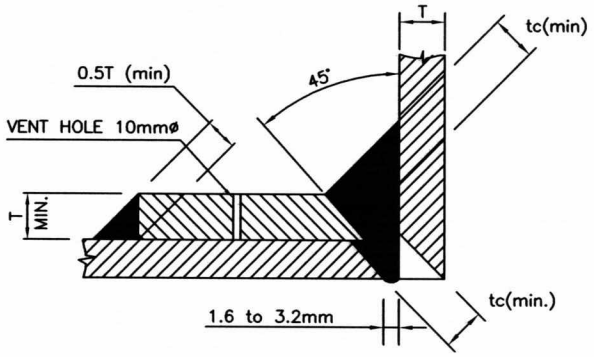
SECTION X AT CROTCH OF RIGHT ANGLE BRANCH



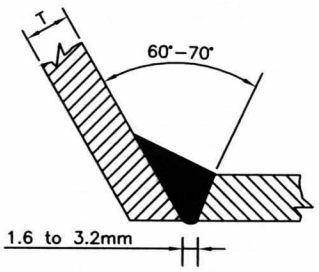
SECTION Y AT FLANK

NOTES :

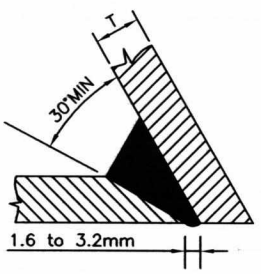
1. THIS STANDARD IS TO BE USED WHEN BRANCH IS MORE THAN ONE SIZE SMALLER OF MAIN PIPE.
2. FOR LINES COMING UNDER THE PURVIEW OF IBR RECOMMENDED MIN. ANGLE SHALL BE 60°.
3. MAXIMUM INTERNAL PROTRUSION OF WELD SHALL BE AS PER ASME B31.3



tc=MIN OF 0.7T & 6.4mm



SECTION Q AT BACK OF SLOPING BRANCH



SECTION Z AT CROTCH OF SLOPING BRANCH

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
6	22.01.24	REVISED & ISSUED AS STANDARD	PK	SH	GB	MN
5	31.12.18	REVISED & ISSUED AS STANDARD	PK	SH	MI	RKT

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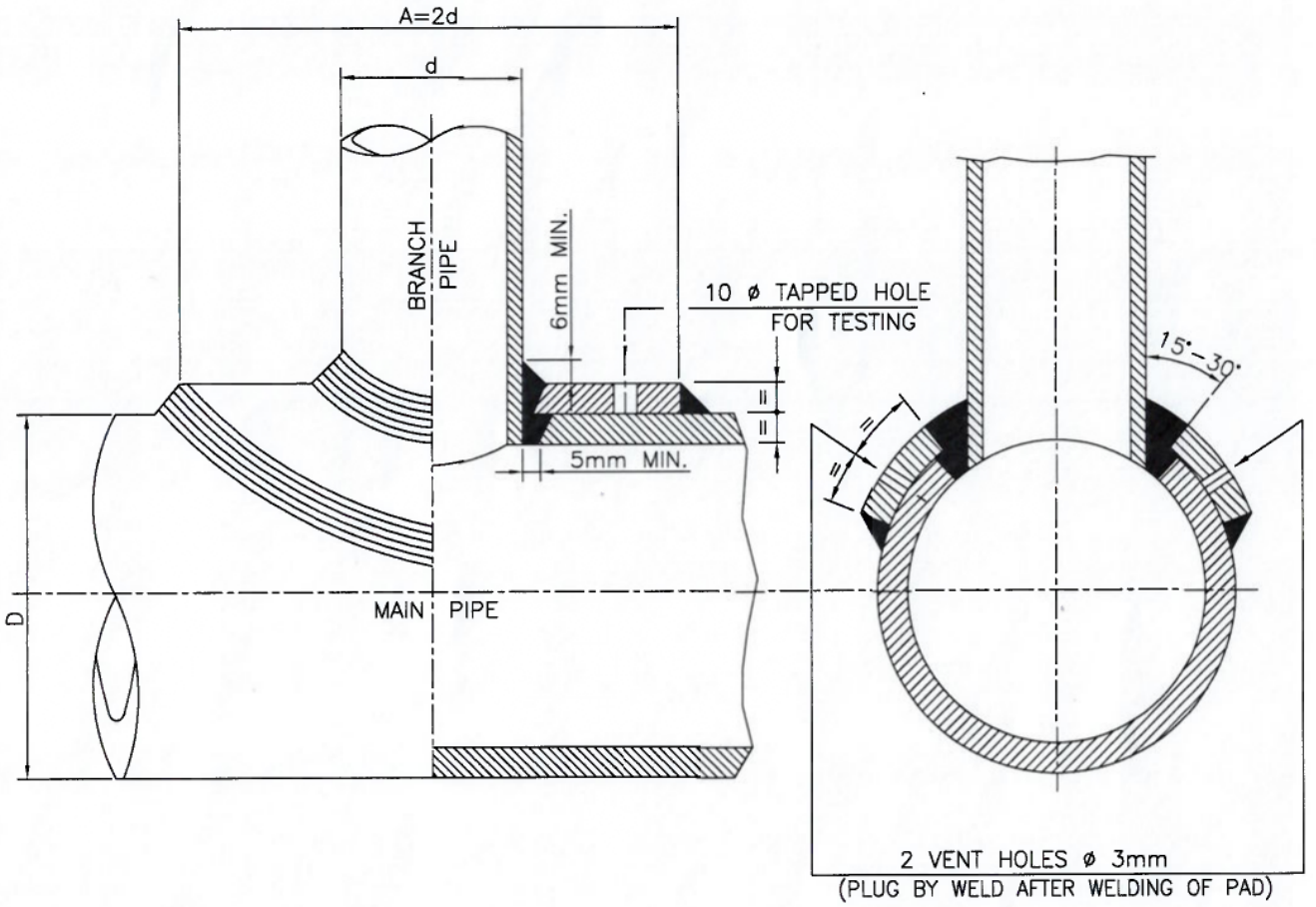
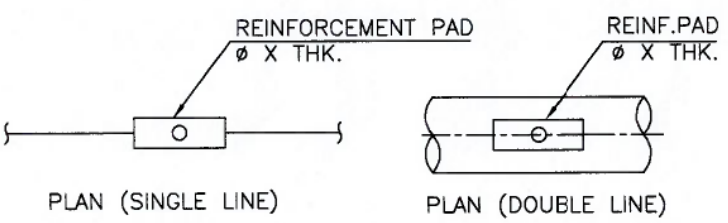
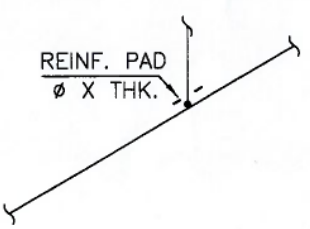


SYMBOL ON GADS



SYMBOL ON ISOMETRICS



NOTES :-

1. THIS TYPE IS TO BE USED WHEN BRANCH IS MORE THAN ONE SIZE SMALLER THAN MAIN PIPE
2. THE REINFORCING PLATE SHALL BE OBTAINED FROM PIPE SPOOL OF THE SAME SCHEDULE AND DIAMETER OF MAIN PIPE UNLESS OTHERWISE STATED
3. THE REINFORCEMENT PAD CALCULATIONS SHALL BE MADE IN ACCORDANCE WITH ASME B31.3 OR IBR AS APPLICABLE
4. ALSO SEE STANDARD 7-44-0484
5. REFER STANDARD 7-44-0482 FOR WELDING DETAILS.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
6	10.10.25	REAFFIRMED & ISSUED AS STANDARD	SRG	PK	SH	MN
5	23.09.20	REAFFIRMED & ISSUED AS STANDARD	SG	SH	GB	SM

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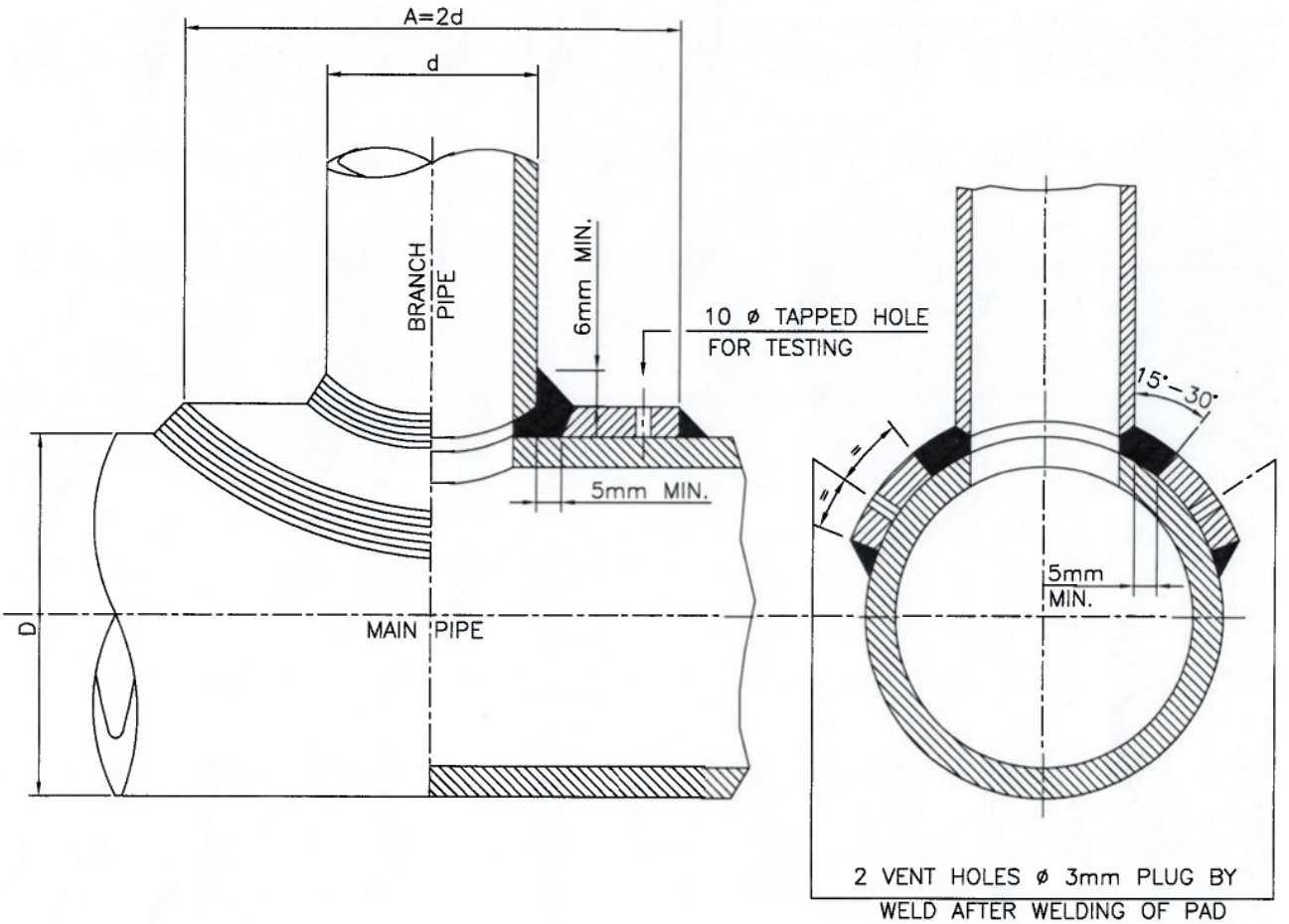

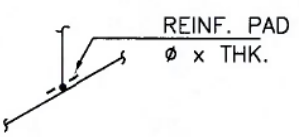

SYMBOL ON ISOMETRICS

NOTES :

1. THIS TYPE IS TO BE USED WHEN BRANCH IS EQUAL TO OR ONE SIZE SMALLER THAN MAIN PIPE
2. THE REINFORCEMENT PAD SHALL BE OBTAINED FROM PIPE SPOOL OF THE SAME SCHEDULE AND DIAMETER AS MAIN PIPE UNLESS OTHERWISE STATED.
3. THE REINFORCEMENT PAD CALCULATION SHALL BE MADE IN ACCORDANCE WITH ASME B31.3 OR IBR AS APPLICABLE.
4. ALSO SEE STANDARD 7–44–0483
5. REFER STANDARD 7–44–0479 FOR WELDING DETAILS.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
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5	23.09.20	REAFFIRMED & ISSUED AS STANDARD	SG	SH	GB	SM

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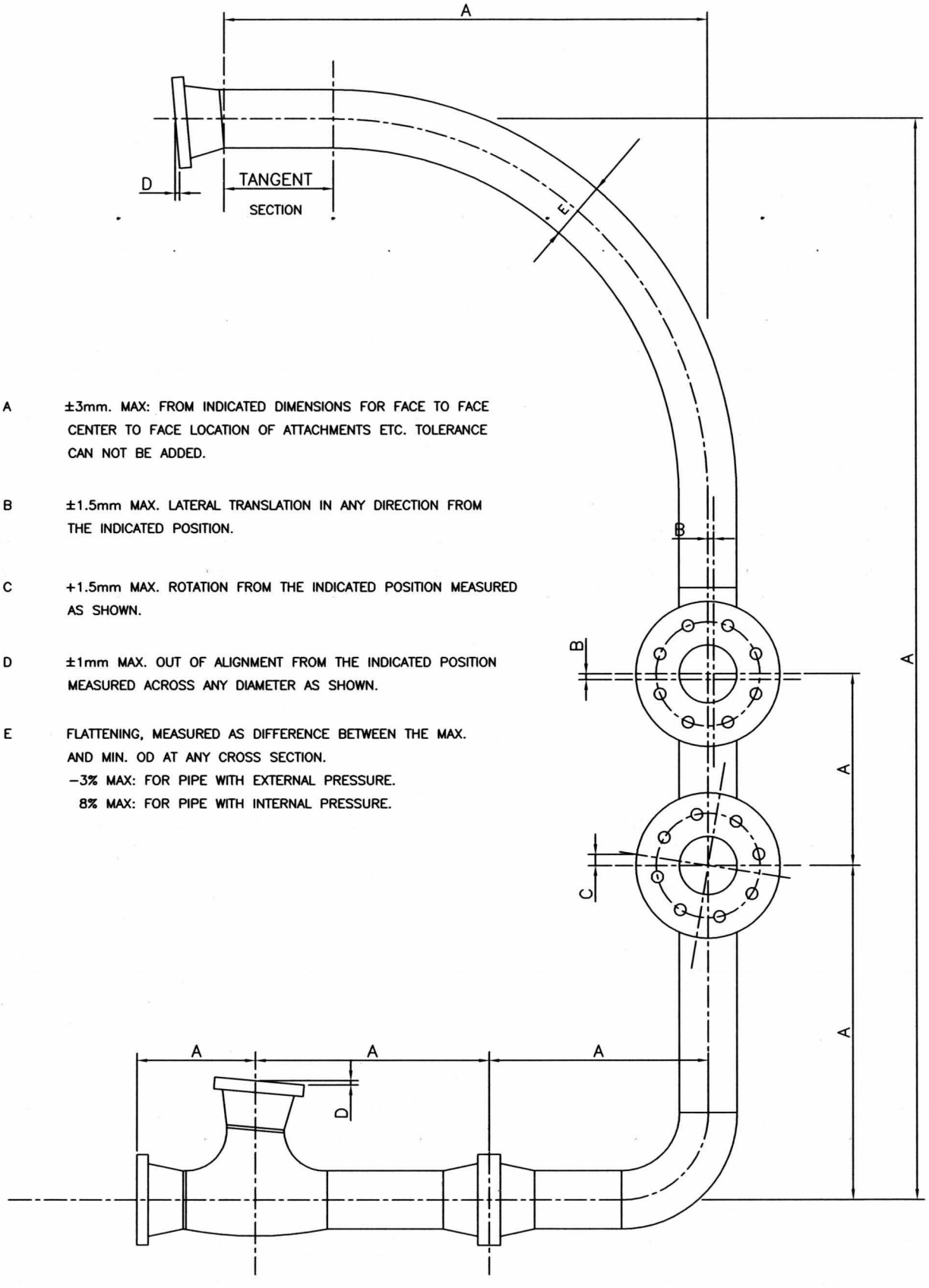

A ±3mm. MAX: FROM INDICATED DIMENSIONS FOR FACE TO FACE CENTER TO FACE LOCATION OF ATTACHMENTS ETC. TOLERANCE CAN NOT BE ADDED.

B ±1.5mm MAX. LATERAL TRANSLATION IN ANY DIRECTION FROM THE INDICATED POSITION.

C +1.5mm MAX. ROTATION FROM THE INDICATED POSITION MEASURED AS SHOWN.

D ±1mm MAX. OUT OF ALIGNMENT FROM THE INDICATED POSITION MEASURED ACROSS ANY DIAMETER AS SHOWN.

E FLATTENING, MEASURED AS DIFFERENCE BETWEEN THE MAX. AND MIN. OD AT ANY CROSS SECTION.
−3% MAX: FOR PIPE WITH EXTERNAL PRESSURE.
8% MAX: FOR PIPE WITH INTERNAL PRESSURE.

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
6	22.01.24	REAFFIRMED & ISSUED AS STANDARD	PK	SH	GB	MN
5	31.12.18	REAFFIRMED & ISSUED AS STANDARD	PK	SH	MI	RKT

Approved by

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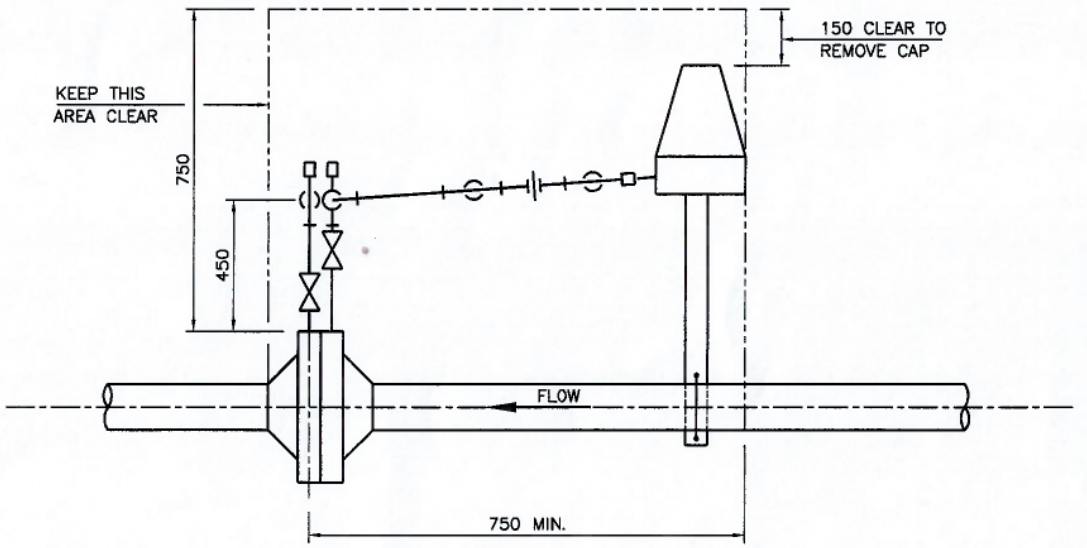
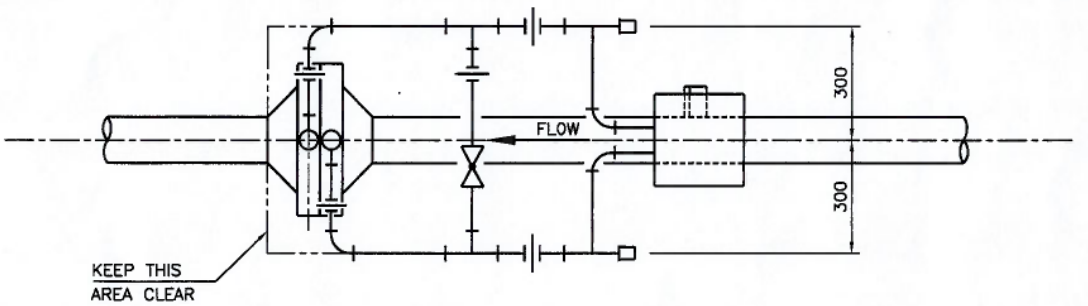


ELEVATION



PLAN

90° TOP TAPS – DRY GAS SERVICE / HF ACID

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
5	10.10.25	REAFFIRMED & ISSUED AS STANDARD	SRG	PK	SH	MN
4	23.09.20	REAFFIRMED & ISSUED AS STANDARD	SG	SH	GB	SM

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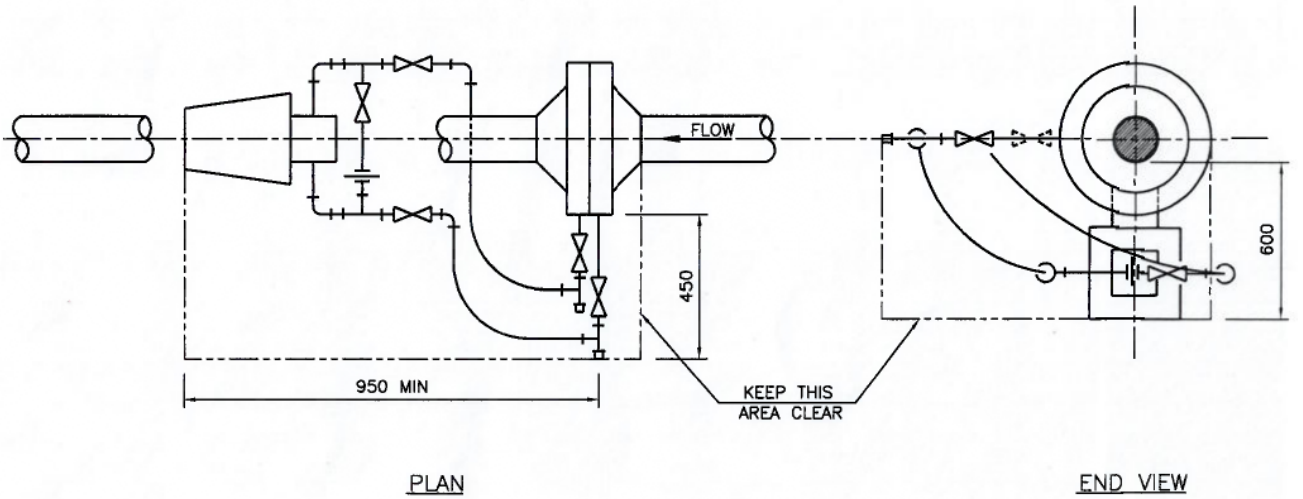
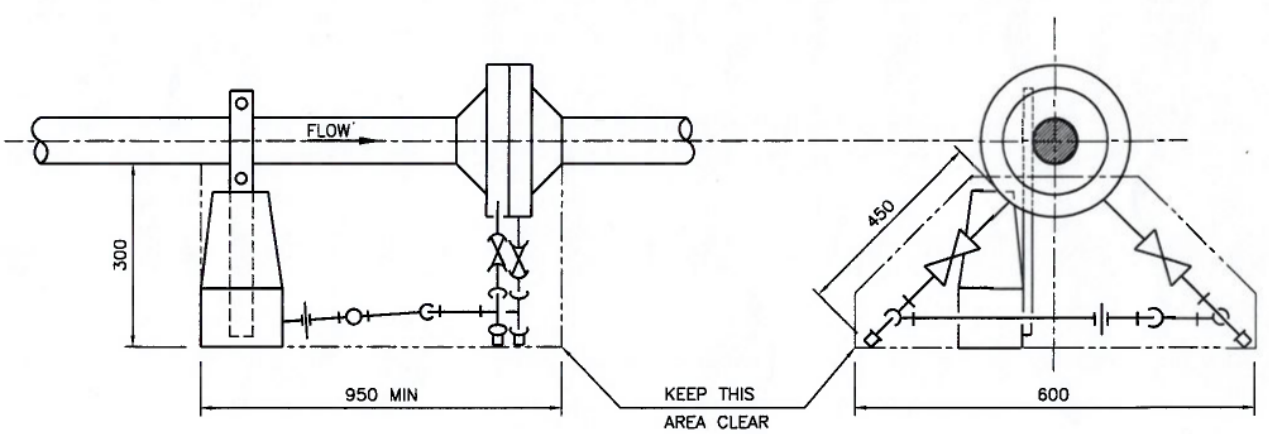


PLAN END VIEW

90° HORIZONTAL TAPS – LIQUID OR STEAM SERVICE



ELEVATION END VIEW

45° BOTTOM TAPS – HORIZONTAL LIQUID SERVICE

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
5	10.10.25	REAFFIRMED & ISSUED AS STANDARD	SRG	PK	SH	MN
4	23.09.20	REAFFIRMED & ISSUED AS STANDARD	SG	SH	GB	SM

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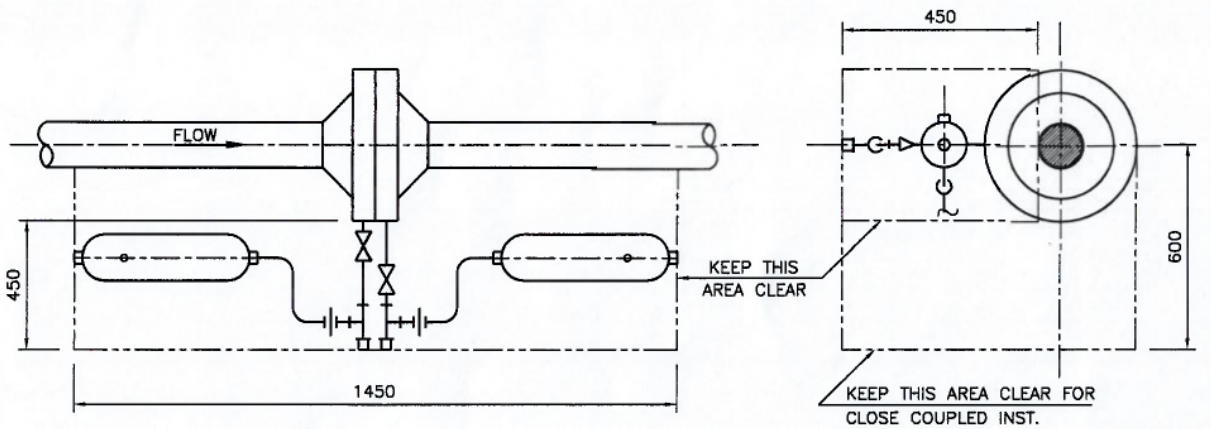
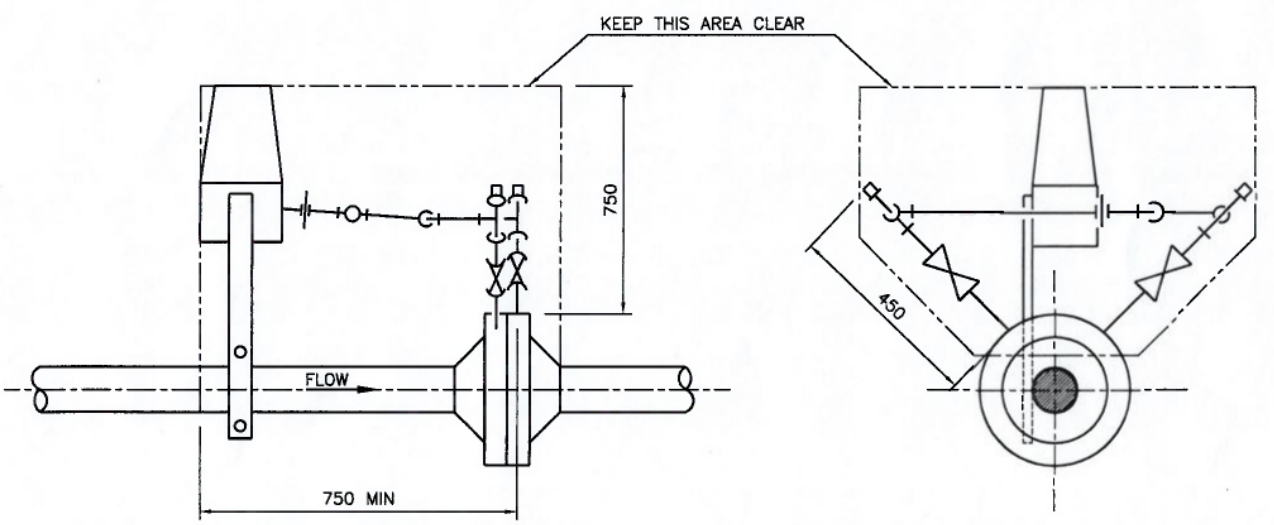


PLAN END VIEW

90° HORIZONTAL TAPS STEAM SERVICE W/CONDENSATE POTS



ELEVATION END VIEW

45° TOP TAPS – HORIZONTAL GAS SERVICE

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
5	10.10.25	REAFFIRMED & ISSUED AS STANDARD	SRG	PK	SH	MN
4	23.09.20	REAFFIRMED & ISSUED AS STANDARD	SG	SH	GB	SM

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TABLE OF BASIC SPAN

NOM. PIPE SIZE	SCH/ THK	INSULATED PIPE						BARE PIPE				NOM. PIPE SIZE
		VAPOR			LIQUID			EMPTY		WATER FILLED		
		BASIC SPAN (L)M			BASIC SPAN (L)M			SPAN (L) M	WEIGHT KG/M	SPAN (L) M	WEIGHT KG/M	
		UP TO 175°C	176°C TO 315°C	316°C TO 400°C	UP TO 175°C	176°C TO 315°C	316°C TO 400°C	UP TO 175°C		UP TO 175°C		
3/4	40	3.5	3.5	2.5	3.5	3.0	2.0	4.5	1.68	4.0	2.04	3/4
1	40	4.5	4.0	3.0	4.5	3.5	3.0	5.0	2.52	4.5	3.07	1
1 1/2	40	5.0	5.0	4.5	5.0	4.5	3.5	6.0	4.08	5.0	5.4	1 1/2
2	40	5.5	5.0	3.5	5.0	4.5	3.5	6.5	5.47	5.5	7.65	2
2 1/2	40	6.5	6.0	3.0	6.0	5.5	4.5	7.5	8.7	6.5	11.79	2 1/2
3	40	7.5	6.5	5.5	6.5	6.0	5.0	8.0	11.35	6.5	16.15	3
4	40	8.0	7.5	6.5	7.5	7.0	6.0	9.0	16.2	7.5	24.45	4
6	40	10.0	9.5	8.5	9.0	8.0	7.5	10.5	28.3	9.0	46.7	6
8	40	12.0	11.0	10.0	10.0	10.0	9.0	12.0	42.84	10.0	75.22	8
10	40	13.5	13.0	12.0	11.5	10.5	10.5	14.0	60.74	11.5	111.9	10
12	9.53mm	14.5	13.5	13.0	12.0	11.5	11.0	15.0	74.40	12.0	147.5	12
14	9.53mm	15.0	14.5	13.5	12.0	12.0	11.5	16.0	82.5	12.5	172.05	14
16	9.53mm	16.0	15.5	14.5	13.0	12.5	12.0	17.0	94.5	13.0	213.15	16
18	9.53mm	17.0	16.5	15.0	13.5	13.0	12.0	18.0	106.5	13.5	258.3	18
20	9.53mm	18.0	17.5	16.0	14.0	13.5	12.5	19.0	118.5	14.0	307.5	20
24	9.53mm	20.0	19.0	17.5	14.5	14.5	13.0	21.0	142.5	15.0	418.2	24
3/4	80	3.5	3.5	2.5	3.5	3.0	2.0	4.5	2.20	4.0	2.49	3/4
1	80	4.5	4.0	3.0	4.5	3.5	3.0	5.0	3.25	4.5	3.72	1
1 1/2	80	5.0	5.0	4.5	5.0	4.5	4.0	6.0	5.45	5.0	6.60	1 1/2
2	80	6.0	5.0	4.5	5.5	5.0	4.0	6.5	7.53	6.0	9.45	2
2 1/2	80	6.5	6.0	5.5	6.0	6.0	5.0	7.5	11.49	6.5	14.25	2 1/2
3	80	7.5	6.5	6.0	6.5	6.5	6.0	8.0	15.37	7.0	19.66	3
4	80	8.0	8.0	7.0	7.5	7.5	6.5	9.0	22.47	8.0	29.94	4
6	80	10.5	10.0	9.0	9.5	9.0	8.5	10.5	42.90	9.5	59.85	6
8	12.7mm	12.0	11.5	10.5	10.5	10.0	10.0	12.0	65.10	11.0	94.8	8
10	12.7mm	13.5	13.0	12.0	11.5	11.5	10.5	14.0	82.20	12.0	130.69	10
12	12.7mm	14.5	13.5	13.0	12.5	12.0	11.5	15.0	98.13	13.0	168.64	12
14	12.7mm	15.0	14.5	13.5	13.0	12.5	12.0	16.0	108.15	13.5	194.4	14
16	12.7mm	16.0	15.5	15.0	13.5	13.0	13.0	17.0	124.2	14.0	240.0	16
18	12.7mm	17.5	17.0	16.0	14.5	14.0	13.5	18.0	140.25	14.5	286.64	18
20	12.7mm	18.0	17.5	17.0	15.0	14.5	14.0	19.0	157.5	15.0	341.8	20
24	12.7mm	20.0	19.0	18.5	16.0	15.0	15.0	21.0	188.25	16.0	458.44	24
1	10 S	4.0	3.5	3.0	4.0	3.0	2.5	4.5	2.08	4.0	2.7	1
1 1/2	10 S	5.0	4.5	3.0	4.5	4.0	3.0	5.5	3.12	5.0	4.57	1 1/2
2	10 S	5.0	4.5	3.5	4.5	4.0	3.0	6.0	3.94	5.5	6.33	2
2 1/2	10 S	6.5	5.5	4.5	5.5	5.0	4.5	7.0	5.26	6.0	8.85	2 1/2
3	10 S	7.0	6.0	5.0	6.0	5.5	5.0	7.5	6.45	6.0	11.91	3
4	10 S	7.5	7.0	6.0	6.5	6.0	6.0	8.0	8.34	7.0	17.67	4
6	10 S	9.5	9.0	8.0	8.0	7.5	7.5	10.0	13.82	8.5	34.54	6
8	10 S	11.0	10.5	10.0	10.9	9.5	8.5	11.5	19.94	10.0	55.5	8
10	10 S	12.5	12.0	11.0	10.5	10.0	9.5	13.0	27.83	11.0	83.4	10
12	10 S	14.0	13.0	12.0	11.0	11.0	10.0	14.5	36.00	11.5	114.6	12
14	10 S	14.5	14.0	13.0	11.5	11.0	11.0	15.5	41.18	11.5	132.6	14
16	10 S	16.5	14.5	14.0	12.0	11.5	11.5	16.5	47.33	12.5	172.2	16
18	10 S	16.5	15.5	14.5	12.5	12.5	11.5	17.5	53.18	13.0	212.1	18
20	10 S	17.5	16.5	15.5	13.0	13.0	12.0	18.5	68.50	13.0	264.5	20
24	10 S	19.0	18.0	17.0	14.0	13.5	12.5	20.5	94.37	14.0	376.8	24

Rev. No.	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
5	17.12.25	REVISED & ISSUED AS STANDARD	SRG	PK	SH	MN
4	23.09.20	REAFFIRMED & ISSUED AS STANDARD	SH	AK	RN	SC

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MAXIMUM SPACING OF GUIDES FOR VERTICAL & HORIZONTAL PIPES

NOM PIPE SIZE	VERTICAL SPACING (M)	HORIZONTAL SPACING (M) (NOTE-1)
1	6.0	6.0
1 ½	6.0	6.0
2	6.0	6.0
3	8.0	12.0
4	8.0	12.0
6	8.0	12.0
8	8.0	12.0
10	12.0	18.0
12	12.0	18.0
14	12.0	18.0
16	12.0	18.0
18	12.0	18.0
20 and above	16.0	18.0

Notes :

1. These spacings may be varied to suit column spacing of rack. The above spacing is for straight runs of pipe & does not include guides which are used for control of thermal movements, as decided by stress group.

2. The guide spacings given in the above table are indicative only.

Rev. No	Date	Purpose	Prepared by	Checked by	Stds. Committee Convenor	Stds. Bureau Chairman
					Approved by	
5	10.10.25	REAFFIRMED & ISSUED AS STANDARD	SRG	PK	SH	MN
4	23.09.20	REAFFIRMED & ISSUED AS STANDARD	PK	SH	GB	SM

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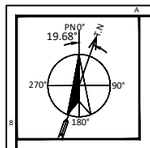
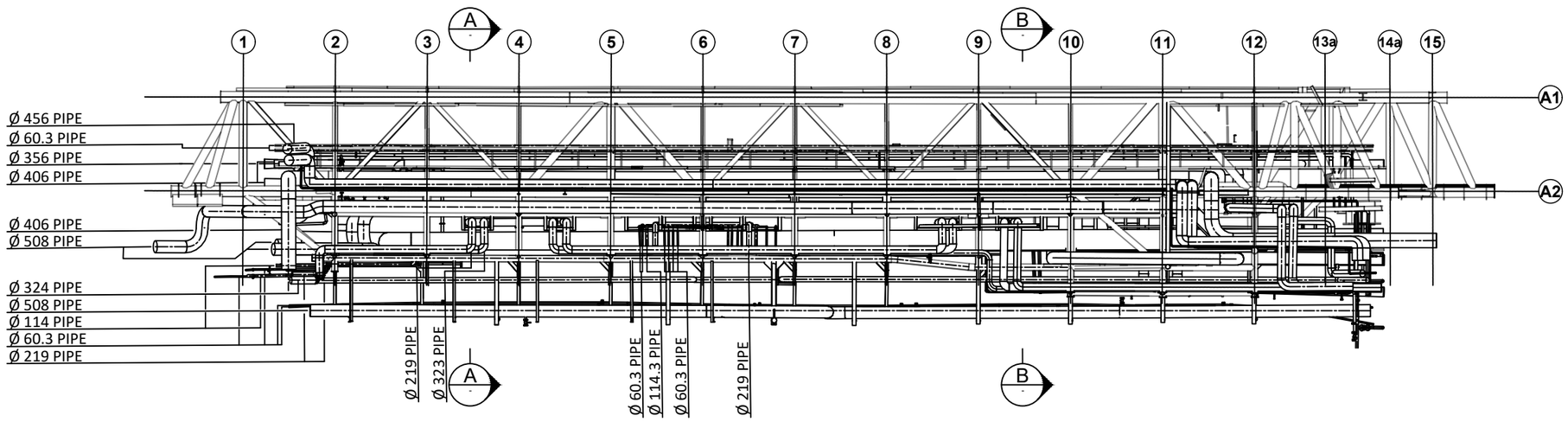
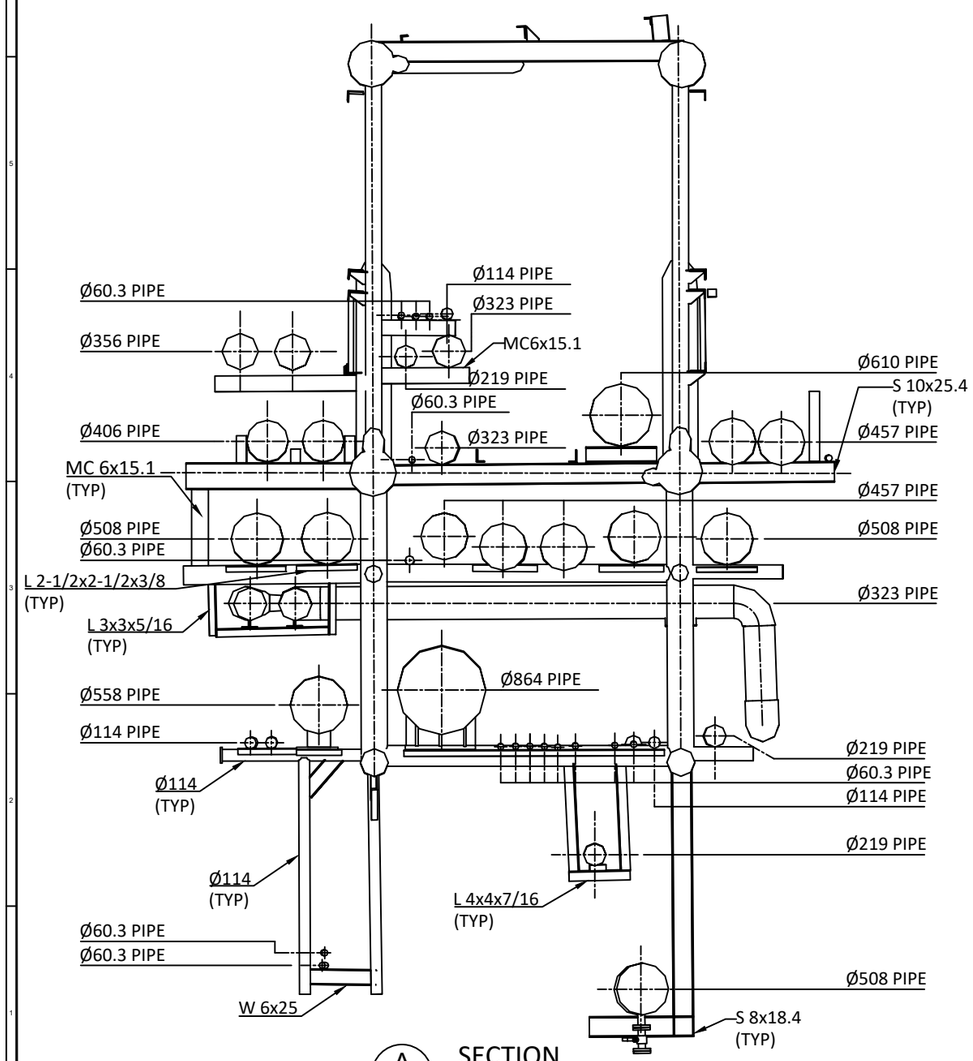


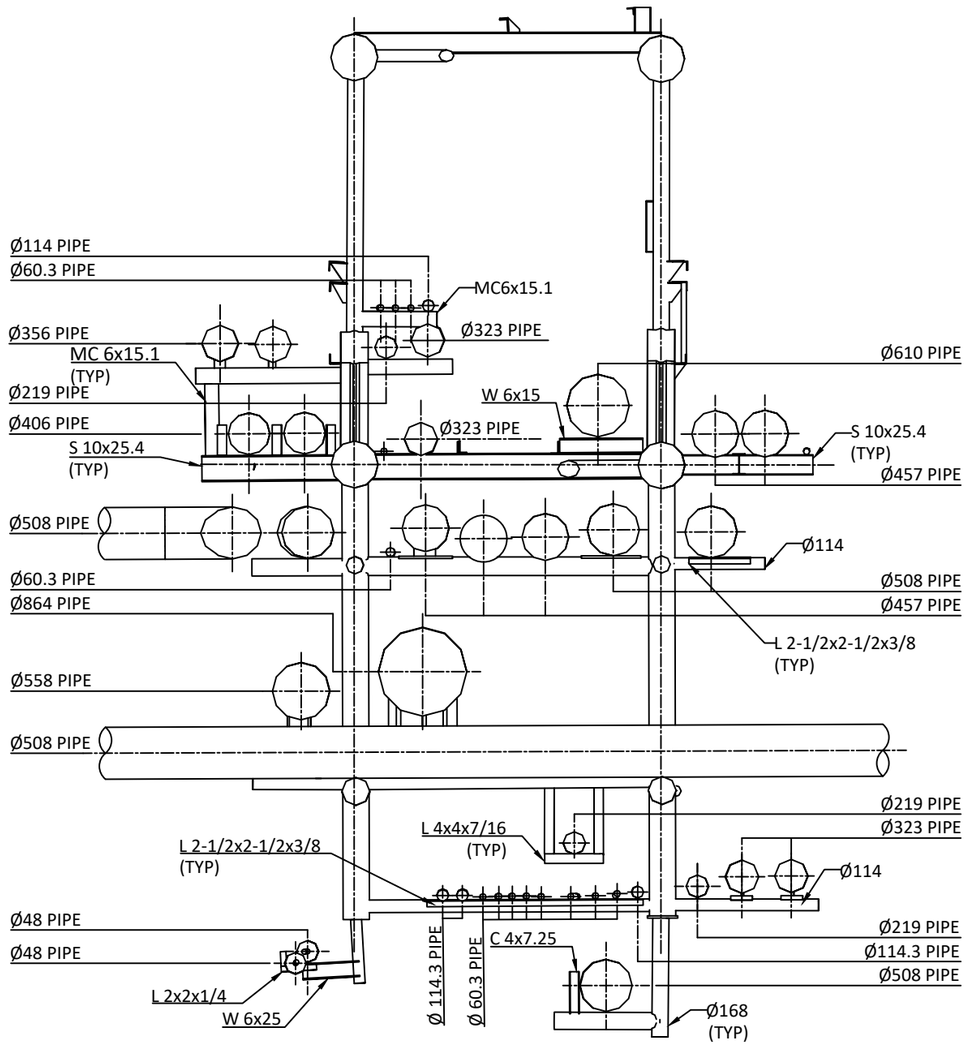


PIPING ELEVATION LAYOUT AT GRID-A

SCALE 1:75



A SECTION
SCALE 1:25



B SECTION
SCALE 1:25

GENERAL NOTES

1. ALL DIMENSIONS ARE IN MM & ELEVATIONS ARE IN METERS UNLESS OTHERWISE STATED.
2. THIS DRAWING IS EXTRACTED USING LASER SCAN MODEL.

Rev.	Source Drawing No.	Reference Drawing Title
00	SAPL0034-STRL-L-071	KEY PLAN
00	SAPL0034-STRL-L-072	FRAMING PLAN
00	SAPL0034-STRL-L-073	FRAMING ELEVATION
00	SAPL0034-STRL-L-074	GRATING AND HANDRAIL
00	SAPL0034-STRL-L-075	SLIDING & ROOFING DETAILS
00	SAPL0034-PL-L-076	PIPING PLAN AND ELEVATION
00	SAPL0034-E&I-L-078	CABLETRAY PLAN , ELEVATION
		SECTION & DETAILS

Rev	Date	Description	Drawn	Checked	SAP-APP	EIL-APP
00	13.03.25	ISSUED FOR REVIEW	PS	KINK	SM/TAP	

Client : ONGC — OIL AND NATURAL GAS CORPORATION LIMITED
MH ASSET - MUMBAI

Engineering Consultants : Engineers India Limited; Oil India Limited

3D SCAN Contractor :

Area : ICP PLATFORM COMPLEX | ICG

Project : HIRING OF SERVICES FOR NDT AND LASER SCANNING OF BRIDGES IN MH ASSET

Drawing Title : **ICP TO ICG BRIDGE PIPING SECTIONS AND DETAILS**

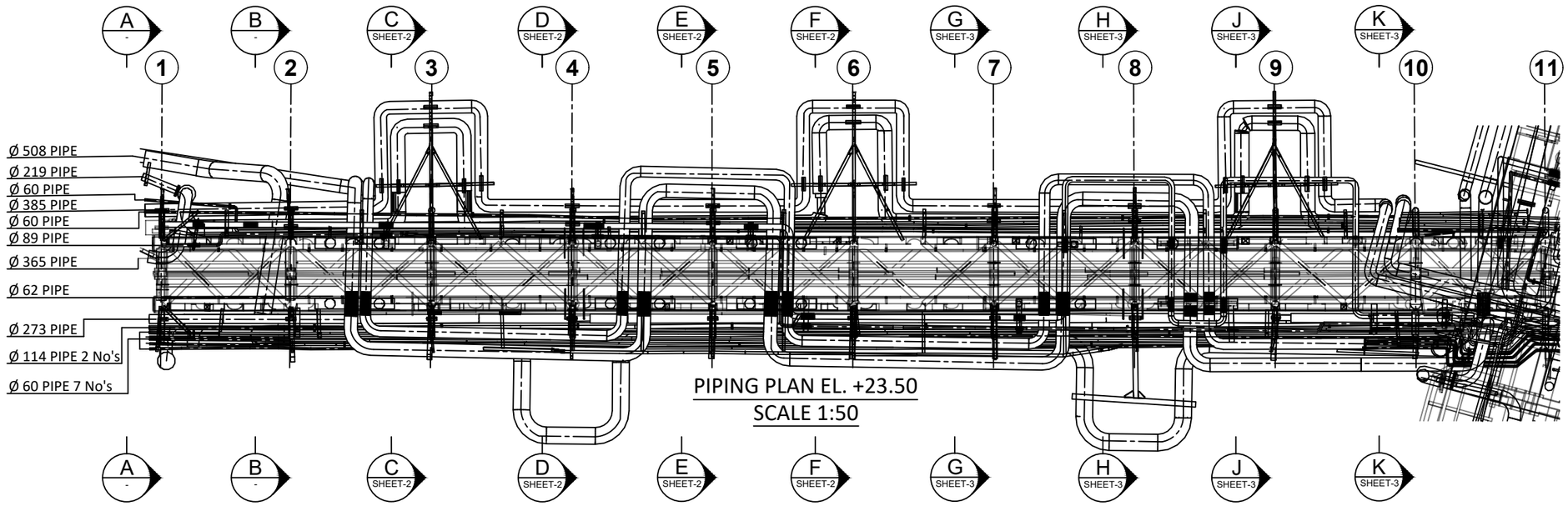
Scale : 1:50 | Dwg Size : A0 | File Name : SAPL0034-PL-L-077.dwg

Source / Project No.	Sequence No.	Sub.
-	SAPL0034-PL-L-077	1/1

K V SATYA VARMA

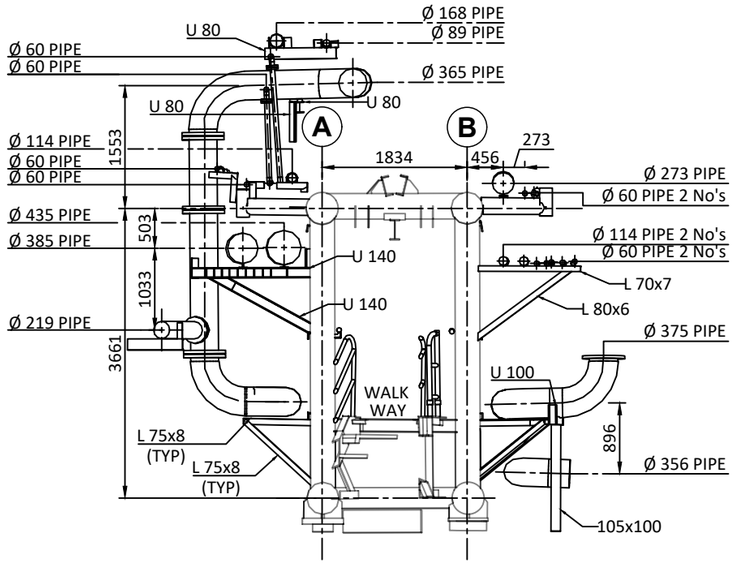
08 March 2025 11:02:12

DATE & TIME / USER NAME / FOLDER LOCATION / DRAWING FILE NAME



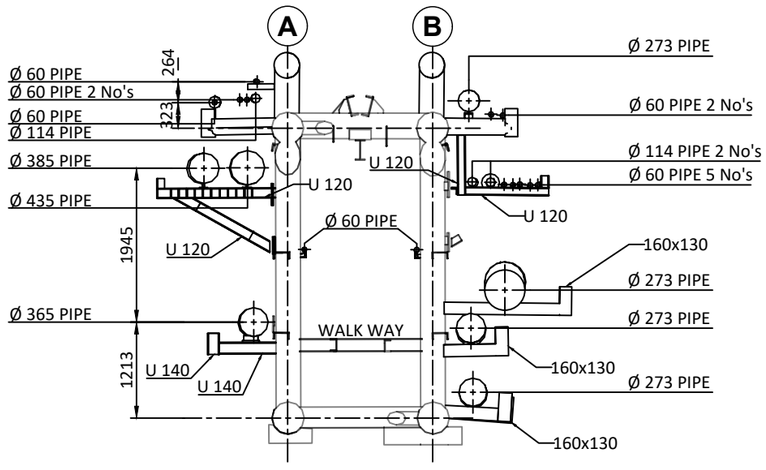
PIPING PLAN EL. +23.50

SCALE 1:50



A - SECTION

SCALE 1:25



B - SECTION

SCALE 1:25

GENERAL NOTES

1. ALL DIMENSIONS ARE IN MM & ELEVATIONS ARE IN METERS UNLESS OTHERWISE STATED.
2. THIS DRAWING IS EXTRACTED USING LASER SCAN MODEL.

Rev.	Source Drawing No.	Reference Drawing Title
00	SAPL0034-STRL-L-001	KEY PLAN
00	SAPL0034-STRL-L-002	FRAMING PLAN
00	SAPL0034-STRL-L-003	FRAMING ELEVATION
00	SAPL0034-STRL-L-004	GRATING AND HANDRAIL
00	SAPL0034-STRL-L-005	SLIDING & ROOFING DETAILS
00	SAPL0034-PL-L-006	PIPING PLAN AND ELEVATION
00	SAPL0034-E&I-L-008	CABLETRAY PLAN LAYOUT
00	SAPL0034-E&I-L-009	CABLETRAY SECTION & DETAILS

Rev.	Date	Description	Drawn	Checked	SAP-APP	EIL-APP
00	17.02.26	ISSUED FOR REVIEW	ASR	KINK	SMTAP	

Client: OIL AND NATURAL GAS CORPORATION LIMITED
MH ASSET - MUMBAI

Engineering Consultant: STRUCTURAL ENGINEERS & ENGINEERS ASSOCIATES

3D SCAN Contractor:

Area: BHS PLATFORM COMPLEX | SLQ

Project: HIRING OF SERVICES FOR NDT AND LASER SCANNING OF BRIDGES IN MH ASSET

Drawing Title: **BHS TO SLQ BRIDGE PIPING SECTIONS AND DETAILS**

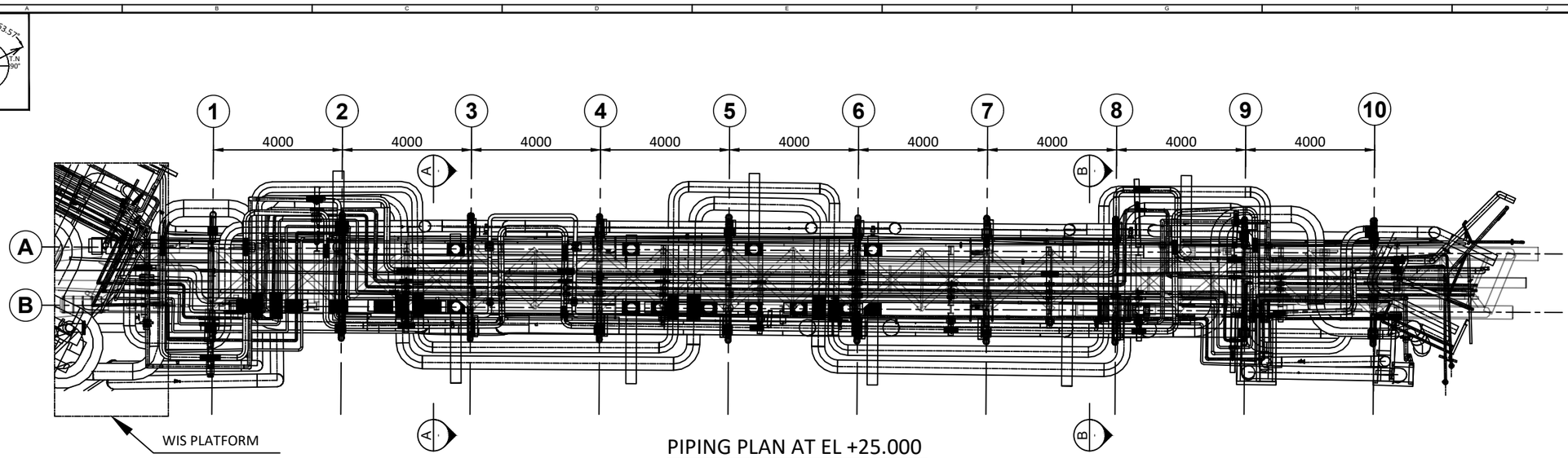
Scale: 1:50 | Dwg Size: A0 | File Name: SAPL0034-PL-L-007_Sheet1.dwg

Source / Project No.	Sequence No.	Sub.
1980-43-565	SAPL0034-PL-L-007	1/3

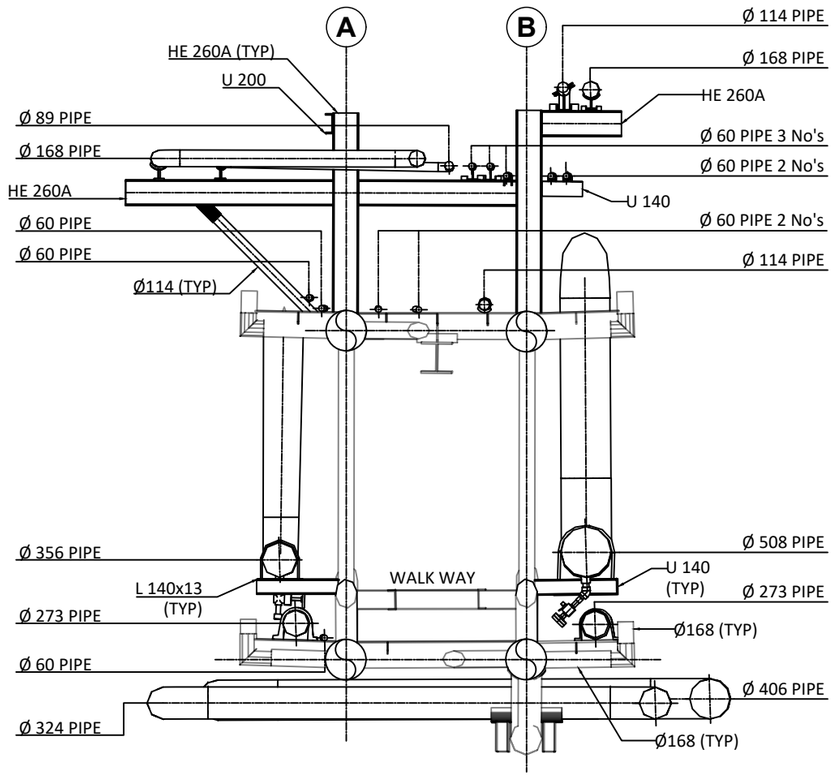
K V SATYA VARMA

17 February 2026 16:26:56

DATE & TIME / USERNAME / FOLDER LOCATION / DRAWING FILE NAME



PIPING PLAN AT EL +25.000
SCALE 1:50



A
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SECTION
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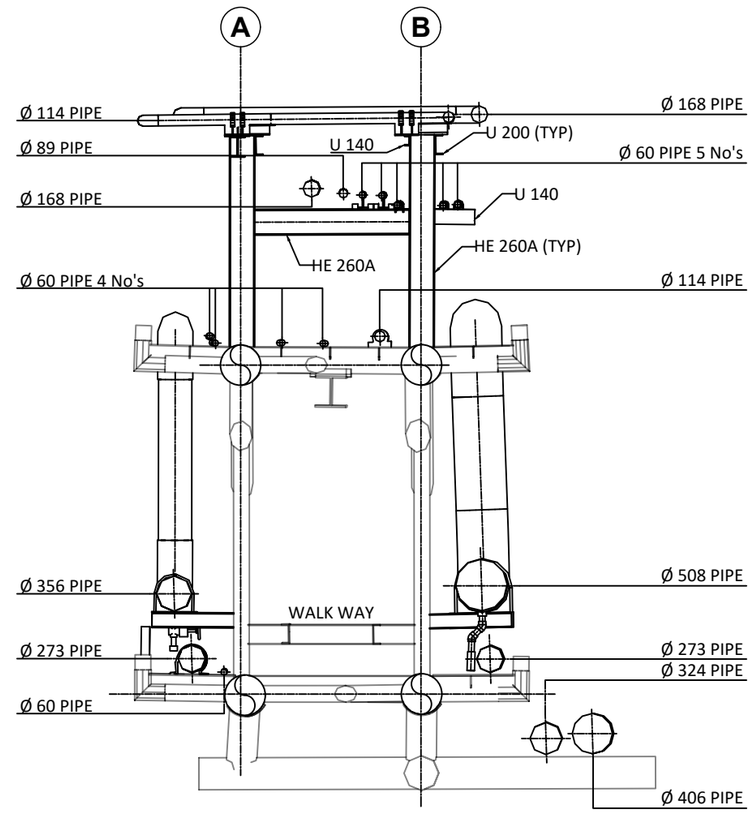


B
-
SECTION
SCALE 1:20

GENERAL NOTES

1. ALL DIMENSIONS ARE IN MM & ELEVATIONS ARE IN METERS UNLESS OTHERWISE STATED.
2. THIS DRAWING IS EXTRACTED USING LASER SCAN MODEL.

Rev.	Source Drawing No.	Reference Drawing Title
00	SAPL0034-STRL-L-011	KEY PLAN
00	SAPL0034-STRL-L-012	FRAMING PLAN
00	SAPL0034-STRL-L-013	FRAMING ELEVATION
00	SAPL0034-STRL-L-014	GRATING AND HANDRAIL
00	SAPL0034-PL-L-015	PIPING LAYOUT
00	SAPL0034-E&I-L-017	CABLE TRAY PLAN, ELEVATION AND SECTION DETAILS

Rev	Date	Description	Drwn	Checked	SAP-APP	EIL-APP
00	03.03.25	ISSUED FOR REVIEW	ASR	KINK	SMKTAP	

Client: OIL AND NATURAL GAS CORPORATION LIMITED
MH ASSET - MUMBAI

Engineering Consultant: STUP CONSULTANTS PVT. LTD. | INCHEM ENGINEERS
3D SCAN Contractor:

Area: WIS PLATFORM COMPLEX | SLQ

Project: HIRING OF SERVICES FOR NDT AND LASER SCANNING OF BRIDGES IN MH ASSET

Drawing Title: **WIS TO SLQ BRIDGE PIPING SECTIONS AND DETAILS**

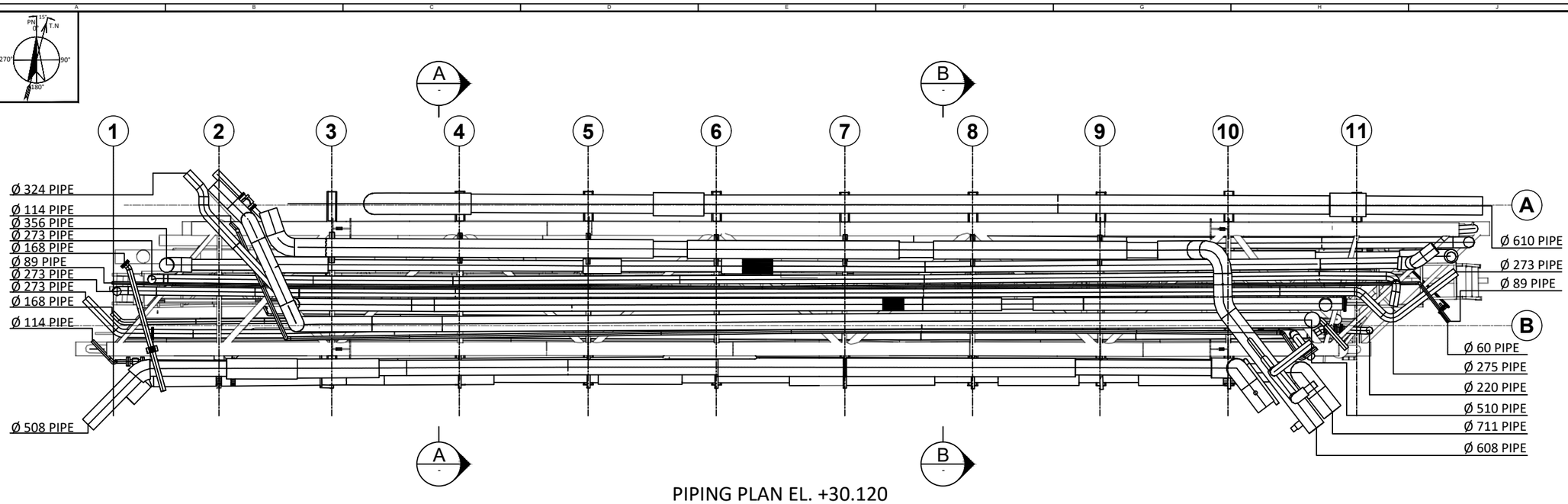
Scale : 1:50 | Dwg.Size : A0 | File Name : SAPL0034-PL-L-016.dwg

Source / Project No.	Sequence No.	Sch.
1980-43-565	SAPL0034-PL-L-016	1/1

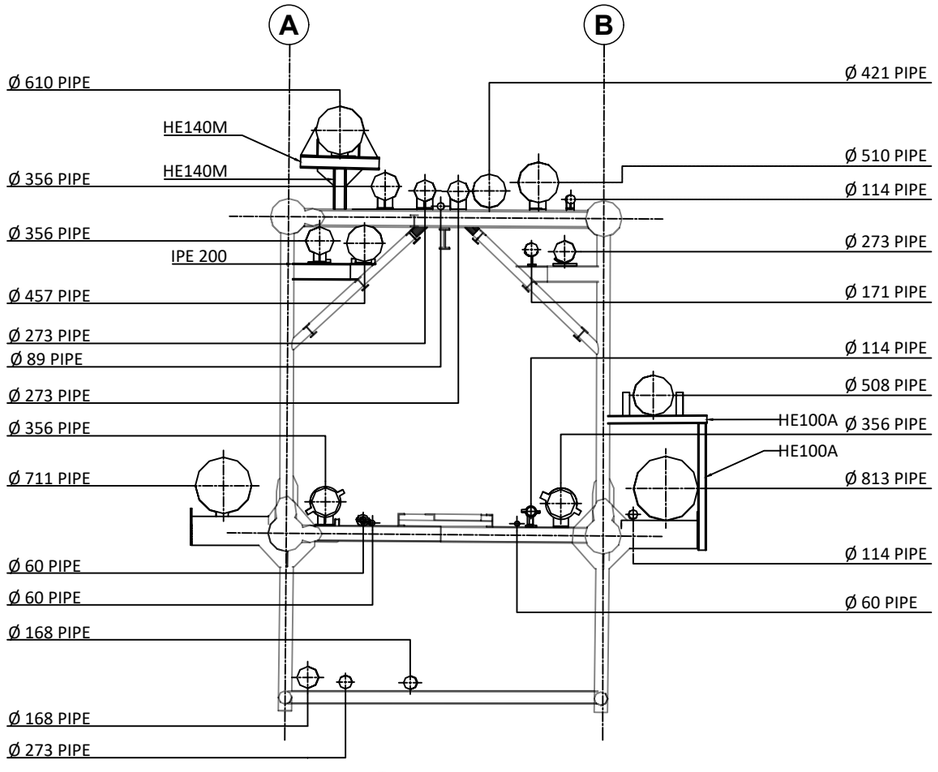
K V SATYA VARMA

17 February 2025 14:26:06

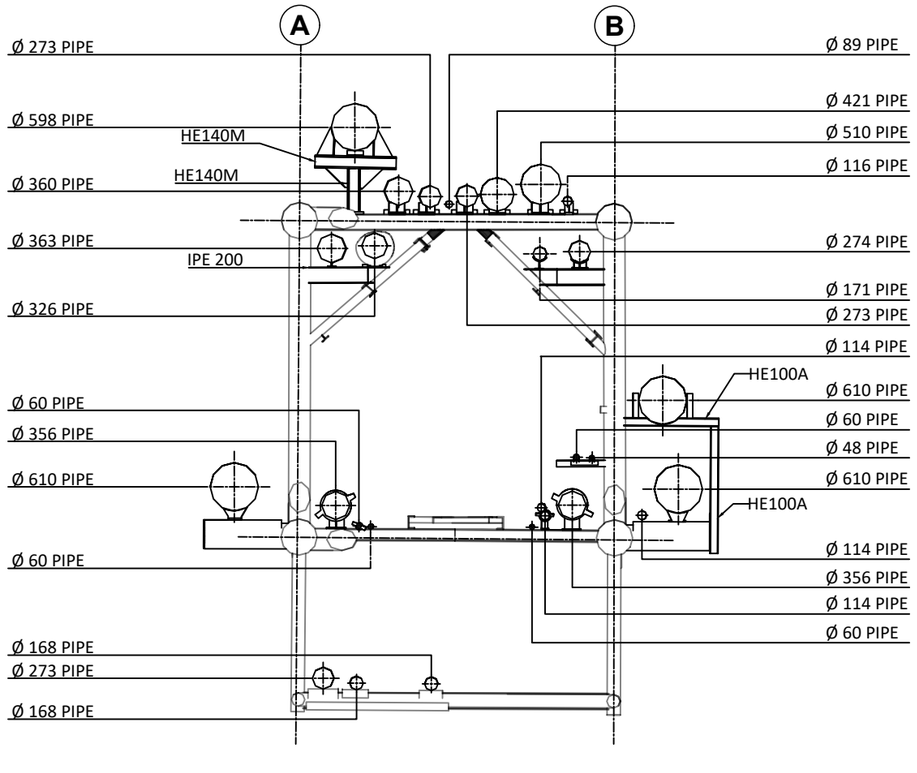
DATE & TIME : \\EIL\HANME / FOLDER LOCATION / DRAWING FILE NAME



PIPING PLAN EL. +30.120
SCALE 1:50



A SECTION
SCALE 1:25



B SECTION
SCALE 1:25

GENERAL NOTES

1. ALL DIMENSIONS ARE IN MM & ELEVATIONS ARE IN METERS UNLESS OTHERWISE STATED.
2. THIS DRAWING IS EXTRACTED USING LASER SCAN MODEL.

Rev.	Source Drawing No.	Reference Drawing Title
00	SAPL0034-STRL-L-051	KEY PLAN
00	SAPL0034-STRL-L-052	FRAMING PLAN
00	SAPL0034-STRL-L-053	FRAMING ELEVATION
00	SAPL0034-STRL-L-054	GRATING AND HANDRAIL
00	SAPL0034-PL-L-055	PIPING PLAN AND ELEVATION
00	SAPL0034-E&I-L-057	CABLETRAY PLAN,ELEVATION & SECTION DETAILS

Rev	Date	Description	Drawn	Checked	EM/AP	EIL-APP
00	06.03.25	ISSUED FOR REVIEW	ASR	KNK	EM/TAP	

Client: ONGC — OIL AND NATURAL GAS CORPORATION LIMITED
MH ASSET - MUMBAI

Engineering Consultant: ENGINEERS INDIA LIMITED

3D SCAN Contractor:

Area: NQO PLATFORM COMPLEX | NQD

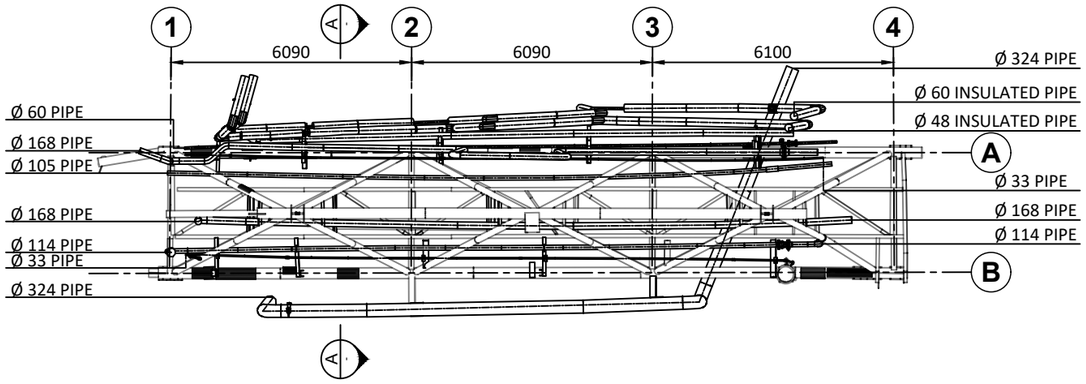
Project: HIRING OF SERVICES FOR NDT AND LASER SCANNING OF BRIDGES IN MH ASSET

Drawing Title: NQO TO NQD BRIDGE PIPING SECTIONS AND DETAILS

Scale: 1:50 | Dwg.Size : A0 | File Name : SAPL0034-PL-L-056.dwg

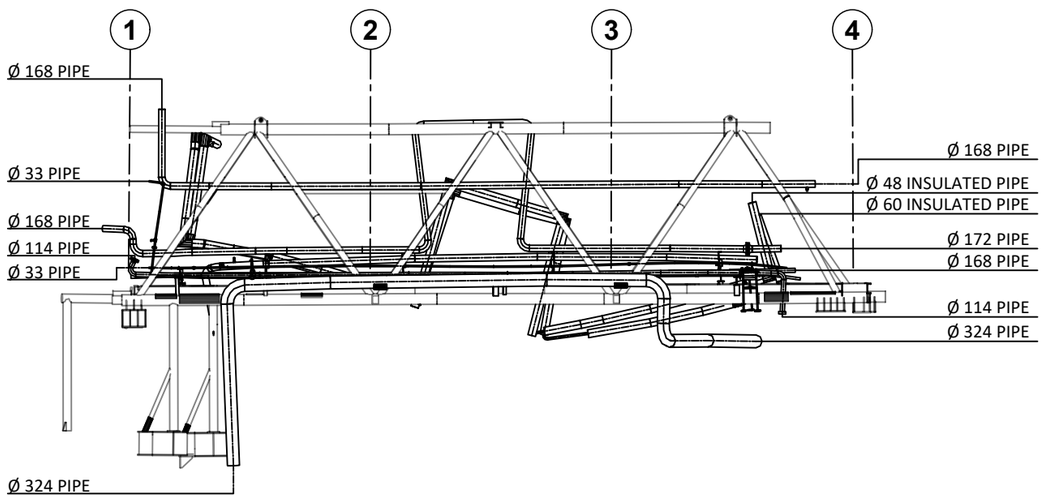
Source / Project No.	Sequence No.	Sub.
-	SAPL0034-PL-L-056	1/1

K V SATYA VARMA
17 February 2026 16:26:04
DATE & TIME / USERNAME / FOLDER LOCATION / DRAWING FILE NAME



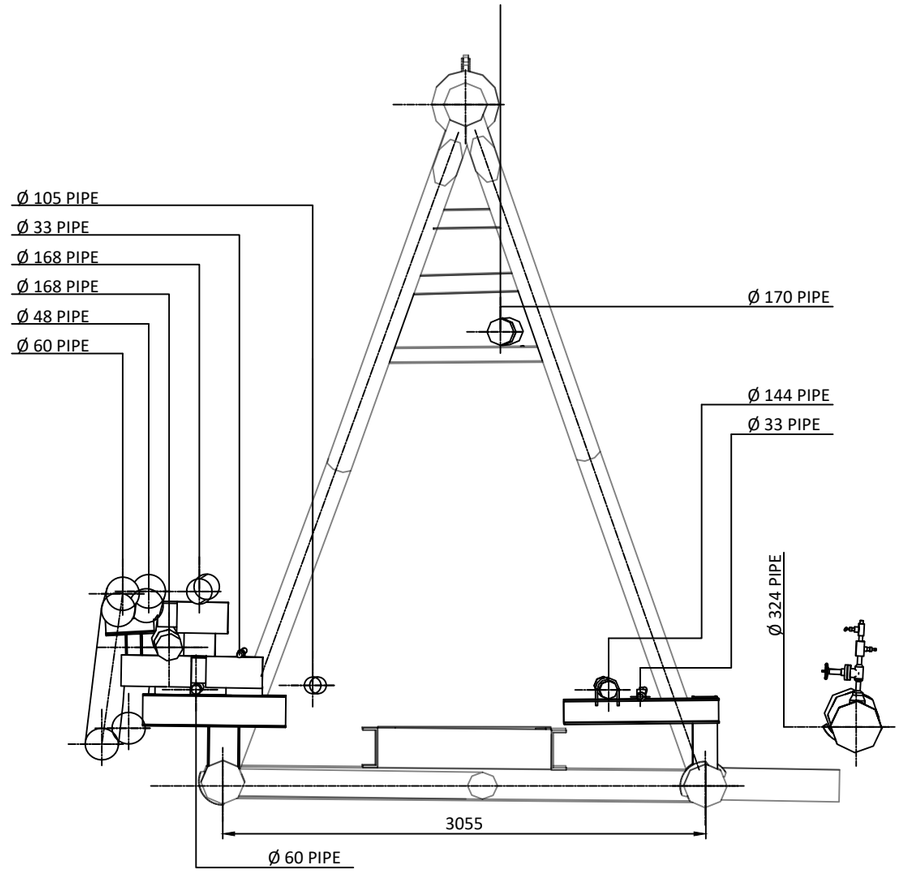
PIPING LAYOUT

PLAN VIEW



PIPING LAYOUT

VIEW LOOKING TOWARDS NORTH



SECTION-AA

GENERAL NOTES

1. ALL DIMENSIONS ARE IN MM & ELEVATIONS ARE IN METERS UNLESS OTHERWISE STATED.
2. THIS DRAWING IS EXTRACTED USING LASER SCAN MODEL.

Rev.	Source Drawing No.	Reference Drawing Title
00	SAPL0034-STRL-L-031	KEY PLAN
00	SAPL0034-STRL-L-032	FRAMING PLAN
00	SAPL0034-STRL-L-033	FRAMING ELEVATION
00	SAPL0034-STRL-L-034	GRATING AND HANDRAIL
00	SAPL0034-PL-L-036	CABLE TRAY PLAN, ELEVATION AND SECTION DETAILS

Rev	Date	Description	Drawn	Checked	SM/TAP	ES_APP
00	17.02.25	ISSUED FOR REVIEW	KVSV	KINK	SM/TAP	

Client : OIL AND NATURAL GAS CORPORATION LIMITED
MH ASSET - MUMBAI

Engineering Consultant : ENGINEERS INDIA LIMITED

3D SCAN Contractor :

Area : WIN PLATFORM COMPLEX | NC

Project : HIRING OF SERVICES FOR NDT AND LASER SCANNING OF BRIDGES IN MH ASSET

Drawing Title : **WIN TO NC BRIDGE PIPING LAYOUT**

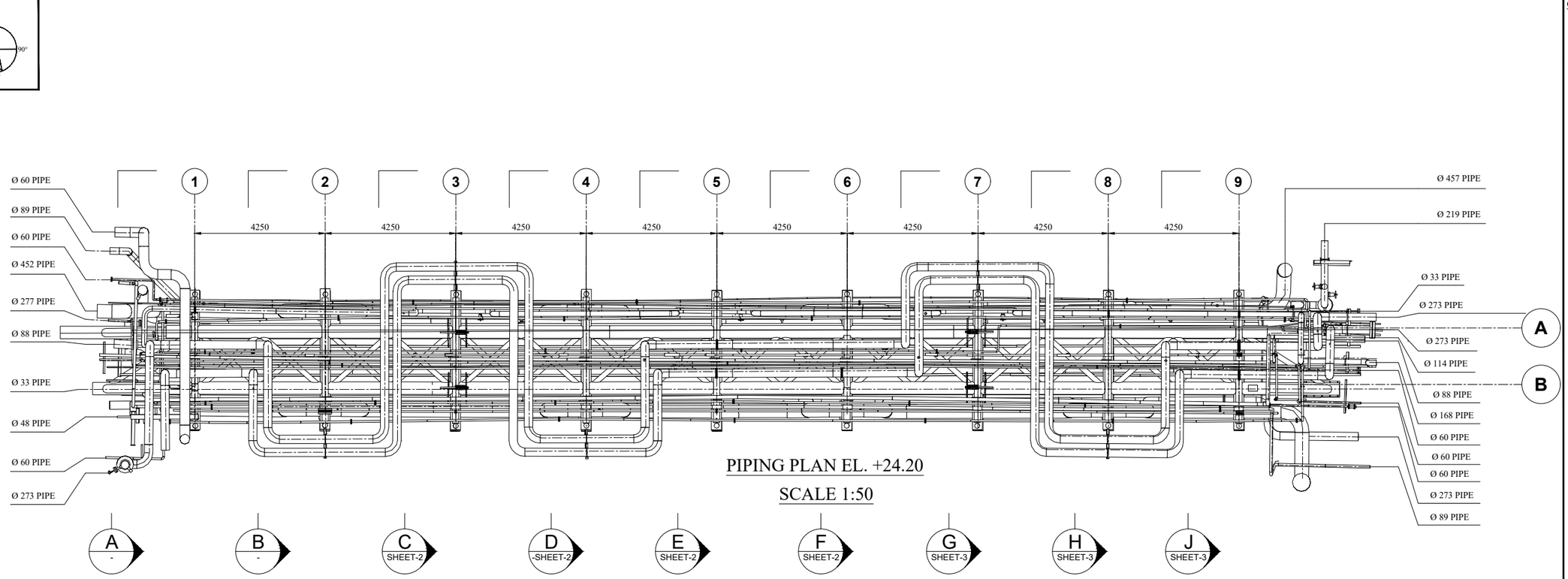
Scale : 1:50 | Dwg Size : A0 | File Name : SAPL0034-PL-L-035.dwg

Source / Project No.	Sequence No.	Sub.
-	SAPL0034-PL-L-035	1/1

K V SATYA VARMA

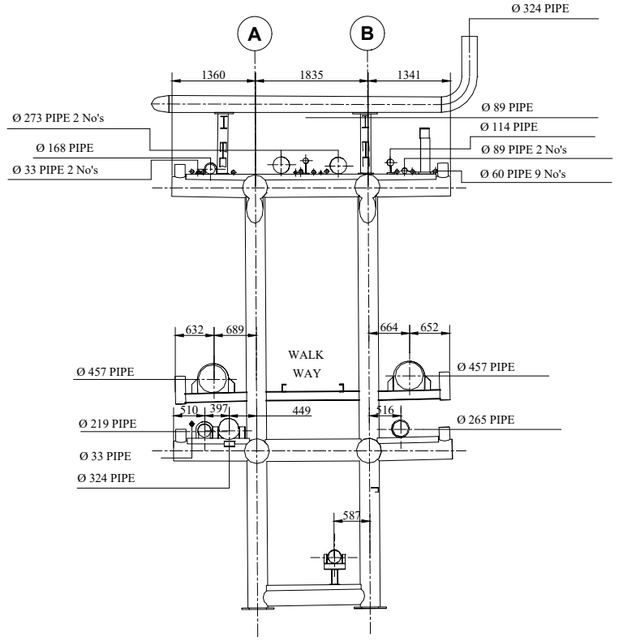
17 February 2025 16:29:06

DATE & TIME / USERNAME / FOLDER/LOCATION / DRAWING FILE NAME



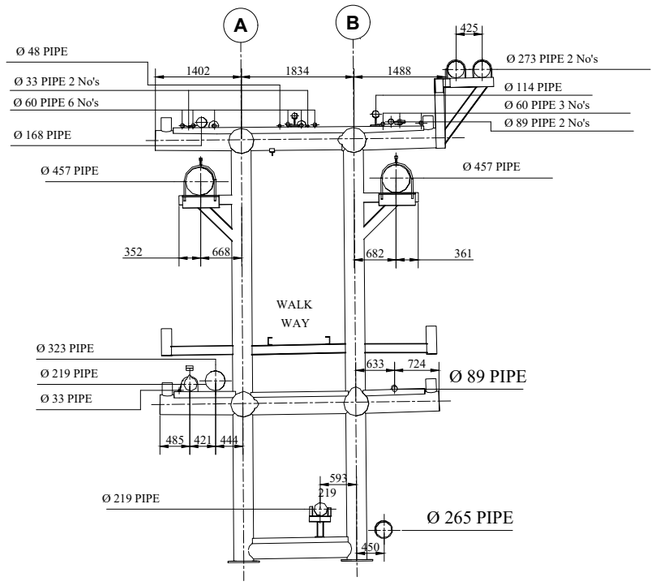
PIPING PLAN EL. +24.20

SCALE 1:50



A - SECTION

SCALE 1:33



B - SECTION

SCALE 1:33

GENERAL NOTES

1. ALL DIMENSIONS ARE IN MM & ELEVATIONS ARE IN METERS UNLESS OTHERWISE STATED.
2. THIS DRAWING IS EXTRACTED USING LASER SCAN MODEL.

Rev.	Source Drawing No.	Reference Drawing Title
01	SAPL0038-STRL-L-311	KEY PLAN
01	SAPL0038-STRL-L-312	FRAMING PLAN
01	SAPL0038-STRL-L-313	GRATING AND HANDRAIL
01	SAPL0038-E&I-L-411	CABLE-TRAY PLAN & ELEVATION
01	SAPL0038-MP-L-211	PIPE LAYOUT PLAN & ELEVTION
01	SAPL0038-MP-L-213	PIPE LAYOUT PLAN & ELEVTION
01	SAPL0038-MP-L-214	PIPE LAYOUT PLAN & ELEVTION

Rev	Date	Description	Drawn	Checked	SAP-APP	EIL-APP
01	14.08.25	ISSED FOR APPROVAL	ANK	KNP	SM/TAP	
00	18.07.25	ISSUED FOR REVIEW	GKM	KINK	SM/TAP	

Client : OIL AND NATURAL GAS CORPORATION LIMITED
MH ASSET - MUMBAI

Engineering Consultant : Projects & Development India Limited

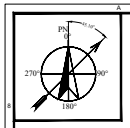
3D SCAN Contractor : Scantech

Area : SC1-SCA BRIDGE | SCA-SC1

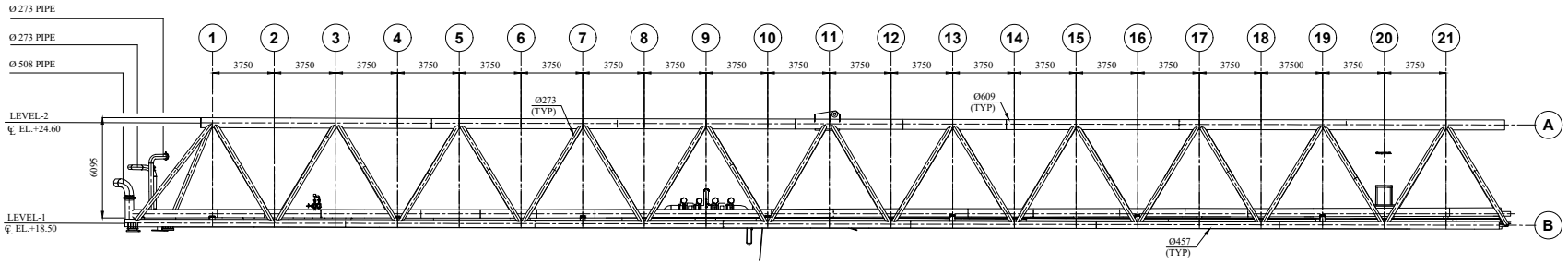
Project : HIRING OF SERVICES FOR NDT AND LASER SCANNING OF BRIDGES IN MH ASSET

Drawing Title : **SC1-SCA BRIDGE PIPING SECTIONS AND DETAILS**

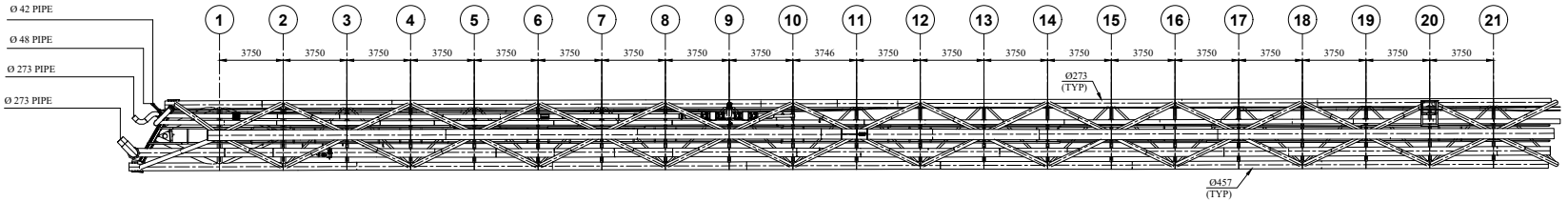
Scale : 1:33	Dwg.Size : A0	File Name : SAPL0038-MP-L-212.dwg	
Source / Project No.		Sequence No.	Sub.
-		SAPL0038-MP-L-212	1/1



SIDE VIEW
TOWARD EAST
SCALE 1:100



ELEVATION VIEW LOOKING NORTH
SCALE 1:100



PIPING PLAN VIEW
SCALE 1:100

GENERAL NOTES

1. ALL DIMENSIONS ARE IN MM & ELEVATIONS ARE IN METERS UNLESS OTHERWISE STATED.
2. THIS DRAWING IS EXTRACTED USING LASER SCAN MODEL.
3. ALL STRUCTURAL STEEL PROFILE SIZES ARE INDICATED AS FLANGE WIDTH x WEB HEIGHT UNLESS OTHERWISE SPECIFIED.

Rev.	Source Drawing No.	Reference Drawing Title
01	SAPL0038-STRL-L-321	KEY PLAN
01	SAPL0038-STRL-L-322	FRAMING PLAN
01	SAPL0038-STRL-L-323	GRATING AND HANDRAIL
01	SAPL0038-STRL-L-421	CABLE-TRAY PLAN& ELE.

Rev	Date	Description	Drawn	Checked	EIC-APP	CL-APP
01	14.08.22	ISSUED FOR APPROVAL	ANK	KNP	SM/TAP	
00	19.07.22	ISSUED FOR REVIEW	ANK	KNK	SM/TAP	

Client : OIL AND NATURAL GAS CORPORATION LIMITED
MH ASSET - MUMBAI

Engineering Consultant : MC SCAN Contractor

Area : SCA-SCF BRIDGE | SCA-SCF

Project : HIRING OF SERVICES FOR NDT AND LASER SCANNING OF BRIDGES IN MH ASSET

Drawing Title :

SCA-SCF BRIDGE PIPING LAYOUT PLAN & ELEVATION

Scale : 1:100 | Dwg.Size : A0 | File Name : SAPL0038-MP-L-221.dwg

Source / Project No.	Sequence No.	Sub.
-	SAPL0038-MP-L-221	1/1

JOB SPECIFICATION FOR PIPING DRAWINGS FOR OFFSHORE APPLICATION

PROJECT	**:**	**ENGINEERING / TECHNICAL CONSULTANCY SERVICES FOR PREPERATION OF EXECUTION METHODOLOGY, SOW AND COST ESTIMATES FOR REPLACEMENT /REFURBISHMENT OF BRIDGES IN MH ASSET**
OWNER	**:**	**OIL AND NATURAL GAS CORPORATION LTD. (ONGC), MUMBAI**
LOCATION	**:**	**MH ASSET (NORTH AND SOUTH FIELD)**
CONSULTANT	**:**	**ENGINEERS INDIA LTD, DELHI**
EIL JOB NO.	**:**	**B774**

Rev. No	Date	Purpose	Prepared by	Checked by	Approved by
0	14-11-2025	ISSUED FOR BIDS	AG	SKS	AKK

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Abbreviations:

GAD	**:**	General Arrangement Drawings
NB	**:**	Nominal Bore
P & ID	**:**	Piping & Instrumentation Diagram

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CONTENTS

1.0 SCOPE…….....4

2.0 MINIMUM REQUIREMENTS TO BE COVERED IN PIPING GENERAL ARRANGEMENT DRAWINGS (GAD)……...4

3.0 GENERAL LAYOUT REQUIREMENTS…….4

4.0 MINIMUM REQUIREMENTS TO BE COVERED IN PIPING ISOMETRICS……..……6

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1.0 SCOPE

This Specification contains minimum requirements for the preparation of piping drawings, line numbering system and line schedule. This document shall be read in conjunction with "ONGC Functional Specification for Piping Design" (Spec No.: - 2004 A).

2.0 MINIMUM REQUIREMENT TO BE COVERED IN PIPING GENERAL ARRANGEMENT DRAWINGS (GADs)

The following shall be minimum to be shown on the piping general arrangement drawing wherever applicable.

2.1 Line Number

2.2 Instrument Tag Number

2.3 Instrument Connections

2.4 Orientation of North Arrow

2.5 Battery/Area limit as per area division/skid

2.6 Title block, drawing nos. as per area division/drawing schedule

2.7 Key plan

2.8 Reference drawings

2.9 Legends and Notes

2.10 Equipment name, Tag number, Reference elevation i.e. Centerline elevation/ Bottom Tangent Line elevation

2.11 Table of Nozzle orientation and location

2.12 Match line, continuation drawing number

2.13 Location of equipment as per equipment layout

2.14 P&ID's special notes to be listed

2.15 Mark slope and working point (WP) elevations at grid crossing and at every change in direction/elevation.

2.16 Dimension and equipment elevations with reference to data sheets, vendor drawing/data & other reference drawings.

2.17 Hook-up/ Tie-in point numbers

3.0 GENERAL LAYOUT REQUIREMENTS

3.1 All piping drawings shall be drawn to scale 1:33 1/3 and in metric system only.

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3.2 Area division plan where necessary (key plan) shall be prepared prior to preparation of piping general arrangement drawings (GAD).

3.3 All deck levels shall be divided into two parts (for clarity) as explained hereunder: e.g. for Cellar Deck there shall be two piping GADs:

i) Above Cellar Deck

ii) Below Cellar Deck

Title of each drawing shall be marked with the elevation (i.e. From….. To …..) covered in the drawings

3.4 Piping 12 Inch NB and below shall be shown in single line for sake of clarity while piping 14 Inch NB and above shall be shown in double lines.

3.5 All the information regarding instrument, equipment and line identification, direction of flow for lines, elevations and dimensions shall be clearly marked on piping plans. Efforts shall be made to provide maximum possible information including operating platforms, ladders, monorails, other material handling equipment's, electrical and instrument cable trays, etc. shall be shown on piping plans to avoid reference to other drawings.

3.6 Dimensioning shall be done with equipment centerlines or main structural columns only.

3.7 For any portion of a line where complete details i.e. dimensions, elevations etc. cannot be furnished in piping plan, a detailed isometric for that portion shall be indicated at a suitable place in the same GAD only. As far as possible all such details shall be shown together at a particular space provided for the purpose.

3.8 The bracing structural members up to 2.3 meters above the deck level shall be shown on equipment layouts and piping plans.

3.9 Nozzles on equipment shall be shown in double line (thin & firm).

3.10 Actuators, valve hand wheels shall be shown to scale for all valves.

3.11 Piping plans shall show all lines indicated in piping and instrument diagrams.

3.12 Pipe supports shall be marked on separate piping plans and support drawings shall be furnished separately for company's review.

3.13 Piping arrangement drawings for skid mounted equipment/ vessels, shall be furnished separately for review and approval.

3.14 Piping isometrics shall be prepared for all lines.

3.15 Piping material specification breaks shall be shown on all GAD's and Isometrics.

3.16 All the future line shown in P&IDs shall be routed and shown as chain dotted on GADs with full details indicating dimensions, line numbers, location etc.

3.17 Escape route and Safety Equipment Layouts shall be prepared separately showing all the escape routes, Fire Safety and Life Saving equipments etc.

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3.18 Piping under other double line pipe, structure and any other items shall be shown in dotted.

3.19 Monorails with Tag number, Bottom of rail elevation and capacity etc. shall be shown in the GAD's

4.0 MINIMUM REQUIREMENTS TO BE COVERED IN PIPING ISOMETRICS:

The following shall be minimum to be shown on the isometric drawing wherever applicable:

4.1 Line number

4.2 Instrument connection with tag number.

4.3 Orientation of north direction in the left-hand top corner. North arrow should be either left side upwards or right side upwards.

4.4 Location of origin and destination w.r.t. structural column grids along with equipment numbers.

4.5 All connected lines with line numbers and continuation references.

4.6 All special fittings & specialty items e.g. strainers, traps, shutdown valves, control valves, pig detectors, hinged closures, scraper tees etc. shall be marked with their tag numbers.

4.7 High point vents and low point drains.

4.8 Out of rating items shall be called out.

4.9 Working point elevations of slope lines as per the slope requirements marked in the P&ID's.

4.10 Nozzle number, size, rating and facing, center line elevation.

4.11 Centerline elevations for valves.

4.12 Isometrics shall be properly dimensioned from fabrication and erection point of view.

4.13 Item wise bill of material.

4.14 Process parameters regarding pressure, temperature & hydrotesting.

4.15 Insulation type and thickness.

4.16 NDT & stress relieving requirements.

4.17 Items/Instruments mounted on the equipments shall be shown on Vessel trim drawing.

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FUNCTIONAL SPECIFICATION

FOR

PIPING DESIGN

Prepared / Revised By	Reviewed By	Approved By	Total No. of Pages	Date	Rev. No.
PS	AKM	SJ	75	09.05.16	9

REVISION STATUS RECORD

PS	SJ	AR	74	04.04.14	8
PS	GJ	GRP	78	21.07.09	7
PS	GJ	JSS	84	07.12.06	6
BLS	GJ	JSS	83	20.09.06	5
BLS	MP	JSS	83	04.02.06	4
PS	MP	JSS	83	21.06.05	3
Prepared By/Revised By	Reviewed By	Approved By	Total No. of Pages	Date	Rev. No.

CONTENTS

1. INTRODUCTION

1.1 General

1.2 Purpose of this document

1.3 Contractor's Responsibilities

2. ENVIRONMENTAL DESIGN CRITERIA AND UTILITIES

2.1 Basic Climatic Condition

2.2 Seismic and Transportation Loads

2.3 Design Life

2.4 Dimensions

3. DESIGN REQUIREMENTS

3.1 General

3.2 Design Load Cases

3.3 Design Stress

3.4 Drawings and Calculations

4. MECHANICAL REQUIREMENTS

4.1 Minimum Thickness

4.2 Corrosion Allowances

5. MATERIALS

5.1 General Specification

5.2 Sour Service Requirement

5.3 Corrosion Allowance

5.4 Carbon Steel

5.5 Stain less Steel

5.6 Duplex Stain less Steel

5.7 Incoloy /Inconel Cladded piping

5.8 Cu-Ni (90-10)

5.9 Copper

5.10 GRE

5.11 PVC

5.12 CPVC
5.13 TITANIUM
5.14 Material Identification
5.15 Supports and Miscellaneous
5.16 Bolting
5.17 Gaskets
5.18 Certification Documents

6. TECHNICAL NOTES FOR PIPING COMPONENTS
6.1 Pipe
6.2 Tubing
6.3 Fittings
6.4 Branch Connections
6.5 Flanges
6.6 Bolting
6.7 Valves
6.8 Specialty Items

7. PIPING SYSTEM DESIGN
7.1 General
7.2 Pipe Routing
7.3 Pipe Supports
7.4 Pipe-ways
7.5 Piping Clearances
7.6 Flanged Connections
7.7 Threaded Pipe work
7.8 Changes in Materials
7.9 Vents, Drains & Bleeds
7.10 Corrosion Inhibition & Monitoring Pipe work
7.11 Relief Valves
7.12 Control Valves
7.13 Isolations
7.14 Fire Suppression System
7.15 Cupro-Nickel Pipe work
7.16 Copper Piping

7.17 GRE Piping Systems
7.18 Piping on the Bridge
7.19 Piping at Equipment
7.20 Piping layout requirements
8. PROTECTIVE COATINGS & INSULATION
9. PIPING SPECIFICATION INDEX
10. GENERAL REQUIREMENTS
10.1 Pre qualification criteria and tests for manufacturers (New Vendor)
10.2 Suggested Third Party Inspection Agencies
10.3 Quality Assurance
10.4 Safety Provisions
10.5 Receipt & Storage
10.6 Spares
10.7 Warranties
10.8 Documentation

GLASS REINFORCED EPOXY (Annexure-1)
1.1 Scope
1.2 Design / Operating Data
1.3 Design & Manufacture of GRE piping
1.4 Qualification & Quality Control of GRE Piping
1.5 Fabrication & Installation of GRE Piping
1.6 Certification Requirements of GRE Piping.
1.7 List of Codes & Standards for (GRE) Composite Material Piping

1. INTRODUCTION

1.1 GENERAL

This refers to the minimum and mandatory requirements of designs & materials for piping & piping components. All piping assemblies, equipment & materials supplied or installed under these specifications shall be in accordance with sound engineering principles. Any omission from this specification shall not relieve the contractor from his responsibility of furnishing equipment or materials to meet the specific process parameters, environmental parameters, safety parameters and any other applicable statutory laws or relevant codes & standards. Substitution or changes from this specification must be accompanied with sufficient information/justification and written approval shall be obtained from the Company.

All piping, piping components, piping specialties and vessels shall be painted in accordance with Spec. No. 2005.

All piping, piping components, piping specialties and vessels shall be insulated in accordance with Spec. No. 2006.

All welding and NDT shall be performed as per Spec. 2009 F.

1.2 PURPOSE OF THIS DOCUMENT

This document shall form the basis upon which the Engineering Procurement and Construction Contractor can develop a detailed specification. This document should be considered as a general specification. The Contractor should update this document with the detailed design information that is developed during the detailed design phase.

1.3 CONTRACTOR'S RESPONSIBILITIES

The Contractor shall be responsible for the selection and design of piping, including full compliance with all applicable project specifications and design Codes / Standards, including those listed in Piping Design Criteria.

The Contractor shall provide and follow detailed pipe work specifications, covering materials, classes, fittings, valves, branches and special items. These specifications shall generally comply with this Specification.

This Specification covers all pipe work on the offshore production facilities at New platforms and for modification jobs as per the bid package and approved P&ID except the following: -

a. All instrument piping downstream of the last piping block valve, as defined on the P & ID's.

The Contractor shall provide the Company with all drawings, specifications and detailed pipe stress calculations for approval.

2. ENVIRONMENTAL DESIGN CRITERIA AND UTILITIES

2.1 BASIC CLIMATIC CONDITIONS

Refer structural design criteria, Vol-II, Section-3.4 attached in bid package.

2.2 SEISMIC AND TRANSPORTATION LOADS

All equipment supports and braces, pipe supports and other support steel work, including temporary braces, shall be designed to withstand seismic loads applicable to the present location. Refer to the Structural Basis of Design for seismic design considerations.

All equipment supports and braces, pipe supports and other support steel work, including temporary braces, shall be designed to withstand the operating, lifting, transport (by road and by sea) and hydro-test loads specified in Project Specification.

2.3 DESIGN LIFE

The process facilities design life requirement is 25 years.

2.4 DIMENSIONS

S1 units shall be used. Dimensions shall be in mm and be related to the Platform datum's or reference lines.

3. DESIGN REQUIREMENTS

3.1 GENERAL

Design requirement shall be as per Cl. 4 of piping design criteria, Vol-II section 3.3.

3.2 DESIGN LOAD CASES

Pipe work, its supports and anchors, shall be designed to withstand the results of the following applicable combinations of loads and forces within the limits of stress set by ASME B31.3:

a) Hydro-test Condition (The empty weight plus weight of water to fill the piping).

b) Operating and Design Conditions (The empty weight plus the weight of operating fluid).

c) Wind loading condition

d) Dynamic Loading Condition

e) Periodic Site Test Condition

f) Any other condition that would affect the safety of the pipe work, e.g. cyclic loading and slug forces, when identified on the Data Sheet.

The pipe work shall be analyzed in its corroded state for each load combination.

3.3 DESIGN STRESS

Allowable stress shall be the maximum stresses permitted by ASME B31.3.

3.4 DRAWINGS AND CALCULATIONS

The Isometric drawings submitted by contractor shall contain information not limited to the following:

a) Material of construction.
b) Design pressure and design temperature.
c) Pressure rating of piping/piping components.
d) Hydro-test pressure.
e) Stress relieving requirements.
f) Stress analysis requirements.
g) Insulation requirements.
h) Supports details.
i) Line number.
j) NDT requirements.

Note:

In addition to above isometric drawing shall contain all pertinent information relating to the Standards, Codes and Specifications used in the design, fabrication, and inspection and testing of the pipe work, including the materials used.

The Contractor shall submit detailed calculations establishing the compliance of their design with applicable codes and standards specified in Piping Design Criteria.

All calculations shall be complete, giving all references and showing all working methods. The Contractor shall be able to provide proof of software verification for any software used. Computer printouts will not be accepted without input data and complete printout.

Wherever relevant, additional calculations shall be undertaken regarding the effects of slug forces.

Contractor to prepare the piping material specification for each class & shall provide the complete information such as pipe size, thickness, fittings, flanges branch connection tables, valve selection & tag numbers, designed pressure- temperature ratings and applicable codes/standards etc. Contractor shall submit the piping class data sheets for company's approval.

Contractor to prepare the data sheets & specifications for piping specialties based on good engineering practice & applicable codes/standards and obtain company's approval.

Approval of drawings, calculations and other documents by the Company does not relieve the Contractor of their responsibility for the correctness of the design to suit the stated conditions.

4. MECHANICAL REQUIREMENTS

4.1 MINIMUM THICKNESS

The minimum thickness of material other than carbon steel shall be based on requirements of pressure & other mechanical loading. However, the minimum thickness of high alloy steel vessels (austenitic) & their components shall not be less than 3 mm.

Contractor to provide thickness calculations for each size & piping class. Minimum wall thickness of carbon steel & low alloy pipes, including corrosion allowance, shall be as following:

- DN 40 ($1^1/_2$") or less – schedule 80
- DN 50 (2") through DN 150 (6")- schedule 80
- DN 200 (8") through DN 600 (24")- standard wall.

4.2 CORROSION ALLOWANCES

Unless otherwise specified on the Pipe Specification Index & other project specifications, carbon steel pipes shall have 3 mm corrosion allowance & carbon steel (NACE) shall have 6

mm corrosion allowance applied to all pressure retaining parts and all surfaces of non-removable internals exposed to the process fluid.

When corrosion protection is provided by a corrosion resistant metallic lining, a minimum thickness of 3 mm of lining material shall be used.

Pipe work that is subjected to erosion e.g. due to impingement by the process stream, shall be protected with extra wall thickness.

5. MATERIALS

5.1 GENERAL SPECIFICATION

Materials shall be as per ASTM, BS or API specifications referenced in Piping Design Criteria.

Materials shall be new and unused, clean and free from rust, pits and obvious defects. Material older than one year from date of manufacturing shall not be permitted.

Cast, ductile or malleable iron, aluminum, copper, or copper-bearing alloys shall not be used in hydrocarbon service.

For carbon/carbon manganese steel vessels operating below 0ºC and requiring impact testing, all pressure parts and direct attachment materials shall be manufactured with fully killed, fine grain materials.

All pipe fittings specified as galvanized shall be hot dip galvanized with a minimum of 763 grams of galvanizing material per square meter of surface area, in accordance with ASTM A123M/ASTM A153.

5.2 SOUR SERVICE REQIREMENTS

5.2.1 CARBON STEEL (NACE)

This shall be in accordance with NACE Standard MR-01-75/ISO 15156-1/2/3 and shall meet the following special requirements: -

SPECIAL REQUIREMENTS FOR MANUFACTURE, INSPECTION, TESTING AND SUPPLY OF CARBON STEEL (NACE) MATERIAL:

5.2.1.1 MANUFACTURE

The following treatments during steel making are mandatory:

a) All steel shall be fully killed and fine grained.

b) All steel shall be produced using Basic oxygen or Electro Furnace Process.

c) Steel shall be made by low sulphur and low phosphorous refining process and shall be vacuum degassed while molten by means of an approved procedure.

d) Effective sulfide shape control by calcium treatment shall be carried out if sulphur level is in excess of 0.002% for CS-NACE plate material and piping finished product fabricated from plate material.

e) Specific treatment to control non-metallic inclusions likes Aluminum oxide clusters, silicates and magnesium sulphide etc.

The manufacturer shall take particular care to control the rolling and heat treatment conditions so as to eliminate low temperature transformation microstructures associated with segregation such as bainite band or islets of martensite in order to reduce the propagation of HIC.

5.2.1.2 INSPECTION AND TESTING

The following tests shall be conducted in addition to relevant codes and standards. Test certificates shall be duly witnessed & certified by a Company approved third party inspection agency.

A. HYDROGEN INDUCED CRACKING (HIC) TESTING

This test shall be conducted only for pre-qualification of vendors who are not listed in suggested vendor list. This test shall be carried out on one finished product of each heat (material wise and type of construction wise, i.e. seamless and welded separately) irrespective of size/thickness. The test shall be carried out as per NACE-TM-02-84 standards.
HIC testing for casting is not required.

The test shall be performed on a set of three test specimens. The test shall be performed, as per NACE-TM-02-84 and the acceptance criteria shall be as Crack sensitivity ratio (CSR) < 1% & Crack Length Ratio (CLR) < 10.00%

In case any one of the above samples fails to meet the acceptance criteria, three more additional specimen from the product from which the first set of specimen were taken, shall be retested and results reported.

In case of failure of any of the samples in above, two additional products shall be selected from the same heat and size specimens shall be tested (three from each product).

In case of failure of any one of the six-samples, the particular heat will be rejected.

B. SULPHIDE STRESS CORROSION CRACKING (SSCC) TEST

This test shall be conducted only for pre-qualification of vendors who are not listed in suggested vendor list. This test shall be carried out on one finished product of each heat (material wise and type of construction wise i.e. seamless and welded separately) irrespective of size/thickness. The test shall be carried out as per NACE-TM-01-77.

Reporting of test result: Curve shall be reported as per NACE-TM-01-77 for various stress level between 72%and 90% of SMYS.

Acceptance Criteria: At72%SMYS, time of failure shall not be less than 720 hrs.
Sampling for test shall be same as indicated clause no. 5.2.1.2 A

C. HARDNESS TEST

This shall be carried out on finished product of each heat irrespective of size/thickness.

This test shall be carried out as per ASTM E-18/ ASTM E-92/ASTM E 10.

The product/heat for which hardness values are found in excess of specified value shall be rejected. Maximum hardness value shall be limited to HRC-22.

D. MICROSCOPIC TEST

This shall be carried out on finished product of each heat (material wise and type of construction wise i.e. seamless and welded separately) irrespective of size/thickness.

This test shall be carried out as per ASTM E-45 Method D. Steel shall be calcium treated for inclusion morphology control & there shall not be elongated manganese sulphide inclusions.

E. CHEMICAL ANALYSIS

This shall be carried out on finished product & raw material for each heat (material wise and type of construction wise i.e. seamless and welded separately) irrespective of size/thickness. Test results shall meet the relevant codes & standards.

The acceptable level of Sulphur is as follows:

i) 0.003 % Max for Flat rolled Products
ii) 0.01 % Max for Seamless products.
iii) 0.025 % Max for Forging.

If the Sulphur level is higher than these limits, HIC test as per clause no. **5.2.1.2 A** shall be carried out.

Carbon equivalent (CE) shall be less than 0.40 & PCM shall be less than 0.21 and shall be computed as per one of the following formulae depending upon.

a) If C $\geq$ 0.12%, CE= C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15
b) If C< 0.12%, PCM= C+ Si/30 + (Mn+Cu+Cr)/20 + Ni/60 + Mo/15 + V/10 +5B

F. MECHANICAL TEST

The mechanical properties shall be as per the specified material specification.

F. THROUGH THICKNESS TENSION TEST (FOR PLATES)

Plates 25mm and above in thickness shall be tested perpendicular to the rolled, surface with frequency and test procedure as per ASTM A 770 and determination of reduction of area as per ASTM A370. Minimum reduction area shall not be less than 35%.

G. ULTRASONIC TESTING (FOR PLATES)

Test shall be performed for thickness 12.7mm and above according to ASTM A578 level – B. No repaired welding shall be permitted on plates.

5.2.2 STAINLESS STEEL (NACE)

This shall be in accordance with NACE Standard MR-01-75/ ISO 15156-1/2/3 and test requirement as per applicable material standard.

A. INTER GRANULAR CORROSION CRACKING (IGC)

For all austenitic stainless steel Piping, Piping components and Valves, Inter-granular Corrosion Test (IGC) shall be conducted as per following ASTM A 262 practice "B" with acceptance criteria of 60 mills per year (max) for all materials- forged, rolled, wrought and casting.

5.2.3 DUPLEX STAINLESS STEEL

This shall be in accordance with NACE Standard MR-01-75/ ISO 15156-1/2/3 and test requirement as per applicable material standard.

The following tests shall be conducted in addition to relevant codes and standards. Test certificates shall be duly witnessed & certified by a Company approved third party inspection agency.

A. CHLORIDE STRESS CORROSION CRACKING TEST

This test shall be conducted only for pre-qualification of vendors who are not listed in suggested vendor list.

Either of the two tests given below can be carried out to assess the chloride Stress Cracking Resistance of the Material.

Test: 1

This test shall be conducted on the material at 155 Deg C with aeration in 45% $MgCl_2$ solution as per ASTM G36. Stress to cause rupture in 500 hours shall exceed 0.35 times the ultimate tensile strength. This test shall be carried out on one finished product of each heat irrespective of size & thickness.

Test: 2

This test shall be conducted on specimen as per ASTM G 36 at 100 degree Celsius with aeration in boiling 40% $CaCl_2$ solution & pH shall be at 6.5. Stress to cause rupture in 500 hours shall exceed 0.85 times the ultimate tensile strength. This test shall be carried out on one finished product of each heat irrespective of size & thickness.

B. PITTING CORROSION TEST

This test shall be conducted on specimen as per **ASTM G48** using ferric chloride solution (10% $FeCl_3.6\ H_2O$) as per method A at 22 ± 2^0 C for 72 hours and at 50^0 C for 24 hours. The acceptance criteria shall be no pitting shall occur on surface of tested material for the test at 22 ± 2^0 C and for the test at 50^0 C weight loss shall be reported to company for review. In addition to that, another test shall also be carried out to establish the critical pitting temperature as per method E of ASTM G 48 and report the critical pitting temperature to company for review. Preferably minimum critical pitting temperature shall be 30^0 C.

C. INTERGRANULAR CORROSION TEST (HUEY TEST)

The test shall be carried out, as per ASTM **A 262 Practice C.** weight loss shall not exceed 4 mils/month. This test shall be carried out on one finished product of each heat irrespective of size & thickness. The entire test reports shall be from the product specimen drawn from the heat bearing same number.

D. CHEMICAL TEST

This test shall be carried out on finished product of each heat irrespective of size & thickness as per relevant codes & standards with following limitation.

Pitting index (P.I) shall be computed as per following formulae.
P.I. = %Cr + 3.3(% Mo) + 16(% N_2)
The Pitting index of the material shall be greater than 35.
Carbon content shall be 0.03% (max.)
Sulpher content shall be 0.02%(max.)

E. MICRO STRUCTURAL EXAMINATION AND FERRITE MEASUREMENT

Ferrite content measurement of the material shall be carried out as per ASTM E 562 for each heat by metallographic examination. Ferrite content shall be in the range of 45% to 55%.

F. HARDNESS TEST

This test shall be carried out on finished product per heat, manufacturing method wise as per ASTM E 18 irrespective of size & thickness. Maximum hardness of the product shall be limited to HRC 25 (max.).

G. SULPHIDE STRESS CORROSION CRACKING (SSCC) TEST

This test shall be conducted only for pre-qualification of vendors who are not listed in suggested vendor list.

Resistance of the materials to sulphide stress corrosion cracking shall be tested using:

a) NACE –TM- 01-77 test solution (test temp.24^{o} C) minimum stress for cracking in 720 hrs. is 350 N/mm^2.

b) NACE-TM-01-77 test solution (test temp. 90^{o} C and total pressure of H_2S =16 bars) minimum stress for cracking in 720 hrs. is 325 N/mm^2.

H. MECHANICAL TEST

The following mechanical test shall be carried out: -

(I) Tensile Test

The material in solution-annealed condition shall conform to the mechanical properties requirements specified below-

a) Ultimate Tensile Strength : 680-900 N/mm^2

b) Yield Strength (0.2% offset) : 450 N/mm^2 (min)

c) Yield Strength (1.0% offset) : 500 N/mm^2 (min)

d) Elongation : 25% minimum

e) Hardness : HRC-25 (max.)

(II) Impact Test

Fracture Toughness (Charpy V-Notch Energy)

- For Wrought Materials

At +20 ^{o}C	(68 ^{o}F)	120 Joule (Avg)
		90 Joule (Min)
At -30 ^{o}C	(-22 ^{o}F)	100 Joule (Avg)
		75 Joule (Min)

- For Cast Materials

At +20 ^{o}C	(68 ^{o}F)	75 Joule (Avg)
		55 Joule (Min)
At -30 ^{o}C	(-22 ^{o}F)	60 Joule (Avg)
		40 Joule (Min)

I. CREVICE CORROSION

This test shall be conducted on specimen as per **ASTM G48** using ferritic chloride solution (10% FeCl 3.6 H2O) at 25 degree Celsius for 24 hours. The acceptance criteria shall be no crevice corrosion shall occur on surface of tested material at specific temperature. This test shall be carried out on finished product of each heat irrespective of size & thickness.

5.2.4 Incoloy 825/Inconel 625 AND Incoloy 825/ Inconel 625 CLAD CARBON STEEL PIPE MATERIALS

This shall be in accordance with NACE Standard MR-01-75 and test requirement as per applicable material standard.

A TECHNICAL REQUIREMENTS FOR CLAD PLATES

1. Clad plates shall be made by using standard practice suitable to obtain metallurgical bonding between carbon steel and corrosion resistant alloy. A detailed procedure for production of clad plates shall be submitted for company's approval and shall include the following as a minimum:

a) Material traceability system

b) Steel and plate making process
c) Heating technique, rolling reductions, and working temperatures.
d) Cladding process
e) Procedure for heat treatment
f) Ultrasonic testing of plates and bond integrity check.

2. Each finished plate shall be subjected to automatic ultrasonic examination for checking bonding integrity and soundness of carbon steel backing plate. This test shall be carried out on 100% plate surface area on both sides, in accordance with ASTM A 578 and its supplementary requirement S3. The acceptance criteria for the finished clad plates shall be as per ASTM A 578 level B besides specific technical requirements mentioned below.

3. Discontinuity, causing complete loss of back reflection, which cannot encompass within 75-mm diameter circles, shall be unacceptable.

4. Clad plates having a total area of discontinuity greater than 6,000 sq. mm. shall be rejected. Also where the lamination exceeds 6.35 mm. in length measured along the two longitudinal edges of the plate, shall be rejected. A Laminar defect in the backing plate is also unacceptable. Unbounded area, which cannot be encompassed within 25-mm. diameters circles, shall be unacceptable.

5. Acceptance criteria for integrity of bond of the cladding material shall be as per ASTM A578, level S7.

6. No defect in corrosion resistant alloy is permitted. Minor defects, if any as mentioned in (a) and (b) below may be repaired with company's prior approval by welding and shall be subsequently tested by dye-penetrant examination. This manufacturer shall submit welding procedure for company's approval prior to undertaking such repair works.

a) Finished clad plates having defects lower than 500-cm2 area and total defect area lower than 1.5% of the total plate surface area is only allowed to be repaired.

b) The plates having defects lower than 500 cm^2 area and total defects greater than 1.5% but lower than 3% of the total surface are may be repaired with company's prior approval.

c) Plates having defective area greater than 3% or single defect greater than 500 cm^2 shall be rejected.

B. TECHNICAL REQUIREMENTS FOR CLAD PIPES

1. A detailed procedure for manufacture of pipes from clad plates shall be submitted for company's approval and shall include the following as a minimum.

    a. Pipe making procedure
    b. Welding procedure specification including chemical composition of filler wire and flux
    c. Flux handling, Heat treatment
    d. Visual and dimensional checks
    e. Inspection and testing, chemical, mechanical, corrosion, microstructure etc.
    f. Non destructive testing etc.
    g. Hydrostatic testing
    h. Finishing treatment
    i. Marking

2. The tolerances on dimensions and weights of the finished clad pipes shall be as per API Spec. 5L with the following modifications:

Permissible tolerances on nominal wall thickness shall be as follows:

a) On Carbon Steel backing material + 10%, -5%
b) On Clading (Incoloy-825/Inconel-625) material + 10%, negative tolerance NIL
c) Cumulative tolerance: The cumulative tolerance resulting from all the individual variation shall be such that the maximum radial mismatches between the inside diameter of any two-pipe cross-sections is limited to 0.5mm.

3. Pipe ends shall be beveled to suit API-5L.

C. HEAT TREATMENT

After finish rolling for bonding, all clad plate produced shall be normalized at a temperature of 950 + 50 deg.C followed by air cooling and water quenching. This will be followed by tempering treatment at 620/650 deg.C followed by air cooling.

The detailed heat treatment procedure (including any alternatives proposed) shall be submitted for Company's approval.

D. PHYSICAL PROPERTIES AND TESTS

Test tabs of adequate length shall be attached to an adequate number of pipes so as to facilitate carrying out the specified mechanical and corrosion tests.

The clad pipe shall be subjected to the following tests to determine the physical properties of the finished product. The test shall be carried out on the full thickness of carbon steel, after removal of corrosion resistant alloy Incoloy-825 Inconel 625, on the finished pipe. While removing corrosion resistant alloy, care must be taken not to reduce the wall thickness of carbon steel backing metal as far as possible.

1. MECHANICAL PROPERTIES

The carbon steel backing metal shall meet the requirements of mechanical properties set forth in API spec. for Line pipe and shall in addition have fracture toughness of minimum 47 Joules (average)/38 Joules (individual), when subjected to Charpy V-notch test at 0 deg.C, in accordance with ASTM A370. Testing frequency shall be as required for guided bend test mentioned below. Orientation, location, and number of specimens per pipe shall be as per supplementary requirements SR 5 of API Spec.5L.

2. WELD TENSILE TEST

The specimen shall represent full wall thickness of carbon steel with corrosion resistant alloy removed from specimen. The ultimate tensile strength shall be higher than the minimum specified for the base material. Testing frequency shall be as required for guided bend test mentioned below.

3. GUIDED BEND TESTS

The guided bend test shall be carried out on specimens taken from the each of the two representative pipes of each lot. The lot shall be defined as a maximum of 50 pipes belonging to the same heat of carbon steel and manufactured with the same process.

4. MICROSCOPIC EXAMINATION AND HARDNESS TEST

Test specimen for microscopic examination shall be taken transverse to the longitudinal weld, from the finished pipe. The width of the specimen shall be minimum of three times the width of the weld. The specimen shall then be ground suitably and etched to reveal the microstructure. The specimen shall be visually examined using a magnification profile with smooth transition to base material. The microstructure shall also be examined at a suitable, higher magnification to ensure favorable distribution of micro-constituents. Chemical composition shall be ascertained by appropriate methods to verify that in the top 2mm of the cladding it is within the specified limits. Hardness test HV 10 according to ASTM A370 and at suitable locations as per MR 01 75/ISO 15156-2, Fig.2 (Along with hardness at Parent

material, weld metal and HAZ of Cladded layer) shall be carried out on the test pieces already submitted to micro structural examination. In the heat affected zone indentations shall start as close to the fusion line as possible. The maximum Vickers hardness value for each single point shall not exceed 248 HV 10 on carbon steel portion and 285 HV 10 corrosion resistant alloys.

E. BOND EFFICIENCY TEST

1. SHEAR TEST

Shear test shall be performed according to ASTM A265 and minimum shear strength in the longitudinal direction shall be 140 Mpa.

2. DUCTILITY TESTS

Ductility test in longitudinal direction shall be performed according to ASTM A 265. No crack and no unbounding shall be allowed.

3. CHARPY IMPACT TEST

This test shall be performed in accordance with ASTM A 370. From each finished pipe per lot representing same heat of carbon steel and manufactured with the same process, five samples shall be taken transverse to the principal rolling direction. From each sample three specimens shall be machined. The specimens shall be notched as follows: -

- Weld Metal
- Fusion line + 2MM
- Base metal

At the test temperature of 0^0.C, the fracture toughness values obtained shall be consistent with those indicated in clause C-1 above.

F. CORROSION TESTS

The specimen shall be machined only to remove the carbon steel portion and shall contain the full weld on the corrosion resistant alloy in as welded condition. Corrosion resistant alloy shall not be ground or picked and surface of specimen shall approximate the same roughness as the finished pipe. Specimen shall be machined transverse to the weld with following dimensions:-

Width (transverse to the weld)	:	50 mm
Length (parallel to the weld)	:	25 mm
Thickness	:	1.5 mm (Min.)

The following corrosion tests shall be carried out on the specimens prepared as above.

1. INTERGRANULAR CORROSION TEST

(a) For Incoloy 825, this test shall be carried out in accordance with ASTM A-262 practice C. Maximum permitted weight loss shall not exceed 0.0040 inch/month.

(b) For Inconel 625, this test shall be carried out in accordance with ASTM G28 Method A. Maximum permitted weight loss shall not exceed 0.0040 inch/month

2. PITTING CORROSION TEST

This test shall be conducted on specimen as per ASTM G48 using ferric chloride solution (10% FeCl3.6 H2O) as per method A at 22 ± 20^0 C for 72 hours and at 50^0 C for 24 hours. The acceptance criteria shall be no pitting shall occur on internal pipe surface of tested material for the test at 22 ± 2^0 C and for the test at 50^0 C weight loss shall be reported to company for review. In addition to that, another test shall also be carried out to establish the critical pitting temperature as per method C of ASTM G 48 and report the critical pitting temperature to company for review. Preferably minimum critical pitting temperature shall be 30^0 C for Incoloy 825 and 85^0 C for Inconel 625.

3. SULPHIDE STRESS CORROSION CRACKING (SSCC) TEST

This test shall be conducted only for pre-qualification of new vendors who are not listed in suggested vendor list.

Sulphide Stress Corrosion cracking test shall be carried out on the cladding alloy as follows:

a) Test as per NACE-TM-01-77 at test temperature.

i) Room/24^0C

b) Test as per NACE-TM-01-77 except that the test solution shall have H_2S partial pressure of 5bars and CO_2 partial pressure of 20bars and test temperature shall be:

i. At 90^0C

ii) At 120^0C

Note:- Test solution – 5% NaCl + 0.5% Acetic Acid

Each test result shall be evaluated based on acceptance criteria of Minimum Stress for no cracking to occur after 720 hrs equals to 95% of SMYS of cladding material used

G. HYDROSTATIC TEST

All the finished pipes shall be subject to mill hydrostatic test pressures corresponding to 90% of the specified minimum yield strength for Carbon Steel and the test pressure shall be held for at least 30 seconds. Permanent test records (Pressure time chart) shall be maintained by the manufacturer and submitted for review of Company.

H. NON-DESTRUCTIVE INSPECTION

The following non-destructive inspection shall be performed on each pipe after hydrostatic testing. A detailed non-destructive testing specification shall be developed and submitted to Company for qualification and approval.

1. RADIOGRAPHIC EXAMINATION

After hydrostatic test, the weld seams shall be checked for detection of longitudinal and transverse defects to the weld by radiographic method on the full length of the pipe. Acceptance criteria for the radiographs shall be as per ASME BPV-VIII, Division I, clause UW 51.

The film used for radiographic inspection of pipe welds shall be class 1 and 2 referring to Table –II of ASTM E-94, with film density of 1.8-3.8 that allow the sensitivity of at least 1.5% of the nominal total thickness of pipe weld bead.

2. SOUNDNESS OF BONDING

After hydrostatic test, each pipe shall be inspected by ultrasonic method for the detection of lamination and for bonding check on a circumferential band of 50mm along each side of the longitudinal weld.

No lamination and unbonding are allowed on pipes. Pipe ends of each pipe shall also be inspected by Ultrasonic examination with angle probe to detect hair line crack defects on pipe body, perpendicular to the surface, for a length at least 50 MM. No defect shall be allowed.

3. MAGNETIC PARTICLE EXAMINATION

After hydrostatic testing, all pipes shall be subject to Magnetic particle examination. This shall include full length of external surface of the weld bead of backing material to be examined by Magnetic particle method. The test procedure shall be in accordance with

ASTM E 709 and acceptance criteria shall be according to ASME Sec. VIII, Div.2 Part 7 Clause no. 7.5.6.2.

5.3 CORROSION ALLOWANCE

The corrosion allowance for various service of piping systems are shown below. MOC shall be as indicated in the P & ID.

(Wherever more than one material is indicated against the same service, the MOC shall be as per the class and grade indicated in the P&ID.)

SERVICE	Min CA (mm)	PIPES	FITTINGS	FLANGES
Hydrocarbon (Sour service) for a) Gas Lift b) Well Fluid	6.0	CS (NACE)	CS (NACE)	CS (NACE)
	1.5	SS316L (NACE)	SS316L (NACE)	SS316L (NACE)
	0.0	DSS	DSS	DSS
	0.0	CS with INCOLOY	CS with INCOLOY	CS with INCOLOY
INJECTION WATER	3.0	CS	CS	CS
PRODUCE WATER	0.0	GRE (Glass reinforced Epoxy)	GRE	GRE
INSTRUMENT GAS	1.5	SS316L (NACE)	SS316L (NACE)	SS316L (NACE)
	6.0	CS (NACE)	CS (NACE)	CS (NACE)
CLOSED DRAIN	6.0	CS (NACE)	CS (NACE)	CS (NACE)
OPEN DRAIN	3.0	CS	CS	CS
	0.0	GRE	GRE	CS
SEA WATER a)Raw Sea Water • Utility Water	0.0	90-10Cu-Ni	90-10 Cu-Ni	90-10Cu-Ni

• Cooling Water • Fire water	0.0	GRE	GRE	Al-Bronze
b)TreatedSeaWater	0.0	GRE	GRE	CS
• Fire water	3.0	CS	CS	CS
INSTRUMENT AIR	1.5	CS(G)	CS(G)	CS(G)
CHEMICAL	0.0	SS316L	SS316L	SS316L
FERRITIC CHLORIDE CHEMICALS	0.5	TITANIUM	TITANIUM	TITANIUM
Potable Water	0.0	Cu	Cu	BRONZE
(Drinking)	0.0	GRE	GRE	BRONZE
Sodium Hypo chlorite	0.0	CPVC	CPVC	CPVC
SEWAGE	0.0	GRE	GRE	GRE
ACIDISATION	3.0	CS	CS	CS

ABBREVATIONS:

AL-NI	-	ALUMINIUM NICKEL
CA	-	CORROSION ALLOWANCE
CPVC	-	CHLORINATED POLYVINYL CHLORIDE
CS	-	CARBON STEEL
CS(G)	-	CARBON STEEL (GALVANISED)
DSS	-	DUPLEX STAINLESS STEEL
GRE	-	GLASS REINFORCED EPOXY
MM	-	MILIMETER
NACE	-	NATIONAL ASSOCIATION OF CORROSION ENGINEERS
PVC	-	POLYVINYL CHLORIDE
SS	-	STAINLESS STEEL

5.4 CARBON STEEL

5.4.1 CARBON STEEL (NON-NACE)

Carbon Steel materials used and specified as per American specifications. Bolting shall be ASTM A193 Grade B7/194 Grade 2H and it shall be Hot dip galvanized as per ASTM 153.

PIPES:

ASTM A 106 Gr. B (seamless), API 5L Gr. B (seamless / SAW),
API 5 L Gr. X 52 (Seamless/SAW), API 5L Gr. X 60 (Seamless / SAW), ASTM 333 Gr.6 (Seamless)

FITTINGS:

ASTM A 234 Gr. WPB, ASTM-A105,
MSS SP Gr. WPHY 52 MSS SP Gr. WPHY 60
ASTM A 350 Gr. LF2 ASTM A 420 Gr. WPL6

FLANGES:

ASTM A105, ASTM A694 Gr. F52,
API 6A Type 2, API6A Type 4
ASTM A 350 Gr. LF2

VALVES:

BODY- ASTM 216 GR WCB, ASTM A 105, ASTM A 350 Gr. LF2, ASTM A 352 Gr. LCC
TRIM- 11-13% CROME (MIN.)/ ASTM 182 GRADE F316

PLATE : ASTM A-516 Grade 70

5.4.2 CARBON STEEL (NACE)

Material shall be same as clause no. 5.4.1. Sour service requirement shall be in compliance with clause no. 5.2.1 of this document.

STUD BOLTS: ASTM A-193 Grade B7M, For Low temperature use ASTM A 320 Gr. L7,(22 HRC max. hardness)
NUTS: ASTM A-194 Grade 2HM, For low temperature use ASTM 194 Gr.4 (22 HRC max. hardness)

Stress relieving of rolled plates, formed heads and pipe fittings shall be in accordance with NACE Standard MR0175.

5.5 STAINLESS STEEL

5.5.1 STAINLESS STEEL (Non-NACE)

Stainless steel materials are suitable for inner/wet parts of valves, instrumentation & vessels. Only following American specifications as specified against each product are permitted. MOC of stud bolt and nut shall be same as that of CS NON-NACE (cl. 5.4.1). Necessary considerations shall be followed to avoid galvanic corrosion.

PIPES : ASTM A 312 Gr. TP 316
ASTM A 358 TP 316

FITTING : ASTM A 403 Gr. WP 316
ASTM A 182 Gr. F316

FLANGES : A 182 Gr. F 316

VALVES :
BODY : ASTMA182GRADEF316/ASTM-A-351GRCF8M SS316
TRIM : ASTM A 182 GRADE F 316/ SS 316

Note:
For all austenitic stainless steel Piping, Piping components and Valves, Inter-granular Corrosion Test (IGC) shall be conducted as per following ASTM A 262 practice "B" with acceptance criteria of 60 mills per year (max) for all materials- forged, rolled, wrought and casting.

Or

ASTM A 262 practice "E" with acceptance criteria of "No cracks as observed from 20X Magnifications" for all materials other than castings "Microscopic structure to be observed from 250-X magnification" in addition.

5.5.2 STAINLESS STEEL (NACE)

Sour service requirement shall be in compliance with clause no. 5.2.2 of this document. MOC of stud bolt and nut shall be same as that of CS NACE (cl. 5.4.2).

PIPES : ASTM A 312 Gr. TP 316 L
ASTM A 358 TP 316 L

FITTING: ASTM A 403 Gr. WP 316 L
ASTM A 182 Gr. F316L

FLANGES: A 182 Gr. F 316 L

VALVES: BODY : ASTM A 182 GRADE F 316L / ASTM-A-351 GR CF8M SS316L
TRIM : ASTM A 182 GRADE F 316L / SS 316L

5.6 DUPLEX STAINLESS STEEL MATERIAL

Two grades are indicated in this MOC. However, MOC shall be as per the class and grade indicated in the P&ID.

PIPES : a) ASTM A790 UNS S 32205
b) ASTM A790 UNS S 31803

Note:
On case to case basis for higher thickness if manufacturing of seam less pipe is not possible, ASTM A 928 Class1 & 3 may be used.

FITTINGS: a) ASTM A790 UNS S 32205,
ASTM A182 Gr. F60 UNS S 32205,
ASTM A815 Gr. WP-S/WX UNS S 32205

b) ASTM A790 UNS S 31803,
ASTM A182 Gr. F51 UNS S 31803,
ASTM A815 Gr. WP-S/WX UNS S 31803

FLANGES: a) ASTM A182 Gr. F60 UNS S 32205
b) ASTM A182 F51 UNS S 31803

BOLTING: ASTM A 453 GR 660 Class A

GASKETS: Duplex stainless steel spiral wound with CNAF, Octagonal ring duplex stainless steel UNS S 32205/ UNS S 31803; Maximum hardness 22HRC.

VALVES: BODY-DSS, TRIM –DSS

NOTES:

1. Maximum hardness shall be limited to 25 HRC. Material shall meet the requirement of NACE MR 01-75 along with additional requirements

indicated in clause no. 5.2.3 of this document. This specification is in addition to relevant codes & standards.

2. Maximum temp. Limit shall be limited to 121^0C. For max temp greater than 121^0C, company's approval shall be obtained.
3. Stub end connections are not permitted for branch connection.
4. For butt weld fittings thickness shall be same as pipe
5. DSS shall have approximately equal proportion of Austenitic & ferritic matrix phase (ferritic content: 45-55%) for DSS.

5.7 (A) INCOLOY CLAD PIPING (Incoloy 825)

PIPES: ≤ 6" Seamless Pipe as per ASTM B423 (UNS N 08825) cold finished annealed.

>6" but ≤ 12" Seamless Pipe as per ASTM B423 (UNS N 08825) hot finished annealed. SMYS of hot finished annealed pipe shall as be established considering it as unlisted component of ASME B 31.3 for the temperature range based on actual evaluation after testing.

≥ 14" – API 5L Gr. B / API 5L Gr X52, Cladding Material – Incoloy 825.

FITTING: ≤ 1 ½ " Forged ASTM B564 (UNS N 08825)

2" – 12" Butt welded fitting ASTM B366 (WP NIC MC S)

≥ 14" – from clad steel with single weld seam (joint factor 1), Cladding Material Incoloy 825.

FLANGE: ≤ 12" -- ASTM B564 (UNS N 08825)

≥ 14" -- ASTM A105 with Cladding Material Incoloy 825.

BOLTING: ASTM A453 Gr. 660 Class A.

GASKET: Spiral wound duplex stainless steel with CANF, Octagonal ring duplex stainless steel, Hardness 22HRC

VALVE: ≤ 11/2" Body Forged ASTM B 425/ASTM B 564 (UNS N08825)

Trim Incoloy -825

≥ 2" but ≤ 6" Body Incoloy 825/DSS and Trim Incoloy 825/DSS

≥ 8" Body = (ASTM A 105 + 3mm Weld overlay Incoloy 825)/Solid Incoloy 825/DSS and Trim Incoloy 825/DSS. (See note-4)

5.7 (B) INCONEL CLAD PIPING (Inconel 625)

PIPES: ≤ 12" Seamless Pipe as per ASTM B444 UNS N 06625Gr.1 annealed.
≥ 14" – API 5L Gr. B / API 5L Gr X52, Cladding Material – Inconel 625.

FITTING: ≤ 1 ½ " Forged ASTM B564 (UNS N 06625)
2" – 12" Butt welded fitting ASTM B366 (WP NC MC S) UNS N 06625
≥ 14" – from clad steel with single weld seam (joint factor 1), Cladding Material Inconel 625.

FLANGE: ≤ 12" -- ASTM B564 (UNS N 06625)
≥ 14" -- ASTM A105 with Cladding Material Inconel 625.

BOLTING: ASTM A453 Gr. 660 Class A.

GASKET: Spiral wound duplex stainless steel with CANF, Octagonal ring duplex stainless steel, Hardness 22HRC

VALVE: ≤ 11/2" Body Forged ASTM B 446 UNS N 06625
Trim Inconel -625
≥ 2" but ≤ 6" Body Inconel 625/DSS and Trim Inconel 625/DSS
≥ 8" Body = (ASTM A 105 + 3mm Weld overlay Inconel 625) /Solid Inconel 625/DSS and Trim Inconel 625/DSS. (See note-4)

NOTES:

1. The corrosion resistant alloy shall meet the requirements of ASTM B424, UNS No N 08825 or ASTM B 443, UNS N 06625 as applicable with the following additional stipulated requirements:
   (i) **For Incoloy 825**
   a) Max carbon content shall be 0.025%
   b) Minimum pitting index shall be 32.
   Pitting index = (% Cr + 3.3%Mo)
   c) The minimum cladding thickness shall be 3mm.
   d) Hardness shall be as per NACE MR-01-75/1SO-15156-3.

   (ii) **For Inconel 625**
   a) Max carbon content shall be 0.05%

b) Minimum pitting index shall be 52.
Pitting index = (% Cr + 3.3%Mo)

c) The minimum cladding thickness shall be 3mm.

d) Hardness shall be as per NACE MR-01-75/1SO-15156-3.

2. This shall be in accordance with NACE Standard MR-01-75 and test requirement as per applicable material standard along with requirements indicated below for solid Incoloy/Inconell:
   (a) Mechanical Properties Test.
   (b) Hardness Test.
   (c) Chemical Properties.
   (d) Corrosion Test as per clause No. 5.2.4, (F), (1) & (2) shall be carried out on each heat and test indicated in clause no. 5.2.4(F), (3) shall be carried out only for qualification of vendor not listed in suggested vendor list of ONGC.

3. For thickness of clad pipe and piping items only base material shall be considered.

4. MOC of valve shall be decided based on service.

5.8 CUPRO-NICKEL (90-10)

Cupro-nickel materials are suitable for firewater, seawater & salt-water services. Only following American/British specifications as specified against each product are permitted. Bolting shall be ASTM A193 Grade B7/194 Grade 2H and it shall be Hot dip galvanized as per ASTM 153.

PIPES: ASTM B 466 Copper alloy no. C-70600 or BS 2871 CN 102

FLANGE: EEMUA 145

FITTINGS: 90-10 Cu-Ni

VALVES Body - 5% Ni-AL BRONZE (BS 1400 AB2C), ASTM B 148 C 95800
Trim - MONEL K 500 for sizes up to and including 6" and K400/K500 above 6"

NOTES:

a) 90-10 Cu-Ni piping material thickness EEMUA PUB 144 to be decided based on 16-bar system

b) Blind flanges shall be A 105 with 3 mm 90-10 Cu-Ni overlays.

c) For composite flanges the outer flange shall be ASTM A105 (galvanized) to #150 of ASME B-16.5 up to 24" and ASME B 16.47 A for 22" and 28" &

inner flange shall be 90/10 CU-NI forged for < 2" S.O, 3" to 24" W.N.(For 3" to 24' S.O. Bossed flanges (suitable for welding)
F F material Cu Ni forged (When used with flanged valve use full faced gasket and S.O. Bossed flange).

d) Stud bolts shall be supplied with insulating sleeves and washers.

5.9 COPPER

Copper materials are suitable for potable water (drinking water) services. Only following American/British specifications as specified against each product are permitted. Bolting shall be ASTM A193 Grade B7/194 Grade 2H and it shall be Hot dip galvanized as per ASTM 153/ ASTM 123.

PIPES: Seam less hard drawn H-80/H55 regular Copper to ASTM B 42(UNS NO. C12200)

FLANGES: ISO 7005-3: ASTM B 61(Leaded tin Bronze)

FITTINGS: For less than or equal to 2" ASTM B 124 (UNS C 11000),
For 3 to 4" ASTM B42 (UNS NO C 12200)

VALVES: BODY, BONNET, TRIM – ASTM B 61(Leaded tin Bronze)
Stem, Wedge disc, Body seat ring - Forged Brass as per ASTM B 124 UNS NO. C37700 (No casting)

NOTES:

a) Stud bolts to be supplied with insulating sleeves & washers.

b) Blind flanges for Cu. piping shall be ASTM A105 with 3 mm copper overlays

c) Composite flange: Outer Flange : ASTM A 105 (Galvanized)
Inner Flange : ASTM B 124 (UNS NO. C 11000)
≤ 2" S.O.
≥ 3 to 4" W.N
For S.O. and W.N. Flange FF valves shall be used for size 2 to 4" only. FF solid slip on class 150 flange to ISO-7005-3: ASTM B 61 shall be used.

d) Threading on pipes is not allowed. Screwed are only for vents and drains.

5.10 GLASS REINFORCED EPOXY

Refer Annexure- 1

Note: Valve material for GRE piping shall be suitable to services for which GRE piping is used.

5.11 POLY VINYL CHLORIDE

PVC materials are suitable for sewage water & chemical services. Only following American specifications as specified against each product are permitted:

PIPES	:	ASTM D 1785
FLANGES	:	ASTM D 2466/ASTM D 2467
FITTINGS	:	ASTM D 2466/ASTM D 2467
VALVES	:	BODY – PVC-ASTM D 1784 TRIM – As per manufacturer Standards.

NOTES:

a) Complete piping system shall be enclosed in Galvanized carbon steel pipe for protection against mechanical damage. The size of cover shall be one size higher than the line size.

b) The hydro test pressure shall be 1.5 times the design pressure.

5.12 CHLORINATED POLY VINYL CHLORIDE

CPVC materials are suitable for sodium hypo chloride services. Only following American specifications as specified against each product are permitted

PIPES	:	ASTM F 441
FLANGES	:	ASTM F 439 up to 3"/ ASTM F 437 for Threaded fittings.
FITTINGS:		ASTM F 439 S.W up to 3", ASTM F437 Union up to 3"/ / ASTM F 437 for Threaded fittings.
VALVES	:	Body ASTM- D-1784 Trim –As per manufacturer standards.

NOTES:

a) Complete piping system shall be enclosed in Galvanized carbon steel pipe for protection against mechanical damage. The size of cover shall be one size higher than the line size.

b) All CPVC material shall be type IV Gr. 1 complying to ASTM-D-1784 (Class-23447-B)

c) Union shall be provided on one side of SCRD valves to facilitate its removal.

d) Maximum test pressure for valve and fabrication assembly shall be 225 psi.

5.13 TITANIUM

Following American specifications as specified against each product are permitted.

PIPES	:	ASTM B 861 Grade 2/ B862 Grade2
FLANGES	:	½" to 6" # 150 ANSI , Lap Joint Flange -Inner Flange ASTM B 363 Gr WPT2, Outer Flange ASTM A 105 or ½" to 6" #150 WN, RF to ASTM B 381Gr F2 ½" TO 6" Blind Flange ASTM B 381Gr F2
FITTINGS	:	ASTM B 363 Gr WPT2 Seam less/ BW (Lap joint stub end with Neo prene insulating ring)
VALVES	:	Body/Trim: ASTM B 381Gr F2/ASTM B 367Gr C2

5.14 MATERIAL IDENTIFICATION

All piping materials (pipe, fittings, flanges, valves, piping specialties etc., shall be supplied with mill certified test reports and certificates to identify the type of steel, composition, heat number and any special testing.

The Contractor shall furnish to the Company with one (1) copy of all mill certificates for all the materials purchased by the Contractor duly certified by company approved third party inspection agency.

The contractor shall mark with double blue stripes for NACE materials.

5.15 SUPPORTS AND MISCELLANEOUS

Any material shall not be welded directly on process piping. If situation arises then prior approval of Company shall be sought during detail Engineering & shall note that the material welded directly to pressure retaining pipe work shall be of similar quality as the pipe work, including impact requirements, if any, for a length measured from the vessel wall of at least 150 mm. The material of such items beyond this point may be structural

quality A283 Gr. C. or equal/superior (contractor to substantiate with sufficient information & documentary evidence).

5.16 BOLTING

For carbon steel and stainless steel flanged piping systems, stud bolts shall comply with ASTM A193-B7 and ASME B1.1. Nuts shall be heavy hexagon semi-finished per ASTM A194-2H and ASME B18.2.2. Carbon steel studs and nuts shall be hot dip spun galvanized in accordance with ASTM A153.

For carbon steel (NACE) flanged piping systems, stud bolts shall comply with ASTM A193-B7M and ASME B1.1. Nuts shall be heavy hexagon semi-finished per ASTM A194-2HM and ASME B18.2.2. Carbon steel (NACE) studs and nuts shall be hot dip spun galvanized in accordance with ASTM A153.

Duplex stain less steel flange, studs and nut shall be ASTM A 453 Gr 660 Cl A.

5.17 GASKETS

Gasket materials must be such that the internal fluids shall have no harmful or corrosive effects on them.

Gaskets for raised face flanges shall be spiral wound, non-asbestos filled, with 316 stainless steel in accordance with ASME B16.20, with the exception that compressed fibre gaskets complying with ASME B16.21 are acceptable for cooling water service.

Full-face gaskets for flat face flanges shall be made from 3-mm (1/8") thick compressed, oil-resisting non-asbestos sheets, or neoprene. Materials shall be specified in the pipe class Data Sheets.

Ring gasket for ring type joint shall be octagonal and shall be as per ASME B 16.20.
Rings for API 5000# flanges shall conform to API 6A. Type RX rings shall be used.

5.18 CERTIFICATION DOCUMENTS

All pressure parts material certification shall be traceable to heat numbers. Certificates, including all material certificates, Mechanical test certificates, welding qualification certificates, heat treatment certificates and hydrostatic test certificates duly certified by

company recognized third party inspection agency shall be available at final inspection and for counter signature by the certification authority and stored by the Contractor for a minimum of 5 years after acceptance of the piping by the Company. Pressure retaining parts shall be clearly marked to allow verification of tractability.

6. TECHNICAL NOTES OF PIPING COMPONENTS

6.1 PIPE

Pipe dimensions shall be in accordance with ASME B36.10 for carbon steel pipe and ASME B36.19 for stainless steel pipe and BS2871 Part 2 for 90/10 Cu-Ni pipe work up to DN 500 (20").

Nominal pipe sizes DN 30 (1¼"), DN65 (2½"), DN85 (3½") and DN 125 (5") shall not be used except where they are required for connections to equipment of standard design or where specific velocities must be maintained. When these sizes are used on equipments, the connecting piping shall be increased or decreased to standard sizes as close to equipment as practical.

Company approval shall be sought prior to using such components.

The minimum nominal pipe size shall be DN20 (¾") except for air, instrument air, water and manufacturer's standard equipment piping.
All nipples shall be made from pipe of same quality as that of parent piping

Carbon steel pipe DN40 (1½") and smaller used for process lines and other lines carrying flammable or toxic fluids shall have wall thickness at least Schedule 80.

Galvanized pipe shall not be bent or welded. Piping requiring bending, welding or threading shall be galvanized after fabrication.

All CS Pipes shall be seamless up to 16". Above 16" pipes shall be SAW with 100% radiography. Weld joint factor to be considered shall be as per ASME B 31.3/ASME B 31.4/ASME B 31.8 (As applicable).

All 90/10 Cu-Ni pipes shall be seamless up to 16". Above 16" pipes shall be SAW with 100% radiography. Weld joint factor to be considered shall be as per ASME B 31.3.

All stainless steel including DSS piping up to 8" shall be seamless. Above 8" shall be EFW with 100% radiography. Weld joint factor to be considered shall be as per ASME B 31.3.

All pipes shall have bevel end for size equal to and above 2" and plain end below 2".

6.2 TUBING

All tubing, unless otherwise noted, shall be 316 SS/316L SS seamless with wall thickness appropriate for the service & piping class.
Refer to the Specification FS 3507 for Instrumentation Bulk item requirements.

6.3 FITTINGS

All unions DN25 (1") and larger shall comply with BS 3799.

No street elbows or threaded bushings shall be used in piping. Hexagonal bushings (but no flush bushings) may only be used with tubing fittings for connection to instruments, or as otherwise specifically approved by the Company.

The thickness of reducing fittings shall match the wall thickness of the higher schedule pipe wall. The fitting wall thickness shall be tapered on a 1:4 gradient to ensure that the pipefitting wall thickness matches the lower schedule pipe wall.

All 90º-weld elbows shall be long radius, unless restricted by available space. Short radius (SR) welding elbows shall not be used unless absolutely required for clearance purpose with prior approval from the company. If short radius weld elbows are used, they shall be de-rated to 80% of the calculated allowable working pressure if subject to pulsations.
Thickness for fitting shall be same as that of corresponding pipe. Short radius elbows, which have been de-rated as specified above, may require a wall thickness greater than that of the connecting pipe.

Welded fittings materials shall be compatible with the piping material.

Fittings DN40 (1-½") and smaller shall be socket weld except as dictated.

Fittings DN50 (2") and larger shall be butt welded except as dictated.

Mitred joints shall not be used.

6.4 BRANCH CONNECTIONS

Branch connections shall be in accordance with API RP 14E. Branch connections for non-ferrous materials shall be as per manufactures standard with prior approval from Company. Contractor to prepare and list the branch tables in piping class data sheets. The lists shall show requirements for branches at 90º angles to the header & branching component with sufficient details.

6.5 FLANGES

Flanges shall be in accordance with ASME B16.5 for DN50 to DN600 and with ASME B16.47 Series A for flanges DN650 and larger. They shall be raised face unless otherwise shown on the individual vessel data sheets and/or drawings. Non-standard size flanges shall be calculated in accordance with ASME Code Rules & prior approval shall be sought from company.

Flanges shall be raised face up to #600 rating and shall be RTJ above #600 rating.

API ring joint 5000-psi flanges shall comply with API 6A.

ASME ring joint (RTJ) flanges shall have octagonal grooves conforming to ASME B 16.5.

API ring joint flanges shall conform to API specification 6A.

The bolt-hole pitch circle diameter for orifice flanges DN50, DN80 and DN100 shall be 1.6 mm smaller than specified in ASME B16.5.

Flange bolt-holes shall straddle the piping vertical and horizontal centerlines.

Flanges in the piping shall be kept to a minimum. Flanges shall be installed only to facilitate construction, maintenance and inspection and in cases where process conditions dictate.

Spectacle blinds rather than spade blinds shall be provided where required. Thickness of blinds shall be calculated in accordance with ASME B31.3. Pairs of spacers and blinds shall be used instead of spectacle blinds of size DN 350 and larger.
Class 400 flanges shall not be used unless required to match nozzles of compressors, pump, turbines, etc. of standard design supplied by equipment manufacturers.

Smooth finish on flanges shall be 32-63 AARH.

Wherever face finish is not mentioned, it shall be serrated spiral/concentric

6.6 BOLTING

Flange bolting of nominal size M40 (1-½") and above shall be subject to bolt tensioning. Flange bolting shall be a full threaded alloy steel stud bolt, each with two heavy hexagonal nuts. Stud bolts shall have full continuous threads and have lengths in accordance with B16.5 with the provision that a minimum of one (1) thread and a maximum of three (3) threads outside each nut and complete with 2 nuts to facilitate bolt tensioning.

Stud bolts shall be used for all piping closures except where tapped wafer valves dictate the use of machine bolts.

6.7 VALVES

Valve bodies, seals, etc., shall be in accordance with the design pressure and design temperature of the applicable project Specification. Valves may be supplied with higher design pressure or design temperature trims in order to meet the service requirements.

Each valve shall be supplied with a stainless steel tag, attached to the gland bolting, or hand wheel, with stainless steel wire, bearing the applicable valve identification, Tag Number and Purchase Order number.

Ball valves shall comply with API 6D or BS EN ISO 17292/BS 5351 (below 2"). All ball valve body patterns shall be long pattern to ASME B16.10 and shall be quarter-turn design. Soft seals and seats for ball valves shall be suitable for the maximum applied differential pressure, the service fluid and the specified pressure and temperature ratings.

As a general philosophy, ingress of sand particle is envisaged in well fluid and injection water services unless indicated otherwise elsewhere in the bid hence all valves are required to be metal seated in well fluid and injection water service. Trim of the metal seated Ball Valves as a minimum shall be Tungsten Carbide coated. Minimum surface hardness shall be 1050 Vickers and minimum coating thickness of finished and machined surface of Tungsten Carbide shall be 150µ to 250µ. Uses of soft seated valves are allowed only when no erosive effect is envisaged due to abrasive particle. Contractor is required to establish the same during detail engineering.

Check valves shall comply with BS 1868 or BS 5352/BS EN ISO 15761 or API 602. Swing type check valves shall have bolted bonnets. Where check valves are placed in vertical runs, valves shall be equipped with flapper stops. The stops shall not be connected to bonnet taps in any way.

In pulsating, turbulent or high velocity flow, to avoid possibility of slamming in check valve tilting disc slam check valve shall be preferred.

For heavy check valves, provisions shall be available for lifting by way of lugs, eyebolts and such standard devices

Gate valves shall comply with API 600, 602 or 603 as applicable. Gate and butterfly valves shall be used in “clean” non-hydrocarbon services like firewater services only.
In hydro carbon services for #900 & above, gate valves may be used in hydrocarbon service in place of ball valves for sizes up to 1.5”.

Globe valves shall comply with API 602/ BS 1873/ BS 5352/BS EN ISO 15761.

Plug valves shall comply with BS 1873/ BS 5353.

Butterfly valves shall comply with BS 5155/BS EN 593/ API 609

Steel and alloy valves shall be designed and tested in accordance with the applicable codes as per type & class of valves are as under: -

1. ASME 150# - Designed and examined in accordance with ASME B16.34 and tested in accordance with API 598.

2. ASME 300# through ASME 2500# - Designed and tested in accordance with API 6D for ball valves & other applicable codes as per type of valves.

3. API 2000# through API 5000# - Designed and tested in accordance with API 6A.

4. All hydrocarbon valves with non-metallic seats and seals including ball valve and butterfly valve shall be fire-safe, tested in accordance with API 607, API 6FA, ISO 10497 or BS 6755 PART II. The ball valves shall be fire safe in accordance with the requirements of either API 6FA (for trunnion ball valves) or API 607/ISO 10497 (for floating ball valves). All valves shall be witnessed & certified by a company approved third-party agency.

5. Metal Seated Ball valve shall be tested as per API 6D or API 6A as applicable. Leakage rate shall not be more than Rate – "B" as per API 6D/ISO 5208.

Valves i.e. Ball, Gate, Globe, Choke and Needle used in service with more than 230ppm of H2S shall be designed for Fugitive Emission from external leakage paths of Valve stem seals and body joints/Seals. Each type of valve design shall be certified by type testing as per BS EN 1SO 15848-1. Each valve as a minimum shall comply with:

(a) Tightness Class as per Table -1 (Based on type of stem seal) and Leakage from body seals as per Table-2 of BS EN ISO 15848-1.

(b) Endurance Class - CO1 of BS EN ISO 15848-1

(c) Temperature Class – (t-RT) & (t200^{0}C) of BS EN ISO 15848-1

The type test for each design of Valve shall be witnessed and certified by ONGC approved Third Party Inspection Agency.

During production of Valves, each valves which has been type tested as per BS EN ISO 15848-1 shall be revalidated by testing for tightness class as per BS EN ISO 15848-2. Tightness class test for each Valve shall be witnessed and certified by ONGC approved Third Party Inspection Agency.

Gate, globe, angle, ball and check valves shall be supplied with replaceable seats. Where replaceable seats are not available, the valve seat shall be stellited and welded into the valve body.

Valves designated "LO" (Locked Open) or "LC" (Locked Closed) on drawings, shall be provided with locking devices. Valves shall be furnished with the locking tab hardware installed.

Open-ended valves shall be equipped with threaded plugs or blind flanges.

a) Ball shall be solid type unless otherwise specified.
b) All valves shall be forged type for 1 ½'' & below.
c) All valves shall be supplied with position indicator.
d) All trunnion mounted ball valves shall be supplied with double block & bleed arrangement

Every block valve shall be provided with a lever, handle, or hand wheel as necessary to operate the valve.

Gear operators shall be heavy-duty lubricated type and shall be completely housed in a weatherproof enclosure.

Socket-weld valves shall be bolted body or top entry design, allowing removal of seats/seals for heat protection, prior to welding, without loss of assembly orientation. Single piece valve bodies, or valves bodies assembled by screwed-together components, shall not be used with socket-weld ends.

If an overlay weld-deposit is used for the body seat ring, seating surface, the seat ring base material shall be at least equal to the corrosion resistance of the material of the shell.
All valves shall be reduced bore unless & otherwise specified. Full bore valves shall be provided for all piggable lines.

Valve body thickness shall be as per relevant valve standard, shall be in accordance with ASME B16.34.

Valve body of Cast material of Construction shall be 100% radiographed irrespective of rating in accordance with ASME B16.34 Annexure B.

Valves body other than Cast material of Construction shall be radio graphed to the following extents: -

ASME 150#, DN600 or smaller	25%
ASME 150#, DN650 or larger	100%
ASME 300#, DN400 or smaller	25%
ASME 300#, DN450 or larger	100%
ASME 600# and higher	100%
Carbon steel to NACE requirements	100%
Stainless and high alloy steel (Cast or Forged)	100%
Other alloys (Cast or Forged)	100%

UT in place of radio grapy is acceptable for forged valve in accordance with ASME B16.34.

Socket-weld-end valves with non-metallic seats or seals shall be provided with 80mm long nipples having materials and thickness equivalent to those specified in the relevant piping specifications. These nipples shall be welded to the valves on both ends before the packing, seats and seals are fitted.

Stem protection is required for all carbon steel gate and globe valves where 13% Chromium trims are specified. The stems shall be totally enclosed in sleeves, which shall be packed with grease.

By-pass requirement for gate valves shall be provided as per following. Basic design of bypass shall be MSS-SP-45 & ASME B-16.34

ASME 150 class	on sizes 26” and above.
ASME 300 class	on sizes 16” and above.
ASME 600 class	on sizes 6” and above.

ASME 900 class	on sizes 4" and above.
ASME 1500 class	on sizes 4" and above.
ASME 2500 class	on sizes 3" and above.

By-pass valve shall be a globe valve. The sizes shall be as under:

On main valve	≤ 4"	½" or more
On main valve	> 4" but <10"	¾" or more
On main valve	≥10"	1" or more.

Valves with by-pass shall have the direction of flow marked on the main valve. By-pass attachment to the main valve body shall not be screwed. All fillet welds for by-pass installation shall be 100% examined by DP/MP test.

The by-pass piping arrangement shall be such that clearance between main valve body & bypass assembly shall be the minimum possible for layout reasons.

Material of construction of yoke shall be minimum equivalent to body/bonnet material.

Stelliting / hard facing by deposition shall be minimum 1.6 mm.

Soft-seated Ball, plug & butterfly valves shall be supplied with antistatic devices.

Soft seated Ball valves shall be floating ball type/trunnion mounted type as per following:

150#	8" & below	floating ball
	10" & above	trunion mounted
300#	4" & below	floating ball
	6" & above	trunion mounted
600#	1.5" & below	floating ball
	2" & above	trunion mounted

For metal seated ball valve, trunion mounted ball valve in place of floating ball valve is acceptable. However, vise-versa is not acceptable.

Generally the valves are hand wheel or lever operated. Gear operation shall be provided as under:

VALVE TYPE	CLASS	SIZE REQUIRING GEAR OPER.
Gate valve & diaphragm	150 class	14" and larger
	300 class	14" and larger
	600 class	12" and larger
	900 class	6" and larger
	1500 class	3" and larger
	2500 class	3" and larger
Globe valve	150 class	10" and larger
	300 class	8" and larger
	600 class	6" and larger
	900 class	6" and larger
	1500 class	3" and larger
	2500 class	3" and larger
Ball/plug Valve (other than pressure balance plug valve)	150 class	6" and larger
	300 class	6" and larger
	600 class	4" and larger
	900 class	3" and larger
	1500 class	3" and larger

Where gear operator is not called for as per above but vendor recommends a gear operator, he shall highlight such cases & obtain company's approval. For basis of finalization of gear operator vendor shall follow the guide line as mentioned below:

Torque and Gear Ratio required shall be based on guide line as indicated below:

Valve hand wheel diameter shall not exceed 750mm and lever length shall not exceed 500mm on either side. Effort shall not exceed 35 Kg at hand wheel periphery. Failing to meet above criteria vendor shall offer gear operated Valves even if valve does not fall into the category indicated in the above table

The following guide line shall be followed for gear box/hand wheel/lever:

Hand wheel -operated:
Max. Single hand forces: 200 N.
Max. Diameter of hand wheel: 750 mm.
(Max. torques 150 Nm).

Lever-operated:
Max. Force: 350 N.
Max. Length of lever: 600 mm.
(Max. torques 270 Nm).

Manual and gear operated valve shall be operated without use of any extension device. The torque/force value shall be recorded during the test. Torque value recorded shall not be more than indicated above.
Valve hand wheel/Lever shall be solid malleable Iron to ASTM A 47 (or any other applicable standard)/ Solid SS-316 material as per suitable MOC of the valve for required torque. In case of Malleable Iron, hand wheel shall be galvanized with thickness suitable to marine environment and subsequently painted as per FS 2005.

Vendor shall declare the operating torque value of each valve based on above guide line and carryout the required test as indicated in clause no. 6.7.1.1 through 6.7.1.7

6.7.1 TEST REQUIREMENTS OF VALVES

This clause elaborates minimum mandatory tests required to be carried out by the vendor at his premises, at fabrication yard of LSTK Contractor and offshore location. The tests indicated against this clause are related to the valve only, other material tests as required by relevant clause of this specification shall be complied by the vendor in addition to the tests indicated under this clause.

6.7.1.1 SHUT DOWN VALVE

(a) Material tests as per clauses no. 5 and sub clauses of FS 2004 A.
(b) Radiography of shell/body of the valve.
(c) DPT or MPT (as applicable) of the items/parts of the valve.
(d) Fire type test with respect to design of the valve.
(e) Hydrostatic Shell Test.
(f) Hydrostatic Seat Test.
(g) Pneumatic Seat Test.
(h) Check Torque of the Valve against zero differential pressure (as indicated in relevant para. of clause 6.7 of FS 2004A).
(i) Check Torque of the Valve against rated full differential pressure (as indicated relevant para. of clause 6.7 of FS 2004A).
(j) Check Torque of the Actuator as per FS 2004D.
(k) Type test of Fugitive Emission as per BS EN ISO 15848-1(If applicable).
(l) Fugitive Emission test as per BS EN ISO 15848-2 (If applicable).
(m) Check surface preparation and all the stages of protective coating as per FS 2005.
(n) Functional Test of the Valve – Shut Down Valve be pressurized to full differential rated pressure with liquid same as used for hydro test and thereafter shut down valve be operated by giving full rated pressure to actuator and rated voltage to Solenoid Valve. Full opening

and closing shall be checked. The operation of the valve shall be smooth and jerk free. This activity shall be repeated 5 times at manufacturer's location.

Sequence of operation in the entire test shall be alternate with one cycle with zero differential pressure across the valve and one cycle with full differential pressure across the valve.

Same activity shall be repeated 5 times with zero differential pressure across the valve on installation at fabrication yard and 5 times at offshore.

(o) Valve shall be API Monogrammed which are designed under API Codes and Standards, as applicable.

6.7.1.2 MOTORIZED BALL VALVE

a) Material tests as per clauses no. 5 and sub clauses of FS 2004 A.
b) Radiography of shell/body of the valve.
c) DPT or MPT (As applicable) of the items/parts of the valve.
d) Fire type test with respect to design of the valve.
e) Hydrostatic Shell Test.
f) Hydrostatic Seat Test.
g) Pneumatic Seat Test.
h) Check Torque of the Valve against no differential pressure zero differential pressure (as indicated in relevant para. of clause 6.7 of FS 2004A).
i) Check Torque the Valve against rated full differential pressure (as indicated relevant para. of clause 6.7 of FS 2004A).
j) Check Torque of the Actuator as per FS 2004D.
k) Type test of Fugitive Emission as per BS EN ISO 15848-1 (If applicable).
l) Fugitive Emission test as per BS EN ISO 15848-2 (If applicable).
m) Check surface preparation and all the stages of protective coating as per FS 2004D.
n) Functional Test of the Valve – Motor Operated Valve (MOV) shall be pressurized to full rated differential pressure with liquid same as used for hydro test and thereafter MOV shall be operated by giving full rated power to actuator. Full opening and closing shall be checked. The operation of the valve shall be smooth and jerk free. There shall be not abrupt fluctuation of current except during brake to open and Brake to close. The operation of the valve shall be smooth and jerk free. This activity shall be repeated 5 times at manufacturer's location

Sequence of operation in the entire test shall be alternate with one cycle with zero differential pressure across the valve and one cycle with full differential pressure across the valve.

Same activity shall be repeated 5 times with zero differential pressure across the valve on installation at fabrication yard and 5 times at offshore.

(o) Valve shall be API Monogrammed which are designed under API Codes and Standards, as applicable.

6.7.1.3 BALL VALVE.

(a) Material tests as per clauses no. 5 and sub clauses of FS 2004 A.
(b) Radiography of shell/body of the valve.
(c) DPT or MPT (As applicable) of the items/parts of the valve.
(d) Fire type test with respect to design of the valve.
(e) Hydrostatic Shell Test.
(f) Hydrostatic Seat Test.
(g) Pneumatic Seat Test.
(h) Check Torque of the Valve against no differential pressure zero differential pressure (as indicated in relevant para. of clause 6.7 of FS 2004A).
(i) Check Torque of the Valve against rated full differential pressure (as indicated relevant para. of clause 6.7 of FS 2004A).
(j) Not Applicable.
(k) Type test of Fugitive Emission as per BS EN ISO 15848-1 (If applicable).
(l) Fugitive Emission test as per BS EN ISO 15848-2 (If applicable).
(m) Check surface preparation and all the stages of protective coating as per FS 2005.
(n) Functional Test of the Valve – Ball Valve shall be pressurized to full rated differential pressure with liquid same as used for hydro test and thereafter shall be operated by single hand. Full opening and closing shall be checked. The operation of the valve shall be smooth. The operation of the valve shall be smooth and jerk free. This activity shall be repeated 5 times at manufacturer's location

Sequence of operation in the entire test shall be alternate with one cycle with no differential pressure across the valve and one cycle with full differential pressure across the valve.

Same activity shall be repeated 5 times with zero differential pressure across the valve on installation at fabrication yard and 5 times at offshore.

(o) Valve shall be API Monogrammed which are designed under API Codes and Standards, as applicable.

6.7.1.4 GATE VALVE.

(a) Material tests as per clauses no. 5 and sub clauses of FS 2004 A.
(b) Radiography of Shell/Body of the valve.
(c) DPT or MPT (As applicable) of the items/parts of the valve.
(d) Hydrostatic Shell Test.
(e) Hydrostatic Seat Test.
(f) Pneumatic Seat Test.

(g) Check Torque of the Valve against no differential pressure zero differential pressure (as indicated in relevant para. of clause 6.7 of FS 2004A).
(h) Check Torque of the Valve against rated full differential pressure (as indicated relevant para. of clause 6.7 of FS 2004A).
(i) Type test of Fugitive Emission as per BS EN ISO 15848-1 (If applicable).
(j) Fugitive Emission test as per BS EN ISO 15848-2 (If applicable).
(k) Check surface preparation and all the stages of protective coating as per FS 2005.
(l) Functional Test of the Valve – Gate Valve shall be pressurized to full rated differential pressure with liquid same as used for hydro test and thereafter shall be operated by single hand. Full opening and closing shall be checked. The operation of the valve shall be smooth. The operation of the valve shall be smooth and jerk free. This activity shall be repeated 5 times at manufacturer's location

Sequence of operation in the entire test shall be alternate with one cycle with no differential pressure across the valve and one cycle with full differential pressure across the valve.

Same activity shall be repeated 5 times with zero differential pressure across the valve on installation at fabrication yard and 5 times at offshore.

6.7.1.5 GLOBE VALVE

(a) Material tests as per clauses no. 5 and sub clauses of FS 2004 A.
(b) Radiography of Shell/Body of the valve.
(c) DPY or MPT (As applicable) NDT of the items/parts of the valve.
(d) Hydrostatic Shell Test.
(e) Hydrostatic Seat Test.
(f) Pneumatic Seat Test.
(g) Check Torque of the Valve against no differential pressure zero differential pressure (as indicated in relevant para. of clause 6.7 of FS 2004A).
(h) Check Torque of the Valve against rated full differential pressure (as indicated relevant para. of clause 6.7 of FS 2004A).
(i) Type test of Fugitive Emission as per BS EN ISO 15848-1 (If applicable).
(j) Fugitive Emission test as per BS EN ISO 15848-2 (If applicable).
(k) Check surface preparation and all the stages of protective coating as per FS 2005.
(l) Functional Test of the Valve – Globe Valve shall be pressurized to full rated differential pressure with liquid same as used for hydro test and thereafter shall be operated by single hand. Full opening and closing shall be checked. The operation of the valve shall be smooth. The operation of the valve shall be smooth and jerk free. This activity shall be repeated 5 times at manufacturer's location

Sequence of operation in the entire test shall be alternate with one cycle with no differential pressure across the valve and one cycle with full differential pressure across the valve.

Same activity shall be repeated 5 times with zero differential pressure across the valve on installation at fabrication yard and 5 times at offshore.

6.7.1.6 CHECK VALVE

(a) Material tests as per clauses no. 5 and sub clauses of FS 2004 A.
(b) Radiography of Shell/Body of the valve.
(c) DPT or MPT (As applicable) of the items/parts of the valve.
(d) Hydrostatic Shell Test.
(e) Hydrostatic Seat Test.
(f) Pneumatic Seat Test.
(g) Type test of Fugitive Emission as per BS EN ISO 15848-1 (If applicable).
(h) Fugitive Emission test as per BS EN ISO 15848-2 (If applicable).
(i) Check surface preparation and all the stages of protective coating as per FS 2005.
(j) Functional Test of the Valve – Check Valve shall be pressurized gradually with liquid same as used for hydro test in the direction of flow and the minimum pressure at which the valve allows flow to take place shall be recorded.

6.7.1.7 NEEDLE VALVE.

(a) Material tests as per clauses no. 5 and sub clauses of FS 2004 A.
(b) Radiography of Shell/Body of the valve.
(c) DPT or MPT (As applicable) of the items/parts of the valve.
(d) Hydrostatic Shell Test.
(e) Hydrostatic Seat Test.
(f) Pneumatic Seat Test.
(h) Check Torque of the Valve against no differential pressure zero differential pressure (as indicated in relevant para. of clause 6.7 of FS 2004A).
(i) Check Torque of the Valve against rated full differential pressure (as indicated relevant para. of clause 6.7 of FS 2004A).
(k) Type test of Fugitive Emission as per BS EN ISO 15848-1 (If applicable).
(l) Fugitive Emission test as per BS EN ISO 15848-2 (If applicable).
(m) Check surface preparation and all the stages of protective coating as per FS 2005.
(n) Functional Test of the Valve – Needle Valve shall be pressurized to full rated differential pressure with liquid same as used for hydro test and thereafter shall be operated by single hand. Full opening and closing shall be checked. The operation of the valve shall be smooth. The operation of the valve shall be smooth and jerk free. This activity shall be repeated 5 times at manufacturer's location

Sequence of operation in the entire test shall be alternate with one cycle with no differential pressure across the valve and one cycle with full differential pressure across the valve.

Same activity shall be repeated 5 times with zero differential pressure across the valve on installation at fabrication yard and 5 times at offshore.

6.8 SPECIALTY ITEMS

Specialty Items shall be supplied, designed, tested & installed as specified in the Project Specification for specialty items spec.2004D.

7. PIPING SYSTEM DESIGN

7.1 GENERAL

Design calculations for pressure/temperature wall thickness requirements, vibrations, thermal expansion and contraction, pipe weights and flexibility shall be carried out in accordance with ASME B31.3 and ASME SEC VIII and submitted to the Company for acceptance.

Piping components shall be located where they can safely be operated (where necessary) and maintained. Access shall be provided to such components, which are located out of reach from the platform deck. The use of extended hand wheel stems or chain wheels shall be avoided.

Dead ends on distribution and collection headers shall be blind flanged.

Long radius elbows shall generally be used, but for pigged lines long radius 5D bends are required. Short radius elbows shall be avoided unless essential for clearances. Cold-formed bends are not permitted. Fabricated mitre bends can only be used on gas turbine exhausts.

7.2 PIPE ROUTING

Piping shall be routed so as to have the short runs and minimize pipe supports whilst providing sufficient flexibility for thermal expansion and contraction and mechanical movement. Expansion and swivel joints shall be avoided.

Large bore piping shall be designed to minimize pressure drops. Smaller bore piping shall be routed in groups where practical along main pipe racks.
Piping shall be kept within the deck boundaries.

The number of flanges and unions shall be minimized. The piping carrying hydrocarbon/hazardous chemicals in the safe area shall be of continuous length with welded joints such that no valves, regulators, flanges etc. are located in the safe area. One breakable joint in the safe area at the consumer points is permitted, with adequate safeguards.

Piping shall be routed to avoid trip and overhead hazards.

Consecutive elbows in different planes shall be avoided.

Pipe routing shall allow sufficient space for bolting up flanges or for line-up clamps to be used for field welds. Refer “Piping Clearances” described in piping fabrication (spec 2004-B).

Piping passing through firewalls shall be sealed with fire-retarding sleeves.

Primary utility (air, steam and water) connections on piping shall be DN20 (¾") or larger through the first block valve.

All instrument air and fuel gas connections shall be from the top of the associated headers.

The angle between any branch and its header shall not be less than 45º.
Scraper tees shall be provided for lines, which are to be pigged as per approved P&IDs.

Dimensional rules for piping design are as follows:

Min pipe dia. for thermo-well connection on straight run pipe	DN 100.
Minimum pipe diameter for thermo-well connection on 90º elbow	DN 80
Pipe racks to be sized to allow for the future equipment	+20%.
Minimum run size of piping in racks	DN 50
Spacing of instrument air take-offs along pipe rack headers in process areas.	≥3000 mm
Minimum slope of HP and LP flare headers	1:100
Minimum slope of open drain header	1:100
Minimum slope of closed drain header	1:100
Minimum slope of pump suction lines where vapour may be present	1:50

7.3 PIPE SUPPORTS

Refer piping design criteria (Vol II, sec 3.3). Piping shall be suitably supported to prevent sagging, mechanical stresses vibrations and consequent fatigue, while allowing for thermal and structural movement. Piping shall be adequately supported for the weight of piping filled with water, with attached components unsupported, subject to wind, seismic, insulation and any other applicable loads. The supports shall prevent excessive stresses in the piping and in the nozzles of the equipment to which it is connected.

Small bore instrument tubing and piping shall be adequately supported and protected from impact damage.

Bracing shall be provided for small-bore branches in piping adjacent to vibrating machinery.

Supports used for Cu-Ni piping shall be lined with soft packing strip or pad free from ammoniac components (EG: NEOPRENE ASTM D 2000) to prevent chafing and undue stressing and also to permit free expansion/contraction of Cu-Ni pipes between anchors (Selection of Grade & Type of Neoprene Rubber will be based on Design Temperature & as per ASTM D-2000)

7.4 PIPE WAYS

Generally the piping should be routed in either platform north/south or east/west in established pipe ways. All lines running platform north/south shall be a different elevation from lines running platform east/west, as far as practical. Where possible, a minimum of 400 mm or more between changes in elevation of pipe runs in pipe ways shall be maintained.

7.5 PIPING CLEARANCES

The minimum clearance for maintenance access shall be 750 mm. The following design constraints shall also apply.

Minimum height from underside of pipe or insulation to high point of deck level or platform	200 mm
Control valve arrangement: Preferred bottom of pipe (BOP) of control valve above deck level or platform.	400 mm
For meter runs, the minimum clearance between BOP and deck is	760 mm
Pipe spacing: Minimum space between pipes without flanges (after allowing for insulation and lateral movement)	100 mm
Minimum space between pipes with flanges (largest flange to pipe) (after allowing for insulation and Lateral movement)	100 mm
- Minimum distance from pipe to face of steel work (after allowing for insulation)	50 mm
- Minimum distance from flange to face of steel work, etc.	50 mm
Valve installations and access: Preferred height of hand wheel from deck or platform:	
* horizontally mounted valves * vertically mounted valves	1000/1350 mm 1100/1300 mm

Maximum height from local deck or platform level to center line of horizontal hand wheel without platform on chain wheel	
Vertically-mounted valves (DN100 and larger) (DN 80 and smaller)	2000 mm 2250 mm
* maintenance or isolation (except where temporary platforms can be used and at pipe racks)	3000 mm
Use of chain wheels and extension stems shall be kept to a minimum. Chain shall clear operating level by:	1000 mm

7.6 FLANGED CONNECTIONS

Flanged connections shall be minimized, being used only where frequent dismantling will be required, where specific flanged spools are needed, where needed to provide clearances for equipment removal, or for piping class or material changes.

7.7 THREADED PIPE-WORK

Threaded piping shall not be used to carry hydrocarbons.

Piping DN40 (1½") and smaller, when used for services up to 1900 kPa (g) (275 psig), may be threaded with prior approval of Company.

Screwed fittings shall be rated at least 20.7 Mpa (3000 psi).

Bushings, close nipples and street elbows shall not be used.

Pipe threads shall conform to ASME B1.20.1.

Cu-Ni pipe work shall not be threaded. Adapters can be used at valves or equipment.

7.8 CHANGES IN MATERIALS

Where dissimilar piping materials are connected insulation gaskets shall be provided to prevent galvanic corrosion.

7.9 VENTS, DRAINS AND BLEEDS

High points on all lines shall be provided with DN20 (¾") minimum plugged or flanged connections for venting during hydrostatic tests. For lines carrying hydrocarbons or other toxic fluids, the vents shall be piped to the nearest vent header.

All ODD cups shall be provided with temporary (welded/bolted) plates as cover so as to prevent the clogging/chocking of drain lines during drilling operation in addition to grid plates.

Low points in lines shall be provided with drain connections of nominal sizes as follows: -

Line Size	**Drain Size**
DN15 (½")	DN15 (½")
DN20 (¾") TO DN 100 (4")	DN20 (¾")
DN150 (6") to DN250 (10")	DN25 (1")
DN300 (12") and larger	DN25 (1") to DN 40 (1½")

Drains on lines other than firewater shall be provided with valves and plugged. Firewater drains do not need valves.

All hydro-test vents and drains in hydrocarbon service shall be DN 20 with valves and steel plugs unless noted otherwise.
A hydrostatic vent and drain philosophy shall be developed during detail design and shown on the isometrics.

7.10 CORROSION INHIBITION AND MONITORING PIPE WORK

Corrosion monitoring fittings shall be located as close as practical to the pipe work being monitored, where servicing access is easy and away from sources of vibration. Clearance shall be provided for the removal of the injection quills and monitoring probes, suiting each nozzle orientation and the length of the associated probe. Corrosion probe nozzles shall be mounted on the underside of the pipes.

7.11 RELIEF VALVES

Relief valve assemblies shall be installed in the vertical position, as close to the pressure source as possible, and shall be provided with permanent platform access. Relief valves shall be bolted directly to vessel and equipment nozzles.

Relief valve piping shall be designed to withstand reaction forces and moments caused by the valve discharging.

Piping from relief valves to closed systems shall slope toward the headers and enter them from above, or, where that is not practical, shall have DN20 (¾") drains in safe areas. Headers shall have at least 1:100 slopes toward downstream.

Refer to the Project Specification for General Instrumentation.

7.12 CONTROL VALVES

Control valves shall preferably be installed in horizontal lines, with the actuator in the vertical position. Each valve shall be located as close as possible to the item of plant under control and shall be easily accessible from the deck or permanent platform.

Where control valves are less than line size, reducing spools shall be made long enough to permit bolt removal.

Consideration shall be made for removal or withdrawal of valves or part of valves for maintenance.

Refer to the Project Specification for Instrumentation for additional requirements.

7.13 ISOLATIONS

Piping shall be designed, so the connections to equipment and vessels can be isolated for safe maintenance. This may be accomplished by providing for the insertion of blind flanges at strategic points or removable spools if blinding is not practical due to line size. All vessels containing hydrocarbons or other hazardous fluids and which would involve personnel entry during maintenance require such blinds.

Blinds shall be located so that insertion can be made from the deck or permanent platforms or access ways. Permanent hook eyes shall be provided above blinds, which weigh more than 25 kg. Where blinds are not required for isolation, valves must be provided for safe isolation.

Double block and bleed isolation shall be provided wherever shown in P&ID.

7.14 FIRE SUPPRESSION SYSTEM

Refer design criteria of the Mechanical and safety & life saving equipment (Vol II sec 3.7) and functional safety specification (spec 5102) for fire suppression attached elsewhere in bid package.

7.15 CUPRO-NICKEL PIPE WORK

Cupro-Nickel or Copper-Nickel (Cu-Ni) piping shall be used for firewater deluge systems.

All Cu-Ni piping shall be designed such that velocity in piping does not exceed the values given in project specifications. However, in special circumstances such as downstream of control valves, orifice plate, etc, where fluid velocity may exceed those given in the design criteria special material such as DSS/SDSS/Monel subject to company's approval may be used to withstand higher velocities. When one side is ASME Class 150 Cu Ni and other side is of higher rating carbon steel, a spool of carbon steel having one end of higher rating CS flange and other of lower rating CS flange shall be inserted which also acts as a sacrificial piece

Supports for Cu-Ni piping shall be lined with soft packing pads, neoprene or similar, which shall be free of ammonia compounds.

7.16 COPPER PIPING

Copper (Cu) piping shall be used for potable and other clean water systems.

7.17 GRE PIPING SYSTEMS

Glass-Reinforced Epoxy (GRE) piping shall be used for water services where there is little risk of physical impact, typically for overboard lines.

When a GRE pipe penetrates a fire rated wall or floor, the GRE shall be substituted by a flanged metallic spool piece, fabricated from a material suitable for the proposed service.

GRE piping shall be enclosed in one size higher metallic material when there is risk of moderate physical impact.

7.18 PIPING ON AREAS SUBJECT TO DIFFERENTIAL MOVEMENT

The design of the piping on such areas (if any) shall take into account the differential movement between the two structures under extreme (100 years) storm conditions and thermal expansion

and contraction. Flexibility analysis of the piping systems shall be carried out. The Contractor shall provide any necessary pipe loops and supports.

7.19 PIPING AT EQUIPMENT

Piping at equipment shall be arranged so that the equipment can be removed easily by dismantling / adjusting the piping adjacent to the equipment without the need to dismantle any component of the equipment.

Equipment will not be used to anchor piping. Forces transmitted to equipment at tie-in points will be within the Equipment Supplier's recommended limits.

Piping connected to rotating equipment shall be designed and supported to minimize the transmission of vibrations from the machines. The Contractor shall carry out flexibility analysis of this pipe work to prevent exceeding, allowable nozzle loads as defined by the equipment manufacturer.

7.19.1 PIPING AT HEAT EXCHANGERS

Cooling water piping to shell & tube exchangers shall be arranged so that water will not drain from the outlet when water supply fails.

Exchanger piping shall be arranged so that the exchanger can be removed as one unit and so that the tube bundle can be pulled after disconnecting channel piping.

Piping connections to exchangers shall be designed and properly aligned to allow for hot & cold service and to limit the stress on exchanger nozzles to within allowable levels.

Filters shall be provided in lines to cooling fluid inlets.

7.19.2 PIPING AT PUMPS

Piping at pumps shall be designed and supported so that equipment can be dismantled or removed with a minimum number of temporary supports and without dismantling valves and piping other than the spool that connects to the pump. Clearances shall permit installation of blind flanges against block valves when the service is hazardous and the removal of pump rotating elements.

Pump suction lines shall end with at least four diameters of straight pipe with the same nominal size as the suction flanges. If reducers are required on suction lines, they shall be eccentric and installed flat side up.

Recycle lines shall be provided to allow minimum flows required for pumps.

Pressure relief lines shall be provided for positive displacement pumps.
Valves shall be located as close as possible to the pump nozzles as practical. Isolation valves on pump suction lines shall be full-bore ball type. Isolation valves on discharge lines shall be located downstream of check valves.

Pump suction lines in which, vapour may be present shall be inclined downward towards the pumps with slopes of at least 1:50.

Strainers shall be provided in all pump suction lines. Permanent Y-type or basket strainers shall be provided for reciprocating and rotary pumps. The open area of strainers shall be at least 300% of the cross-sectional area of the pipe. The piping shall be arranged so that the filter or strainer element can be removed from the flanged joints without altering the piping, supports or pump alignment.

7.19.3 PIPING AT TURBINES

Air Inlet and Exhaust duct termination shall be positioned away from hazardous areas, and areas frequented by personnel or any open ended or filtered inlet ducts.

Fuel gas piping within the turbine enclosure shall be subject to strict control with respect to the number and type of flanged joints, fully welded being preferred. Flanged joints shall be provided only for connection to the equipment and for isolation and shut down valves.

7.19.4 PIPING AT DIESEL ENGINES

All piping connected to diesel engines shall be arranged in such a manner that adequate flexibility is maintained so as to effectively isolate the piping from any engine vibration. Piping shall not be routed directly over diesel engines.
Fuel lines shall not be run over exhaust piping or any location where leaks would cause fuel to impinge on to hot surfaces. Fuel lines shall incorporate local isolation valves.

The fuel oil header shall not be dead-ended.
Silencers, where installed in suction and discharge piping, shall be located as close as possible to the engine.

Air intake openings shall be located away from any hazardous area, face toward the prevailing wind direction and be in such a position as to limit the ingress of dust (e.g. salt crystals) and moisture.

7.19.5 LAUNCHERS/RECEIVERS

The "barrel" of the launchers/receivers shall be designed as per ASME-B31.4/31.8 (as applicable) using a design factor of 0.5 and "end closure" shall be designed as per ASME Section VIII, Division-I. Corrosion allowance as per applicable piping class & Material of construction for intended service should be considered in design of barrels. Weld efficiency factor shall be as per ASME-B31.4/31.8 (as applicable)

Launchers and receivers if installed horizontally shall have 5º slopes in the direction of flow. The internal diameter of launcher/ receiver shall be at least 2" higher than the pipe diameter.

Barrel length shall be at least 1-½ times and 2 ½ times as long as the pig selected/specified for launchers and receivers, respectively.

Eccentric type reducers shall be used on all horizontal launchers and receivers with flat side down while concentric type shall be used for vertical launchers/receivers. A spool piece of sufficient length of line pipe size shall be provided with launchers/receivers to ensure an effective seal before the pig enters the first block valve.

Scraper barrel shall be equipped with quick opening hinged type closures. Closures shall be rated for intended service and shall be provided with suitable interlock safety devices to prevent opening under pressure.

Launching/receiving stations shall be provided with an indicating device to detect and indicate the passage of a scraper or sphere.

Launchers and receivers of 600 mm (24 inch) OD and larger shall be equipped with loading/unloading trays to facilitate handling of heavy scrapers and spheres.

All branches from the barrel shall be taken by means of weldolets / sockolets.

Suitable handling system for inserting/retracting the scraper/pig from the barrel shall be provided for each launcher/receiver. The system shall be self contained complete with handling and lifting devices.

7.19.6 PIPING FOR OFFSHORE HOOK-UP

Hook-up piping shall be minimized as far as possible. Piping systems shall be designed so that the contractor in his fabrication yard can perform the maximum level of testing, mechanical completion and commissioning work.

7.20 PIPING LAYOUT REQUIREMENTS

The layout of equipment and piping shall be based on following principles to provide neat and economical layout, allowing for easy supporting and adequate flexibility to meet equipment allowable nozzle load.

1. To locate all equipments identified on equipment list.
2. To comply with standards, regulations, codes, piping specifications and sound engineering practices.
3. To maximize safety of personnel, equipment and facilities.
4. To ensure operability & maintainability of equipment.
5. To provide means of escape & access for fire fighting.
6. To satisfy all requirements indicated in process documents (P&ID's)
7. To minimize shutdown duration.

All piping drawings shall be prepared to scale 1:331/3 with accuracy of +75mm and in metric system only. All piping for new platform shall be modeled using3D modelling software as indicated elsewhere in bid package. Isometrics, Piping GADS, supports GAD's, material take off, interference checking, colour coding, structural module, instrument module, Design review model etc. shall be generated from the 3D model. The contractor shall make interim delivery of 3D model on the completion of each miles stone activity and final delivery on completion of the project. Contractor shall furnish the equipment lay out, safety route drawing, piping layout drawing (GAD), for company's review & approval. Contractor shall furnish one 3D modeling software terminal (dual type) along with dedicated modeling engineer at design center for online review of piping/equipment modeling to ONGC.

However, for modification jobs, piping plan shall be prepared in latest version of MICROSTATION/AUTOCAD.

All the information regarding instrument, equipment and line identification, direction of flow for lines, elevations and dimensions shall be clearly marked on piping plans. Efforts shall be made to provide maximum possible information including operating platforms, ladders, monorails, other material handling equipment, electrical cable and instrument trays, etc. shall be shown on piping plans to avoid reference to other drawings.

For any portion of a line where complete details i.e. dimensions, elevations etc. cannot be furnished in piping plan, a detailed isometric for that portion shall be indicated at a suitable place in the same GAD only. As far as possible all such details shall be shown together at a particular space provided for the purpose.

Piping plans shall show all lines indicated in Piping and Instrument diagrams. Separate line list as per format shall be prepared & submitted to the Company.

Pipe support standard drawings shall be submitted to company for review and separate pipe support drawings shall be generated by marking pipe supports on piping plans.

Piping arrangement drawings for skid-mounted equipment/vessels shall be furnished separately for review and approval. Piping arrangement inside skids need not be shown on piping GAD's.

All distribution/collection headers shall have their dead ends blind flange. Piping Isometrics shall be prepared for all lines and furnished for information.

All the future lines shown in P&ID's shall be routed and shown as chain dotted on GAD's with full details indicating dimensions, line numbers, etc. All future/spare piping connections envisaged in the existing scope of work will be terminated with blind flange. Requirement of spectacle blind and valve at the termination shall be provided if indicated in P&ID. The following shall be the minimum to be shown on the piping drawings wherever applicable:

Line Number

Instrument/specialties/equipment Tag Number

Instrument Connections

Orientation of North arrow

Battery/Area limit as per area division/skid

Title block and drawing nos. as per area division/drawing schedule

Key plan

Reference drawings

Legends and Notes

Table of Nozzle data orientation and location

Match line, continuation drawing number

Location of equipment as per equipment layout; P&ID's special notes.

Dimension & equipment elevations with reference to data sheets.

8. PROTECTIVE COATINGS & INSULATION

Painting and Protective Coatings:

Painting, protective coatings and the procedures used for the preparation of surfaces shall be specified in the Project Specification for Protective Coatings spec 2005.

Flanges shall be painted on the flange edges, inside bolt-holes, and up to the gasket surface.

Insulation:

Piping shall be insulated where indicated on drawings. The piping to be insulated shall be grit blasted and painted as per FS 2005 except for finish coat and only then, insulation to be applied as per the Project Specification for Insulation of Piping and Equipment spec 2006.

9. PIPING SPECIFICATION INDEX

DESIGN CODE: ASME B31.3

SN	CODE	RATING	MATERIAL	SERVICE
1	A1	150#	CS	NOTE-1
2	B1	300#	CS	NOTE-1
3	D1	600#	CS	NOTE-1
4	E1	900#	CS	NOTE-1
5	A2	150#	CS	Glycol
6	B2	300#	CS	Glycol
7	D2	600#	CS	Glycol
8	E2	900#	CS	Glycol
9	A1N	150#	CS-NACE	Hydrocarbon liquid & vapour

10	B1N	300#	CS-NACE	Hydrocarbon liquid & vapour
11	D1N	600#	CS-NACE	Hydrocarbon liquid & vapour
12	E1N	900#	CS-NACE	Hydrocarbon liquid & vapour
13	F1N	1500#	CS-NACE	Hydrocarbon liquid & vapour
14	G1N	2500#	CS-NACE	Hydrocarbon liquid & vapour
15	XG1	API-5000	CS	Hydrocarbon liquid & vapour
16	XG1N	API-5000	CS-NACE	Hydrocarbon liquid & vapour
17	A3	150#	CS-Galv./SS	Instrument Air
18	B3	300#	CS-Galv./SS	Instrument Air
19	D3	600#	CS-Galv./SS	Instrument Air
20	A4	150#	CS-Galvanized	Potable Water-Non Drinking
21	A5	150#	90/10 Cu-Ni	Fire water/Raw sea water
22	A6	150#	SS	Exhaust
23	A8	150#	CS (GALVANISED)	Overboard line
24	A9	150#	SS-316	Lube oil & service oil
25	B9	300#	SS-316	Lube oil & service oil
26	D9	600#	SS-316	Lube oil & service oil
27	E9	900#	SS-316	Lube oil & service oil
28	A10	150#	SS-316L-NACE	Hydrocarbon liquid & vapour
29	B10	300#	SS-316L-NACE	Hydrocarbon liquid & vapour
30	D10	600#	SS-316L-NACE	Hydrocarbon liquid & vapour
31	E10	900#	SS-316L-NACE	Hydrocarbon liquid & vapour
32	F10	2500#	SS-316L-NACE	Hydrocarbon liquid & vapour
33	A12	150#	Titanium	Ferric Chloride/Chemical
34	A13	150#	COPPER	Potable water- Drinking
35	A1LN	150#	INCOLOY 825	Hydrocarbon liquid & vapour
36	A22	150#	SS-316L	Corrosion Inhibitor/Chemical
37	B22	300#	SS-316L	Corrosion Inhibitor/Chemical
38	D22	600#	SS-316L	Corrosion Inhibitor/Chemical
39	E22	900#	SS-316L	Corrosion Inhibitor/Chemical
40	A23	150#	DSS (UNS S 32205)	Hydrocarbon liquid & vapour
41	B23	300#	DSS (UNS S 32205)	Hydrocarbon liquid & vapour
42	D23	600#	DSS (UNS S 32205)	Hydrocarbon liquid & vapour
43	E23	900#	DSS (UNS S 32205)	Hydrocarbon liquid & vapour

44	F23	1500#	DSS (UNS S 32205)	Hydrocarbon liquid & vapour
45	G23	2500#	DSS (UNS S 32205)	Hydrocarbon liquid & vapour
46	A15	150#	INCOLOY 825	Hydrocarbon liquid & vapour
47	B15	300#	INCOLOY 825	Hydrocarbon liquid & vapour
48	D15	600#	INCOLOY 825	Hydrocarbon liquid & vapour
49	E15	900#	INCOLOY 825	Hydrocarbon liquid & vapour
50	XG15	API 5000	INCOLOY 825	Hydrocarbon liquid & vapour
51	XH15	API 5000	INCOLOY 825	Hydrocarbon liquid & vapour
52	A15A	150#	INCONEL 625	Hydrocarbon liquid & vapour
53	B15A	300#	INCONEL 625	Hydrocarbon liquid & vapour
54	D15A	600#	INCONEL 625	Hydrocarbon liquid & vapour
55	E15A	900#	INCONEL 625	Hydrocarbon liquid & vapour
56	PE1	900#	CS	Hydrocarbon liquid & vapour
57	PB1N	300#	CS-NACE	Hydrocarbon liquid & vapour
58	PD1N	600#	CS-NACE	Hydrocarbon liquid & vapour
59	PE1N	900#	CS-NACE	Hydrocarbon liquid & vapour
60	A7	150#	CPVC	Sodium hypochlorite
61	A3S	150#	GRE	Fire Water (raw)
62	A3S-1	150#	GRE	Fire Water-Raw sea water (Dry)
63	A5S	150#	GRE	Produced water, sewage water/Gray Services, Cooling water, Open drain, Utility water,
64	A11	150#	DSS (UNS S 31803)	Hydrocarbon liquid & vapour
65	B11	300#	DSS (UNS S 31803)	Hydrocarbon liquid & vapour
66	D11	600#	DSS (UNS S 31803)	Hydrocarbon liquid & vapour
67	E11	900#	DSS (UNS S 31803)	Hydrocarbon liquid & vapour
68	F11	1500#	DSS (UNS S 31803)	Hydrocarbon liquid & vapour
69	A16	150#	SDSS	Hydrocarbon liquid & vapour
70	B16	300#	SDSS	Hydrocarbon liquid & vapour
71	D16	600#	SDSS	Hydrocarbon liquid & vapour
72	E16	900#	SDSS	Hydrocarbon liquid & vapour
73	A14	150#	PVC	Sewage Water

74	F1	1500#	CS	Note-1
75	F10	1500#	SS-316L	Hydrocarbon liquid & vapour
76	F 15	1500#	Incoloy-825	Hydrocarbon liquid & vapour

NOTES:

1. This standard is used for injection water, vent gas, diesel fuel, process drain water, process drain hydrocarbon, blow down, nitrogen services. However, MOC shall be as per P & ID.
2. Corrosion allowance is indicated in Spec.2004A Clause no.5.3.
3. Material description & special requirements are listed in piping design criteria.

4. Contractor shall prepare the piping class data sheet and shall furnish to Company for review & approval.

GENERAL REQUIREMENTS

10 GENERAL REQUIREMENTS

10.1 PRE-QUALIFICATION CRITERIA AND TESTS FOR MANUFACTURERS (NEW VENDORS)

Suggested vendor list is listed elsewhere in bid package. The vendors who are not listed in suggested vendor list shall be required to furnish the following documents for Company's review for the purpose of pre-qualification.

1. The item/equipment should be in satisfactory service in offshore oil & gas operations for a minimum period of 2 years. This shall be supported with performance feedback from the end-users (The firm using the item/equipment for its offshore oil and gas operations) or repeat purchase orders from the same end user with a minimum time gap of two years of delivery.

2. Catalogue and Drawings for the product.

3. The name(s) & location of Manufacturer/Forging source/Raw material source (Mill) etc.

4. The manufacturer shall be qualified based on results of these tests & information furnished and Company's decision in this regard shall be final and binding on the bidder.

5. The list of manufacturing facilities and quality control equipment.

6. The tests certificates specified in clause 5 shall be conducted by the mill/supplier for the purpose of initial qualification. The test results shall be submitted to the company for approval prior to placement of order. In case the supplier has already carried out all these tests and the test results are certified by a recognized independent inspection agency (viz. Lloyds, BV, DNV etc.) all such test reports/certificates/document shall be forwarded to the company for review and approval. The Company shall assess the capability of the supplier upon scrutiny and verification of the documents submitted. Company's decisions in this regard shall be final and binding on contractor.

7. Qualification for NACE material shall be made product wise, material wise, type of product wise (Seamless or welded), type of valve wise irrespective of thickness/size/type of fittings/pressure rating. Following test certificates/documents to be required:

    1. Steel making procedure from raw material supplier including specific details and confirmation to requirements as per relevant codes & standards.

2. Following test certificates, duly witnessed and certified by any of the Company approved third party inspection agency (as listed in clause 10.2) as per relevant codes & standards and requirements listed below, shall be furnished for Company's approval.

2.1 Chemical composition of heat & finished product. This shall meet the requirements of chemical composition indicated in relevant codes & standards.

2.2 Mechanical Tests on heat & finished product as per relevant codes & standards.

2.3 Hardness test on the heat & finished product as per relevant codes & standards.

2.4 Corrosion tests (HIC, SSCC & inclusion count check) on the finished product of past supplies as per relevant codes & standards. The test shall be applicable to all materials for sour service application (viz: CS (NACE), DSS, INCOLOY CLAD MATERIAL).

2.5 Other test for DSS & INCOLY cladded material: Other test as indicated in clause 5.7 & 5.8 (sour service requirements for DSS & INCOLY CLADDING MATERIAL)

2.6 The requirements stated in relevant codes & standards shall also be fully complied in order to meet the qualification of the product on each heat & finished product.

3. Each certificate shall indicate product name, heat number, raw material source/forge shop/cast shop/manufacturer (as applicable) and shall clearly indicate witnessed and certified with proper signature and seal of the inspecting agency. Any certificate not meeting these requirements shall be rejected.

4. Over and above these certificates, a certificate from the Company approved third party inspection agency shall be furnished containing the following:

4.1 Scope to clarify the extent of involvement of the Inspection Agency for the job order.

4.2 Certification that the sample for the test was taken from the finished product (e.g. pipe, fittings, flange, valve or plate etc.) manufactured from the heat indicated in the certificates and indicate the names of raw material source (Mill)/forge shop cast shop/manufacturer (as applicable) employed in manufacture of the finished product.

5. Quality Assurance Manual shall be furnished for Company's review, which should include the schemes for raw material quality assurance, on line process quality assurance,

product quality assurance, storage and traceability of products, personal quality assurance and quality control procedures.

6. Bidder/manufacturer shall note that for manufacturing of the quoted items, the same sources (i.e. forging source, raw material source, manufacturer etc.) shall be utilized which was employed to manufacture the product on which prequalification tests are carried and approved. Subsequent to qualification, a change of manufacturer/forging source/raw material source (mill) shall not be permitted. In the event that the manufacturer desires to effect any change in any of the above, the manufacturer shall seek fresh qualification as indicated above.

10.2 SUGGESTED THIRD PARTY INSPECTION AGENCIES

M/s Lloyd registers of Shipping (LRS)

M/s Bureau Veritas (BV)

M/s Det Norske Veritas (DNV)

M/s Certification Engineers India ltd (CEIL)

M/s Societe Generate De Surveillance (SGS)

M/s Moody's international

M/s TUV International

M/s ABS

If any of the party is nominated or specified for the job work, then supplier must follow the Company's Requirements.

10.3 QUALITY ASSURANCE

The manufacturer, supplier & his sub supplier shall operate a quality system satisfying the applicable provisions of ISO 9000 (SERIES) or Company agreed equivalent standard, commensurate with the goods & services provided.

10.4 SAFETY PROVISIONS

It is the intent of the company that operational hazards to be reduces to a minimum. Contractor shall use sound engineering judgments to complete an installation that will perform the required process functions without compromising this aim. The platform shall be designed in accordance with API RP 14C, Recommended practice for analysis, design, installation & testing of basic surface safety system on the offshore production platform (latest edition). The contractor shall also follow the safety provisions indicated elsewhere in bid package.

10.5 RECEIPT & STORAGE

The contractor shall develop the comprehensive plan for receipt, storage and release of all equipments released in contractor's care including item supplied by others or free issued by the company. This plan shall provide means to immediately establish the status & locations of any items, which has been issued to the contractor.

The contractor shall be fully responsible for equipment issued & shall provide secure facilities to provide a storage environment, which shall protect the equipment to the vendor's requirement. Any equipment damaged after issue to the contractor shall be repaired, reinstated or replaced by the contractor at company's decision.

10.6 SPARES

Manufacturer shall recommend and provide the spares required for start up and commissioning for all specialties/ vessels or any other item.

Manufacturer shall recommend and provide the spares needed for one-year operation and maintenance for all specialties.

10.7 WARRANTIES

A. The contractor shall agree to warrant all the pipes, fittings, flanges, gaskets, bolting, valves, specialties & vessels furnished by him to be free from defects in design, material, & workmanship.

B. The contractor shall agree to warrant all the pipes, fittings, flanges, gaskets, bolting, valves, specialties & vessels furnished by him will satisfy the requirements of intended use and to be suitable for application.

C. The contractor shall agree to warrant all the pipes, fittings, flanges, gaskets, bolting, valves, specialties & vessels furnished by him for any repair or replace under present tender, which proves to be defective within a period of 12 months from date of completion and handing over to the operation group of company.

D. The contractor shall be solely responsible for obtaining manufacture's performance warranty for all pipes, fittings, flanges, gaskets, bolting, valves, specialties & vessels purchased by him and shall produce the warranty certificate to the company.

E. Where materials of construction or the application of items recommended by contractor do not confirm with the manufacture's recommendation, the contactor shall assume the

equivalent of manufacturer's warranty and in the event of malfunction or damage resulting from this misapplication, the contractor will not be released from this warranty.

10.8 DOCUMENTATION

The contractor shall provide one set of soft copies on CD/Hard disc for approved drawings/ documents & materials for all pipes, fittings, valves, flanges, piping specialties & vessels documents to the piping engineer designated for the project in additions to the company's project requirement. Further, hardcopy any of the above documents shall be submitted as and when required by company. The contractor shall submit the "ORIGINALS" on demand by company to verify the duplicate (Xerox) copies for test certificates review. On placement of order, the manufacturer/mill shall carry out all the production tests as indicated in relevant clauses of this document and submit reports to Company for review and approval. The test reports shall be submitted to Company before dispatch of material from the mill. The approved certificates from company shall be submitted to the piping engineer of company at fabrication site.

GLASS REINFORCED EPOXY Annexure – 1

1.1 Scope

This specification contains minimum requirements for design, manufacture, qualification, quality control, testing, site fabrication, installation and certification of (GRE) composite material piping for facilities on offshore platforms.

1.2 Design / Operating Data

The design / operating data such as pressure, temperature, flow rate, structural load, etc. shall be provided by the user section or the selection / design of GRE pipelines for the relevant services.

1.3 Design & Manufacture of GRE piping

i. The piping and fittings shall be manufactured with epoxy resin as per ASTM D 2996 with thickness to suit the design pressure, taking into account the hydrostatic test pressure at the time of installation i.e. 1.5 times the design pressure. The manufacturer shall give full details of this methodology of arriving at the quoted thickness, including lower confidence limit static (LCLs). The hydrostatic design stress shall be calculated as : 0.5 X LCLs. However, all the components of piping system shall have minimum 3.0 mm reinforced wall thickness. Sea water piping shall be designed for a nominal velocity of 5 m/s. The fire water piping system shall also be designed for blast over pressures.

ii. The piping and fittings shall be manufactured by using epoxy resin and the structural glass fiber shall be E-glass. The fittings shall be manufactured by the same piping manufacturer and compatible adhesives shall also be supplied by the manufacturer.

iii. Inner surface of piping and fittings shall have at least 0.5 mm thick reinforced resin rich liner with C-glass or synthetic fibres and with resin content at least 65%.

iv. For UV protection, UV stabilizer will be added in the resin used for manufacturing GRE piping.

v. Filler shall not be used in the resin during manufacturing process of GRE piping.

vi. The piping and fittings shall be joined by the bell & spigot joint using adhesives or butt and wrap joint.

vii. Flanges shall be heavy-duty filament wound as per ASTM D 4024 and ASME B 16.5.

viii. Stress analysis shall be carried out for estimation of stresses in circumferential and longitudinal direction. Stress analysis shall include, but not limited to, all anchors, guides and supports. The piping system shall be designed in such a way to minimize stresses resulting from surge or transient pressures on account of valve closures or other condition resulting in charge in fluid velocity and those resulting from movements such as expansion and contraction from pressure and temperature differences.

ix. Epoxy resins and additives used in the manufacture of potable water piping and fittings shall satisfy the requirements of FDA / NSF / WHO or other Indian regulatory Authorities for potability of water.

1.4 Qualification & Quality Control of GRE Piping

i. The piping and fittings shall be qualified for pressure rating as per ASTM D 2992. Manufacturer shall enclose the test report and calculations.

ii. For quality control, the short term hydrostatic pressure (STHP) of at least one sample per production lot shall be determined in accordance with ASTM D 1599 at standard laboratory temperature and the value should not be less than mean of STHP for the representative product established in the qualification test. The test report for qualification and quality control shall be certified by reputed certifying agencies like LRS, DNV, BV, ABS, TUV, Vetco Tuboscope.

iii. Test certificates shall be provided for use of GRE piping, fittings, O-rings, gaskets and adhesives in potable water applications as per FDA /NSF / SHO or other Indian regulatory Authorities.

iv. The fire water piping shall qualify endurance testing requirements as per level 3 requirements of IMO Resolution A 753(18) or as latest version of ISO standard 14692-2 for glass reinforced plastics piping in Petroleum and Natural Gas Industries.

v. The GRE piping including all its components for dry portion of fire water system shall be designed such that line is capable of withstanding initially fire up to maximum of 5 minutes by jet fire, followed by flowing water from deluge system by which time the line gets filled with water due to actuation of deluge system. The heat flux of impinging flame shall be decided based upon relevant codes and standard.

vi. The GRE piping shall conform to the fire endurance tests for dry condition in accordance with appendix I of IMO resolution A. 753 (18), 1993.

1.5 Fabrication & Installation of GRE Piping

i. The piping and fittings shall be fabricated and installed as per UKOOA's "Specifications and Recommended Practice for the use of GRP Piping Offshore", March 1994, ISO 14692-1/2/3/4(Petroleum and Natural Gas Industries-Glass Reinforced Plastic Piping).

ii. All GRE/GRP piping shall be fabricated and installed under the supervision of manufacturer/Manufacturer's representative certified by the manufacturer.

iii. All GRE piping shall be conductive and grounded through metallic parts in the system such as valves, pumps, etc. and as per manufacturer's recommendations.

iv. GRE piping & components which remain dry during normal operation of a platform; such as drain lines shall be applied intumescent coating for fire insulation / protection as per applicable standard/application. While deciding type of intumescent coating toxicity of the smoke shall also be considered. Fire insulation material should be such a type that does not require the use of reinforcing mesh.

1.6 Certification Requirements of GRE Piping.

Offered product shall have valid class and type certificate from reputed certifying agencies like LRS, DNV, BV, ABS

1.7 List of Codes & Standards for (GRE) Composite Material Piping

ASTM D 2992	Standard Practice for Obtaining Hydrostatic or Pressure Design Basis for "Fiberglass" (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe and Fittings.
ASTM D 2996	Standard Specification for Filament-Wound "Fiberglass" (Glass-Fiber-Reinforced Thermosetting Resin) Pipe
ASTM D 2105	Standard Test Method for Longitudinal Tensile Properties of "Fiberglass" (Glass Fiber Reinforced Thermosetting Resin) Pipe and Tube
ASTM D 2143	Standard Test Method for Cyclic Pressure Strength of Reinforced, Thermosetting Plastic Pipe.
ASTM D 2517	Standard Specification for Reinforced Epoxy Resin Gas Pressure Pipe and Fittings.
ASTM D 2997	Standard Specification for Centrifugally Cast "Fiberglass" (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe details the specifications for machine made glass fiber reinforced thermosetting resin pressure pipe.
ASTM C 581	Standard Practice for Determining Chemical Resistance to Thermosetting Resins.
ASTM D 570	Test Method for Water Absorption of Plastics.
ASTM D 2290	Standard Test Method for Apparent Hoop Tensile Strength of Plastic or Reinforced Plastic Pipe by Split Disk Method.
ASTM D 2412	Standard Test Method for Determination of External Loading Characteristics of plastic pipe by parallel – plate Loading sets forth methods.
ASTM D 2924	Standard Test Method for External Pressure Resistance of "Fiberglass" (Glass Fiber Reinforced Thermosetting Resin) Pipe.
ASTM D 5685-01	Standard Specification for "Fiberglass" (Glass Fiber Reinforced Thermosetting Resin) Pressure Pipe Fittings.
ASTM D 638	Standard Test Method for Tensile Properties of Plastics.
ASTM D 1598	Standard Test Method for Time to Failure of Plastic Pipe Under Constant Internal Pressure.
ASTM D 1599	Standard Test Method for Resistance to short time Hydraulic

	Pressure of Plastic Pipe, Tubing and Fittings.
UKOOA	Specification and Recommended Practice for the Use of FRP Piping.
API Spec 15 LR	Low pressure fiberglass line pipe.
API Spec 15 HR	High pressure fiberglass line pipe.
API RP 15 TL 4	Recommended Practice for Care and Use of Fiberglass Tubulars.
ASME B 16.5	Pipes, Flanges and Flange Fittings.
ASME B 31.3	Process Piping.
ASME BPVC Section X	Boiler & Pressure Vessel Code.
ASTM D 695	Test Method for Compressive Properties of Rigid Plastic.
ASTM D 696	Test Method for Coefficient of Linear Thermal Expansion of Plastic.
AWWA C 950	Standard for Fiberglass Pressure Pipe.
BS 5480	Specification for Glass reinforced plastic (GRP) pipes, joints and fittings for use for water supply or sewerage.
BS 6464	Specification of reinforced plastic pipes, fittings and joints for process plants.
ASTM D 648	Test for deflection temperature of plastics under flexural loads.
ASTM D 2310	Classification for machine made reinforced thermosetting resin pipe.
ASTM D 1652	Test Method for epoxy content of epoxy resins.
ASTM D 2343	Test Method for Tensile Properties of Glass Fiber Strands, Yarn and Roving use in Reinforced plastics.
ASTM D 2344	Apparent Inter Laminar Shear Strength of Parallel Fiber Composites by Short Beam Method.
ASTM D 2393	Standard Test Method for Viscosity of Epoxy Resin and Related Components.
ASTM D 2471	Standard of Test Method for Gel Time and Peak Exothermic Temperature of Reacting Thermosetting Resin.
ASTM D 2563	Standard Test Method for Classifying Visual Defects in Glass Reinforced Plastics Laminate Parts.
ASTM D 2583	Standard Test Method for Indentation Hardness of Rigid Plastics by mean of Barcol Impresser.
ASTM D 2584	Test Method for Ignition Loss of Cured Reinforced Resins.
ASTM G 53	Weathering QUV Weatherometer.
UKOOA	Guidelines for Fiber Reinforced Use Offshore.
USCG PFM 1-98	US Coast Guard Guide on Fire test requirements for plastic pipe.
IMO A 753(18)	Guidelines for application of plastic pipes on ships.
ISO 14692 Part 1-4	Petroleum & Natural Gas Industries – Glass Reinforced Plastics Piping.

FUNCTIONAL SPECIFICATION

FOR

PIPING FABRICATION AND INSTALLATION

TAMBADE NARENDRA SATISH Digitally signed by TAMBADE NARENDRA SATISH Date: 2022.02.09 11:54:58 +05'30'

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CONTENTS

1. **INTRODUCTION**
1.1 General
1.2 Application of this Document
1.3 Contractors Responsibility
1.4 Sub-Contractors

2. **CODES & STANDARDS**
2.1 Codes, standards & regulations
2.2 Project specifications
2.3 Mandatory Indian Statutory Requirements

3. **SCOPE OF WORK**
3.1 General
3.2 Scope of supply

4. **ENVIRONMENTAL DESIGN CRITERIA & UTILITIES**
4.1 Seismic & transportation loads
4.2 Design life

5. **GENERAL REQUIREMENTS**
5.1 General requirements
5.2 Dimension
5.3 Documentation

6. **MATERIALS**
6.1 General specifications
6.2 Material identification
6.3 Supports & miscellaneous

6.4 Bolting
6.5 Gaskets

7. **FABRICATION**
7.1 General
7.2 Flanging & bolting
7.3 Threaded & socket weld piping
7.4 Brazed piping
7.5 Pipe supports
7.6 Valves
7.7 GRE piping system
7.8 Penetrations
7.9 Piping at equipment
7.10 Piping at instrumentation

8. **TOLERANCES**

9. **TESTING REQUIREMENTS**

10. **IDENTIFICATION & REALEASE OF DELIVERY**

11. **INSTALLATION**

12. **CLEANING**

13. **PROTECTIVE COATING**

14. **SHIPPING & HANDLING**

1. INTRODUCTION

1.1 GENERAL

This specification gives the general requirements for the fabrication, installation, inspection and testing of piping materials, fittings and valves for this project.

This specification covers all pipe work pertaining to this project except the following:

Heating, ventilating, air and water pipe work serving non-process areas.

Instrument piping downstream of the last piping block valve, as normally defined on the approved P&ID's.

1.2 APPLICATION OF THIS DOCUMENT

The scope of this document is limited to piping, piping components & vessels fabrication of this project. This document shall not be used for any other purpose.

1.3 CONTRACTOR'S RESPONSIBILITIES

The Contractor shall be responsible for the complete design, manufacture, supply, inspection and testing, including full compliance with all applicable design codes and standards, including those listed in Section 2.0 of this specification.

1.4 SUB-CONTRACTORS

Approval shall be obtained from the Company before any work is subcontracted.

2. CODES AND STANDARDS

2.1 CODES, STANDARDS AND REGULATIONS

The production piping shall be designed, manufactured and tested in accordance with the requirements of this specification, and the latest edition of the following Codes, Standards and Statutory Regulations (where applicable):

1	API RP 14E	Recommended Practice for Design & Installation of Offshore Production Platform Piping.
2	ASME VIII Division 1	Pressure Vessels and Boiler Code.
3	ASME II Part A	Materials – Ferrous Materials Specifications.
4	ASME V	Non-Destructive Examination.
5	ASME IX	Welding and Brazing Qualifications.
6	ASME B1.20.1	Pipe Threads, General Purpose (Inch).
7	ASME B16.20	Metallic Gaskets for Pipe Flanges.
8	ASME B31.3	Chemical Plant and Petroleum Refinery Piping Code.
9	ASME B31.4	Oil Transportation Piping.
10	ASME B31.8	Gas Transmission and Distribution Piping.
11	ASME B16.25	Butt Weld Ends for Pipes, Valves, Flanges & Fittings.
12	ASTM A283	Carbon steel Plates.
13	ASTM A370	Standard Test Methods & Definitions for Mechanical Testing.
14	ASTM A435	Ultrasonic Examination of Steel Plates.
15	AWS	Welding Handbook.
16	NACE MR 01 75	Sulphide Stress Cracking Resistant Metallic Materials for Oil field Equipment.
17	ASTM E1002-11	Standard Practice for Leak using Ultrasonics

In addition to above refer the codes and standards specified in Cl.2 of Piping Design Criteria.

2.2 PROJECT SPECIFICATIONS

The piping specifications covered by this specification shall be designed, manufactured and tested in accordance with the requirements of these Project Specifications:

Spec. 2004-A	Piping Design
Spec. 2004-D	Piping specialties
Spec. 2005	Protective Coatings
Spec. 2006	Piping insulations
Spec.2009F	Welding & NDT

2.3 MANDATORY INDIAN STATUTORY REQUIREMENTS

This documents has been prepared to the International Standards detailed within. The EPC Contractor shall ensure that the Scope of Work is executed in accordance with all mandatory Indian Statutory requirements.

3. SCOPE OF WORK

3.1 GENERAL

The Contractor shall be responsible for supply of materials, fabrication, coating, installation testing and insulation of the pipe work, all sub-contractor coordination and guarantees as required and provision of certificates, quality control records and as-built drawings, notwithstanding any omissions from this specification.

This specification covers all pipe work on this project except Heating & ventilating areas and Instrument piping downstream of the last piping block valve, as defined on the approved P&IDs.

3.2 SCOPE OF SUPPLY

The Contractor shall provide and install the entire pipe work shown on the Project approved P&IDs outside package boundaries, including the necessary supports, connections, valves, a packaging and nozzle protection required.

The Contractor shall furnish all labour, materials and equipment necessary to perform the piping fabrication and installation.

The Contractor shall also supply the documentation required by this specification.

4. ENVIRONMENTAL DESIGN CRITERIA AND UTILITIES

4.1 BASIC CLIMATIC CONDITIONS

Refer clause no. 2 of piping design (Spec. no. 2004A)

4.2 SEISMIC AND TRANSPORTATION LOADS

Refer clause no. 2 of piping design (Spec. no. 2004A)

5. GENERAL REQUIREMENTS

5.1 GENERAL REQUIREMENT

Fabrication, installation and testing of pipe work shall conform to this Specification and ASME B31.3 and API RP 14E.

All equipment used, and materials used, are subject to inspection by the Company. The Company reserves the right to approve all material, tools and fabrication procedures, prior to Contractor's proceeding with the work.

5.2 WELDING REQUIREMENT

Welding shall be as per spec. 2009F.

5.3 DIMENSIONS

Dimensions shall be in mm and be related to the Platform datums or reference lines.

SI units shall be used, with the sole exception of pressure, for which kg/cm^2 shall be used.

5.4 DOCUMENTATION

The contractor shall comply all the bid specifications. The Contractor shall provide the Company with all fabrication drawings; weld procedures and fabrication procedures for approval. Shop work shall not start until the relevant drawings and weld procedures are received & approved by the Company. Weld procedures shall be accompanied by a weld procedure index and weld map for each line.
After construction, the Contractor shall supply project isometric drawings and P&ID's annotated to show as built information i.e. any change made to the pipe work design during fabrication and installation.

The Contractor shall be responsible for compiling and handing over all material certificates, welder qualification certificates, weld records, signed inspection & test plans, mechanical clearance report, installation check lists, charts of weld heat treatments, test and inspection records applicable to the competed work. The contractor shall furnish these certificates/reports & relevant documents for company's review & approval.

6. MATERIALS

6.1 GENERAL SPECIFICATION

Material shall comply with the Piping design criteria (Vol. II sec 3.3) and Project Specification for Piping Design, Spec. 2004 A.

6.2 MATERIAL IDENTIFICATION

For all pressure parts material certification shall be traceable to heat numbers. Certificates, including all material certificates, fully catalogued and NDT test records, mechanical test certificates, welding qualification certificates, heat treatment certificates and hydrostatic test certificates shall be available at final inspection and for counter signature by the certification authority and stored by the Contractor for a minimum of 5 years after acceptance of the pipe work by the Company. Pressure retaining parts shall be clearly marked to allow verification of traciability. Double blue stripes shall be marked for NACE material for clear identification & traceability.

The contractor shall provide one set the hard copies for approved drawings/ documents & materials for all pipes, fittings, valves, flanges, piping specialties & vessels documents to the piping engineer designated for the project in additions to the company's project requirement. The contractor shall submit the "ORIGINALS" on demand by company to verify the duplicate (Xerox) copies for test certificates review. On placement of order, the manufacturer/mill shall carry out all the production tests as indicated in relevant clauses of piping design criteria & project specification of piping design and submit reports to Company for review and approval. All test certificates & relevant documents shall be properly bound & submitted to company for review. Loose certificates & relevant documents submission is not considered acceptable. The test reports shall be submitted to Company before dispatch of material from the mill. The approved certificates from company shall be submitted to the piping engineer of company at fabrication site.

6.3 SUPPORT AND MISCELLANEOUS

Any material shall not be welded directly on process piping. If situation arises then prior approval of Company shall be sought during detail Engineering & shall note that the material welded directly to pressure retaining pipe work shall be of similar quality as the pipe work, including impact requirements, if any, for a length measured from the vessel wall of at least 150 mm. The material of such items beyond this point may be structural quality A283 Gr. C. or equal/superior (contractor to substantiate with sufficient information & documentary evidence).

6.4 BOLTING

Please refer to the Project Specification for Piping design criteria & project specification of piping Design (Spec. 2004- A).

6.5 GASKETS

Please refer to the Project Specification for Piping design criteria & project specification of piping Design (Spec. 2004- A).

7. FABRICATION

7.1 GENERAL

Piping shall be fabricated in accordance with the requirements of this Specification, with the piping codes and standards specified herein and under the individual material specification classes.

Piping design drawings and isometrics shall govern dimensions and materials.

The requirements of this specification shall apply to shop and field fabrication and to piping within packaged units.

Piping shall be prefabricated to the most practical extent to minimize offshore fabrication or installation.

Galvanized pipe shall not be bent or welded. Any such piping requiring bending, welding or threading shall be galvanized after fabrication.

7.2 FLANGING & BOLTING

Flange bolt-holes shall straddle the established horizontal and vertical centerlines of the pipe except where connecting to equipment dictates otherwise and any deviation shall be shown on the piping design drawings and/or isometrics.

Company approved lubricants such as colloidal molybdenum disulphide shall be applied to all flange bolts before the joints are assembled.

Flange bolts shall be tightened evenly and sequentially to impose equal pressure on the gasket and to avoid distortion or overstressing of equipment (As per Contractor's recommendations).

Bolts in gaseous hydrocarbon service and liquid service over ANSI 150# rating shall be torque-checked.

Flange make up shall be made with clean new gaskets. The used gasket sealing compounds or sealing agents is prohibited except for anti-stick release agent.

Prior to installation, stud bolts shall be smeared with API thread compound containing molybdenum disulphide, on that portion of the stud bolt, which will be in the holes of the flanges and under the nut. Bolts shall be progressively tightened to full bolt torque using sequence tightening as per the gasket manufacturer's recommendation.

7.3 THREADED & SOCKET WELD PIPING

Threads shall be concentric with outside of pipe, per ASME B 31.3, tapped and cleaned out.

The inside of each pipe spool shall be de-burred by Reaming.

Threaded connections shall be gauge checked or chased after welding or galvanizing.

When socket weld fittings are used, pipe shall be spaced approximately 1.5 mm away from the "bottom" of the socket.

Threaded joints, which are not to be seal welded shall be thoroughly cleaned, and the male threads coated with sealant.

Sealant shall be Company approved.

Seal welding is required as follows however while seal welding in threaded piping, maintenance aspect of equipment/component connected to threaded piping consequent upon seal welding shall be evaluated and if required, break out spool shall be provided:

1. All services where subject to vibrations (such as lines to/from reciprocating compressors, pumps etc.), Hydrocarbons or other flammables, all pyrophoric, toxic, lethal and glycol services.

2. Seal welding of threaded connections, when specified, shall include the first block valve, cover all threads and be done with electrodes not to exceed 3 mm in diameter.

3. Thermo wells shall not be seal welded.

4. Seal welded and threaded piping shall not be substituted for socket weld piping connections without prior written approval by the Company.

5. Immediately before erection of piping, all threads of the pipe and fittings shall be thoroughly cleaned of cuttings, dirt, oil and any other foreign matter. The male threads shall be coated with thread sealant and the piping made up sufficiently for the threads to full thread engagement.

6. Threaded connections shall be protected from oxidation during heat treatment and be gauge checked after welding or heat treatment.

7.4 BRAZED PIPING

Brazing temperatures shall be achieved as quickly as possible using a suitably sized torch.

The inside of brazed fittings and the outside of the associated tube shall be cleaned with sand paper. Flux paste is to be evenly spread over the joint.

7.5 PIPE SUPPORT

Piping shall be suitably supported to prevent sagging, mechanical stresses and vibrations, while allowing for thermal and structural movement. Piping shall generally be fastened to pipe racks with hot dip galvanized or cadmium plated U bolts and double nuts.

All pipes in hazardous service shall have 10 mm thick wear pads between the pipe and the support when the pipe would otherwise be resting directly on the support steel.

In locations where potentially severe vibration can occur, such as near control valves, pressure relief valves, pumps, compressors, and in high velocity streams, all small branch connections shall be braced by means of welded gussets or brackets in two (2) planes.

7.6 VALVES

Consideration shall be made for removal or withdrawal of valves or part of valves for maintenance.

7.7 GRE PIPING SYSTEMS

Refer Cl. 7.17 of Piping design (Spec. no. 2004A)

7.8 PENETRATIONS

Final location and dimensional checks of the pipe penetrations are the responsibility of the Contractor.

Deck or skid penetrations shall provide a minimum of 25 mm radial clearances around the piping.

All open deck drain cups shall be properly covered with suitable hinged caps in order to restrict the blockage due debris and unwanted mud / dirt.

7.9 PIPING AT EQUIPMENT

Piping at equipment shall be arranged so the equipment can be removed without the need to dismantle the equipment, adjacent equipment or piping.

Equipment will not be used to anchor piping. Forces transmitted to equipment at tie-in points will be within the recommended limits.

Piping connected to rotating equipment shall be supported to minimize the transmission of vibrations from the machines.

Piping connections to exchangers shall be properly aligned to allow for hot & cold service and to limit the stress on exchanger nozzles to be within allowable levels.

Piping at pumps shall be supported so that equipment can be dismantled or removed with a minimum number of temporary supports and without dismantling valves and piping other than the spool that connects to the pump. Clearances shall permit installation of blind flanges against block valves when the service is hazardous and the removal of the pump impeller without removing the pump. Where reducers are required on suction lines, they shall be eccentric and installed flat side up.

Valves shall be located as close as possible to the pump nozzles as practical. Isolation valves on pump suction lines shall be full-bore ball type. Isolation valves on discharge lines shall be located downstream of check valves.

Pump suction lines in which vapors may be present shall be inclined with sufficient slopes.

Fuel gas piping within the turbine enclosure shall be subject to strict control with respect to the number and type of flanged joints, fully welded being preferred.

All piping connected to diesel engines shall be arranged in such a manner that adequate flexibility is maintained so as to effectively isolate the piping from any engine vibration. Piping shall not be routed directly over diesel engines. Fuel lines shall not be run over exhaust piping or any location where leaks would cause fuel to impinge on to hot surfaces.

7.10 PIPING AT INSTRUMENTS

Thermometer and thermocouple well connections, as well as vents and drains, shall be tapped after welding or stress relieving operations, using a plug tap. Verification gauges shall be used to prove the suitability of the threads and adequacy of instrument fitting-clearance.

8. TOLERANCES

Dimensional tolerances at Annex. -1 shall apply to in-line items and connections or branches for other lines.

Flange alignment for piping connected to rotating machinery must not cause machine-to-driver misalignment to exceed the equipment manufacturer's tolerances.

Tolerances may be exceeded to a maximum of 15 mm for items such as vents, drains, dummy support lengths, field supports and temperature and pressure connections where the deviation does not affect another line.

Material, which has been locally stress-relieved, shall be rechecked for tolerances in accordance with the drawings. Small residual deviations remaining after heat treatment may only be corrected by cold straightening. Hot straightening after heat treatment, expect for non-pressure attachments, will not be approved.

9. TESTING REQUIREMENTS

9.1 PNEUMATIC TESTING

Pneumatic checks of branch reinforcing pads shall be made on completion of the pipe spool and before hydrostatic testing. The air pressure used for this check shall not be greater than 1.75 kg/cm^2 nor less than 0.70 kg/cm^2. Pressure shall be increased gradually and soap leak tests carried out on the outside welds of the pads, the inside welds between pads and pipes, as well as inside the branch pipes. Testing shall be witnessed by a Company representative.

9.2 HYDROSTATIC TESTING

Piping shall be hydrostatically pressure tested in accordance with ASME B 31.3 and this Specification where called for on the piping design isometric drawings. Workshop hydrostatic testing is not required the pipe is to be field-tested. Pressure test shall be performed to ASME B31.3.

9.3 PREPARATION FOR TESTING

The Contractor shall be responsible for properly preparing facilities for pressure testing.

The Contractor shall review the drawings and piping to ensure that adequate temporary supports, vents, drains, and blinds (as necessary to allow testing) are provided to test the facility in accordance with this specification. The Contractor shall report to the Company the findings from this review.

The Contractor shall provide all high points in the piping with vents and all low points with drains, regardless of whether or not they are specifically shown on the drawings.

The Contractor shall prepare, for the purpose of the pressure tests, a procedure and schedule for testing all process and utility piping within the limits of their responsibilities. The procedure and schedule shall be presented to the Company for approval at least sixty (60) days prior to the scheduled start of pressure tests. Such procedures shall include, but are not limited to, details of the following:

1. Mark-ups of piping and instrument diagrams (P&IDs) showing the limits of piping and equipment to be included in each test.

2. Test media to be used.

3. Location of all isolation blinds, high point vents, low point drawings, and temporary piping supports.

4. Procedures for flushing lines including fill points and equipment to be removed for flushing.

5. Procedures for filling lines including addition of oxygen scavenger and bactericide in proper quantities.

6. Procedures for pressurization of the system, including the location of the injection points and specified test pressures.

7. The forms, which shall be used to record hydrostatic test data. In this form calibration details of gauges, chart, etc., shall be indicated.

The testing operations shall be conducted diligently, thoroughly, and in a safe workman-like manner in accordance with accepted piping testing practices & codes.

The Contractor shall correct any work by the Contractor, or existing conditions, which in the opinion of the Company are deemed to be unacceptable. The testing operations may not continue until the unsafe conditions have been corrected to the

satisfaction of the Company. The Contractor shall not receive additional compensation for time lost due to unacceptable conditions.

If testing is subcontracted, the Contractor shall provide an experienced supervisor who will be responsible for supervising testing operations.

The Company, or the Company's Representative, will witness and approve all tests.

9.3.1 MATERIALS AND EQUIPMENT

The Contractor shall furnish all materials and equipment necessary to pressure test all piping. This includes, but is not limited to, pumps, compressors, valves, calibrated pressure gauges, thermometers and chart recorders, filters, plugs, blinds, caps, strainers, flanges, bolts, gasket, all piping incidental to the filling, flushing, transferring and disposing of the test medium, bactericides, oxygen scavengers, dyes (if required), and the test medium.

Prior to hydrotest, calibration certificates for pressure gauges, thermometer, chart recorder, etc., shall be submitted to the company for review. Further, these instruments shall be recalibrated in the presence of company during which calibration certificates for recalibration instruments like dead weight indicator, master gauges shall be submitted to the company for review.

The Contractor shall provide a suitable pressure-relieving device between the pressuring pump discharge nozzle and the fill point, calibrated to prevent excess pressure in any of the piping and equipment being tested.

9.3.2 PRE-TEST FLUSHING

Prior to commencing any flushing, the Contractor shall verify that all required non-destructive testing (NDT) such as radiography, ultrasonic, etc, has been completed.

The Contractor shall also verify that all required post-weld heat-treating has been completed.

Upon verification of the above requirements, Company's Representative shall authorize in writing the Contractor to proceed with flushing.

All completed piping systems shall be flushed prior to testing.

All lines DN50 (2") and larger, other than instrument and utility air, shall be flushed with potable water so as to completely clean them of any loose mill scale, rust, or various extraneous materials.

All lines smaller than DN50, and air lines of any size, shall be cleaned out in like manner using compressed air.

The flushing shall be done at a high a velocity as practical (minimum of 1.5 m/sec.) in order to flush out any loose material in the line.

Flanges shall be opened and spool pieces shall be removed as necessary to flush the piping thoroughly.

Flushing shall be performed until a clear stream of flushing medium exits the piping, with minimum flush volume being equal to five (5) times the volume of the piping being flushed.

In case flushing as required above is not practical, contractor shall submit alternate proposal to company for approval along with justification for not adopting procedure as mentioned above. Alternate proposal for flushing submitted by contractor must completely clean any loose mill scale, rust, or various extraneous materials inside piping system.

All flushing shall be done in the presence of the Company's Representative.

Orifice plates, thermo wells, flow straightening vanes, positive displacement (PD) and turbine meters and other vulnerable equipment, if already installed, shall be removed prior to flushing.

All instrument leads shall be disconnected prior to flushing and shall remain disconnected during pressure test.

All control valves shall be removed.

Spring type supports shall have stops installed to prevent overloading of the support. All vessels included within a piping system being flushed are to be filled and drained through the vessel drain system.

All check valve flappers shall be removed in the presence of the Company's Representative and attached to the outer body of the valve for the flushing. The flappers shall be reinstated after the flushing in the presence of the Company's Representative.

9.3.3 PIPING ACCESSORIES & EQUIPMENT EXCLUDED FROM HYDROSTATIC TESTING

The following piping accessories and equipment shall be excluded or isolated from hydrostatic testing of piping:

1. Rotating machinery, pumps, compressors, etc.
2. Strainers and filters.
3. Pressure-relieving devices such as pressure relief valves and rupture disc.
4. All instruments, including level controllers and gauge glasses.
5. Pressure gauges.
6. Expansion joints.
7. Flow nozzles.
8. Control valves.
9. Corrosion probes.
10. Orifice plates, thermo wells, flow straightening vanes and instrumentation such as positive displacement (PD) and turbine meters, corrosion probes, displacement-type level instruments, carioles flow meters.
11. Any other piping such as open drain lines, atmospheric vents, etc., or equipment designated by the Company.

9.3.4 TEST FLUID

Except as otherwise specified, all piping shall be hydrostatically tested with water. The hydro test water shall be clean and free from suspended matter. The hydro test water may be analyzed by the Company to determine its suitability for use in the testing operations.

The hydro test water temperature shall be within 7°C - 25°C.
Suspended matter in the hydro test water shall be removed before injection into the piping by a filter capable of removing 99% of all particles 92 microns in diameter and larger, or equivalent to using a 100 x 100 wire-mesh screen.

If foreign material is pumped into the piping, the Contractor shall again flush the line with clean water.

Only potable water shall be used in piping systems containing stainless steel components. No seawater can be used in these line tests. The lines shall be drained immediately after tests.

When air is specified as the testing medium, the air shall have a maximum dew point of -7°C.

Water used for hydrostatic testing shall be clean and contain corrosion inhibitors in the concentrations recommended by the Manufacturers. The Contractor is required to verify with the chemical contractor, current requirements regarding proper disposal procedures for the chemically inhibited test water.

Potable water used to hydrostatically test piping or vessels fabricated from, or containing parts of austenitic stainless steel, ferritic stainless or high nickel alloy, shall be free from undissolved solids and have less than 200-ppm chlorides. The system shall be flushed after testing.

Instrument airlines and utility airlines shall be tested with compressed air of instrument quality.

Testing of vessels or other equipment with air shall not be allowed except by written Company approval.

9.3.5 HYDROSTATIC PRESSURE TEST

The Contractor shall hydrostatically test piping, vessels and equipment for which he has fabrication or installation responsibility. Hydrostatic pressure testing of piping and piping systems shall be in accordance with ASME B31.3.

Piping and vessel test pressures and the Hydro test schedules shall be provided by the Contractor in accordance with this Specification.

Upon verification of the pre-flushing requirements, the Company's Representative shall authorize the Contractor to proceed with pressure testing.

Hydrostatic testing may be performed for completed systems or for individual components of the system, as approved by the Company.
The lines shall be drained immediately after tests.

Vents shall be open during test water filling so that air is vented prior to applying pressure to the system. Vents shall then be closed to permit pressurization.

Hydrostatic testing shall not be done against closed valves. Where it is impractical to hydrostatically test through a valve, that the valve shall be isolated by blinds or disconnected from the piping system.

The hydrostatic test pressure for each section tested shall be contained by suitably placed spectacle blinds, spades, skillets, blank insert chokes, blind flanges or other means of positive closure. The Contractor's test procedures shall indicate the thickness of blinds to be used for isolation during hydrostatic tests.

Hydrostatic test pressure connections shall be on the upstream side of check valves.

Thermo wells shall be included in hydrostatic tests.

All hydrostatic pressure tests will be recorded on a two-pen recorder (24-hour chart type) with a range suitable for recording the test pressure. The second pen shall be used to record the ambient temperature.

Hydrostatic test pressure shall be achieved and relieved gradually so as to allow sufficient time for pressure equalization throughout system under test.

Hydrostatic test pressure shall be maintained for a minimum of two hours duration.

Hydrostatic test pressure readings shall be taken from the test gauge at fifteen (15) minute intervals during the test, and recorded on form provided by the Contractor.

During hydrostatic pressure test, the test medium may be subjected to thermal expansion.

The Contractor shall take proper precautions to avoid excessive pressure caused by thermal expansion.
The Company's inspectors will begin a visual examination of each joint after the hydrostatic test pressure has been maintained for thirty (30) minutes.

After completion of hydrostatic test, the pressure shall be reduced to the design pressure and all valves including check valves in the section of the line shall be tested by being closed for ten (10) minutes with pressure contained by specific valve at one end to determine whether each valve is in turn positively shutting off and holding pressure.
Hydrostatic test pressure shall be removed only with the permission of the Company's representative.
All joints, including welds, are to be left un-insulated and exposed (joints may be primed prior to hydrostatic test) during the hydrostatic test.

Only hydrostatic tests with ending pressures equal to or above starting pressure will be accepted.

The Contractor shall schedule hydrostatic tests for that portion of the day when thermal contraction will not result in ending pressures lower than starting pressure.

Hydrostatic pressure tests on each major section of piping shall be carried out with all pressure relief valves removed and their respective nozzles or connections blinded off or plugged.

Upon completion of piping or vessel hydrostatic tests, pressure relief valves shall be reinstalled in each system.

Drain, instrument, relief, and blow-down lines are to be air and soap bubble tested at 1.76 kg/cm^2 unless otherwise noted.

9.3.6 AFTER-TEST FLUSHING, DRAINING, DRYING & REINSTATEMENT

Piping and equipment fabricated from or containing parts of austenitic or ferritic stainless steels, or high nickel alloy shall, immediately after hydrostatic test completion, be drained and then flushed with distilled or demineralised water.

Piping and equipment that are not fabricated from or do not contain parts of austenitic steel, ferritic stainless steel, or high nickel alloy may be hydrostatically tested with inhibited sea water, per this specification, provided the system is, immediately after testing, drained of all seawater and then flushed with potable water.

Potable water so used for flushing shall be subject to analysis and approval by the Company and shall be treated per this specification.

All after-test flushing shall be performed so that all pipe surfaces previously wetted during the pressure test are flushed.

All piping shall be thoroughly dried.

The piping, equipment and instrumentation shall be reinstated.

9.3.7 HYDROSTATIC TEST RECORDS

The Contractor shall provide all hydrostatic test recording instruments with all necessary supplies, and shall maintain accurate and permanent records. The hydrostatic testing and pressure recording equipment shall be subject to the written approval of the Company.
All hydrostatic pressure test records, data, and charts shall be clearly marked with the following information and furnished to the Company:

1. The Company name and authorized representative.
2. The hydrostatic testing Contractor's name and authorized representative.
3. Line identification of piping and/or equipment tested.
4. Date, start time, and stop time of hydrostatic test.
5. Hydrostatic test pressure and duration.
6. Hydrostatic test medium used.
7. Post-test flushing medium used and its source.
8. Ambient air temperature.
9. The recorder element range.
10. Explanation of any hydrostatic pressure discontinuities that may appear on the chart.
11. Listing of items excluded from the hydrostatics test.
12. Marked up P&IDs depicting valves, which passed and failed their respective service test.
13. Signature of the Company's and the testing Contractor's authorized representative.

9.3.8 RE-TESTING

Should piping failure occur during hydrostatic testing, the Contractor shall locate the failure, for repair purposes.

The Contractor shall, at their expense, repair all defects resulting from inferior workmanship or defective materials furnished by the Contractor.

After repairs have been made and after the items have been removed, the Contractor shall perform a new hydrostatic test until a successful test result is obtained to the satisfaction of the Company's Representative.
The Contractor shall then clean and dry the piping including the reinstatement of the items and produce the records.

9.3.9 COMPLETION OF HYDROSTATIC PRESSURE TESTS

Upon completion of all hydrostatic tests, the Contractor or Sub-Contractor shall:

Dispose of all hydrostatic test medium to a location designated by the Company, exercising proper caution to avoid accidental creation of a vacuum during draining of water from equipment and piping.

Restore any equipment excluded from hydrostatic tests to a fully operational condition.

Remove all temporary facilities installed by the Contractor, including temporary supports and bracing.

Remove and temporary plugs or blinds.

Replace any used gaskets or damaged nuts and bolts.

Air purges all piping and equipment, unless specifically deleted by the Company in writing, to remove low-point pockets of water. This air purging shall be preformed in the presence of the Company's Representative and last a minimum of thirty (30) minutes.

Re-torque all flange bolts on flanges in hydrocarbon service with bolts 25 mm (1") or larger and all flanges in gas services over ANSI-150# which were remade after hydrostatic testing. These must be re-torque to their stated torque valves in the presence of a Company Inspector.

9.3.10 FINAL PIPING SYSTEM LEAK TEST

The Piping System Leak Test is the final pressure test, used to prove the integrity of the installed flanged and threaded joints and connections of piping and equipment systems. It shall be designed to demonstrate that the process and utility piping systems are properly prepared to operate at their specified design pressure.

The leak test shall be performed after the final accepted hydrostatic tests, upon completion of final assembly and prior to startup.

The Contractor shall submit a detailed written procedure describing the complete piping system leak test for Company approval. The piping systems shall each be pressurized to 1 kg/cm^2. The test may be carried out using either the bubble or acoustic emission methods like Ultrasonics as per ASTM E1002 -11 or preferably combination of both. Piping system leak testing shall be done only to the approved procedure and with the Company monitoring personnel present.

The piping system leak test shall be performed on all completed systems, including equipment that may have been previously tested by the Contractor or Contractors.

All valves, control valves, filters, strainers, turbine meters, PD meters, flow nozzles, level gauges, level switches, level controls, rupture disks, pressure relief valves, and pressure gauges shall be included in the piping system leak test. Some instruments or equipment may be excluded from the piping system leak test at the Company's discretion.

The inert atmosphere remaining following the piping system leak testing is intended to eliminate the need for further pre-startup inerting of piping equipment.

10. IDENTIFICATION & RELEASE FOR DELIVERY

Piece marking of prefabricated piping for identification during installation shall be the responsibility of the Contractor.

Piece marks shall bear the line number with a suffix painted or tagged on each piece.

After inspection, testing of piping and application of protective coating, identification numbers shall be stenciled in several places along with length of the spool.

Marking paint or ink containing harmful metal salts such as zinc, lead or copper shall NOT be used to mark alloy, stress relieved or Charpy test materials.

Stamping of identification on pipe or flanges is not permitted.

11. INSTALLATION

Piping shall be installed in accordance with the piping drawings.

Piping installation includes the supply and installation of supports shown in the drawings and piping schedules together with those supports necessary to adequately and properly secure the piping.
All lines shall be installed true and plumb except where specifically noted on the drawings.

Installation of piping shall be scheduled so that no undue strain will be inflicted on equipment due to lack of support, dead weight or misalignment.

Piping shall be made, broken, and re-made at all rotating equipment connecting points, with alignment checks made with and without pipe in order to prove piping does not affect alignment.

Flange bolts shall be tightened evenly and sequentially to avoid distortion or overstressing of equipment.

If any piping does not meet alignment checks, the Contractor shall cut and refit that pipe.

In special cases, when approved by the Company, heating and stress relieving of the pipe may be allowed to make final fit-up.

No heating or bending may be done except by the using Company approved procedures.

Pipe shall be installed beginning at the equipment connection then moving away as the piping is assembled.

Where required, field welds shall be used when installing pipe to equipment.

Field modification of prefabricated piping shall be performed in strict accordance with this specification and the applicable codes, standard or recommended practice.

All weld burrs and all foreign matter shall be removed from the piping prior to closure.

Field fabricated carbon steel seal oil and lube oil piping shall be “pickled” in accordance with this specification.

Construction supports required during installation shall be provided by the Contractor and shall be clearly marked by the Contractor for removal before commissioning.

At all times when piping fabrication or installation is not in progress and at other times when openings in pipe work are not required to be left exposed, they shall be protected against entry of foreign matter by reinstalling shipping protector caps and covers or other suitable means.

Tack welded fit-ups lacking a completed root pass at the cessation of a work period shall be covered at the seam with polyethylene tape.

Before installing pre-fabricated piping, the inside of the pipe shall be inspected by the Contractor to ensure that all foreign matter has been removed.

Austenitic stainless steel pipe and fittings and all valves and special items shall be stored out of contact with the ground. Valves and special items shall also be covered to prevent entry of moisture or foreign matter.

The Contractor shall take all possible care to avoid contamination of stainless steel materials. Stainless steel materials shall be lifted with nylon or other approved slings, which are free from dirt or ferritic particles. Under no circumstances shall chains or wire ropes be used.

Shoes and cradles shall preferably be located and attached to the pipe work after the line has been installed. They may be attached to pipes prior to installation provided that when the pipe is installed the shoes and cradles are central over the pipe supports.

To prevent damage to valve seal and seat surfaces, valves, when installed, shall be, and shall be left in, the following open/close positions:

Ball and plug valves - fully open

Globe and gate valves - fully closed.

Piping shall not be connected to a valve of any description unless the joints can be made without inducing stress in either the valve or the pipe work.

Temporary strainers, slip-plates, blind flanges and the like shall be properly installed in the location manner and orientation shown in the drawings.

12. CLEANING

After installation, each pipeline shall be thoroughly cleaned to remove all moisture and loose scale, sand, paint and any other foreign matter.
Special cleaning procedures may be required for specific services. Particular care shall be taken to ensure that lines downstream of the dehydration system are thoroughly internally dried.

After fabrication and cleaning, assembly ends shall be protected for handling and shipment.

If the pipe is not immediately installed and is to be stored or barged to the platforms, all ends and opening shall be sealed in a suitable and approved method so as to prevent the entrance of moisture, dirt or other foreign matter.
The Contractor shall furnish all materials for field flushing, cleaning or protection during shipment and handling.

13. PROTECTIVE COATING

Painting, protective coatings and the procedures used for the preparation of surfaces shall be specified in the Project Specification for Protective Coatings (Spec. 2005).

Piping shall be insulated where indicated on drawings. The piping to be insulated shall be grit blasted and given one coat of primer only then insulation applied per the Project Specification (Spec. 2006), Insulation of Piping & Equipment.

During fabrication, sandblasting, painting and installation, care shall be exercised to keep blast grit, dirt and other debris out of piping systems, instruments, and equipment. The interior of piping systems shall be thoroughly cleaned by the Contractor prior to assembly and after blasting/coating is completed.

Flanges shall be painted on the flange edges, inside bolt holes, and up to the gasket surface.

Corrosion inhibited grease with wraps shall be provided to exposed spectacle blind sealing surfaces.

The Contractor shall be responsible for painting and/or insulation will also be responsible for cleanup after their respective work is completed. The Company will inspect/examine all such work and then provide a written acceptance prior to release of Contractor.

14. SHIPPING & HANDLING

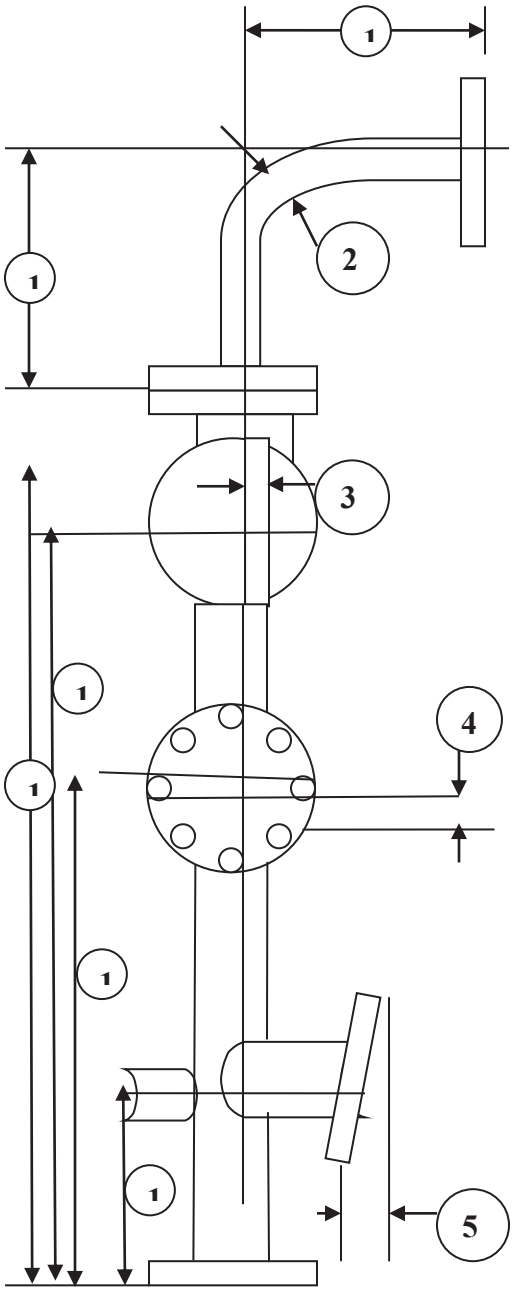
Protection for pipe, fitting and valve openings shall be applied after inspection as follows:

Flange faces (gasket contact surfaces) shall be adequately protected by coating with a removable rust preventive and attaching a bolted or steel strapped 20 mm thick wooden or metal blank. Unpainted steel blanks shall not be used where rust would discolors the coated surface of skid-assembled components. Gasket contact surfaces shall not be painted. If the surface is painted, it shall be restored to its original specification condition before application of protection.

Threaded connections shall be closed with a steel pipe plug.
Beveled ends shall be closed with metal or a plastic covers.

Each spool piece shall be clearly marked on both ends with the line number. Marking shall withstand exposure during shipping.

Extra protection from impact shall be provided for Cu-Ni / GRE pipe.



ANNEXURE - 1

1. +/- 3mm MAX. FROM INDICATED DIMENSIONS FOR FACE TO FACE, CENTER TO FACE LOCATION OF ATTACHMENTS ETC.
2. 10% MAX. FOR INTERNAL PRESSUER 3% MAX FOR EXTERNAL PRESSURE FLATTENING MEASURED AS DIFFERENCE BETWEEN THE MAXIMUM AND MINIMUM O.D. AT ANY CROSS SECTION.
3. +/- 3mm MAX. LATERAL TRANSLATION OF BRANCHES OR CONNECTIONS.
4. +/- 1.5 mm MAX. ROTATION OF FLANGES FROM THE INDICATED POSITION, MEASURED AS SHOWN.
5. 0.5mm MAX. OUT OF ALIGNMENT OF FLANGES FROM THE INDICATED POSITION MEASURED ACROSS ANY DIAMETER.

FUNCTIONAL SPECIFICATION

FOR

PIPING SPECIALTIES

Prepared / Revised By	Reviewed By	Approved By	Total No. of Pages	Date	Rev. No.
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CONTENTS

1.0 **INTRODUCTION**
1.1 General
1.2 Contractor's Responsibilities

2.0 **PROJECT SPECIFICATIONS**

3.0 **ENVIRONMENTAL DESIGN CRITERIA AND UTILITIES**
3.1 Seismic and Transportation Loads
3.2 Design Life

4.0 **GENERAL REQUIREMENTS**
4.1 Company's Requirements
4.2 Dimensions

*5.0 **DESIGN REQUIREMENTS**
5.1 General
5.2 Design Load Cases
5.3 Design Stress
5.4 Drawings and Calculations

6.0 **MATERIALS**
6.1 General Specification
6.2 Material Identification
6.3 Sour Services
6.4 Certification Documents

7.0 **SPECIALTY ITEMS**
7.1 Strainers
7.2 Scrapper tees
7.3 Shutdown valves
7.4 Utility hose reel
7.5 5D bends
7.6 CP/BP/SP & Retrieval tool kit
7.7 Pig detector
7.8 Hinged closure
7.9 Choke valves
7.10 Spray nozzles

7.11 Fire & foam hose reel
7.12 Continuous drainer
7.13 Chemical utility, hoses & hose connections
7.14 Flame Arrestor
7.15 Hand control valve
7.16 Double Block & Bleed Valve
7.17 HIPPS Valve
7.18 Sample Bomb

8.0 **SPARES**

9.0 **WARRANTY**

1.0 INTRODUCTION

1.1 GENERAL

This refers to the minimum and mandatory requirements of designs & materials for piping specialties & components as per approved P& ID , approved Data-Sheet and Bid-Package. All piping assemblies, specialties & materials supplied or installed under these specifications shall be in accordance with sound engineering principles. Any omission from this specification shall not relieve the contractor from his responsibility of furnishing equipment's or materials to meet the specific process parameters, environmental parameters, safety parameters and any other applicable statutory laws or relevant codes & standards. Substitution or changes from this specification must be accompanied with sufficient information/justification and written approval shall be obtained from the Company.

All welding and NDT shall be performed as per Spec. 2009 F.

1.2 CONTRACTOR'S RESPONSIBILITIES

The Contractor shall be responsible for the selection and design of piping specialties, including full compliance with all applicable project specifications and design Codes / Standards, including those listed in Section 2.0 of this Specification.

2.0 PROJECT SPECIFICATIONS

2.1 The piping specialties covered by this Specification shall be designed, manufactured and tested in accordance with the requirements of Piping Design Criteria and the following Project Specifications:

1. Piping Design (spec. 2004-A)
2. Piping fabrication (spec. 2004-B)
3. Protective Coatings (spec. 2005)
4. Insulation of piping and equipment (spec. 2006)

2.2 The Contractor shall be responsible for the design, engineering, supply of materials, fabrication, coating, installation and testing of the pipe work, supply of drawings and data, all sub-contractor co-ordination and guarantees as required, notwithstanding any omissions from this Specification.

3.0 ENVIRONMENTAL DESIGN CRITERIA AND UTILITIES

3.1 SEISMIC AND TRANSPORTATION LOADS

Refer relevant clause of piping design (Spec. no. 2004A)

3.2 DESIGN LIFE

The process facilities design life requirement is 25 years.

4.0 GENERAL REQUIREMENTS

4.1 COMPANY'S REQUIREMENTS

The contractor shall prepare detailed datasheet for each specialty item based on this bid package and approved P&ID and submit to the company for approval.

The Datasheet shall contain the following data, but not limited to:

Design Pressure and design temperature.
Material of construction.
Rating.
Governing Codes and Standards and Specification.
Relevant technical notes.
Tag no.
Size.
Reference to approved P&ID.
Relevant calculations.
Relevant drawings.

4.2 DIMENSIONS

SI units shall be used. Dimensions shall be in mm and be related to the Platform datum's or reference lines.

5.0 DESIGN REQUIREMENTS

5.1 GENERAL

Materials shall conform to this Specification with the attached piping specification index and the identified API, ASME, ASTM, BS and NACE codes and Standards.

Refer piping design criteria (Vol II sec 3.3) & piping design (spec 2004-A).

Pressure and temperature ratings of ASME B16.5 shall apply for the design conditions.

Requests for substitutions of any kind shall be complete with all pertinent engineering information required for the Company's evaluation of the proposed substitution.

5.2 DESIGN LOAD CASES

Specialties, its supports and anchors, shall be designed to withstand the results of the following applicable combinations of loads and forces within the limits of stress set by ASME B31.3:

1. Hydro-test Condition (The empty weight plus weight of water to fill the piping).

2. Operating and Design Conditions (The empty weight plus the weight of operating fluid).

3. Wind loading condition

4. Dynamic Loading Condition

5. Periodic Site Test Condition

6. Any other condition that would affect the safety of the pipe work, e.g. cyclic loading and slug forces, when identified on the Data Sheet.

5.3 DESIGN STRESS

Allowable stress shall be the maximum stresses permitted by ASME B31.3

5.4 DRAWINGS AND CALCULATIONS

Vendor drawings shall contain all pertinent information relating to the Codes, Standards and Specifications used in the design, fabrication, inspection and testing of all the specialties, the pipe work, including the materials used.

The Contractor shall submit detailed calculations establishing the compliance of their design with the requirements of ASME B31.3.

Contractor to prepare the data sheets for each specialty & shall provide the information such as item description, pipe size, thickness, valve selection & tag Nos, designed pressure- temperature ratings, hydrostatic pressure, list of spares and applicable codes/standards. Contractor to submit the data sheets and all other information required for design of each specialty for company's review & approval.

Contractor to prepare the specifications, data sheets based on the functional parameters, relevant piping class as per approved P&ID & material of construction and submit to company for review & approval.

All calculations shall be complete, giving all references and showing all working methods. The Contractor shall be able to provide proof of software verification for any software used. Computer printouts will not be accepted without the program flow chart, input data and complete printout, and then only by prior written agreement with the Company at the quotation stage.

Where relevant, additional calculations shall be undertaken regarding the effects of slug forces.

The pipe work shall be analyzed in its corroded state for each load combination.

Approval of drawings, calculations and other documents by the Company does not relieve the Contractor of their responsibility for the correctness of the design to suit the stated conditions.

6.0 MATERIALS

6.1 GENERAL

Materials shall be as per ASTM, BS or API specifications referenced in the project piping specifications (SPEC 2004- A & B) and piping design criteria.

6.2 MATERIAL IDENTIFICATION

All piping specialties except those with threaded connections, shall be supplied with mill certified test reports and certificates to identify the type of steel, composition, heat number and any special testing.

The Contractor shall furnish to the Company Inspector (OD- Rep) with one (1) copy of all mill certificates for all the materials purchased by the Contractor duly certified by company approved third party inspection agency.

6.3 SOUR SERVICE

All sour service materials shall conform to piping design criteria and Spec 2004A.

6.4 CERTIFICATION DOCUMENTS

All pressure parts material certification shall be traceable to heat numbers. Certificates, including all material certificates, mechanical test certificates, welding qualification certificates, heat treatment certificates and hydrostatic test certificates shall be available at final inspection and for counter signature by the certification authority and stored by the Contractor for a minimum of 5 years after acceptance of the piping by the Company. Pressure retaining parts shall be clearly marked to allow verification of tractability.

7.0 SPECIALTY ITEMS

Specialty items shall be supplied with a stainless steel tag. Specialty Items shall be supplied, designed, tested & installed as specified in the Project Specification for specialty items. This specialties refers to items like Strainers, Scrapper Tees, Hinged Closures, Pig Detector, CP/BP/SP, 5D radius Bends, SDV, HCV, Choke Valves, Strainers, Spray Nozzle, Fire water & foam hose reel, Chemical and Utility hose reel, Continuous drainer etc. used in the Offshore Platforms. The materials of construction, piping class, governing codes and standards, quality assurance shall be followed as per relevant clauses of this specification. All specialties shall be hydro tested as per respective piping class and 1.5 times the design pressure. All pressure welds shall be 100% radio graphed. All piping specialties shall be painted in accordance with Spec. 2005. However, piping speciality supplied by the vendor with primer as per Spec. 2005 followed by subsequent coating (With touch up primer coating wherever required as per Spec. 2005) by main contractor is also acceptable.

7.1 STRAINERS

All basket type strainers shall be designed as per ASME Section – VIII. Div. 1 and others as per ASME B 31.3,

MOC of the body, nozzle etc shall be same as that of the connected piping.

Screen material shall be as follows:

Body Material		Screen Material
C.S	-	S 316
C.S. (NACE)	-	SS 316 (NACE)
Other than C.S.	-	Compatible with Piping Class.

Mesh size for all strainers shall be 40, unless otherwise specified.

Drains with ball valve and blind shall be provided.

Installation of strainer shall conform cleaning without dismantling strainer housing and associated piping & instrumentation.

Break out spools shall be installed wherever the temporary strainers are recommended in P&ID.

7.2 SCRAPPER TEES

Scrapper tee (or sphere tee or flow tee as it is called) essentially consists of concentric barrel tees. The purpose of the scrapper tee is not to allow passage of pig sphere/scrapper on to the branch side.

The scrapper tee shall be suitable for pigging operation with process hydrocarbon/raw sea water. The scrapper tee inside diameter shall be same as that of inside diameter of riser in splash zone to maintain constant ID to permit smooth pigging operations.

No bars shall be used.

Maximum possible opening shall be provided through slots to limit the pressure drop i.e. total opening shall be 1.0 to 1.5 times the branch pipe area.

Thickness of run/branch is to be calculated by Vendor to the required design pressure and temperature and it shall match with the corresponding matching pipeline for free passage of pig as per DNV rules for submarine pipeline.

7.3 SHUT DOWN VALVES (XSDV)

1. SHUT DOWN VALVES (XSDV):

Type : Flanged Ball valve with manual override

Size : Contractor to prepare data sheet based on process parameter.

Others: Valves rating; flange facing; body, ball, seat and trim material, packing etc. shall be to the relevant piping specs. All XSDV shall have flanged ends irrespective of line sizes and class rating.

Shutdown time : Contractor to specify.

All hydrocarbon valves are to be supplied as per approved P&ID's to meet NACE-MR-01-75 (latest edition) and Piping Design Criteria.

2. ACTUATOR:

Type Pneumatic piston, spring return, Quarter turn operation

Size. Actuator torque shall be 1.25 times the valve torque required at full rated differential pressure of valve, (Vendor shall indicate actuator model no, valve torque and actuator torque in a tabular form along with the quotation.

Supply Pressure Instrument air/gas at 7 kg/cm2g (10.6 kg/cm2 g max) (min. pressure being 5.5 kg/cm2g)

Pneumatic 3/8" NPT (female) Tubing shall be SS 316 3/8" OD x 0.049" WT

Accessories Filter regulator with gauge.

3. VALVE POSITION SWITCH (XVPS):

Valve positions switches to indicate open/close position for remote operation.

Type Sealed Micro type, lever operated

Rating 24V DC 2 amp suitable for inductive load

Enclosure Weatherproof and explosion proof (NEMA 4/7)
Area classification NEC Class 1, Divison1, Group-D
Housing CS Nickel plated

Contacts DPDT

Cable Entry 2 Nos. ¾" NPT (F)

4. PILOT VALVE (XPV):

Pneumatic Pilot valve for remote operation with manual reset / shut in

Type	Three-way NC SIGMA 11 RSS 83 or equal
Body trim material	SS 316 Valve to be leak proof with 'O' ring seal
Supply	Instrument air/gas at 7.0 kg/cm2g (10.6 kg/cm2g max.) (Min pressure being 5.5 kg/cm2g)
Signal	0 or 3.5 kg/cm2g instrument air
Accessories	Filter regulator with gauge (1 No.)

5 SOLENOID VALVE (XSV)

Electrical solenoid valve for remote operation (If shown in P&ID)

Type	Three-way NC manual reset/shut in
Connection	Universal, inbuilt terminal box
Body/trim material	SS 316 Valve to be leak proof with O ring seals
Enclosures	Weatherproof and explosion proof (NEMA 4/7)
Area Classification	NEC Class-I, Division I, Group-D
Power supply	24V DC (-ve earthed)
Supply Pressure	Instrument air/gas at 10.6 kg/cm2 (max) 7.0 kg/cm2g (normal) 5.5 kg/cm2g (min.)
Cable entry	¾" NPT (F)
Accessories	Filter regulator with gauge.

NOTES:

a) Valves to be designed and tested as per API-6D or BS-5351 as applicable.

b) Actuator shall be mounted on the valve and tested as follows:

- Cycle (open and shut) each valve with its actuator at least five (5) times to ensure smoothness of operation

- Shut valve and apply hydraulic differential pressure across the valve. Open valve with the installed actuator and note valve operation. Valve operation should not be jerky or binding. This shall be repeated at least three (3) times and shutdown time noted at minimum supply pressure.

c) Test shall be witnessed by company's or company's authorized representative.

d) All soft seated valves shall be of fire safe design as per API 607/API6FA/BS 6755 Part 2 (latest edition) except the valves used in services such as injection water, fire water, pot water service etc.

e) Fire safe tests duly witness and certified by recognized third party inspection agency e.g. LLOYD/DNV/BV/EIC shall be furnished.

g) Valve with actuator shall be suitable for installation in horizontal/vertical lines.

h) Shut down valve is a package with valve vendor having single point responsibility and actuator shall be sourced from ONGC SVL.

7.4 UTILITY HOSE REELS

Galvanized, welded steel hose reel drum with single length (15M min.), non-collapsible, non-shrinkable hose with end connections.

Nozzles wherever required shall be of Elkart SFL make (or equal), 95 GPM capacity suitable for required service conditions.

Hose coupling wherever required shall be KAMLOCK Fig 633A and 634B or equivalent.
Hose reel assembly shall be mounted on a frame.

Hosepipe shall have an external coating of oil and abrasion resistant material unless otherwise specified to protect them from sunlight or mild dew damage.

Each hose reel assembly shall be hydrostatically tested to two times the working pressure or at 21.0 kg/cm2g whichever is higher

7.5 5D RADIUS BENDS

Bends shall be manufactured by hot bending of pipe applying heating only. Company's approval for bending procedures shall be obtained prior to start of work. The adopted procedure shall be such that the finished product shall not require any additional heat treatment after bending. If

such a heat treatment is required, the same shall be permitted only after written approval of company/and/or Engineer-in-Charge.
Bends shall not have any circumferential joint.

No repair by welding is allowed on any part of the bends.

Bulges, dents and flats shall not appear within 100 mm from the ends and for remaining part they shall be as per MSS-SP-75.

Wrinkles : Measurement of the outside diameter shall be taken in the plane of bend at the locations where wrinkles are present (O.D. max) and at locations where wrinkles are not present (O.D. min.). The acceptance limit shall be as given below:

$$\frac{\text{O.D. Max} - \text{O.D Min.}}{\text{Nominal dia}} < 2\%$$

CTOD TEST shall be performed on each size, service & piping class as per BS 7448 PART I.

7.6 SCALE/CORROSION/BIOLOGICAL PROBES & RETRIEVAL TOOLS KIT

1. The scale/corrosion/biological unit shall consist of the following assemblies

A) The access fitting assembly consisting of three parts:

a) Access Fitting Body (Non Tee): Access fitting body shall be of a nominal 2" system. Body shall have a weldolet (Flare Weld) base and 3" ACME Threads on the outlet. Weldolet Base shall be 2" Flare weld size.

b) Plug assembly: solid plug assembly for holding scale/corrosion/ biological probe coupon.

c) Protective Cover: A steel protective cover for the ACME Threads.

B) Corrosion Coupons and Coupon Holder.:

Multiple disc coupons and holders shall be supplied for a nominal 2" system. Holder shall be supplied with mounting and insulating hardware. Each holder shall position three discs. one each at top of line, middle of line, and bottom of the line. Coupons shall be multiple disc type. However, in case wake frequency with multiple disc type is above the acceptable value then contractor shall provide corrosion coupon as per following order of priority:

(i) Contractor shall provide two single disc flush coupon, one at 6 o'clock position and another at 12 o' clock position.

In case space does not permit installation of two coupons as above, then contractor shall provide

(ii) Multiple disc coupon and holder having more than 2" nominal system.

In case, this option is also not feasible, then following option shall be provided.

(iii) Multiple strip coupon and number of coupon shall not be less than 2 numbers.

C) Scale Coupons and Coupon Holder:

Strip coupons of 150 mm (6") length shall be supplied for a nominal 2" system. Scale coupons shall have a range of predrilled holes suitable for monitoring scale buildup. Holder shall be supplied with mounting and insulating hardware. Each holder shall position two strips for bottom of line monitoring.

D) Biological Coupon and coupon holder shall be for nominal 2" system. Holders shall be supplied with mounting and insulating hardware.

2. Retrieval Tool Kit:

A retrieval tool kit shall consist of complete kit for inspection and removal of corrosion coupons. The retrieval tool kit shall be for nominal size of 2" system.

The retrieval tool kit shall be sized for the largest line size and pressure rating and shall be suitable for retrieving all scale/corrosion /biological probes during operation on the platform.

Tool shall be complete with full-bore ball valve kit as per the respective piping class.

Demonstration/ commissioning shall be shown to company's or company's authorized representative about retrieval of all probes while in operation.
All probes to be installed in 6 O'clock position.

All probes shall be accessible for maintenance/retrieval while in Operation.

Quantities of retrieval tool kit shall be one (1) each for each controlling process platform suitable for each type of probe.

7.7 PIG DETECTOR

Pig Detector is a device which indicates the presence of a pig at a specific position in a pipeline.

There are two categories of Pig Detecting device – Intrusive Pig Detector and Non-Intrusive Pig Detector.

7.7A Intrusive Pig Detector:

Intrusive pig detector is static and mechanically activated and as a result of activation of the trigger due to presence of pig in a specific position in the pipeline wherever the pig signaler is installed. Actuating mechanism of an intrusive pig detector is a mechanical trigger therefore; the intrusive pig detector is required to intrude into the pipeline. Intrusive pig detector is required to be suitable to pressure temperature rating and medium flowing inside the pipeline.

Pig detector shall be Bi-directional type mechanical sphere/pig Detector with multiple accessory flexibility.

Accessories: Visual indicator with manual reset, Micro switch lever arm operated, DPDT (Double Pole Double Throw) Type rated for 24 volts DC at 2.0 amps. Inductive load or SPST (single pole single throw) proximity type switch. Enclosure shall be weather proof and explosion-proof as per NEMA-4 & 7, suitable for area classification of Class-1, Division-2, and Group-D. Electrical connections shall be ¾" NPT (F) with two (2) entries.

Accessory may be installed or changed without removing unit from line. Isolation Ball Valve shall also be provided for replacement/maintenance of internal parts.

One tool for removal of pig indicators shall be provided for each platform.

Any possibility of detector being operated by line operating pressure shall not be permitted.
All pig detectors shall have:

Internal parts / Enclosures as given below:

MOC of Connecetd Piping	MOC of Internal parts/Enclosures
CS/C.S (NACE)	S.S 316
D.S.S	D.S.S
Incoloy	Incoloy/D.S.S

External parts :
- For C.S /C.S (NACE) -SS-316 except scarfed welding .
- For others it shall be compatible with the MOC of the Piping.

7.7B Non-Intrusive Pig Detector:

Non-intrusive pig detector is device which detects the passage of pig at a specific location. However, design shall be such that the device is not required to intrude in the pipe line on which the detector is installed.

The pig detector shall be bi-directional and suitable for all types of pigs including magnetic as well as non-magnetic pig.

Pig detector shall be such that it can be installed easily on the pipeline. All the required accessories for installation of the detector on to the pipeline shall be provided. Pig detector shall also have facility to detach it easily from its installed position so that same pig detector can be utilized for other pipeline segment.

Pig Detector for well head platform shall have own power/battery source and shall be standard size battery (AA/R6 or C/R 14 or D/R 20) operated. The battery shall not be rechargeable type.

For well head platform, pig detector shall have provision for local display and also remote monitoring at control room of the respective process complex through SCADA interface, separate SPDT volt free contact shall be provided.

Pig detector for process platform shall be 24V DC type.

For process platform, the pig detector shall have provision such that data/information of the detector has local and remote display at control room. The pig detector shall have interface suitable for all type of PLC with smart technology to ensure valid communication is maintained.

The inbuilt Power Source/Battery for system back up (if any) of well head as well as Process Platform shall have a minimum self-life of one year.

Pig detector shall be suitable for salt laden marine environment and highest surface temperature of the pipeline segments under the project.

MOC of the Pig Detector as a minimum shall be Stainless Steel (SS 316).

Design of pig detector shall be such that it can display information related to Pig arrival/departure and shall have the facility to store data for minimum 100 nos. of pig passage events.

Any possibility of detector being operated by line operating pressure shall not be permitted.

Enclosure shall be weather proof as per IP66 and Explosion-proof suitable for ZONE – 1. Electrical connections shall be ¾" NPT (F) with two (2) entries. As a minimum complete enclosure shall be ATEX certified for ZONE-1

In addition to pig detector as specified in clause no. 7.7A, two nos. of non-intrusive pig detectors suitable for the entire pipeline segments (i.e. pipeline segments means pipeline from pig barrel to pig barrel or any part of it) under each project shall be provided.

7.8 HINGED CLOSURES

The hinged closures shall be quick opening type closure with O- ring seats and necessary pressure warning/release devices. These shall be suitable for horizontal or vertical mounting as per layout requirements. Hinge orientation shall be as per approved piping plan.

End closure shall be designed as per ASME Section VIII. Division-I. Quick actuating closure shall be designed that the failure of a single locking component while the vessel is pressurized (or contains a static head of liquid acting at the closure) will not cause the hinged closure to open or leak or result in the failure of any other locking component or holding element or increase the stress in any other component or holding element by more than 50% above the allowable stress of the component.

Corrosion allowance shall be as MOC suitable to respective piping class.

Closure shall be so designed that holding elements of the closure are fully engaged upon closing and that all internal pressure is released before closure can be opened. A vent plug interlocked with clamp ring/operating lever shall be provided to warn the operator if an attempt is made to open the door, before the pig barrel is depressurized.

Clamp type quick closure with bolt tightening at both sides of the door or end closure with exposed screw expander or captive ratchet braces shall not be used.

Lifting Eyes shall be required for size 16" and higher.

The closure shall be approved by ASME- approved third party inspection agency.

The fit up and installation of hinged closure shall be witnessed by the manufacture of hinged closure and a certificate shall be obtained from the manufacturer stating that the hinged closure has been installed correctly and submit to company for approval. And there after a minimum of two closing and opening cycles shall be performed in presence of company approved TPI. The correct operation of both quick opening and safety system shall be established and furnished to the Company.

Wetted Seal Face of Hinged closure of MOC - CS Nace, Non Nace shall be cladded with Incoloy 825 or Inconell 625 by weld overlay as per following:

1. The minimum cladding thickness shall be 3mm.

2. Applicable specification for cladding shall be ASME IX, API 6A and NACE MR01-75/ISO-15156-1/2/3.

3. A complete Overlay Welding Procedure Qualification Record in accordance with ASME IX shall be developed for overlay of nickel based alloy (generally Incoloy 825 or Inconell 625).

4. Testing requirement on cladded seal face:

   (a) Chemical Composition: Chemical composition shall be ascertained by appropriate method for top 2mm of cladding thickness and chemical composition shall be within

limit of nickel base alloy (generally Incoloy 825 or Inconell 625). Iron dilution shall not exceed 10%.

(b) Other tests indicated in FS 2004A for Incoloy 825 or Inconell 625 shall not be applicable for overlay of seal face of hinged closure.

For cladding by over lay, the vendor shall develop their own procedure and testing requirements during detail engineering.

Hinged closure shall be painted as per FS 2005 suitable to marine environment after installation.

7.9 CHOKE VALVES

7.9A Choke valve: (Lift Gas Injection)

Choke is normally used for controlling Flow rate as well as pressure for injection of lift gas into wells.
The body shall be straight line adjustable choke assembly or angle – globe type as indicated in the P&ID.

Trim shall be microform type.

Material of Body shall be compatible with MOC of the connected piping.

Trim material shall be as given below.

MOC of Connected Piping	MOC of Trim
CS/C.S (NACE)	S.S 316
D.S.S	D.S.S
Incoloy	Incoloy/D.S.S

Actuation shall be manual single port design.

Inspection and testing of choke Valve shall be same that of globe Valve indicated in FS 2004A.

7.9B Adjustable Choke Valve: Well fluid flow arm

This part of specification is for adjustable choke valve being used in the well fluid flow arm.

The design and testing of the choke valve shall be as per API 6A. The choke valve shall be angle type. The choke valve shall be generally manually actuated type unless otherwise stated. The actuator design shall also conform the requirement of API 6A. The manually actuated choke valve shall have position indicator as indicated in API 6A in steps of 1/64" and also have locking arrangement.

Body material shall be compatible to connected piping and selected from material indicated in API 6A.

Trim material shall be tungsten carbide to handle fluid velocity at MACH 1 or more.
Trim form shall be cage guided sleeve and designed such that the trim can be changed to multi stage type without change of body.

Trim shall have sufficient stages.

Trim shall meet noise requirement of 85db as per National Institute of Occupational Safety and Health.

Inspection and testing of choke Valve shall be same that of globe Valve indicated in FS 2004A.

7.9C Adjustable Choke Valve: Water Injection

This part of specification is for adjustable choke valve being used in the water Injection Service

The design and testing of the choke valve shall be as per ASME B 16.34. The choke valve shall be angle type unless otherwise dictated by process requirements. The choke valve shall be generally manually actuated type unless otherwise stated. The actuator design shall such that torque requirement is within the limit as indicated in API 6D or any other relevant guide line indicated specification. All-out effort shall be made to keep the torque requirement as low as possible with hand wheel diameter as indicated in API 6D. The manually actuated choke valve shall have position indicator and also have locking arrangement.

Body material shall be compatible to connected piping and high flow velocity fluid.

Trim material shall be tungsten carbide to handle high fluid velocity with abrasive material.

Trim form shall be plug and cage.

Inspection and testing of choke Valve shall be same that of globe Valve indicated in FS 2004A.

7.10 SPRAY NOZZLES

This refers the requirements for the supply of spray nozzles to be installed in a deluge type salt/fire water spray network on the offshore platform.

The spray nozzles shall be attached to piping by means of threaded half couplings or threadolets.

All spray nozzles shall of the Type-Medium velocity, open, non-clog type.

Discharge pattern: Solid cone of dense, far reaching water fog.

Spray nozzles shall meet NFPA-15 requirements and shall be approved by

Underwriters Laboratories, USA/VJTI, Mumbai, India.

Materials: Nozzles (including deflector plate): copper alloy of a grade suitable for salt water service and marine atmosphere, material to 5% Ni-Al-Bronze conforming to BS-1400 AB 2C/, BS EN 1982 C333G/ASTM B 148 C 95800 or equivalent.

7.11 FIRE AND FOAM HOSE REELS

Galvanized, welded steel hose reel with single length (18 M min.) 1½" (ID) all rubber non-collapsible, non-shrinkable hose with brass couplings.
Nozzle shall be an Elkhart SFL (or equal, 95 GPM capacity with straight steam or fog capability, of chrome-plated brass for salt water service. Supply shall also include an AFFF 6% concentrate, tank constructed of 316 SS. Hose reel shall be mounted on the top of tank having capacity of 30 Gal (US). (Minimum).

Eductor system shall be of bypass type with metering valve for foam control 0% to 6% setting and a transparent pick-up tube.

Hose shall be Neoprene rubber lined and have an external coating of oil and abrasion resistant material to protect them from sunlight or mild dew damage.

Hose reels shall have underwriters laboratory/US coast Guard/FM approval. Hose shall be suitable for maximum working pressure of 14 kg/cm2 and tested to 21 kg/cm2g.

Design and construction of hose reel station shall be in accordance with NFPA and relevant Codes and Standards.

7.12 CONTINUOUS DRAINERS

Continuous Drainers shall be direct acting float operated type with MOC compatible with connected piping as per P&ID and with RF flanged ends to ASME 150/300 as applicable for 2"and above, Screwed for 1-1/2 and below.

Contractor to select proper valve and seat to ensure that continuous Drainers shall operate at 3 times the normal flow.

All continuous Drainers shall be suitable for the intended applications.

All continuous Drainers shall be supplied with line size Y-type strainers.

7.13 CHEMICAL UTILITY, HOSES & HOSE CONNECTION

1. HOSES

The manufacturer shall guarantee the suitability of the hoses for the intended service and working conditions .

The requirement of Materials, Construction, Dimensions and Testing of Hoses shall be as per approved applicable codes & relevant piping class of approved P&ID'S.

The nipples shall have male threads and be made from seamless pipe conforming to ASTM A312 TP 316 unless stated otherwise.
Nipples shall be built-in-type i.e. vulcanized with the Hose such that normal flexing and stretching will not subject it to stresses which might cause breakage/damage.

Each hose shall be hydrostatically tested to two times the working pressure, or at 21.0 kg/cm2g whichever is higher.

The length of finished hose assembly shall not differ from nominal length by more than +2% and –1%. For this purpose the hose assembly shall be measured after being subjected to tests. All hoses shall be of single hose lengths.

All hose assemblies shall be suitable for Marine Environment and shall be resistant to aging, abrasion, sunlight and dews.

All hoses shall be non-collapsible type and able to withstand full suction.

All potable water hoses shall be FDA approved types. All hoses for chemical diesel and hydrocarbon service shall be fire-resistant type as per United States Coast Guard (USCG).

2. HOSE CONNECTIONS

A. Description:

a) KAMLOCK type couplers and adapters for use in hose connections of different services is as under:

b) KAMLOCK Type Hose Couplings: KAMLOCK Type Hose Couplings shall be KAMLOCK 633-D Services with NPTF ends with adapter KAMLOCK 633- A Series with NPTF ends and dust plug (KAMLOCK 634A/634B series).

Foster one-way shut-off 6 series industrial interchange quick detachable type plug and socket for use in hose connectors.

B. Materials:

Body (coupler, adaptor and dust plug type) - 316 L (Stainless Steel)
Cam Arms and Pins - 316 (Stainless Steel)
Gasket - PTFE/Buna-N.

C. Each coupler, adopter and dust plug shall be subjected to a hydrostatic test at 21 kg/cm2g or at two times the working pressure whichever is higher.

7.14 HAND CONTROL VALVE

a) Body & Bonnet Material: Carbon Steel 'ENP' Coated

b) Trim Material : Tungsten Carbide/Stainless Steel with stellite facing.

c) Type of Trim : Variable Resistance trim

d) Type of Valve : Multistage Pressure Reducing Valve (FO)

e) Operation: Pneumatic Actuator with manual override available gas with supply pressure 8.0 g/cm2 (g) (Maximum Pressure 10.6 kg/cm2(g) (Minimum Pressure 5.5 kg/cm2(g)

g) Function: To divert derated sea water of 0.02 mg/lit. oxygen (sometimes raw sea water with 7 ppm oxygen) over board during start-up and pre-commissioning operations. Inlet from de-rated seawater header and outlet dumped to sea.

Notes for Valve:-

Valve shall be 100% radio graphed as per the procedure and acceptance criteria specified in ANSI B 16.34 latest edition.

Vendor shall furnish sizing calculations, material of construction etc. along with quotation and shall also furnish predicted noise level calculations with the quotation.

Vendor shall perform the functional test at offshore. To ensure cavitation/vibration free operation, pressure drop and noise level to be within specified/acceptable limits. In case there is a fault/malfunctioning vendor must replace the parts/complete valve or carryout necessary rectification is at his cost.

Trim shall be anti-cavitation, anti-corrosion, anti-erosion and anti-noise type.

An input torque limiter shall be provided in order to close the valve adequately to avoid wiring erosion.

Vendor shall submit the recommended piping installation drawings upstream and downstream of the Hand Control Valve in order to minimize the noise level.

Valve shall be equipped with a seat incorporating Teflon insert to avoid wiring erosion.

A restriction orifice assembly shall be provided downstream of each Hand Control Valve. Hand Control Valve Vendor shall decide intermediate pressure (PA).

A complete restriction orifice plate with matching flanges shall be supplied along with Hand Control Valve to suit working conditions as specified.

All external hardware to be suitable for offshore service where the atmosphere is salt laden and corrosive.

Valves shall be with 'Variable Resistance Trim concept which allows cavitation free operation. Vendor shall provide Pneumatic actuator spring return type.

One pressure regulator with ϕ150mm (0-100%) dial allowing to deliver 3-15 psig to the valve positioner along with adapting parts. Vendor shall supply one pressure regulator with gauge reducing the instrument air pressure from 10.6 kg/cm2g to 20 psig along with adapting parts for mounting on 2" pipe for Pneumatic Actuator Manual holding station.

7.15 FLAME ARRESTORS

Body shall be of 316 SS bank. End connection shall be flanged. Weather- hood shall be provided wherever necessary. Flame arrestor shall be listed by UL/ FM or equivalent, if applicable. Design standard for Flame Arrestor shall be BS EN ISO: 16852 or Equivalent.

7.16 DOUBLE BLOCK AND BLEED VALVE

Double Block and Bleed Valve under this specification means a device having two Ball Valves in the main flow line for isolation and a needle Valve installed in between the two ball Valves to bleed the fluid trapped between the Isolation Ball Valves. The design of two isolation ball valves and the needle Valve shall be same as that of Ball Valve/Needle Valve as described elsewhere in the bid package.

The combination of both the Ball Valve and the needle shall be of integral and single assembly. It is preferred that both the Ball Valves and the Bleed Valve assembly is of compact design so that space requirement is as less as possible.

An assembly having two independent ball valves and an independent needle Valve assembled together is not acceptable wherever Double Block and Bleed Valve are specified in the bid package. End connections can be flanged by flanged or flanged by screwed. Vent connection shall be ½" NPT as standard.

Piping or tubing (as applicable) shall be routed to nearest vent or drain line (as applicable).

Torque requirement and hand wheel/lever of both the Ball Valves and Needle Valve shall not be more than as specified in API 6D or any guide line specified in Bid Package.

Testing requirements of Ball Valves and Needle shall be same as that of respective valve standards under which Valves are designed.

Double Block and Bleed Valve assembly shall be of fire safe design as per API 607/API6FA/BS 6755 Part 2 (latest edition) except the valves used in services such as injection water, fire water, pot water service etc.

7.17 VALVE FOR HIPPS (High Integrity Pressure Protection System)

HIPPS as defined in this specification is Instrumented System of High Reliability for isolation of topside facilities or pipeline from pressure source before allowable pressure is exceeded beyond the design pressure at Down Stream of HIPPS.

This specification of HIPPS is a functional description indicating requirements for design and manufacture of main process isolation Valves used in HIPP System. Since HIPP System calls for high reliability and redundancy, minimum two numbers of isolation Valves shall be installed in each HIPP System. Valves used in HIPPS shall be termed as Primary Process Isolation Valve & Secondary Process Isolation Valve. For items/equipment other than the main process isolation valves refer scope of work and Functional Specification of HIPPS enclosed in the bid package.

Reliability and Safety Integrity level of the main process isolation valve must be same as that of the complete HIPPS including instruments and other items and equipment used along with the HIPPS.
Valves used in HIPPS shall comply with following:

Type: Ball valve or Axial Flow Valve.

Application:
(a) Piggable Pipeline - Full Bore Ball Valve.
(b) Non-Piggable Pipeline or Piping at top Side - Axial Flow Valve or Full Bore Ball Valve.

Size: As per P&ID.

Others: **Full Bore Ball Valve:** Valves rating; flange facing; body, piston, seat and trim material, packing etc. shall be to the Piping Functional Specification, FS 2004A. Valves shall have flanged ends irrespective of line sizes and class rating. Metal seated Ball valves may be considered for HIPPS application.

Axial Flow Valve: Valves rating; flange facing; body, Piston, seat and trim material, Packing etc. shall be to the Piping Functional Specification, FS 2004A. Valves shall have flanged ends irrespective of line sizes and class rating. Flow passage area of the valve shall be equivalent to full bore Ball Valve of same size to ensure pressure across the valve is equivalent to full bore Ball Valve of same size. In case of Axial Flow Valve, there is possibility of kinetic energy of fluid getting dissipated on the piston of the Valve. Hence MOC of part of trim which faces high velocity fluid directly on its surface shall be of a material suitable for the purpose i.e hardness of surface shall be

increased by using material such as tungsten Carbide, stellite, DSS, Incoloy, Inconel etc. based on fluid velocity and fluid characteristics.

Torque: Actuator Torque required to operate the valve shall be at least 1.5 times more than Break away torque of the Valve.

Installation: Valve installation shall be Horizontal unless otherwise established that Horizontal installation is not suitable for the purpose.

Shutdown time: Shut Down time shall be as dictated by Functional Specification of HIPPS enclosed in Bid package but preferred time for shut down shall be less than 0.2 Sec/inch diameters in case Specification of HIPPS does not dictate this parameter.

Valve Design: (a) Full Bore Ball Valve:
Refer FS 2004A for details.

(b) Axial Flow Valve:
Design Standard: API 6D or BS EN ISO 17292 (Up to 2") and design must be fire safe design.

Leakage Rate: Ball Valve:

(i) Acceptable leakage rate for hydro test- No visible leakage (Rate-A for soft seated valve & Rate-B for metal seated Valve).

(ii) Acceptable leakage rate for pneumatic/nitrogen gas test – Tight Shut off leakage rate- VI (ANSI class VI leakage)

Axial Flow Valve:

(i) Acceptable leakage rate for hydro test-No visible leakage (Rate-A).

(ii) Acceptable leakage rate for pneumatic/nitrogen gas test – Tight Shut off leakage rate- VI (ANSI class VI leakage)

PST
(Partial Stroking): Ball Valve: Design of the Valve shall be such that there is provision to partially stroke the valve without effecting shut down of the process.

Axial Flow Valve: Design of the Valve shall be such that there is provision to partially stroke the valve without effecting shut down of the process.

SIL Level: Safety Integrity level of HIPPS Valve shall be same as that of complete HIPPS system described in HIPPS specification i.e. C-104 and Valve shall be proven in use with the required SIL Level as indicated in C-104.

All hydrocarbon valves are to be supplied as per approved P&ID's to meet NACE-MR-01-75 (latest edition) and Piping Design Criteria.

Actuator shall be mounted on the valve and tested as follows:

- Cycle (open and shut) each valve with its actuator at least five (5) times to ensure smoothness of operation.

- Shut valve and apply hydraulic differential pressure across the valve. Open valve with the installed actuator and note valve operation. Valve operation should not be jerky or binding. This shall be repeated at least three (3) times and shutdown time noted at minimum supply pressure.

c) Test shall be witnessed by company's representative or company's authorized representative.

d) All soft seated valves shall be of fire safe design as per API 607 or API6FA or ISO 10497 or BS 6755 PART II (latest edition) except the valves used in services such as injection water, fire water, pot water service etc.

e) Fire safe tests duly witness and certified by recognized third party inspection agency e.g. LLOYD/DNV/BV/EIC shall be furnished.

NOTE: HIPPS system shall be a package item with component vendors as enlisted in SVL. For new PQDs of HIPPS System Package, PTR for the HIPPS package shall be for complete package supplied to various offshore oil and gas projects and shall be submitted with satisfactory Performance feedback for 2 years service OR in the form of repeat PO copies with a minimum time gap of 2 years from the date of delivery. The major components of the package are Logic Solver, valve & Actuator which shall remain same as supplied in package (considered for PTR) above.

7.18 SAMPLE BOMB

Sample bomb shall be designed as per ASME Sec VIII Div I . Hose for sample bomb assembly shall be fire-resistant type as per United States Coast Guard (USCG).

8.0 SPARES

Manufacturer shall recommend and provide the spares required for startup and commissioning for all specialties.

Manufacturer shall recommend and provide a list of spares needed for one-year operation and maintenance for all specialties.

9.0 WARRANTY

Vendor shall warrant all the piping specialties furnished by him in accordance with Piping Design Criteria (Vol II sec 3.3).

FUNCTIONAL SPECIFICATION FOR INSULATION OF PIPING & EQUIPMENT

RITESH KUMAR YADAV Digitally signed by RITESH KUMAR YADAV Date: 2021.11.12 14:45:24 +05'30'

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CONTENTS

1.0 INTRODUCTION
1.1 General
1.2 Exclusions
1.3 Contractor's Responsibilities
1.4 Application of this Document

2.0 CODES AND STANDARDS
2.1 Codes, Standards and Regulations
2.2 Project Specifications
2.3 Mandatory Indian Statutory Requirements

3.0 ENVIRONMENTAL DESIGN CRITERIA
3.1 Design Basis Climatic Conditions
3.2 Seismic and Transportation Loads

4.0 GENERAL REQUIREMENTS
4.1 Items Requiring Insulation
4.2 Exclusions

5.0 MATERIALS
5.1 General
5.2 Classifications
5.3 Materials for Low Temperature Services (Class C)
5.4 Material for Heat Conservation Insulation (Class H)
5.5 Materials for Personnel Protection (Class PP)
5.6 Material for Acoustic Insulation (Class IA)
5.7 Sealers & Accessories
5.8 Materials for Weatherproofing & Ancillaries
5.8.1 General
5.8.2 Pipework & Equipment less than DN610
5.8.3 Piping & Equipment greater than DN610

6.0 INSULATION THICKNESSES

7.0 INSTALLATION

7.1 Application of Low Temperature Insulation (Class C)

7.1.1 Items Requiring Insulation

7.1.2 General Application

7.1.3 Piping Application

7.1.4 Equipment Application

7.1.5 Vapour Barriers

7.2 Installation of High Temperature Insulation (Class H)

7.2.1 General

7.2.2 Piping Application

7.2.3 Vessel & Equipment Application

8.0 INSPECTION, TESTING & CERTIFICATION

8.1 Quality Assurance

8.2 Inspection of Materials

8.3 Pre-Insulation Inspection

8.4 Final Inspection & Release

8.5 Testing

9.0 APPLICATION DETAILS

9.1 Pipe Insulation-Class H

9.2 Pipe Insulation-Class C

9.3 Termination of insulation on pipework

9.4 Valve insulation Class H

9.5 Valve insulation Class C

9.6 Vertical support for Class C insulation

9.7 Expansion/Contraction joints

9.8 Insulated supports for vertical insulated pipe

9.9 Typical detail for flanged joint Class H

9.10 Vessel support, flashing & weatherproofing details Class H insulation

9.11 Insulation support ring detail Class H insulation

9.12 Vessel support, flashing & weatherproofing details Class C insulation

9.13 Typical vessel nozzle flange detail for Class C insulation

9.14 Valve box cover construction

1.0 INTRODUCTION

1.1 GENERAL

This Specification gives the minimum requirements for the thermal and personnel protection insulation of piping and equipment (vessels, heat exchangers, pumps, etc.) and method of application.

Equipment and piping shall be insulated for the purposes of conservation of heat, maintenance of stabilized process temperatures during atmospheric temperature changes, condensation prevention, burn prevention of personnel or limiting noise levels according to the classification nominated.

1.2 EXCLUSIONS

This Specification does not cover:

- Electrical insulation or heat tracing
- Acoustic insulation of equipment
- Acoustic or thermal insulation of HVAC equipment or ducting
- Passive fire protection insulation for structures

1.3 CONTRACTOR'S RESPONSIBILITIES

This document shall form the basis upon which the EPC Contractor can develop a detailed specification. The document presents the findings of the Definition Engineering Study and as such, should be considered as a general specification. The Contractor should update this document with the detailed design information that is developed during the detailed design phase.

Further the contractor shall be responsible for the complete design, selection, supply, installation and inspection of insulation on piping and equipment, including full compliance with all applicable Codes and Standards, including those listed in section 2.0 of this Specification.

1.4 APPLICATION OF THIS DOCUMENT

The scope of this document is limited to this project & associated modifications. This document shall not be used for any other purpose.

2.0 CODES AND STANDARDS

2.1 CODES, STANDARDS AND REGULATIONS

The insulation covered by this Specification shall be designed, manufactured and tested in accordance with the requirements of this Specification, the attached Data Sheets and the following latest edition of Codes, Standards and Statutory Regulations (where applicable)

ASTM C165	Test Method for Measuring Compressive Properties of Thermal Insulation
ASTM C195	Mineral Fibre Thermal Insulating Cement
ASTM C303	Test Method for Density of Preformed Block Type Thermal Insulation
ASTM C335	Test Method for Steady-State Heat Transfer Properties
ASTM C449	Mineral Fiber Hydraulic Setting Thermal Insulation and Finishing Cement
ASTM C518	Test Method for Steady-State Heat Transmission Properties
ASTM C547	Mineral Fibre Pre formed Pipe Insulation

ASTM C552	Cellular Glass Block and Pipe Insulation
ASTM C591	Un-faced Preformed Rigid Cellular Polyurethane Thermal Insulation
ASTM C592	Mineral Fibre Blanket Insulation
ASTM C612	Mineral Fibre Block & Board Thermal Insulation
ASTM D1621	Test Method for Compressive Properties of Rigid Cellular Plastics
ASTM D1622	Test Method for Apparent Density of Rigid Cellular Plastics
ASTM D2126	Test Method for response of Rigid Cellular Plastics to Thermal & Humid Aging
ASTM D6226	Test Method for Open Cell Content of Rigid Cellular Plastics
ASTM D3014	Test Method for Flame Height, Time of Burning & Loss of Mass of Rigid Thermoset Cellular Plastics in a Vertical Position
ASTM E84	Test Method for Surface Burning Characteristics of Building Materials
ASTM E96	Test Method for Water Vapour Transmission of Materials
ASTM C1055	Standard Guide for Heated System Surface Condition that Produce Contact Burn Injuries
ISO 13732-1	Ergonomics of the thermal environment-Methods for the assessment of human responses to contact with surfaces-Part 1: Hot surfaces
ASTM C 1696	Standard Guide for Industrial Thermal Insulation System
IS 3150	Indian Standard-Standard for Hexagonal Wire Netting for General Purposes
API 574	Inspection Practices for Piping System Components

2.2 PROJECT SPECIFICATIONS

The insulation covered by this Specification shall be designed, manufactured and tested in accordance with the requirement of these Project Specifications.

1. Piping fabrication (FS 2004B)
2. Piping Specialties (FS 2004D)
3. Protective coatings (FS 2005)

2.3 MANDATORY INDIAN STATUTORY REQUIREMENTS

This document has been prepared to the international standards detailed within. The EPC Contractor shall ensure that the Scope of Work is executed in accordance with all mandatory Indian Statutory requirements.

3.0 ENVIRONMENTAL DESIGN CRITERIA AND UTILITIES

3.1 DESIGN BASIS CLIMATIC CONDITIONS

The environment conditions & climates under which the Equipment & piping will operate is as below.

Temperature: 16-40°C
Relative Humidity: 90% (max) at 40°C
Air Quality: saline, humid, corrosive

Designs shall be based on outdoor exposure to the conditions described. Conditions specific to the relevant Equipment Package & process parameters related to piping will be detailed by the contractor on individual Equipment Data Sheets as per approved P&ID's or process design criteria attached elsewhere in bid package and the same shall be submitted to the company for review.

3.2 DESIGN LIFE

The process facilities design life requirement is 25 years.

4.0 GENERAL REQUIREMENTS

The Supplier's standard designs (if applicable) which meet the performance requirements and which will operate in the marine environment documented in Section 3.0 for 25 years may be considered with prior approval of Company

4.1 ITEMS REQUIRING INSULATION

- Items shall be insulated for the purpose of conservation of heat, temperature stabilization, personnel protection, condensation prevention (anti-sweat) or limiting noise levels.

- Piping operating with fluid temperature in excess of 54^0C or below 10^0C shall be insulated as burn or freeze protection for personnel to a height of 2.1m above grade and platforms and 0.6m outside the platforms, walkways, etc. with the exception that lines operating above 200^0C shall be fully insulated.

- Heat transfer equipment in hot services, other than coolers and condensers, shall be completely insulated except for supporting members and flanges. Shell flanges, channel flanges and nozzle necks shall not be insulated. Nozzle necks shall be insulated to the nozzle flanges.

- All piping and equipment operating with fluid temperatures below 27^0C shall be insulated for condensation prevention ("anti-sweat"). Nozzles, components and appurtenances shall be insulated and covered with vapor barrier and weatherproof cladding.

- All accessories (such as level controllers, level gauges, pipe bridles, etc.) which are attached to insulated equipment shall be insulated with equivalent thickness and materials required for the equipment. The insulation shall not obstruct operation or necessary viewing of equipment components, drains, vents and sample points.

- Piping that is considered a potential source of noise should be acoustically insulated. A noise and vibration specification shall be as per spec 5002 and 5004 respectively.

4.2 EXCLUSIONS

The following hot parts shall not be insulated, except for burn protection:

- Man ways, manholes and inspection openings on hot vessels.
- Code inspection plates and nameplates
- Connectors and tube unions on fin-tube heat exchangers
- Flanges of piping and equipment in hot service
- Expansion and rotating joints

5.0 MATERIALS

5.1 GENERAL

All new materials shall be used. Materials showing any evidence of containing contamination or moisture shall not be used.
Insulation materials for equipment and piping shall be suitable for the application, operation, and temperature and be compatible with the contained fluids.

Insulation materials shall have the following properties:

- Completely free from all forms of asbestos.
- Water-soluble chloride and fluoride ion concentrations as low as possible, 20 ppm by mass maximum.
- Chemically neutral, with pH preferably between 7 and 8 in the wet condition.
- Non-combustible and odorless at operating temperatures.

- Have sufficient compressive strength that the method of fixing and finish shall not damage the insulation.

Phenolic and polyisocyanurate (polyurethane) foams shall not be used for services, which may exceed 100°C because these produce toxic smoke when burning.

Any wire mesh reinforcement shall be stainless steel grade 316. For size and details of wire mesh refer IS 3150 or any other international Standard as applicable. Size of aperture shall be 19 to 25mm and wire diameter shall not be less than 0.8mm.

Any required painting, protective coatings and the procedures used for the preparation of surfaces shall be as specified in the Project Specification for Protective Coatings spec. 2005.

5.2 CLASSIFICATIONS

The Line schedule and the Piping and Instrument Diagrams shall indicate the thickness and insulation classification required.
The following insulation classifications are described in this Specification:

C Low temperature insulation designated for processes with operating temperatures below 21^0C where heat gain is undesirable, or for the prevention of condensation on pipes and equipment with operating temperatures below 27^0C.

H Heat conservation insulation designated for all processes (except heat transfer fluids) with operating temperatures at or above 25^0C.

PP Personal protection insulation designated for equipment and vessels with operating temperatures in excess of 54^0C or below 10^0C.

IA Acoustic Insulation to restrict noise emanating on account of fluid velocity inside piping and due to vibration in piping and equipment.

5.3 MATERIALS FOR LOW TEMPERATURE SERVICES (CLASS C)

The following materials shall be used for piping and vessel insulation for Class C:

- Preformed or foamed-in-place polyurethane for piping up to and including DN610. However, Polyurethane block or segment may also be used for piping greater than DN610 based on vendor input during detail engineering. Same shall be brought to notice of company.

- Polyurethane blocks, segments or foamed in place sections for piping and vessels greater than DN610.

Polyurethane foam shall conform to ASTM C591, Type 11, with a flame spread rating not exceeding 25 in accordance with ASTM E84. Higher density and higher strength foam may be used as load bearing insulation if the properties are approved.

Insulating material used on austenitic steel surfaces shall contain less than 20 ppm leachable chlorides and a minimum of 20 ppm sodium silicate for each part per million of leachable chlorides.

Foamed in place polyurethane foam shall, as a minimum, conform to the following:

- Density of the core shall be between 30 and 35 kg/m3.

- Thermal Conductivity shall not exceed 0.019 W/mK when the foam is new and cured. This value shall be based on a mean temperature of 25^0C as determined in accordance with ASTM C518.

- Flammability shall be guaranteed to have a flame spread classification not exceeding 75 in accordance with ASTM E84.

- Compressive Strength in accordance with ASTM D1621 shall be not less than 170kN/m.

- Closed Cell Content shall not be less than 90%. The cell structure shall be uniform throughout the core and the average cell size shall be no more than 0.40 mm. Elongated cells shall be no longer than three times the smallest dimension. These requirements are to be verified in accordance with ASTM D6226.

- Dimensional Stability: the maximum volumetric change shall not be more than 6% after 7 days exposure at 75 ± 3^0C and 100+0/-3% relative humidity in accordance with ASTM D2126.

- The permeability shall not exceed 2.5 perm in accordance with ASTM C335.

- Friability: A 6 mm cube, when properly cured, shall not powder when crushed.

- Vapor barrier coatings shall cover insulation completely before weatherproofing is installed and have the following characteristics:

    - They must be compatible with the process piping insulation

    - They must be flexible at the lowest temperature to which they are exposed

    - Permeability shall not exceed 0.03 perm in accordance with Procedure E of ASTM E96 for the thickness used.

    - Minimum dry film thickness shall equal or exceed 0.76 mm

    - Flame spread rating shall not exceed 30 in accordance with ASTM E84.

    - They shall be weatherproof and non-absorbent and shall constitute a vapor barrier

5.4 MATERIAL FOR HEAT CONSERVATION INSULATION (CLASS H)

Thickness shall be in accordance with Section 6 of this specification.

The following materials shall be used for piping and equipment insulation for Class H:

- For equipment and piping which is to be operated intermittently or continuously at temperatures not exceeding 100^0C, the insulation may be rigid urethane foam.
- For equipment operated intermittently or continuously above 100^0C or for acoustic insulation applications, the insulation may be calcium silicate. Calcium silicate insulation which contain asbestos filler shall not be acceptable.
- For equipment and piping which is to be operated intermittently or continuously at temperatures up to 260^0C, primary insulation may be foam glass complying with ASTM C552 and with density of at least 88kg/m3 and a temperature rating of 315^0C.
- For equipment and piping which is to be operated intermittently or continuously at temperatures up to 500^0C, the insulation may be mineral wool with temperature rating of 500^0C. All mineral wool shall be water repellent and tested for water absorption and retention.

Mineral wool blankets for piping and vessels greater than DN610 (24“) shall conform to ASTM C592. They shall have density of at least 128 kg/m3 and shall be faced with wire mesh inside and either wire mesh, expanded metal, or glass mat outside.

Mineral wool preformed sections shall be used for piping larger than DN40 and up to and including DN610. They shall conform to ASTM C547, have density of maximum 160 kg/m3 and shall be faced with wire mesh inside and wire mesh or glass mat on the outside.

Mineral wool blocks and boards shall conform to ASTM C612. Their maximum nominal density shall be 160 kg/m3.

5.5 MATERIALS FOR PERSONNEL PROTECTION (CLASS PP)

Piping, vessels and equipment operating at a temperature of 54^0C and above, or below 10^0C for a cold service and not falling within any other insulation category, shall be protected by Stainless Steel metal mesh guard or similar protective material. The minimum distance between hot or cold surfaces and guarding shall be 32mm.

Where this is not practical, thermal insulation, Class H or C as appropriate, shall be applied.

5.6 MATERIALS FOR ACOUSTIC INSULATION (CLASS IA)

Piping and Vessels which is potential source of noise on account of any reason such as process flow or due to vibration on piping and equipment or any other reason shall be insulated for limiting noise. Requirement of acoustic insulation in piping, vessels and Equipment shall be as per process data and applicable specification i.e. FS 5002.

While finalizing acoustic insulation circumstances of combination insulation system such as cold insulation and acoustic insulation, hot insulation and acoustic insulation as well as Personal protection insulation and acoustic insulation may be required to be addressed depending upon process requirement. Same shall be finalized during detail engineering.

Material for acoustic insulation shall be Mineral wool, glass fiber or calcium silicate. The thickness required for acoustic insulation shall be as indicated in clause no. 6 of this specification.

For General guidance of acoustic insulation, ASTM C 1696: Standard Guide for Industrial Thermal Insulation System, clause no.7.2.8 may be referred.

5.7 SEALERS & ACCESSORIES

Flashing compounds, joint sealers and bedding compounds shall be permanently flexible through a temperature range of -18^{0}C to 500^{0}C, be capable of withstanding repeated expansion and contraction without cracking and experience minimum shrinkage.

All metal accessories shall be stainless steel SS316.

Wire to secure insulation shall be a minimum of 1.2mm annealed stainless steel. Heavier stainless wire or cable shall be used for insulation on vessel heads where floating rings are employed.

Expander bands shall be capable of remaining in tension during the heating and cooling cycles experienced during normal operation.

Breather springs for bands (if used) shall be capable of remaining in tension during the heating and cooling cycles experienced during normal operation.
Screw fasteners shall be stainless steel, self-tapping with pan heads.

Wire mesh for cement reinforcement shall be 25mm hexagonal mesh with 0.8mm diameter Monel or stainless steel wire.

Mineral wool insulating cement shall meet the requirements of ASTM C449, and be suitable for temperatures up to 315^{0}C.

Finishing cement shall be asbestos free, meet the requirements of ASTM C449 and shall be suitable for temperature up to 315^{0}C. It shall be fast drying cement that provides smooth, hard surfaces when dry. The cement shall contain a corrosion inhibitor to prevent corrosion when wet.

5.8 MATERIALS FOR WEATHER PROOFING & ANCILLARIES

5.8.1 GENERAL

Jacketing materials for vessels, equipment and piping shall be stainless steel grade 316 or Aluminum as per ASTM B 209, corrugated in accordance the thickness and corrugation requirements as follows:

- For piping, for vertical vessels with 0.40mm thick with 32mm corrugations and for all horizontal vessels, shall be 0.40mm thick with 5mm corrugations.
- In case aluminum is used, it shall be epoxy coated on exterior surface for all services and also for interior surface when used over calcium silicates.

Jacketing shall be installed and supported to withstand direct fire hose stream impingement, site and climatic conditions.

5.8.2 PIPE WORK & EQUIPMENT LESS THAN DN610

Stainless steel jacketing used for pipework shall be so arranged that it shed water at all times. Overlaps shall be swaged and rendered watertight. Bands for piping shall be at least 12mm wide x 0.4mm thick and secured with wing type seals.

Jacketing for flange, valve and in-line instrument boxes shall be fabricated from stainless steel flat sheet as used for cladding of the adjacent pipework and shall be built in at least two parts. Joints in the jacketing at pipe tees, branches, welds, etc. shall be sealed with stainless steel segmented flashing or preformed sections and metal seam sealant.

5.8.3 PIPING & EQUIPMENT GREATER THAN DN 610

Panels on vessels and large diameter equipment shall be held tight over the insulation by means of 316SS circumferential metal bands and attachment to insulation support rings. Panels shall have minimum overlaps of 100mm for vessels and 50mm or pipes. All overlaps, which are not subject to thermal expansion, shall be rendered watertight by use of a metal seam sealant.

Bands shall be positioned on all horizontal overlaps and at 300mm centers maximum. The bands shall be held in their relative positions with clips fitted to jacketing at suitable centers. Bands for equipment and vessels shall be at least 19mm wide x 0.5mm secured with wing type seals. Breather springs with flat bands shall be used where required for expansion or contraction.

Vessel head jacketing shall consist of preformed flat stainless steel sheets in a 'segmented' construction with all radial seams overlapping. The exact method of construction shall be made available to the Company for review.
The Supplier shall propose details of methods of attachment and support of jacketing.

The vessels, nozzles and piping shall have removable plugs in insulation lining for facilitating of thickness check (ultrasonic) of wall during lifetime of vessel. Typical drawings for the same are enclosed. Plug shall be made with the compatible materials. For nozzle pocket (plug) shall be in four directions at the center of the nozzle.

Insulation plugs for thickness survey shall be as per API 574 or any other international Standard. Insulation Plug shall be in four directions i.e. 6 O'clock, 3 O'clock, 12 O'clock, 9 O'clock position other than piping having branch connection. In location of branch connections, insulation plug shall be installed in three directions depending on orientation of branch connection.

6.0 INSULATION THICKNESS

The following insulation thickness are to be applied to piping and equipment for the various applications:

MINIMUM THICKNESS OF INSULATION (MM)

Nominal Equipment Size or Diameter of Pipe	Cold to 10^0C	Acoustic	Burn* r Freeze Stabili-n	20 to 70	71 to 100	101 to 200	201 to 300	301 to 400	401 to 500
DN40 & smaller	40	75	25	25	25	40	50	75	90
DN50 to DN100	50	75	25	25	40	50	75	100	120
DN150	50	75	25	25	40	65	75	115	140
DN200 & larger	65	75	40	40	40	80	100	145	170

* For line temperature greater than 301 deg. C, the burn protection insulation thickness shall be as specified for conservation of heat.
Where the insulation thickness requirement exceeds 80mm, the insulation shall be applied in not less than two staggered layers.

7.0 INSTALLATION

7.1 APPLICATION OF LOW TEMPERATURE INSULATION (CLASS C)

7.1.1 ITEMS REQUIRING INSULATION

All pipe, valves, flanges and fittings shall be insulated. Valves shall be insulated to the packing gland.

Vessel and exchanger flanges, manhole covers, and all appurtenances shall be insulated. All attachments to the vessel or exchanger such as skirts, supports, ladder and platform clips, etc. shall be covered with insulation for a distance of three (3) times the basic insulation thickness, with the vapour barrier continuing and sealing to the metal.

7.1.2 GENERAL APPLICATION

Insulation shall not be applied until welds have been inspected, the piping system has been pressure tested, and all related piping and equipment has been painted, in accordance with the relevant Project Specifications.

The insulation manufacturer's installation recommendations shall be followed. In the event of conflict between manufacturer's recommendations and this Specification, the former shall govern.

Piping, vessel and equipment insulation thickness shall be as shown in Section 6. The thickness listed is those required to prevent condensation forming on the insulation surface at 200 C and a relative humidity of 85%.

7.1.3 PIPING APPLICATION

Contraction joints shall be provided on straight runs of piping at 10m intervals.

Insulation supports shall be provided on vertical piping immediately above each pair of flanges and at contraction joints.

Pipes, valves and fittings shall be fitted with preformed insulation and secured in place with bands placed on 230mm centers. Joints are to be snugly butted together and sealed with joint sealer. Voids between fittings and fitting covers shall be filled with glass fibre insulation.

Insulation shall be installed around valves and flanges so as to allow removal and re-use without damage to the insulation or jacketing.

Pipe, fittings and valves of insulated lines shall be weatherproofed with a metal jacket. The jacket shall be secured with bands on 230mm centers. A band shall be applied over each circumferential lap joint.

7.1.4 EQUIPMENT APPLICATION

Insulation shall be installed around access hatches, inspection hatches, manholes and exchanger tube sheet flanges, vents, drains and sample points so as to allow removal and re-use without damage to the insulation or jacketing.

For vessels and tubular equipment DN610 and smaller, pre-formed pipe insulation shall be used. Insulation may be blocks, beveled lags, or curved segments for vessels and exchangers DN610 and larger. All such pieces of insulation shall be beveled or shaped to fit closely to the contour of the equipment.

Insulation on vertical vessels and tubular equipment shells shall be secured in place with bands on approximately 300mm centers. Where irregular surfaces make banding impractical, tie wire shall be used. The butt edge surfaces of all segments and block sections of the insulation shall be buttered with approximately 1.6mm coating of joint sealer prior to application. All voids shall be filled with glass fibre. Vertical contraction shall be provided by installing glass fibre in the space below the support ring and covering with additional insulation.

Insulation on all horizontal vessels, tubular equipment and other cold equipment surfaces shall be secured in place with bands on approximately 300mm centers. Where irregular surfaces make banding impractical, tie wire shall be used. The butt edge surfaces of all segments and block sections of the insulation shall be buttered with 1.6mm coating of joint sealer prior to application. All voids shall be filled with mineral wool insulation.

Heads on vessels and tubular equipment shall be insulated with blocks. The butt edges of all segments shall be buttered with approximately 1.6mm coating of joint sealer and segments secured with bands attached to a floating ring in the center and to a stainless steel band installed at the tangent point of

the vessel head. Band spacing shall be 300mm maximum at the circumference of the vessel.

All insulated surfaces shall be weatherproof with a metal jacket. Metal jacketing over the top of any piece of equipment or vessel shall overlap the wall jacket so as to shed water.

7.1.5 VAPOUR BARRIERS

All vapor barriers shall be applied as a minimum of two coats of contrasting colors. Glass cloth shall be imbedded into the first layer whilst still wet. Following the manufacturer's recommended drying time, the second coat shall be applied to the appropriate wet film thickness. No glass mesh shall be visible after the application of the second layer.

Vapor barriers shall be applied as soon as practicable following the application of the insulating material to ensure the insulating material is kept dry.

Self-tapping screws shall not be used on metal jacketing for IL and IC services to prevent puncture of the vapor barrier. Pop rivets with a metal backing strip shall be used on insulation classes requiring a vapor barrier.

7.2 INSTALLATION OF HIGH TEMPERATURE INSULATION (CLASS H)

7.2.1 GENERAL

Insulation shall be applied after all pipe welds have been inspected and pressure tested, and the piping or equipment has been painted in accordance with the relevant Project Specification.

Insulation on equipment and piping shall be stopped far enough from flanges or mechanical joints to permit the removal of bolts without damage to the insulation. Insulation near the flanges or mechanical joints shall be beveled at 45^0 & weatherproofed.

Insulation manufacturers' installation recommendations shall be followed. In the event of conflict between any manufacturer's recommendations and this Specification, the former shall govern.

Piping, vessels and equipment insulation thickness shall be as shown in Section 6.

7.2.2 PIPING APPLICATION

Single thickness sectional pipe insulation shall be applied with longitudinal joints staggered and shall be secured with wire ties on no more than 230mm centers. The insulation shall be applied with the joints tight and with all cracks and voids filled with insulating cement.

Screwed and welded fittings DN40 and smaller shall be insulated with insulating cement applied in 12mm layers to a thickness equal to that of the insulation on adjacent piping or preformed or mitered sectional pipe insulation, securely wired in place. Fittings DN50 and larger shall be insulated with preformed or mitered sectional pipe insulation, securely wired in place. Weatherproofing shall extend a minimum of 80mm under the adjacent pipe weatherproof jacketing joints between cement and jacketing shall be sealed with mastic to prevent water ingress.

Vertical piping shall include insulation supports at a maximum of 3m intervals. The material for the support ring shall be identical, or compatible to the material of the equipment to which they are attached in order to prevent galvanic corrosion.

Straight portions of insulated lines shall have metal jackets. All joints shall be lapped at least 80mm and arranged to shed water. The jacket shall be secured with bands on no more than 230mm centers. A band shall be applied over each circumferential lap joint.

Fittings DN50 and larger shall be covered with pre-formed metal jackets.

Where insulation is required around flanged joints in piping or vessel nozzles, the flanges shall be encased in an insulation-lined box as shown attached. Drainage shall be provided from the base of the box to provide indication of leakage. The Supplier shall ensure leakage of heat transfer fluid will not drain onto walkways, platforms or ladders.

7.2.3 VESSEL & EQUIPMENT APPLICATION

For vessels and tubular equipment DN610 and smaller, single-layer or preformed pipe insulation and cement shall be used as above. For larger diameter vessels and tubular equipment, blanket insulation shall be used.

Where the insulation thickness requirement (as shown in Section 6) exceeds 80mm, the insulation shall be applied in not less than two layers.

Blanket insulation on vertical vessels shall be applied with stagger joint arrangement. Top and bottom edges shall be securely tied over support rings with wire at no more than 300mm centers. Vertical and intermediate horizontal seams shall be laced together with gauge wire or hog rings at no more than 300mm centers. The insulation shall then be secured in place with bands spaced on no more than 300mm centers. Nozzle projections through insulation shall be adequately insulated.

Blanket insulation on horizontal vessels heaters and exchangers shall be applied in a staggered joint arrangement with all edges securely laced together with wire or hog rings at no more than 300mm centers. Nozzle projections through insulation shall be adequately insulated. Further securing shall be with bands spaced at no more than 300mm centers.

Heads of all vessels and tubular equipment shall be covered with preformed or segmented metal jackets. If a surface is so irregularly shaped that it cannot be jacketed, it shall have a topcoat of weather proofing mastic. Total finish coat shall be a minimum of 3mm thick when dry. Weatherproofing shall extend a minimum of 80mm under the adjacent jacketing.

Heavy fillets of mastic shall be applied to all corners and crevices where water is likely to collect. Mastic shall also be used as flashing at all possible sources of moisture penetration, such as intersections of insulation, nozzles and other protrusions through the surface coating.

Straight shells of vertical vessels with diameters over DN610 shall be weatherproofed with metal jacketing as follows:

- Securing of the jacket shall be with banding located over every circumferential lap of the jacket and on 300mm maximum centers. When expansion springs are required, the same number of expansion springs per band shall be used on all banding securing the jacket.

- Corrugated sheets shall be applied with the corrugations positioned vertically with the vertical seams lapped a minimum of two corrugations and the horizontal seams lapped a minimum of 80mm, arranged to shed water.
- Corrugated sheets shall be supported at the horizontal laps with stainless steel S-clips.

Straight shells of vertical vessels DN610 and smaller and all horizontal vessels and exchangers shall be weatherproofed with metal jacketing as follows:

- Securing of jacketing shall be with banding located over girth seams and on no more than 300mm centers.

8.0 INSPECTION, TESTING & CERTIFICATION

8.1 QUALITY ASSURANCE

The contractor shall develop QA/QC procedures for the work, which shall include the inspection of all materials, preparatory works and application, before commencement, during progress and after completion of the insulation work.

Inspection shall be carried out after completion of any one stage and before commencement of the following stage.

In progress fabrication and installation checks shall be conducted at various fabrication and installation phases, starting from material checks to final performance checks.

For foamed-in-place insulation, the Supplier shall utilize control sheets for each pipe spool in order to assess the step-by-step inspection status on a continuous basis.

8.2 INSPECTION OF MATERIALS

Inspection of materials shall be made either at the supplier's work or in the field.

Laboratory test results for various insulating materials carried out by independent authorities must be available prior to commencement of the work together with the "Site Instruction Manual" complied by Supplier.

A "Compliance Certificate" shall be prepared in case of various production batches to certify that the materials are in accordance with the technical specifications of the first production batch.

8.3 PRE-INSULATION INSPECTION

Equipment and piping surfaces shall be clean and dry and coated prior to insulation installation.

Equipment and piping shall be hydrostatically tested prior to insulation installation.

Hangers and supports must be properly located and free from obstruction to allow sufficient space for support insulation application and normal expansion/contraction of the system.

Positions for installation of the expansion/contraction joints in the insulation shall be clearly defined and marked.

Heat tracing, electrical and all wiring tubing for instrumentation shall be installed, tested and accepted before insulation installation.

8.4 FINAL INSPECTION & RELEASE

Final inspection shall be conducted by all the parties involved at the point when installation is considered complete or as soon as possible thereafter. An agreed warranty or official release shall be determined at that time.

The Supplier shall inspect the installation to determine that the insulation is of the proper thickness, that its materials, workmanship and finishes meet the specifications and that all the Sub-Supplier obligations have been met. Inspection checklists shall be completed and signed to record the inspection results.

Infrared inspection techniques may be used after start-up to verify the proper performance of the insulation system.

8.5 TESTING

For each batch of insulation supplied to the job site, the following tests have to be carried out a minimum of three times during the usage of this batch:

- Density as per ASTM D1622.
- Closed cells content as per ASTM D6226.
- Thermal conductivity as per ASTM C516 at 24°C mean temperature
- Compressive strength as per ASTM D1621
- Flammability in accordance with ASTM D3014

Results of the tests shall be recorded for two samples of foam from each batch of chemicals. Batches shall be identified by serial number and date.

9.0 APPLICATION DETAILS

Typical sketch are attached for following details

9.1 Pipe Insulation – Class H

9.2 Pipe Insulation – Class C

9.3 Termination of insulation on pipe work detail

9.4 Typical detail for un-insulated flanged valve class H insulation

9.5 Valve insulation class C

9.6 Vertical support for class cc insulation

9.7 Expansion/contraction joints

9.8 Insulated support for vertical insulated pipe

9.9 Typical detail of flanged joint class H

9.10 Vessel support, flashing & weatherproofing details – class H Insulation

9.11 Insulation support ring detail class H Insulations

9.12 Vessel support, flashing & weatherproofing details – class C Insulation

9.13 Typical Vessel nozzle flanged detail for class C insulation

9.14 Valve box cover construction

9.1 PIPE INSULATION – CLASS H

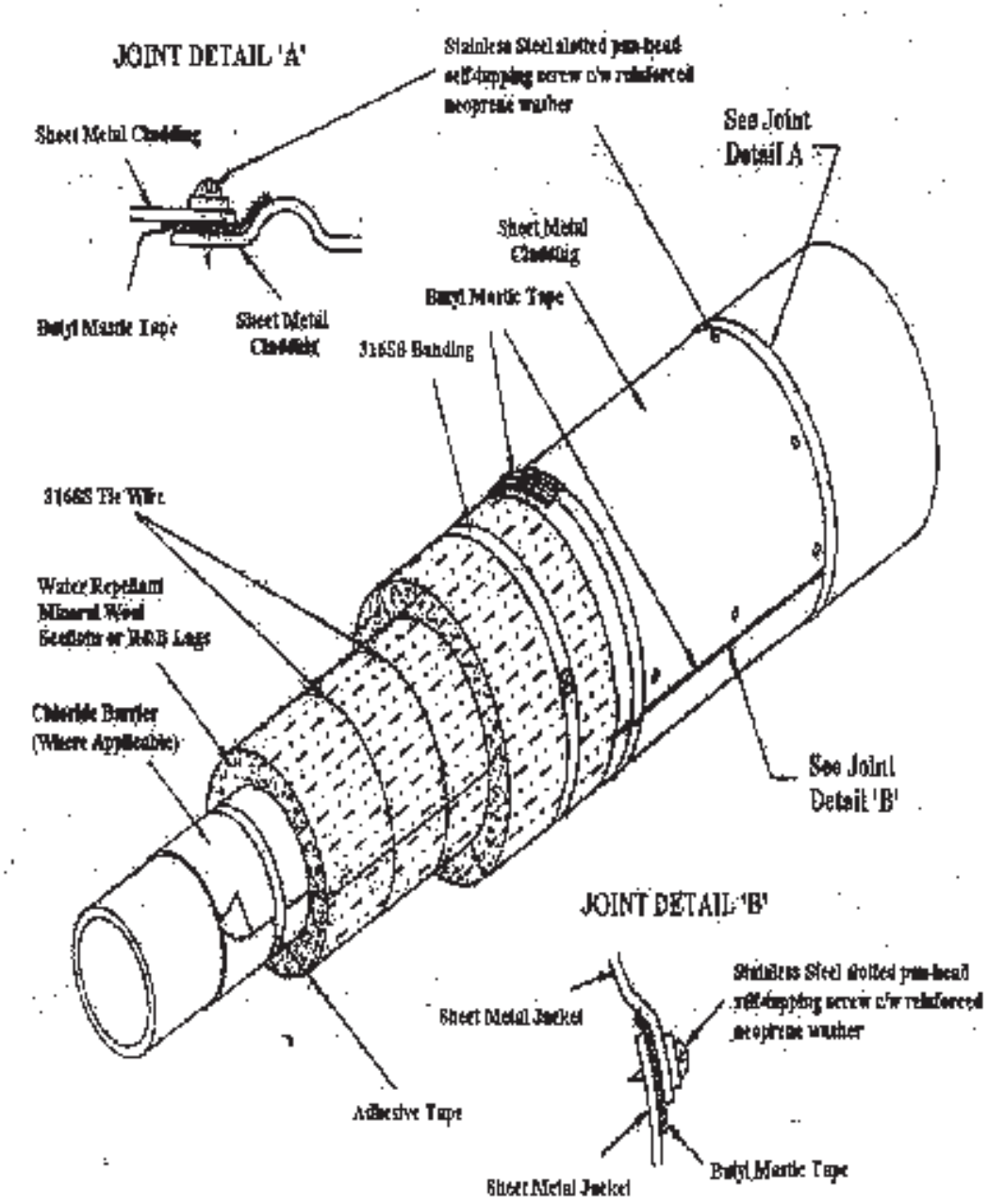


FIGURE 1 – Typical method of installing insulation for Class H

9.2 PIPE INSULATION – CLASS C:

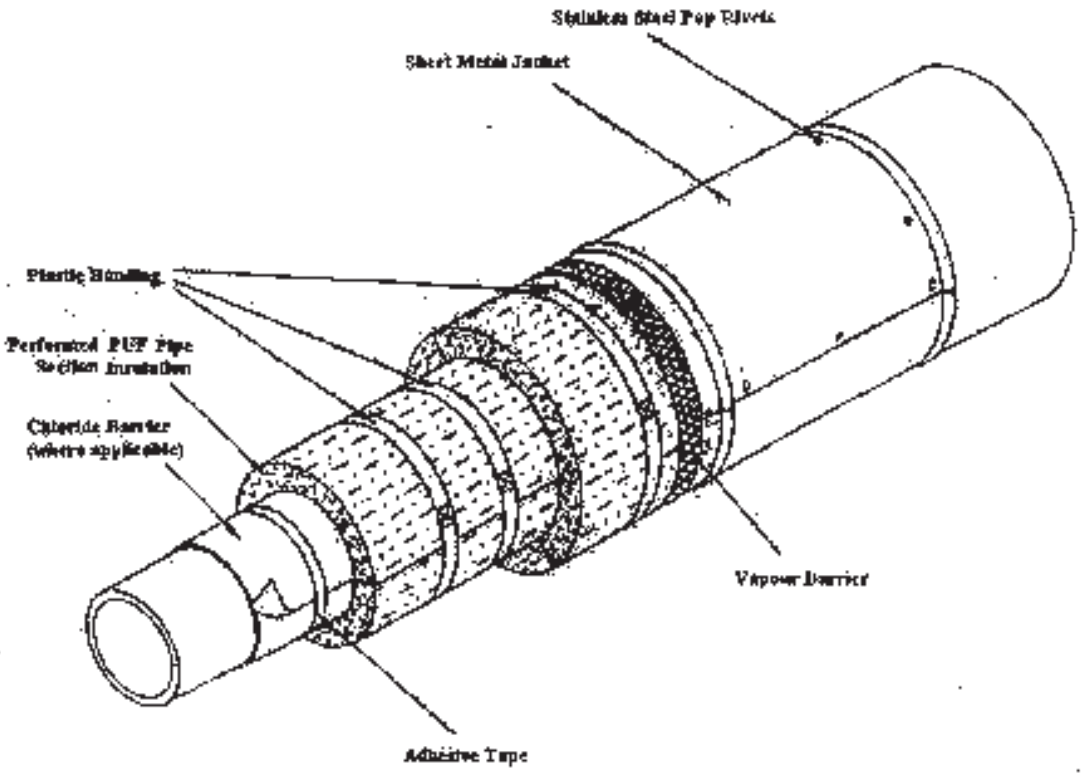


FIGURE 2 – Typical method of installing insulation for Class C

9.3 TERMINATION OF INSULATION ON PIPEWORK

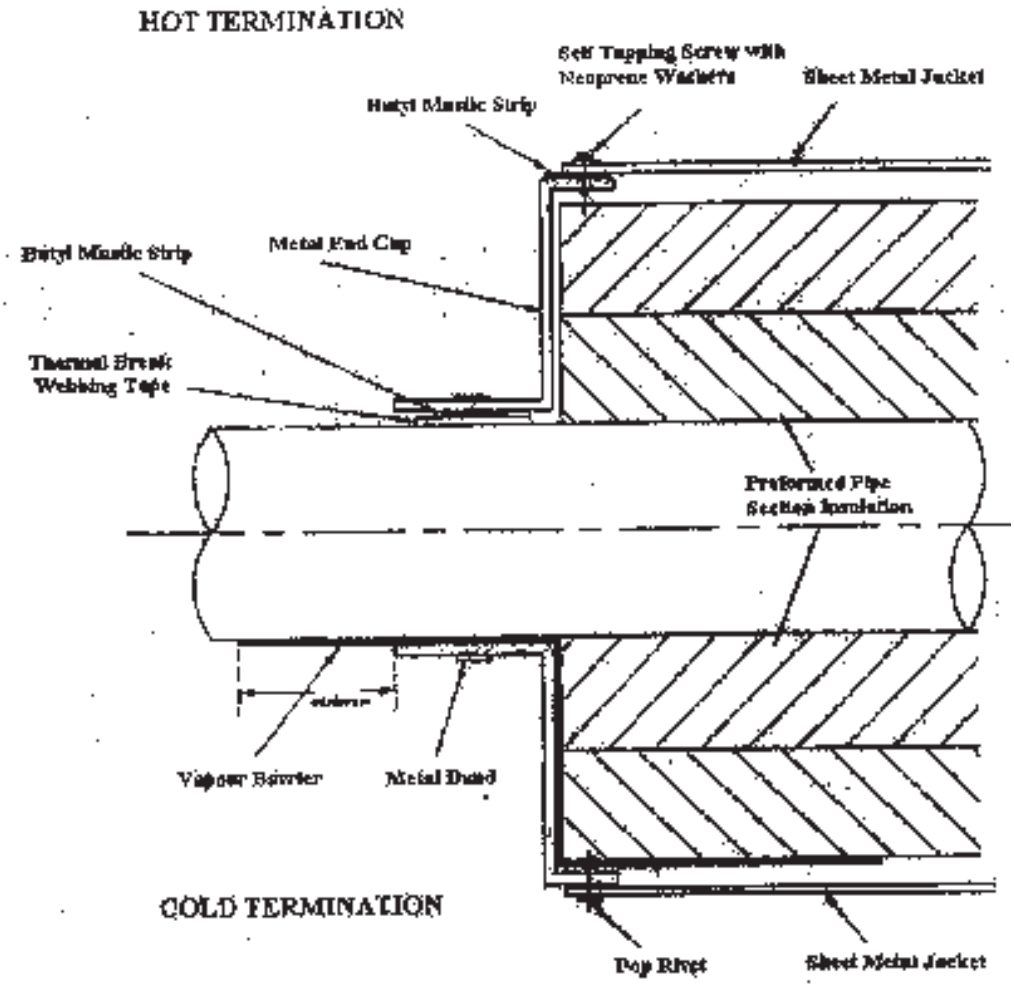


FIGURE 3 – Termination of insulation on pipework detail

9.4 VALVE INSULATION CLASS H:

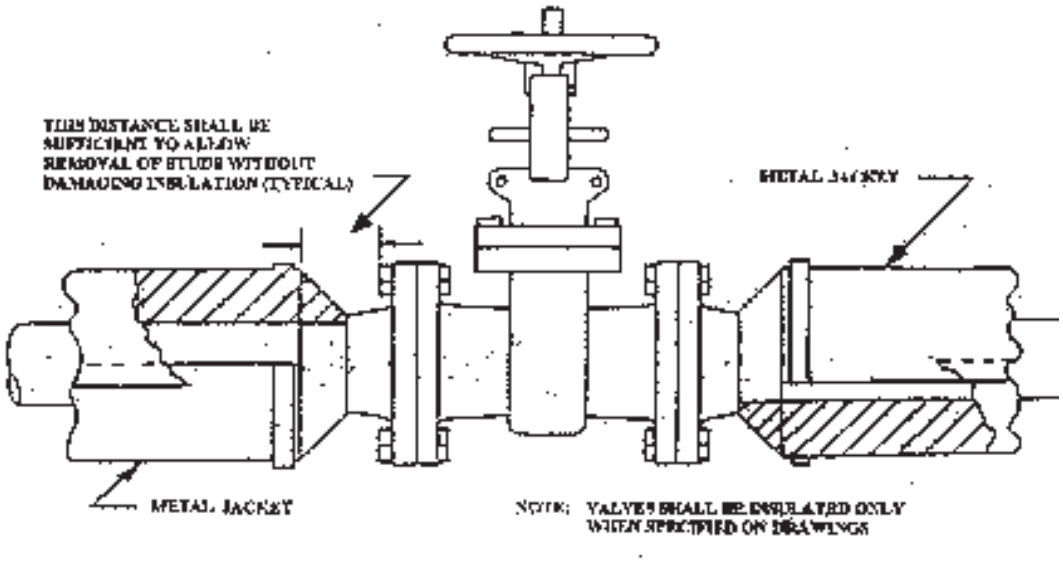


FIGURE 4 – Typical detail for uninsulated flanged valve Class H insulation

9.5 VALVE INSULATION CLASS C:

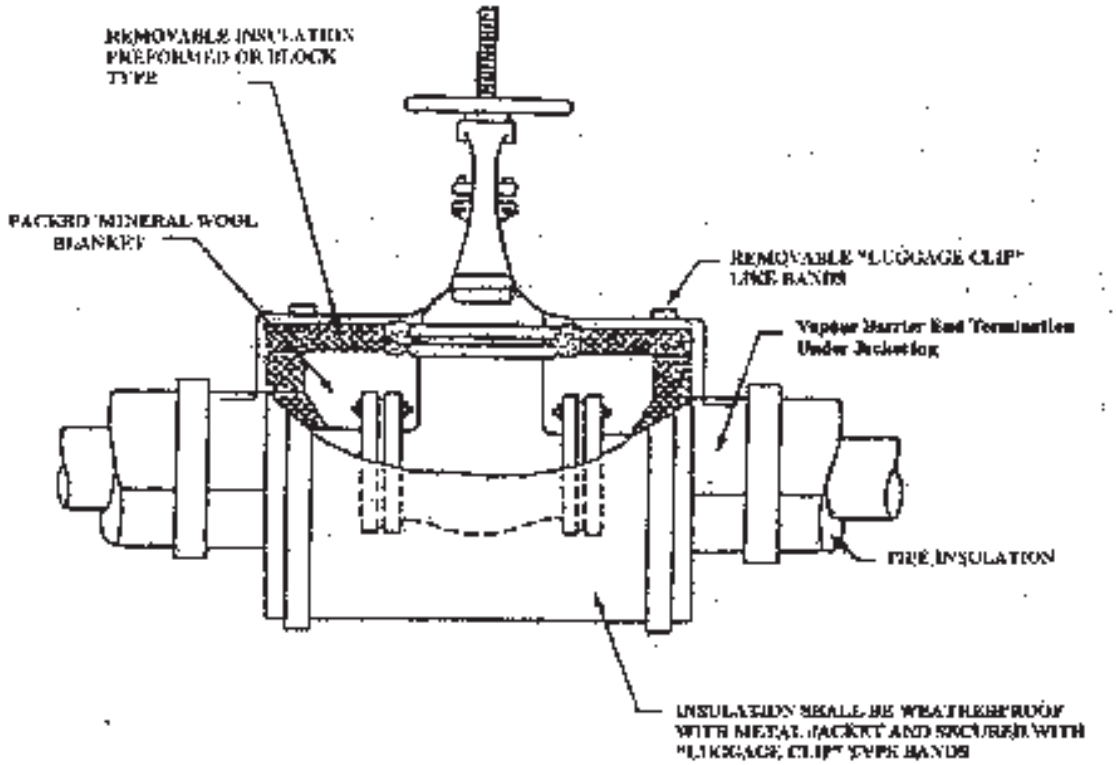


FIGURE 5 - Typical detail of insulated flanged valve for Class C insulation

9.6 VERTICAL SUPPORT FOR CLASS C INSULATION:

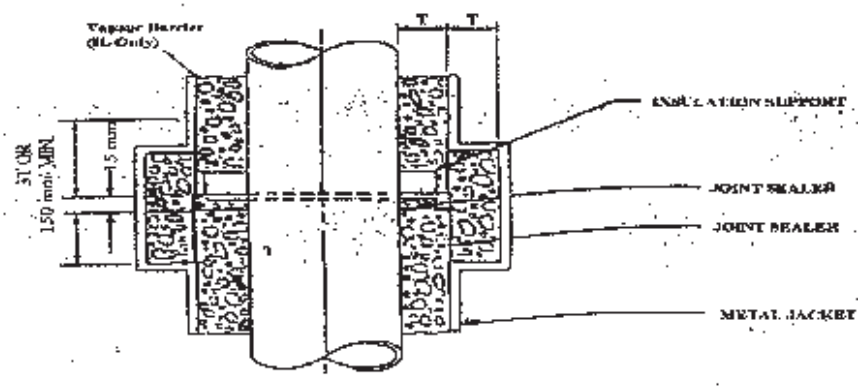


FIGURE 6 – Vertical support for Class C insulation

9.7 EXPANSION/CONTRACTION JOINTS

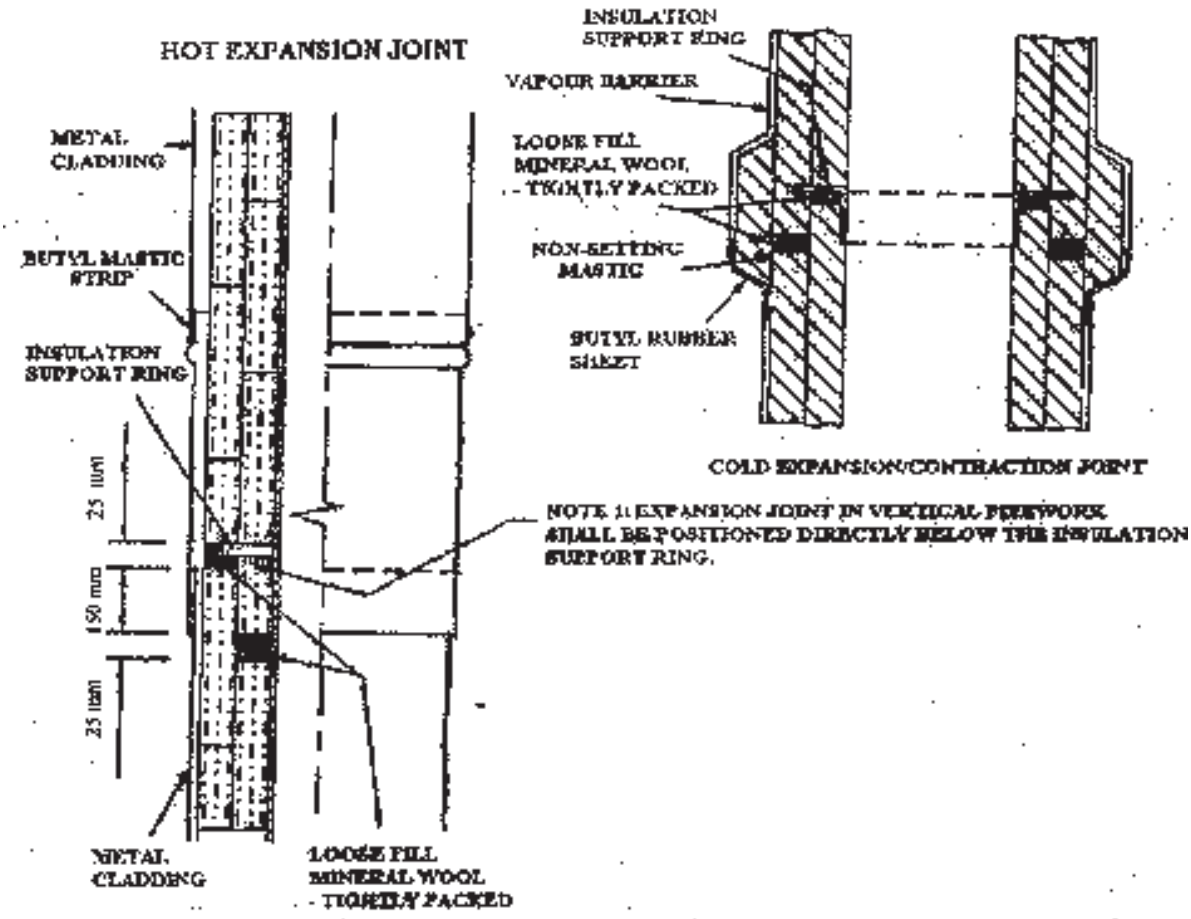


FIGURE 7 – Expansion/Contraction joints

9.8 INSULATED SUPPORTS FOR VERTICAL INSULATED PIPE:

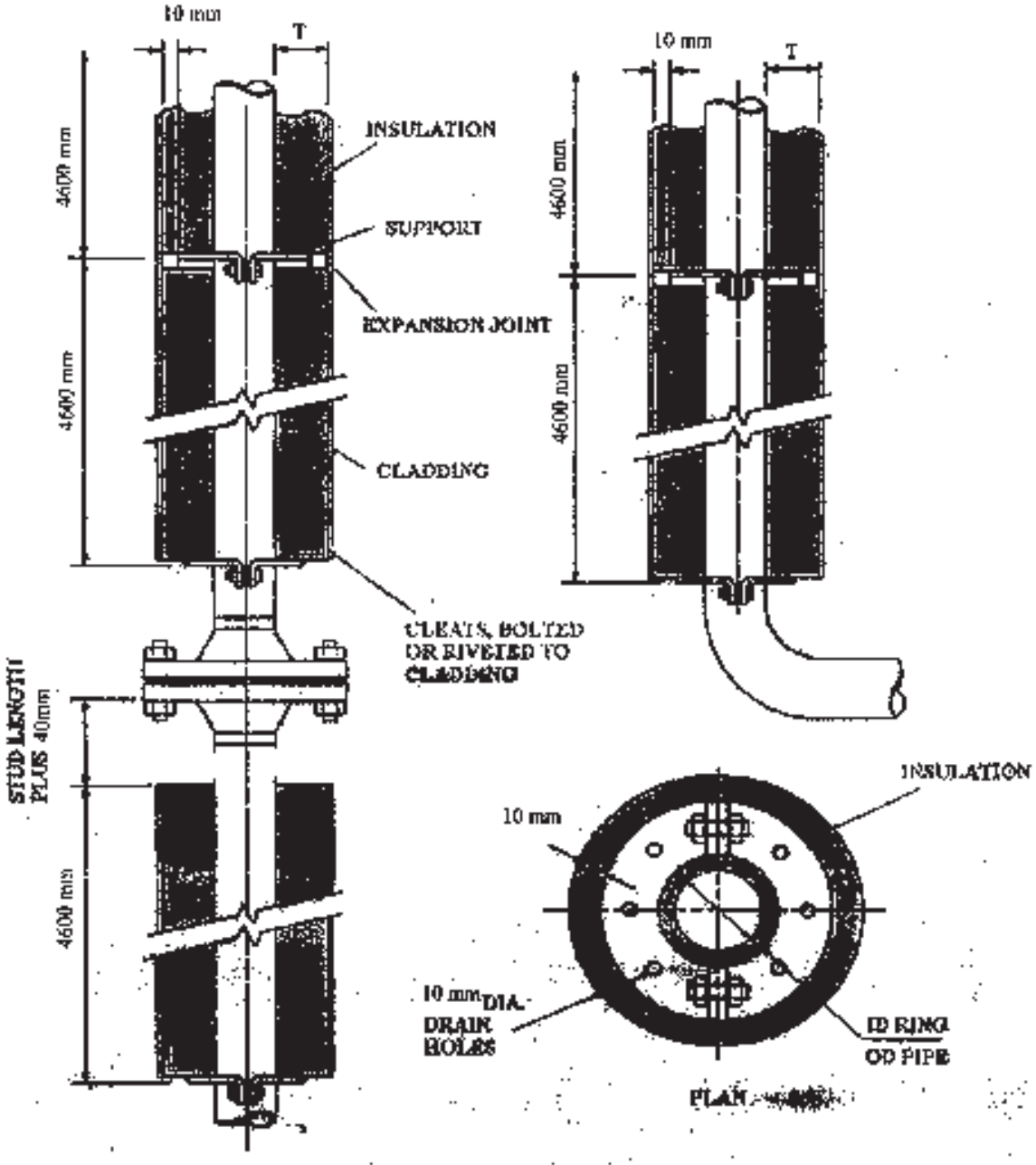
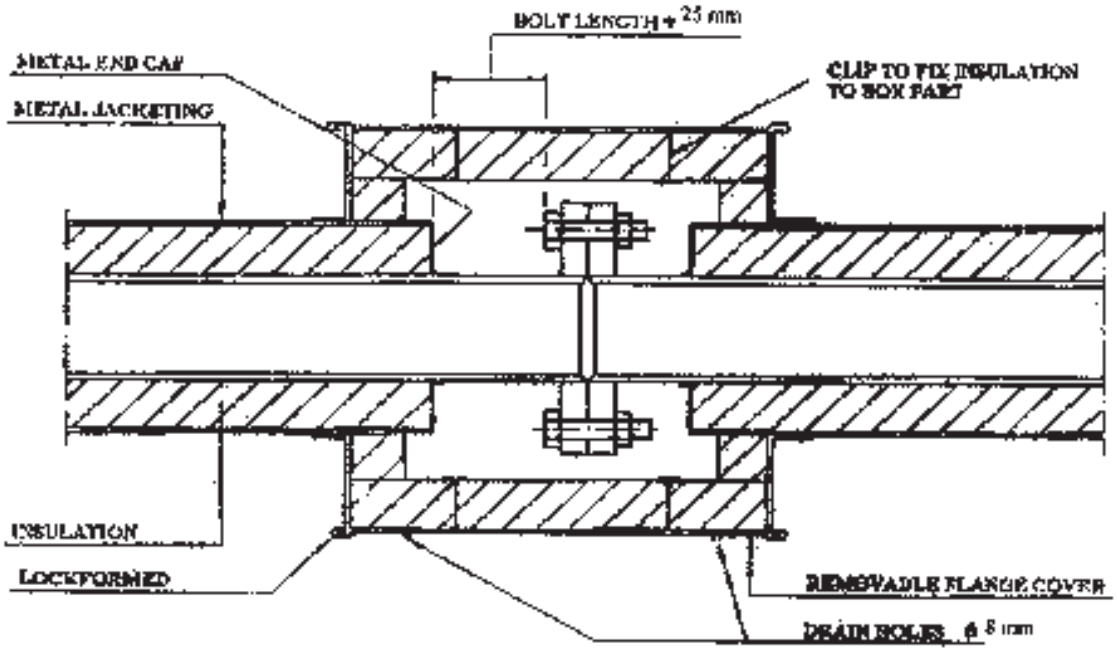


FIGURE 8 – Insulated supports for vertical insulated pipe

9.9 TYPICAL DETAIL FOR FLANGED JOINT CLASS H:



SECTION A-A

FIGURE 9 – Typical detail for flanged joint Class H

9.10 VESSEL SUPPORT, FLASHING & WEATHERPROOFING DETAILS – CLASS H INSULATION

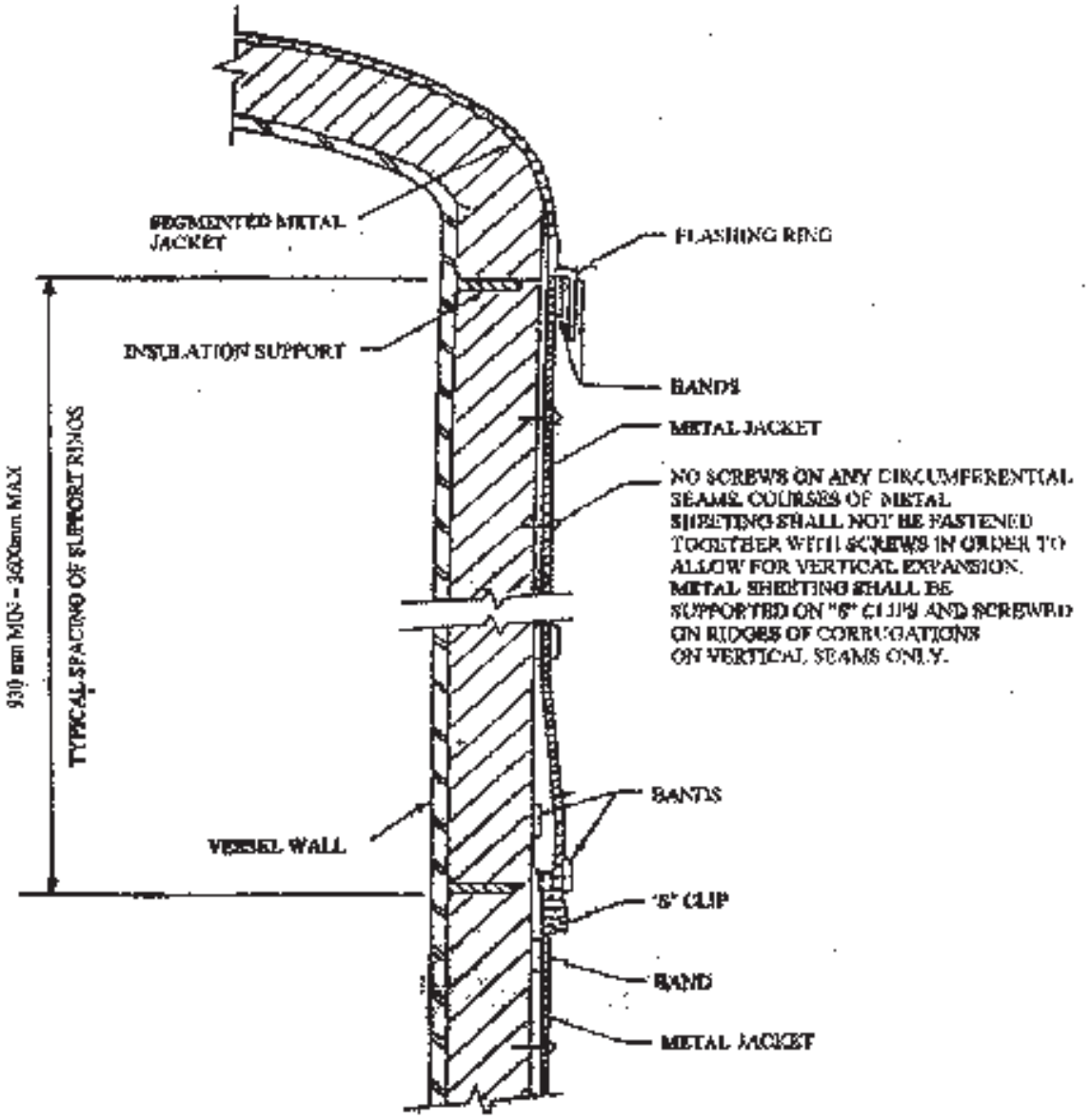


FIGURE 10 – Vessel support, flashing and weatherproofing details – Class H insulation

9.11 INSULATION SUPPORT RING DETAIL CLASS H INSULATION

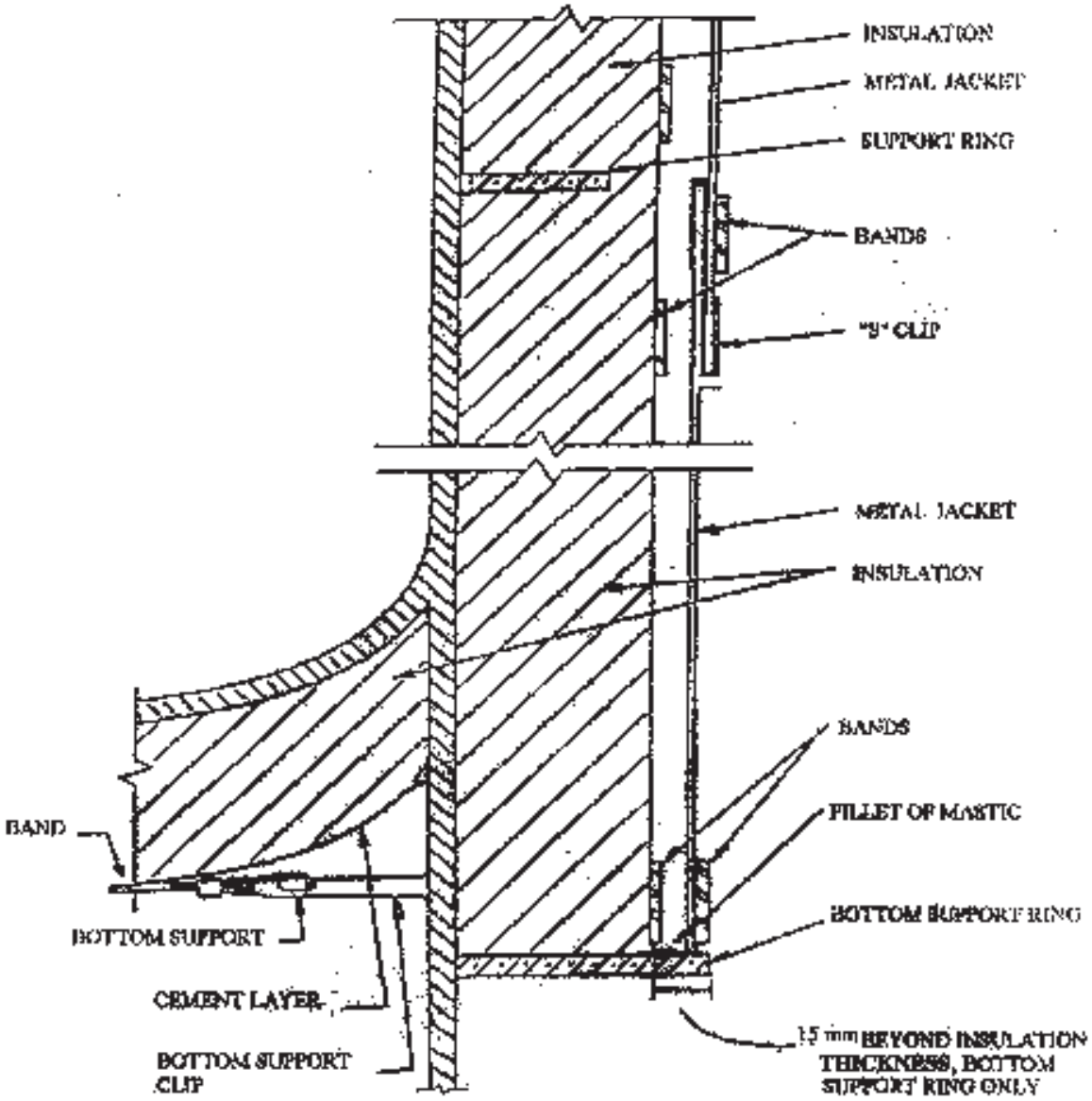
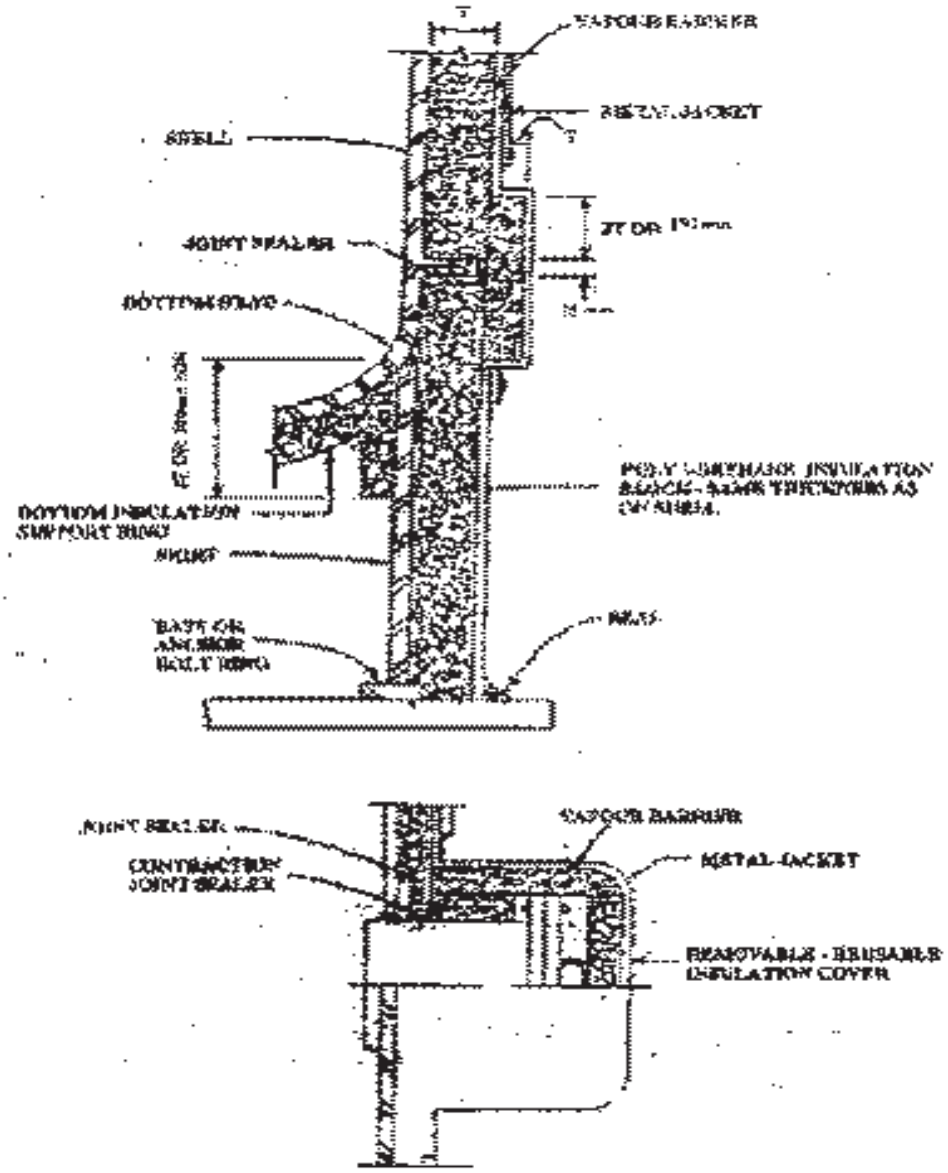


FIGURE 11- Insulation support ring detail Class H insulation



9.12 Vessel support, flashing & weatherproofing details Class C insulation (FIGURE 12)

9.13 TYPICAL VESSEL NOZZLE FLANGE DETAIL FOR CLASS C INSULATION

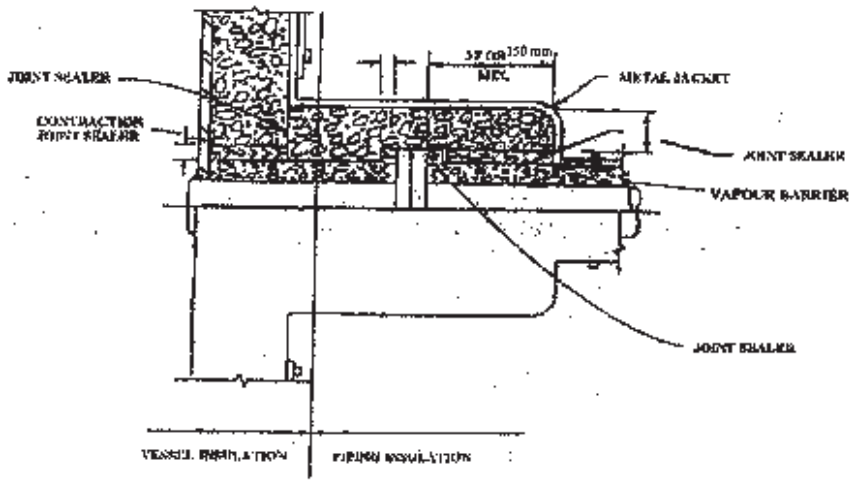


FIGURE 13 – Typical vessel nozzle flange detail for Class C insulation

9.14 VALVE BOX COVER CONSTRUCTION:

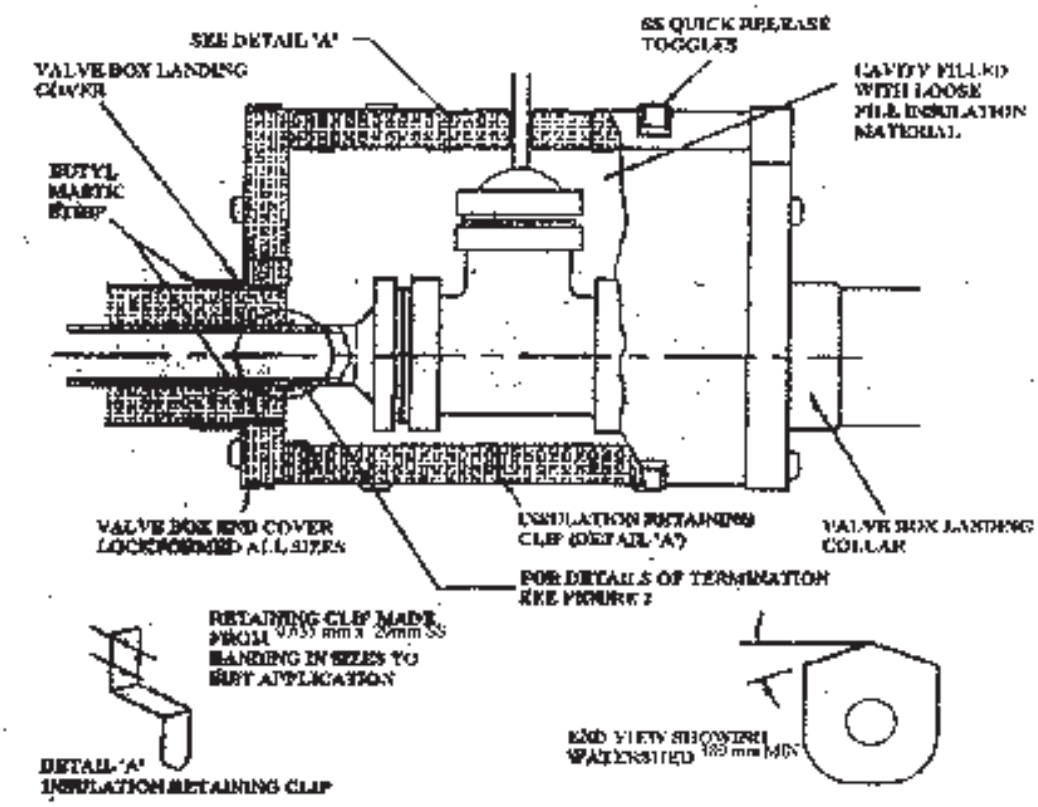
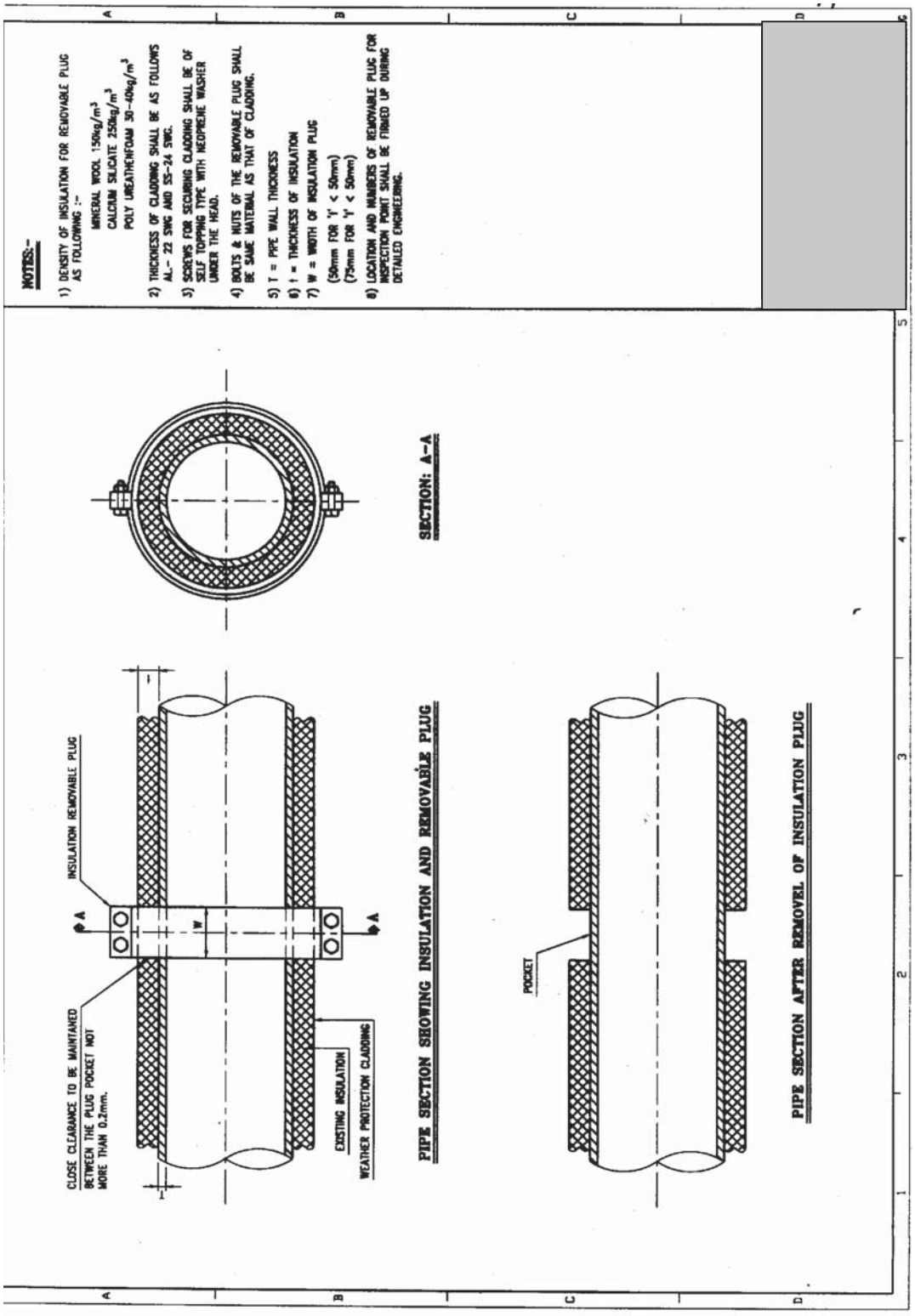
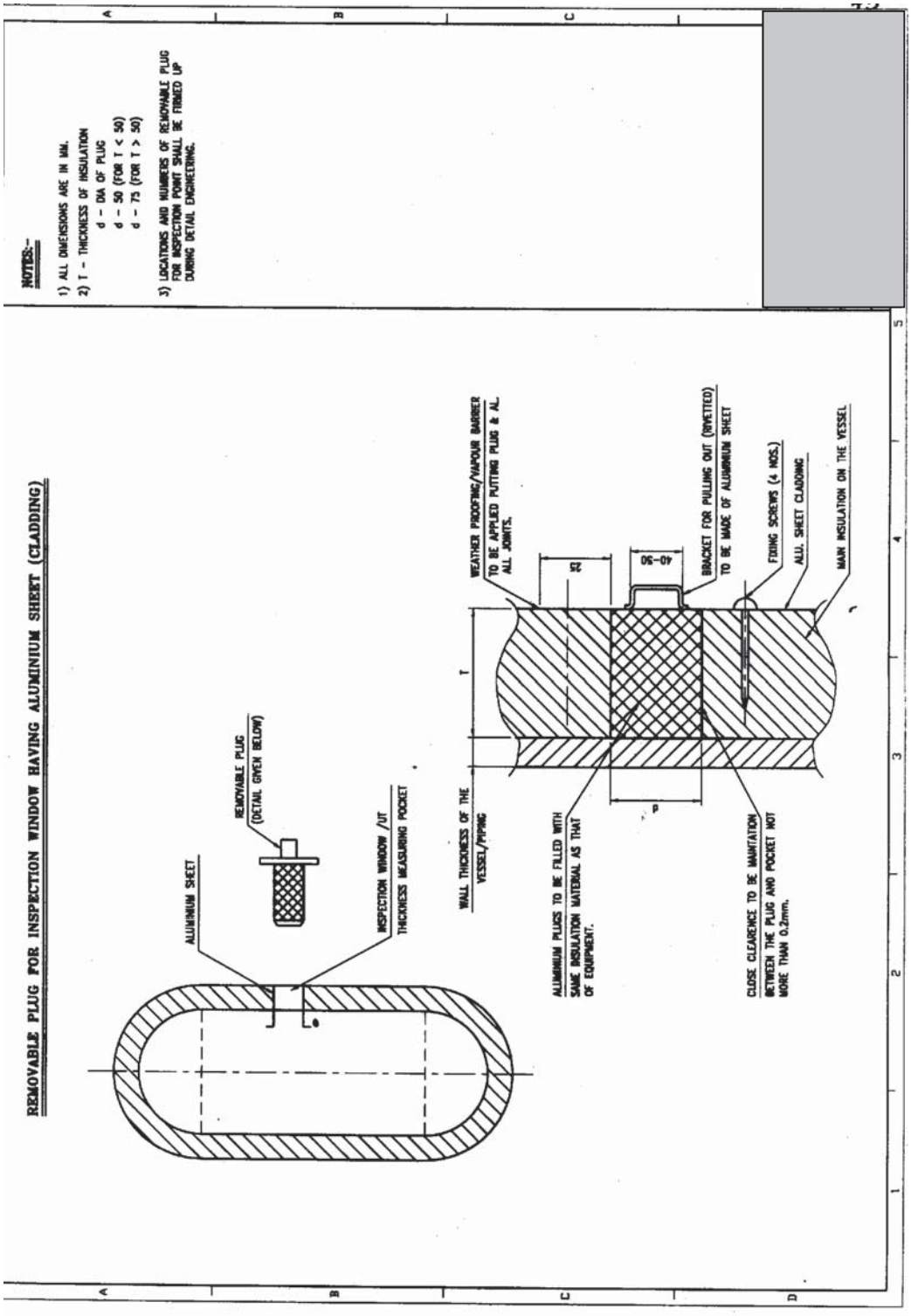


FIGURE 14 – Valve box cover construction


NOTES:-
1) DENSITY OF INSULATION FOR REMOVABLE PLUG AS FOLLOWING :-
MINERAL WOOL 150kg/m³
CALCIUM SILICATE 250kg/m³
POLY URETHENFOAM 30-40kg/m³
2) THICKNESS OF CLADDING SHALL BE AS FOLLOWS AL- 22 SWG AND SS-24 SWG.
3) SCREWS FOR SECURING CLADDING SHALL BE OF SELF TOPPING TYPE WITH NEOPRENE WASHER UNDER THE HEAD.
4) BOLTS & NUTS OF THE REMOVABLE PLUG SHALL BE SAME MATERIAL AS THAT OF CLADDING.
5) T = PIPE WALL THICKNESS
6) t = THICKNESS OF INSULATION
7) W = WIDTH OF INSULATION PLUG
(50mm FOR 't' < 50mm)
(75mm FOR 't' < 50mm)
8) LOCATION AND NUMBERS OF REMOVABLE PLUG FOR INSPECTION POINT SHALL BE FIXED UP DURING DETAILED ENGINEERING.
SECTION: A-A
INSULATION REMOVABLE PLUG
CLOSE CLEARANCE TO BE MAINTAINED BETWEEN THE PLUG POCKET NOT MORE THAN 0.2mm.
EXISTING INSULATION
WEATHER PROTECTION CLADDING
PIPE SECTION SHOWING INSULATION AND REMOVABLE PLUG
POCKET
PIPE SECTION AFTER REMOVEL OF INSULATION PLUG


NOTES:-
1) ALL DIMENSIONS ARE IN MM.
2) T - THICKNESS OF INSULATION
d - DIA OF PLUG
d - 50 (FOR T < 50)
d - 75 (FOR T > 50)
3) LOCATIONS AND NUMBERS OF REMOVABLE PLUG FOR INSPECTION POINT SHALL BE FIRMED UP DURING DETAIL ENGINEERING.
REMOVABLE PLUG FOR INSPECTION WINDOW HAVING ALUMINIUM SHEET (CLADDING)
ALUMINIUM SHEET
REMOVABLE PLUG (DETAIL GIVEN BELOW)
INSPECTION WINDOW /CUT THICKNESS MEASURING POCKET
WALL THICKNESS OF THE VESSEL/PIPING
ALUMINIUM PLUGS TO BE FILLED WITH SAME INSULATION MATERIAL AS THAT OF EQUIPMENT.
CLOSE CLEARENCE TO BE MAINTAINED BETWEEN THE PLUG AND POCKET NOT MORE THAN 0.2mm.
WEATHER PROOFING/VAPOUR BARRIER TO BE APPLIED PUTTING PLUG & AL. ALL JOINTS.
BRACKET FOR PULLING OUT (RIVETTED) TO BE MADE OF ALUMINIUM SHEET
FIXING SCREWS (4 NOS.)
ALU. SHEET CLADDING
MAIN INSULATION ON THE VESSEL
T
d
40-50